

Third Edition (© 2005 McGraw-Hill)

Chapter 1

Note: The first printing has a few odd typos, which are indicated in blue below. These will be corrected in the reprint.

1.1 Virial theorem The Li atom has a nucleus with a $+3e$ positive charge, which is surrounded by a full $1s$ shell with two electrons, and a single valence electron in the outer $2s$ subshell. The atomic radius of the Li atom is about 0.17 nm . Using the Virial theorem, and assuming that the valence electron sees the nuclear $+3e$ shielded by the two $1s$ electrons, that is, a net charge of $+e$, estimate the ionization energy of Li (the energy required to free the $2s$ electron). Compare this value with the experimental value of 5.39 eV . Suppose that the actual nuclear charge seen by the valence electron is not $+e$ but a little higher, say $+1.25e$, due to the imperfect shielding provided by the closed $1s$ shell. What would be the new ionization energy? What is your conclusion?

Solution

First we consider the case when the outermost valence electron can see a net charge of $+e$. From Coulomb's law we have the potential energy

$$\begin{aligned} PE &= \frac{Q_1 Q_2}{4\pi\epsilon_0 r_0} = \frac{(+e)(-e)}{4\pi\epsilon_0 r_0} \\ &= -\frac{(1.6 \times 10^{-19}\text{ C})^2}{4\pi(8.85 \times 10^{-12}\text{ Fm}^{-1})(0.17 \times 10^{-9}\text{ m})} = 1.354 \times 10^{-18}\text{ J or } -8.46\text{ eV} \end{aligned}$$

Virial theorem relates the overall energy, the average kinetic energy $\overline{KE}$, and average potential energy $\overline{PE}$ through the relations

$$\overline{E} = \overline{PE} + \overline{KE} \quad \text{and} \quad \overline{KE} = -\frac{1}{2}\overline{PE}$$

Thus using Virial theorem, the total energy is

$$\overline{E} = \frac{1}{2}\overline{PE} = 0.5 \times -8.46\text{ eV} = -4.23\text{ eV}$$

The ionization energy is therefore **4.23 eV**. Now we consider the second case where electron sees $+1.25e$ due to imperfect shielding. Again the Coulombic PE between $+e$ and $+1.25e$ will be

$$\begin{aligned} PE &= \frac{Q_1 Q_2}{4\pi\epsilon_0 r_0} = \frac{(+1.25e)(-e)}{4\pi\epsilon_0 r_0} \\ &= -\frac{1.25 \cdot (1.6 \times 10^{-19}\text{ C})^2}{4\pi(8.85 \times 10^{-12}\text{ Fm}^{-1})(0.17 \times 10^{-9}\text{ m})} = -1.692 \times 10^{-18}\text{ J or } -10.58\text{ eV} \end{aligned}$$

The total energy is,

$$\overline{E} = \frac{1}{2} \overline{PE} = 5.29 \text{ eV}$$

The ionization energy, considering imperfect shielding, is **5.29 eV**. This value is in closer agreement with the experimental value. Hence the second assumption seems to be more realistic.

1.2 Atomic mass and molar fractions

- a. Consider a multicomponent alloy containing N elements. If $w_1, w_2, \dots, w_N$ are the weight fractions of components 1, 2, ..., N in the alloy and $M_1, M_2, \dots, M_N$ are the respective atomic masses of the elements, show that the atomic fraction of the i -th component is given by

$$n_i = \frac{w_i / M_i}{\frac{w_1}{M_1} + \frac{w_2}{M_2} + \dots + \frac{w_N}{M_N}} \quad \text{Weight to atomic percentage}$$

- b. Suppose that a substance (compound or an alloy) is composed of N elements, $A, B, C, \dots$ and that we know their atomic (or molar) fractions $n_A, n_B, n_C, \dots$. Show that the weight fractions $w_A, w_B, w_C, \dots$ are given by

$$w_A = \frac{n_A M_A}{n_A M_A + n_B M_B + n_C M_C + \dots}$$

$$w_B = \frac{n_B M_B}{n_A M_A + n_B M_B + n_C M_C + \dots} \quad \text{Atomic to weight percentage}$$

- c. Consider the semiconducting II-VI compound cadmium selenide, CdSe. Given the atomic masses of Cd and Se, find the weight fractions of Cd and Se in the compound and grams of Cd and Se needed to make 100 grams of CdSe.
- d. A Se-Te-P glass alloy has the composition 77 wt.% Se, 20 wt.% Te and 3 wt.% P. Given their atomic masses, what are the atomic fractions of these constituents?

Solution

- a. Suppose that $n_1, n_2, n_3, \dots, n_i, \dots, n_N$ are the atomic fractions of the elements in the alloy,

$$n_1 + n_2 + n_3 + \dots + n_N = 1$$

Suppose that we have 1 mole of the alloy. Then it has n_i moles of an atom with atomic mass M_i (atomic fractions also represent molar fractions in the alloy). Suppose that we have 1 gram of the alloy. Since w_i is the weight fraction of the i -th atom, w_i is also the mass of i -th element in grams in the alloy. The number of moles in the alloy is then w_i/M_i . Thus,

$$\text{Number of moles of element } i = w_i/M_i$$

$$\text{Number of moles in the whole alloy} = w_1/M_1 + w_2/M_2 + \dots + w_i/M_i + \dots + w_N/M_N$$

Molar fraction or the atomic fraction of the i -th elements is therefore,

$$n_i = \frac{\text{Number of moles of element } i}{\text{Total number of moles in alloy}}$$

$$\therefore n_i = \frac{w_i / M_i}{\frac{w_1}{M_1} + \frac{w_2}{M_2} + \dots + \frac{w_N}{M_N}}$$

b. Suppose that we have the atomic fraction n_i of an element with atomic mass M_i . The mass of the element in the alloy will be the product of the atomic mass with the atomic fraction, *i.e.* $n_i M_i$. Mass of the alloy is therefore

$$n_A M_A + n_B M_B + \dots + n_N M_N = M_{\text{alloy}}$$

By definition, the weight fraction is, $w_i = \text{mass of the element } i / \text{Mass of alloy}$. Therefore,

$$w_A = \frac{n_A M_A}{n_A M_A + n_B M_B + n_C M_C + \dots}$$

$$w_B = \frac{n_B M_B}{n_A M_A + n_B M_B + n_C M_C + \dots}$$

c. The atomic mass of Cd and Se are $112.41 \text{ g mol}^{-1}$ and 78.96 g mol^{-1} . Since one atom of each element is in the compound CdSe, the atomic fraction, n_{Cd} and n_{Se} are 0.5. The weight fraction of Cd in CdSe is therefore

$$w_{\text{Cd}} = \frac{n_{\text{Cd}} M_{\text{Cd}}}{n_{\text{Cd}} M_{\text{Cd}} + n_{\text{Se}} M_{\text{Se}}} = \frac{0.5 \times 112.41 \text{ g mol}^{-1}}{0.5 \times 112.41 \text{ g mol}^{-1} + 0.5 \times 78.96 \text{ g mol}^{-1}} = \mathbf{0.587 \text{ or } 58.7\%}$$

Similarly weight fraction of Se is

$$w_{\text{Se}} = \frac{n_{\text{Se}} M_{\text{Se}}}{n_{\text{Cd}} M_{\text{Cd}} + n_{\text{Se}} M_{\text{Se}}} = \frac{0.5 \times 78.96 \text{ g mol}^{-1}}{0.5 \times 112.41 \text{ g mol}^{-1} + 0.5 \times 78.96 \text{ g mol}^{-1}} = \mathbf{0.4126 \text{ or } 41.3\%}$$

Consider 100 g of CdSe. Then the mass of Cd we need is

$$\text{Mass of Cd} = w_{\text{Cd}} M_{\text{compound}} = 0.587 \times 100 \text{ g} = \mathbf{58.7 \text{ g (Cd)}}$$

and

$$\text{Mass of Se} = w_{\text{Se}} M_{\text{compound}} = 0.413 \times 100 \text{ g} = \mathbf{41.3 \text{ g (Se)}}$$

d. The atomic fractions of the constituents can be calculated using the relations proved above. The atomic masses of the components are $M_{\text{Se}} = 78.6 \text{ g mol}^{-1}$, $M_{\text{Te}} = 127.6 \text{ g mol}^{-1}$, and $M_{\text{P}} = 30.974 \text{ g mol}^{-1}$.

Applying the weight to atomic fraction conversion equation derived in part (a) we find,

$$n_{\text{Se}} = \frac{w_{\text{Se}} / M_{\text{Se}}}{\frac{w_{\text{Se}}}{M_{\text{Se}}} + \frac{w_{\text{Te}}}{M_{\text{Te}}} + \frac{w_{\text{P}}}{M_{\text{P}}}} = \frac{\frac{0.77}{78.6 \text{ g mol}^{-1}}}{\frac{0.77}{78.6 \text{ g mol}^{-1}} + \frac{0.2}{127.6 \text{ g mol}^{-1}} + \frac{0.03}{30.974 \text{ g mol}^{-1}}}$$

$$\therefore n_{\text{Se}} = \mathbf{0.794 \text{ or } 79.4\%}$$

$$n_{\text{Te}} = \frac{w_{\text{Te}} / M_{\text{Te}}}{\frac{w_{\text{Se}}}{M_{\text{Se}}} + \frac{w_{\text{Te}}}{M_{\text{Te}}} + \frac{w_{\text{P}}}{M_{\text{P}}}} = \frac{\frac{0.2}{127.6 \text{ g mol}^{-1}}}{\frac{0.77}{78.6 \text{ g mol}^{-1}} + \frac{0.2}{127.6 \text{ g mol}^{-1}} + \frac{0.03}{30.974 \text{ g mol}^{-1}}}$$

$$\therefore n_{\text{Te}} = \mathbf{0.127 \text{ or } 12.7 \%}$$

$$n_p = \frac{w_p / M_p}{\frac{w_{Se}}{M_{Se}} + \frac{w_{Te}}{M_{Te}} + \frac{w_p}{M_p}} = \frac{\frac{0.03}{30.974 \text{ g mol}^{-1}}}{\frac{0.77}{78.6 \text{ g mol}^{-1}} + \frac{0.2}{127.6 \text{ g mol}^{-1}} + \frac{0.03}{30.974 \text{ g mol}^{-1}}}$$

$$\therefore n_p = 0.0785 \text{ or } 7.9\%$$

1.3 The covalent bond Consider the H_2 molecule in a simple way as two touching H atoms as depicted in Figure 1.73. Does this arrangement have a lower energy than two separated H atoms? Suppose that electrons totally correlate their motions so that they move to avoid each other as in the snapshot in Figure 1.73. The radius r_o of the hydrogen atom is 0.0529 nm. The electrostatic potential energy PE of two charges Q_1 and Q_2 separated by a distance r is given by $Q_1 Q_2 / (4\pi\epsilon_0 r)$. Using the Virial Theorem as in Example 1.1, consider the following:

- Calculate the total electrostatic potential energy (PE) of all the charges when they are arranged as shown in Figure 1.73. In evaluating the PE of the whole collection of charges you must consider all pairs of charges and, at the same time, avoid double counting of interactions between the same pair of charges. The total PE is the sum of the following: electron 1 interacting with the proton at a distance r_o on the left, proton at r_o on the right, and electron 2 at a distance $2r_o$ + electron 2 interacting with a proton at r_o and another proton at $3r_o$ + two protons, separated by $2r_o$, interacting with each other. Is this configuration energetically favorable?
- Given that in the isolated H-atom the PE is $2 \times (-13.6 \text{ eV})$, calculate the change in PE in going from two isolated H-atoms to the H_2 molecule. Using the Virial theorem, find the change in the total energy and hence the covalent bond energy. How does this compare with the experimental value of 4.51 eV?

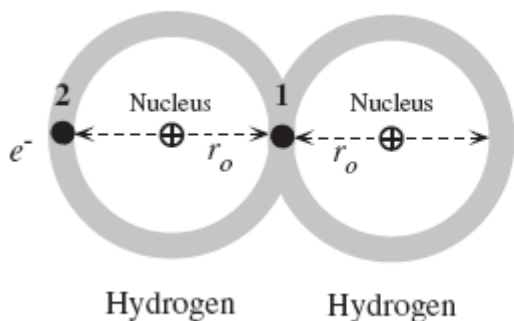


Figure 1.73 A simplified view of the covalent bond in H_2 . A snapshot at one instant.

Solution

a. Consider the PE of the whole arrangement of charges shown in the figure. In evaluating the PE of all the charges, we must avoid double counting of interactions between the same pair of charges. The total PE is the sum of the following:

- Electron 1 interacting with the proton at a distance r_o on the left, with the proton at r_o on the right and with electron 2 at a distance $2r_o$
- + Electron 2 on the far left interacting with a proton at r_o and another proton at $3r_o$
- + Two protons, separated by $2r_o$, interacting with each other

$$PE = -\frac{e^2}{4\pi\epsilon_o r_o} - \frac{e^2}{4\pi\epsilon_o r_o} + \frac{e^2}{4\pi\epsilon_o (2r_o)} - \frac{e^2}{4\pi\epsilon_o r_o} - \frac{e^2}{4\pi\epsilon_o 3r_o} + \frac{e^2}{4\pi\epsilon_o 2r_o}$$

Substituting and calculating, we find $PE = -1.0176 \times 10^{-17}$ J or **-63.52 eV**

The negative PE for this particular arrangement indicates that this arrangement of charges is indeed energetically favorable compared with all the charges infinitely separated (PE is then zero).

b. The potential energy of an isolated H-atom is -2×13.6 eV or -27.2 eV. The difference between the PE of the H_2 molecule and two isolated H-atoms is,

$$\Delta PE = - (63.52) \text{ eV} - 2(-27.2) \text{ eV} = 9.12 \text{ eV}$$

We can write the last expression above as the change in the total energy.

$$\Delta \bar{E} = \frac{1}{2} \Delta PE = \frac{1}{2} (-9.12 \text{ eV}) = -4.56 \text{ eV}$$

This change in the total energy is negative. The H_2 molecule has lower energy than two H-atoms by 4.56 eV which is the bonding energy. This is very close to the experimental value of 4.51 eV. (Note: We used a r_o value from quantum mechanics - so the calculation was not totally classical).

1.4 Ionic bonding and CsCl The potential energy E per $Cs^+ - Cl^-$ pair within the CsCl crystal depends on the interionic separation r in the same fashion as in the NaCl crystal,

$$E(r) = -\frac{e^2 M}{4\pi\epsilon_o r} + \frac{B}{r^m} \quad \text{Energy per ion pair in ionic crystals [1.38]}$$

where for CsCl, $M = 1.763$, $B = 1.192 \times 10^{-104}$ J m⁹ or 7.442×10^{-5} eV (nm)⁹ and $m = 9$. Find the equilibrium separation (r_o) of the ions in the crystal and the ionic bonding energy, that is, the ionic cohesive energy; and compare the latter value to the experimental value of 657 kJ mol⁻¹. Given the *ionization energy* of Cs is 3.89 eV and the *electron affinity* of Cl (energy released when an electron is added) is 3.61 eV, calculate the atomic cohesive energy of the CsCl crystal as joules per mole.

Solution

Bonding will occur when potential energy $E(r)$ is minimum at $r = r_o$ corresponding to the equilibrium separation between Cs^+ and Cl^- ions. Thus, differentiating $E(r)$ and setting it equal to zero at $r = r_o$ we have

$$\left[\frac{dE(r)}{dr} \right]_{r=r_o} = \frac{d}{dr} \left[-\frac{e^2 M}{4\pi\epsilon_o r} + \frac{B}{r^m} \right]_{r=r_o} = 0$$

$$\therefore \left[\frac{e^2 M}{4\pi\epsilon_o r^2} - m \frac{B}{r^{m+1}} \right]_{r=r_o} = 0$$

$$\therefore \frac{e^2 M}{4\pi\epsilon_o r_o^2} - m \frac{B}{r_o^{m+1}} = 0$$

$$\therefore r_o = \left[\frac{4\pi\epsilon_o m B}{e^2 M} \right]^{\frac{1}{m-1}}$$

Thus substituting the appropriate values we have

$$r_o = \left[\frac{4\pi(8.8542 \times 10^{-12} \text{ Fm}^{-1}) \times 9 \times (1.192 \times 10^{-104} \text{ J m}^9)}{1.763 \times (1.6 \times 10^{-19} \text{ C})^2} \right]^{\frac{1}{8}}$$

$$r_o = 3.57 \times 10^{-10} \text{ m or } 0.357 \text{ nm.}$$

The minimum energy is the energy at $r = r_o$, that is

$$E_{\min} = -\frac{e^2 M}{4\pi\epsilon_o r_o} + \frac{B}{r_o^m}$$

which in terms of eV is

$$\begin{aligned} E_{\min} (\text{eV}) &= -\frac{eM}{4\pi\epsilon_o r_o} + \frac{B(\text{eV nm}^9)}{r_o(\text{nm})^9} \\ &= -\frac{(1.6 \times 10^{-19} \text{ C})^2 (1.763)}{4\pi(8.8542 \times 10^{-12} \text{ Fm}^{-1})(3.57 \times 10^{-10} \text{ m})} + \frac{7.442 \times 10^{-4} \text{ eV nm}^9}{(0.357 \text{ nm})^9} \\ &= -6.32 \text{ eV per ion pair, or } 3.16 \text{ eV per ion.} \end{aligned}$$

The amount of energy required to break up $\text{Cs}^+ \text{Cl}^-$ pair into Cs^+ and Cl^- ions = 6.32 eV per pair of ions.

The corresponding *ionic cohesive energy* is

$$\begin{aligned} E_{\text{cohesive}} &= (6.32 \text{ eV})(1.6 \times 10^{-19} \text{ J eV}^{-1})(6.022 \times 10^{23} \text{ mol}^{-1}) \\ &= \mathbf{610 \text{ kJ mol}^{-1} \text{ of Cs}^+ \text{Cl}^- \text{ ion pairs or } 610 \text{ kJ mol}^{-1} \text{ of Cs}^+ \text{ ions and Cl}^- \text{ ions.}} \end{aligned}$$

(Not far out from the experimental value given the large numbers and the high index, $m = 9$, involved in the calculations.)

The amount of energy required to remove an electron from Cl^- ion = 3.61 eV.

The amount of energy released when an electron is put into the Cs^+ ion = 3.89 eV.

Bond Energy per pair of Cs-Cl atoms = 6.32 eV + 3.61 eV – 3.89 eV = **6.04 eV**

Atomic cohesive energy in kJ/mol is,

$$\begin{aligned} E_{\text{cohesive}} &= (6.04 \text{ eV})(1.6 \times 10^{-19} \text{ J eV}^{-1})(6.022 \times 10^{23} \text{ mol}^{-1}) \\ &= \mathbf{582 \text{ kJ mol}^{-1} \text{ of Cs or Cl atom (i.e. per mole of Cs-Cl atom pairs)}} \\ &= \mathbf{291 \text{ kJ mol}^{-1} \text{ of atoms}} \end{aligned}$$

Author's Note: There is a *selected topic* entitled "Bonding" in the Chapter 1 folder in the textbook's CD where the bonding energy is calculated more accurately by taking a more realistic energy curve. The above calculation is similar to that given in Alan Walton, *Three Phases of Matter* (2nd Edition), Oxford University Press, 1983 (pp. 258-259)

Author's Note to the Instructors: Various books and articles report different values for B and m , which obviously affect the calculated energy; r_o is less affected because it requires the $(m-1)^{\text{th}}$ root of mB . Richard Christman (*Introduction to Solid State Physics*, Wiley, 1988) in Table 5-1 gives, $m = 10.65$ and $B = 3.44 \times 10^{120}$, quite different than values here, which are closer to values in Alan Walton's book. The experimental value of 657 kJ mol^{-1} for the ionic cohesive energy (the ionic lattice energy) is from T. Moeller et al, *Chemistry with Inorganic Qualitative Analysis*, Second Edition, Academic Press, 1984) p. 413, Table 13.5.

Some authors use the term molecular cohesive energy to indicate that the crystal is taken apart to molecular units e.g. Cs^+Cl^- , which would correspond to the ionic cohesive energy here. Further, most chemists use "energy per mole" to imply energy per chemical unit, and hence the atomic cohesive energy per mole would usually refer to energy be per Cs and Cl atom pairs. Some authors refer to the atomic cohesive energy per mole as cohesive energy per mole of atoms, independent of chemical formula.

1.5 Madelung constant If we were to examine the NaCl crystal in three dimensions, we would find that each Na^+ ion has

6 Cl^- ions as *nearest* neighbors at a distance r

12 Na^+ ions as *second* nearest neighbors at a distance $r\sqrt{2}$

8 Cl^- ions as *third* nearest neighbors at a distance $r\sqrt{3}$

and so on. Show that the electrostatic potential energy of the Na^+ atom can be written as

$$E(r) = -\frac{e^2}{4\pi\epsilon_0 r} \left[6 - \frac{12}{\sqrt{2}} + \frac{8}{\sqrt{3}} - \dots \right] = -\frac{e^2 M}{4\pi\epsilon_0 r} \quad \text{Madelung constant } M \text{ for NaCl}$$

where M , called **Madelung constant**, is given by the summation in the square brackets for this particular ionic crystal structure (NaCl). Calculate M for the first three terms and compare it with $M = 1.7476$, its value had we included the higher terms. What is your conclusion?

Solution

From Coulomb's law of electrostatic attraction we know that the PE between two charges Q_1 and Q_2 separated by a distance r is given by

$$PE = \frac{Q_1 Q_2}{4\pi\epsilon_0 r}$$

First we consider the interaction between Na^+ ion and 6Cl^- ions at distance r . Applying Coulomb's law we have

$$PE_1 = \frac{Q_1 Q_2}{4\pi\epsilon_0 r} = \frac{(-6e)(+e)}{4\pi\epsilon_0 r} = \frac{-6e^2}{4\pi\epsilon_0 r}$$

Similarly, we now consider 12 Na^+ ions as *second* nearest neighbors at a distance $r\sqrt{2}$

$$PE_2 = \frac{Q_1 Q_2}{4\pi\epsilon_0 r} = \frac{(+12e)(+e)}{4\pi\epsilon_0 r\sqrt{2}} = \frac{12e^2}{4\pi\epsilon_0 r\sqrt{2}}$$

and 8 Cl^- ions as *third* nearest neighbors at a distance $r\sqrt{3}$

$$PE_3 = \frac{Q_1 Q_2}{4\pi\epsilon_0 r} = \frac{(-8e)(+e)}{4\pi\epsilon_0 r\sqrt{3}} = \frac{-8e^2}{4\pi\epsilon_0 r\sqrt{3}}$$

and similarly we can consider the next nearest set of neighbors and so on. Therefore, the overall PE of the Na^+ ion is

$$E(r) = \frac{-6e^2}{4\pi\epsilon_0 r} + \frac{12e^2}{4\pi\epsilon_0 r\sqrt{2}} - \frac{8e^2}{4\pi\epsilon_0 r\sqrt{3}} + \dots$$

or

$$E(r) = \frac{-e^2}{4\pi\epsilon_0 r} \left[6 - \frac{12}{\sqrt{2}} + \frac{8}{\sqrt{3}} - \dots \right] = \frac{-Me^2}{4\pi\epsilon_0 r}$$

where clearly $M = 6 - \frac{12}{\sqrt{2}} + \frac{8}{\sqrt{3}} + \dots$

Considering just the first three terms we have $M = 2.133$. This is considerably different from the value $M = 1.7464$, the value obtained when higher order terms are considered. This implies that the next nearest neighbors have substantial effect on the potential energy.

***1.6 Bonding and bulk modulus** In general, the potential energy E per atom, or per ion pair, in a crystal as a function of interatomic (interionic) separation r can be written as the sum of an attractive PE and a repulsive PE ,

$$E(r) = -\frac{A}{r^n} + \frac{B}{r^m} \quad \text{General PE curve for bonding} \quad [1.39]$$

where A and n are constants characterizing the attractive PE and B and m are constants characterizing the repulsive PE . This energy is minimum when the crystal is in equilibrium. The magnitude of the minimum energy and its location r_o define the bonding energy and the equilibrium interatomic (or interionic) separation respectively.

When a pressure P is applied to a solid, its original volume V_o shrinks to V by an amount $\Delta V = V - V_o$. The bulk modulus K relates the volume strain $\Delta V/V$ to the applied pressure P by

$$P = -K(\Delta V/V_o) \quad \text{Bulk modulus definition} \quad [1.40]$$

The bulk modulus K is related to the energy curve. In its simplest form (assuming a simple cubic unit cell) K can be estimated from Equation 1.39 by

$$K = \frac{1}{9cr_o} \left[\frac{d^2 E}{dr^2} \right]_{r=r_o} \quad \text{Bulk modulus} \quad [1.41]$$

where c is a numerical factor, of the order of unity, given by b/p where p is the number of atoms or ion pairs in the unit cell and b is a numerical factor that relates the cubic unit cell lattice parameter a_o to the equilibrium interatomic (interionic) separation r_o by $b = a_o^3 / r_o^3$

a. Show that the bond energy and equilibrium separation are given by

$$E_{\text{bond}} = \frac{A}{r_o^n} \left(1 - \frac{n}{m} \right) \quad \text{and} \quad r_o = \left[\frac{mB}{nA} \right]^{\frac{1}{m-n}}$$

b. Show that the bulk modulus is given by

$$K = \frac{An(m-n)}{9cr_o^{n+3}} \quad \text{or} \quad K = E_{\text{bond}} \frac{mn}{9cr_o^3}$$

- c. For a NaCl type crystal, Na^+ and Cl^- ions touch along cube edge so that $r_o = (a_o/2)$. Thus, $a^3 = 2^3 r_o^3$ and $b = 2^3 = 8$. There are 4 ion pairs in the unit cell, $p = 4$. Thus, $c = b/p = 8/4 = 2$. Using the values from **Example 1.3**, calculate the bulk modulus of NaCl.

Solution

a. Interatomic separation $r = r_0$ is the distance at minimum $E(r)$, Therefore we differentiate $E(r)$ and set it equal to zero. *i.e.*

$$\left[\frac{dE(r)}{dr} \right]_{r=r_0} = \frac{d}{dr} \left[-\frac{A}{r^n} + \frac{B}{r^m} \right]_{r=r_0} = 0$$

$$\therefore \left[\frac{An}{r_0^{n+1}} - \frac{mB}{r_0^{m+1}} \right]_{r=r_0} = 0$$

$$\therefore \frac{An}{r_0^{n+1}} - \frac{mB}{r_0^{m+1}} = 0$$

$$\therefore \frac{r_0^{m+1}}{r_0^{n+1}} = \frac{mB}{nA} \quad \text{or} \quad r_0^{m-n} = \frac{mB}{nA}$$

$$\therefore r_0 = \left[\frac{mB}{nA} \right]^{\frac{1}{m-n}}$$

The potential energy is minimum at $r = r_0$ and is related with bonding energy $E(r_0) = -E_{\text{bond}}$. From the equation for r_0 we have

$$r_0^{m-n} = \frac{Bm}{An} \quad \text{and isolate for } B,$$

$$B = \frac{Anr_0^{m-n}}{m}$$

Substitute for B in the energy relation

$$\begin{aligned} E(r) &= -\frac{A}{r_0^n} + \frac{B}{r_0^m} = -\frac{A}{r_0^n} + \frac{1}{r_0^m} \left(\frac{Anr_0^{m-n}}{m} \right) \\ &= -\frac{A}{r_0^n} + \frac{Anr_0^{m-n-m}}{m} = -\frac{A}{r_0^n} + \frac{Anr_0^{-n}}{m} = -\frac{A}{r_0^n} + \frac{An}{mr_0^n} \end{aligned}$$

$$\therefore E_{\text{bond}} = -E(r) = -\left[-\frac{A}{r_0^n} \left(1 - \frac{n}{m} \right) \right]$$

$$\therefore E_{\text{bond}} = \frac{A}{r_0^n} \left(1 - \frac{n}{m} \right)$$

b. Show that the bulk modulus is given by

$$K = \frac{An}{9cr_o^{n+3}} (m-n) \quad \text{or} \quad K = \frac{mnE_{\text{bond}}}{9cr_o^3}$$

From the definition of Bulk modulus mentioned in the problem statement above

$$K = \frac{1}{9cr_0} \left[\frac{d^2 E}{dr^2} \right]_{r=r_0}$$

First we find $\frac{d^2 E}{dr^2}$

$$\frac{dE(r)}{dr} = \frac{d}{dr} \left(-\frac{A}{r^n} + \frac{B}{r^m} \right) = \frac{An}{r^{n+1}} - \frac{mB}{r^{m+1}}$$

$$\begin{aligned} \therefore \left[\frac{dE^2(r)}{dr^2} \right]_{r=r_0} &= \left[\frac{d}{dr} \left(\frac{An}{r^{n+1}} - \frac{mB}{r^{m+1}} \right) \right]_{r=r_0} \\ &= \left[\frac{-n(n+1)A}{r^{n+2}} + \frac{m(m+1)B}{r^{m+2}} \right]_{r=r_0} = \frac{-n(n+1)A}{r_0^{n+2}} + \frac{m(m+1)B}{r_0^{m+2}} \end{aligned}$$

Again substituting the value of B in the above relation, i.e. $B = \frac{Anr_0^{m-n}}{m}$ we have

$$\begin{aligned} \left[\frac{dE^2(r)}{dr^2} \right]_{r=r_0} &= \frac{-n(n+1)A}{r_0^{n+2}} + \frac{m(m+1)}{r_0^{m+2}} \frac{Anr_0^{m-n}}{m} = \frac{-n(n+1)A}{r_0^{n+2}} + \frac{An(m+1)}{r_0^{m+2-m+n}} \\ &= \frac{-n(n+1)A}{r_0^{n+2}} + \frac{An(m+1)}{r_0^{n+2}} = \frac{An}{r_0^{n+2}} (-n-1+m+1) = \frac{An(m-n)}{r_0^{n+2}} \end{aligned}$$

Not substitute for the second derivative in the equation for the Bulk modulus

$$K = \frac{1}{9cr_0} \left[\frac{d^2 E}{dr^2} \right]_{r=r_0} = \frac{1}{9cr_0} \left(\frac{An(m-n)}{r_0^{n+2}} \right)$$

or
$$K = \frac{An(m-n)}{9cr_0^{n+3}}$$

From the relationship for bonding energy,

$$E_{\text{bond}} = -\frac{A}{r_0^n} \left(1 - \frac{n}{m} \right) = -\frac{A(m-n)}{r_0^n m}$$

$\therefore K = \frac{A(m-n)}{r_0^n m} \frac{mn}{9cr_0^3}$

or
$$K = E_{\text{bond}} \frac{mn}{9cr_0^3}$$

c. From Example 1.3, the bonding energy for NaCl is $M = 1.748$, $n = 1$, $m = 8$, $r_0 = 0.281 \times 10^{-9}$ m, $c = 2$. Therefore,

$$A = \frac{e^2 M}{4\pi\epsilon_0} = \frac{(1.6 \times 10^{-19} \text{ C})^2 (1.748)}{4\pi(8.85 \times 10^{-12} \text{ F m}^{-1})} = 4.022 \times 10^{-28}.$$

Substitute A in expression for K we have

$$K = \frac{An(m-n)}{9cr_0^{n+3}} = \frac{(4.022 \times 10^{-28})(1)(8-1)}{9 \cdot (2)(0.281 \times 10^{-9})^{1+3}} = 25.1 \times 10^9 \text{ Pa or } 25.1 \text{ GPa}$$

Author's Note: Experimental value is roughly 2.4×10^{10} Pa or 24 GPa. The calculated value is quite close.

1.7 Van der Waals bonding Below 24.5 K, Ne is a crystalline solid with an FCC structure. The interatomic interaction energy per atom can be written as

$$E(r) = -2\varepsilon \left[14.45 \left(\frac{\sigma}{r} \right)^6 - 12.13 \left(\frac{\sigma}{r} \right)^{12} \right] \text{ (eV/atom)}$$

where ε and σ are constants that depend on the polarizability, the mean dipole moment, and the extent of overlap of core electrons. For crystalline Ne, $\varepsilon = 3.121 \times 10^{-3}$ eV and $\sigma = 0.274$ nm.

- Show that the equilibrium separation between the atoms in an inert gas crystal is given by $r_o = (1.090)\sigma$. What is the equilibrium interatomic separation in the Ne crystal?
- Find the bonding energy per atom in solid Ne.
- Calculate the density of solid Ne (atomic mass = 20.18).

Solution

a. Let E = potential energy and x = distance variable between the atoms. The energy E is given by

$$E(x) = -2\varepsilon \left[14.45 \left(\frac{\sigma}{x} \right)^6 - 12.13 \left(\frac{\sigma}{x} \right)^{12} \right]$$

The force F on each atom is given by

$$F(x) = -\frac{dE(x)}{dx} = 2\varepsilon \left[145.56 \frac{\sigma \left(\frac{\sigma}{x} \right)^{11}}{x^2} - 86.7 \frac{\sigma \left(\frac{\sigma}{x} \right)^5}{x^2} \right]$$

$$\therefore F(x) = 2\varepsilon \left[145.56 \frac{\sigma^{12}}{x^{13}} - 86.7 \frac{\sigma^6}{x^7} \right]$$

When the atoms are in equilibrium, this net force must be zero. Using r_o to denote equilibrium separation,

$$F(r_o) = 0$$

$$\therefore 2\varepsilon \left[145.56 \frac{\sigma^{12}}{r_o^{13}} - 86.7 \frac{\sigma^6}{r_o^7} \right] = 0$$

$$\therefore 145.56 \frac{\sigma^{12}}{r_o^{13}} = 86.7 \frac{\sigma^6}{r_o^7}$$

$$\therefore \frac{r_o^{13}}{r_o^7} = \left(\frac{145.56}{86.7} \right) \sigma^{12}$$

$$\therefore r_o = 1.090\sigma$$

For the Ne crystal, $\sigma = 2.74 \times 10^{-10}$ m and $\varepsilon = 0.003121$ eV. Therefore,

$$r_o = 1.090(2.74 \times 10^{-10} \text{ m}) = 2.99 \times 10^{-10} \text{ m for Ne.}$$

b. Calculate energy per atom at equilibrium:

$$E(r_o) = -2\varepsilon \left[14.45 \left(\frac{\sigma}{r_o} \right)^6 - 12.13 \left(\frac{\sigma}{r_o} \right)^{12} \right]$$

$$\therefore E(r_o) = -2(0.003121 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV}) \left[14.45 \left(\frac{2.74 \times 10^{-10} \text{ m}}{2.99 \times 10^{-10} \text{ m}} \right)^6 - 12.13 \left(\frac{2.74 \times 10^{-10} \text{ m}}{2.99 \times 10^{-10} \text{ m}} \right)^{12} \right]$$

$$\therefore E(r_o) = -4.30 \times 10^{-21} \text{ or } -0.0269 \text{ eV}$$

Therefore the bonding energy in solid Ne is 0.027 eV per atom.

c. To calculate the density, remember that the unit cell is FCC, and density = (mass of atoms in the unit cell) / (volume of unit cell). There are 4 atoms per FCC unit cell, and the atomic mass of Ne is 20.18 g/mol. (See Figure 1Q7-1)

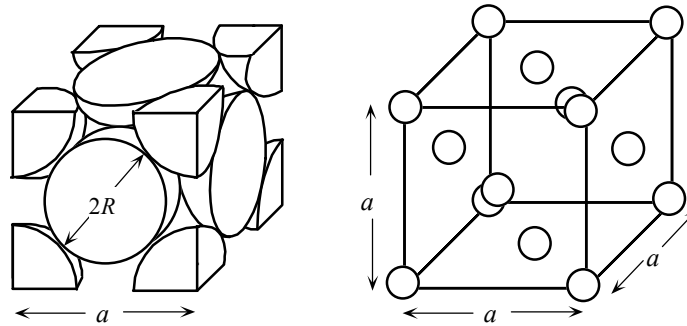


Figure 1Q7-1: Left: An FCC unit cell with close-packed spheres. Right: Reduced-sphere representation of the FCC unit cell. Examples: Ag, Al, Au, Ca, Cu, γ -Fe ($>912^\circ\text{C}$), Ni, Pd, Pt, Rh.

Since it is an FCC crystal structure, let a = lattice parameter (side of cubic cell) and R = radius of atom. The shortest interatomic separation is $r_o = 2R$ (atoms in contact means nucleus to nucleus separation is $2R$ (see Figure 1Q7-1).

$$R = r_o/2$$

and $2a^2 = (4R)^2$

$$\therefore a = 2\sqrt{2}R = 2\sqrt{2} \left(\frac{r_o}{2} \right) = \sqrt{2}(2.99 \times 10^{-10} \text{ m})$$

$$\therefore a = 4.228 \times 10^{-10} \text{ m or } 0.423 \text{ nm}$$

Therefore, the volume (V) of the unit cell is:

$$V = a^3 = (4.228 \times 10^{-10} \text{ m})^3 = 7.558 \times 10^{-29} \text{ m}^3$$

The mass (m) of 1 Ne atom in grams is the atomic mass (M_{at}) divided by N_A , because N_A number of atoms have a mass of M_{at} .

$$m = M_{at} / N_A$$

$$\therefore m = \frac{(20.18 \text{ g/mol})(0.001 \text{ kg/g})}{6.022 \times 10^{23} \text{ mol}^{-1}} = 3.351 \times 10^{-26} \text{ kg}$$

There are 4 atoms per unit cell in the FCC cell. The density (ρ) can then be found by:

$$\rho = (4m) / V = [4 \times (3.351 \times 10^{-26} \text{ kg})] / (7.558 \times 10^{-29} \text{ m}^3)$$

$$\therefore \rho = 1774 \text{ kg/m}^3$$

In g/cm^3 this density is:

$$\tilde{\rho} = \frac{1774 \text{ kg/m}^3}{(100 \text{ cm/m})^3} \times (1000 \text{ g/kg}) = 1.77 \text{ g/cm}^3$$

The density of solid Ne is 1.77 g cm^{-3} .

Author's Note: The experimental value for a at 4 K is 0.44 nm. The calculated value is close.

1.8 Kinetic molecular theory

- In particular Ar-ion laser tube the gas pressure due to Ar atoms is about 0.1 torr at 25 °C when the laser is off. What is the concentration of Ar atoms per cm^3 at 25 °C in this laser? (760 torr = 1 atm = $1.013 \times 10^5 \text{ Pa}$.)
- In the He-Ne laser tube He and Ne gases are mixed and sealed. The total pressure P in the gas is given by contributions arising from He and Ne atoms

$$P = P_{\text{He}} + P_{\text{Ne}}$$

where P_{He} and P_{Ne} are the *partial pressures* of He and Ne in the gas mixture, that is, pressures due to He and Ne gasses alone,

$$P_{\text{He}} = \frac{N_{\text{He}}}{N_A} \left(\frac{RT}{V} \right) \quad \text{and} \quad P_{\text{Ne}} = \frac{N_{\text{Ne}}}{N_A} \left(\frac{RT}{V} \right)$$

In a particular He-Ne laser tube the ratio of He and Ne atoms is 7:1, and the total pressure is about 1 torr at 22 °C. Calculate the concentrations of He and Ne atoms in the gas at 22 °C. What is the pressure at an operating temperature of 130 °C?

Solution

- From the Kinetic molecular theory for gases, we have

$$PV = \left(\frac{N}{N_A} \right) RT$$

where, R is the gas constant constant, T is the temperature. The number of Ar atoms per unit volume is

$$n_{\text{Ar}} = \frac{N}{V} = \frac{PN_A}{RT}$$

We are given $P = \frac{0.1 \text{ torr}}{760 \text{ torr}} \times 1.013 \times 10^5 \text{ Pa} = 13.33 \text{ Pa}$

Therefore the number of Ar atoms per unit volume n_{Ar} will be

$$n_{\text{Ar}} = \frac{(13.33 \text{ Pa})(6.022 \times 10^{23} \text{ mol}^{-1})}{(8.3145 \text{ J K}^{-1} \text{ mol}^{-1})(25 + 273 \text{ K})} = \mathbf{3.24 \times 10^{21} \text{ m}^{-3} \text{ or } 3.24 \times 10^{15} \text{ cm}^{-3}}$$

b. Let $n_{\text{He}} = N_{\text{He}}/V$, the concentration of He atoms; $n_{\text{Ne}} = N_{\text{Ne}}/V$, the concentration of Ne atoms. Given that the total pressure is the sum of the pressure by He and Ne gasses

$$P = P_{\text{He}} + P_{\text{Ne}}$$

$$\therefore P = \left(\frac{N_{\text{He}}}{N_A} \right) \frac{RT}{V} + \left(\frac{N_{\text{Ne}}}{N_A} \right) \frac{RT}{V} = n_{\text{He}} \left(\frac{RT}{N_A} \right) + n_{\text{Ne}} \left(\frac{RT}{N_A} \right)$$

$$\therefore P = 7n_{\text{Ne}} \left(\frac{RT}{N_A} \right) + n_{\text{Ne}} \left(\frac{RT}{N_A} \right) = 8n_{\text{Ne}} \left(\frac{RT}{N_A} \right)$$

$$\therefore n_{\text{Ne}} = \frac{PN_A}{8RT}$$

Thus at $T = 22^\circ \text{C}$ (295 K) for Ne,

$$n_{\text{Ne}} = \frac{(1.013 \times 10^5 \text{ Pa}) \left(\frac{1 \text{ torr}}{760 \text{ torr}} \right) (6.022 \times 10^{23} \text{ mol}^{-1})}{8(8.3145 \text{ J K}^{-1} \text{ mol}^{-1})(273 + 22 \text{ K})}$$

$$= \mathbf{4.09 \times 10^{21} \text{ m}^{-3} \text{ or } 4.09 \times 10^{15} \text{ cm}^{-3}}.$$

Given that n_{He} is 7 times that of n_{Ne} , *i.e.*

$$n_{\text{He}} = 7 \times n_{\text{Ne}} = \mathbf{2.86 \times 10^{22} \text{ m}^{-3} \text{ or } 2.86 \times 10^{16} \text{ cm}^{-3}}.$$

At $T = 130^\circ \text{C}$ (403 K), the atomic concentrations of He and Ne remain unchanged (the tube has the same volume, neglecting the thermal expansion). Thus, the new pressure P' and initial pressure P are related by

$$\frac{P'}{P} = \frac{RT'/V}{RT/V} = \frac{T'}{T} = \frac{(273 + 130 \text{ K})}{(273 + 22 \text{ K})} = 1.366$$

so that the new pressure P' is **1.37 torr**.

1.9 Kinetic Molecular Theory Calculate the effective (rms) speeds of the He and Ne atoms in the He-Ne gas laser tube at room temperature (300 K).

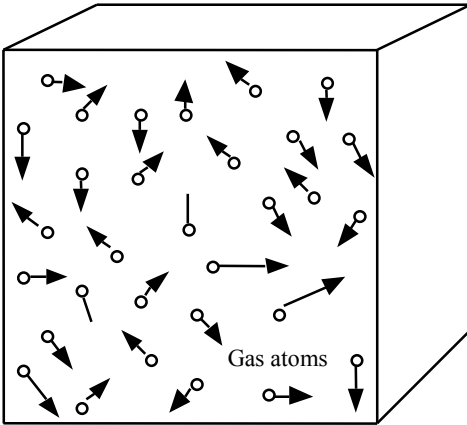
Solution

Figure 1Q9-1: The gas molecule in the container are in random motion.

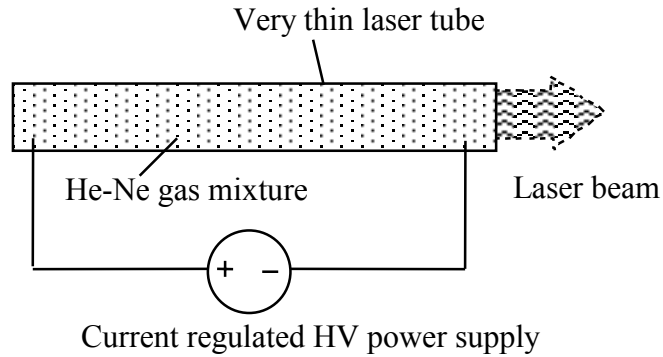


Figure 1Q9-2: The He-Ne gas laser.

To find the root mean square velocity (v_{rms}) of He atoms at $T = 300$ K:

The atomic mass of He is (from Periodic Table) $M_{at} = 4.0$ g/mol. Remember that 1 mole has a mass of M_{at} grams. Then one He atom has a mass (m) in kg given by:

$$m = \frac{M_{at}}{N_A} = \frac{4.0 \text{ g/mol}}{6.022 \times 10^{23} \text{ mol}^{-1}} \times (0.001 \text{ kg/g}) = 6.642 \times 10^{-27} \text{ kg}$$

From kinetic theory (visualized in Figure 1Q9-1),

$$\frac{1}{2} m (v_{rms})^2 = \frac{3}{2} kT$$

$$\therefore v_{rms} = \sqrt{3 \frac{kT}{m}} = \sqrt{3 \frac{(1.381 \times 10^{-23} \text{ J K}^{-1})(300 \text{ K})}{(6.642 \times 10^{-27} \text{ kg})}} = \mathbf{1368 \text{ m/s}}$$

The root mean square velocity (v_{rms}) of Ne atoms at $T = 300$ K can be found using the same method as above, changing the atomic mass to that of Ne, $M_{at} = 20.18$ g/mol. After calculations, the mass of one Ne atom is found to be 3.351×10^{-26} kg, and the root mean square velocity (v_{rms}) of Ne is found to be $v_{rms} = \mathbf{609 \text{ m/s}}$.

Author's Note: Radiation emerging from the He-Ne laser tube (Figure 1Q9-2) is due to the Ne atoms emitting light, all in phase with each other, as explained in Ch. 3. When a Ne atom happens to be moving towards the observer, due to the Doppler Effect, the frequency of the laser light is higher. If a Ne atom happens to be moving away from the observer, the light frequency is lower. Thus, the random motions of the

gas atoms cause the emitted radiation not to be at a single frequency but over a range of frequencies due to the Doppler Effect.

***1.10 Kinetic molecular theory and the Ar ion laser** An argon ion laser has a laser tube that contains Ar atoms that produce the laser emission when properly excited by an electrical discharge. Suppose that the gas temperature inside the tube is 1300 °C (very hot).

- Calculate the mean speed (v_{av}), rms velocity ($v_{rms} = \sqrt{v^2}$) and the rms speed ($v_{rms, x} = \sqrt{v_x^2}$) in one particular direction of the Ar atoms in the laser tube, assuming 1300 °C. (See Example 1.10.)
- Consider a light source that is emitting waves and is moving towards an observer, somewhat like a whistling train moving towards a passenger. If f_o is the frequency of the light waves emitted at the source, then, due to the *Doppler effect*, the observer measures a higher frequency f that depends on the velocity v_{Ar} of the source towards to observer and the speed c of light,

$$f = f_o \left(1 + \frac{v_{Ar}}{c} \right)$$

It is the Ar ions that emit the laser output light in the Ar-ion laser. The emission wavelength $\lambda_o = c/f_o$ is 514.5 nm. Calculate the wavelength λ registered by an observer for those atoms that are moving with a mean speed v_{av} toward the observer. Those atoms that are moving away from the observer will result in a lower observed frequency because v_{Ar} will be negative. Estimate the width of the wavelengths (the difference between the longest and shortest wavelengths) emitted by the Ar ion laser.

Solution

- From Example 1.10 the mean speed is given by

$$v_{av} = \sqrt{\frac{8kT}{\pi m}}$$

m is the mass of atom. $T = 1300$ °C = 1573 K. Atomic mass of Ar is $M_{at} = 39.95$ g mol⁻¹, therefore the mass of the Ar atom is

$$m = \frac{M_{at}}{N_A} = \frac{39.95 \text{ g mol}^{-1}}{6.022 \times 10^{23} \text{ mol}^{-1}} = 6.634 \times 10^{-23} \text{ g or } 6.634 \times 10^{-26} \text{ kg}$$

Mean speed is given by

$$v_{av} = \sqrt{\frac{8kT}{\pi m}} = \sqrt{\frac{8(1.38066 \times 10^{-23} \text{ J K}^{-1})(1300 + 273 \text{ K})}{\pi(6.634 \times 10^{-26} \text{ kg})}}$$

$$\therefore v_{av} = 913.32 \text{ m s}^{-1}.$$

Root mean square RMS velocity is

$$v_{rms} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3(1.3806 \times 10^{-23} \text{ J K}^{-1})(1300 + 273 \text{ K})}{(6.634 \times 10^{-26} \text{ kg})}}$$

$$\therefore v_{rms} = 991.31 \text{ m s}^{-1}.$$

RMS speed, $v_{rms, x} = \sqrt{v_x^2}$ in one particular direction is

$$v_{\text{rms},x} = \frac{v_{\text{rms}}}{\sqrt{3}} = \frac{991.31 \text{ m s}^{-1}}{\sqrt{3}} = \mathbf{572.33 \text{ m s}^{-1}}.$$

b. First we consider the case when the source is moving towards the observer with average speed, v_{av} , the frequency observed is

$$f = f_o \left(1 + \frac{v_{\text{Ar}}}{c} \right)$$

where $f_o = c/\lambda_o = 3 \times 10^8 \text{ m s}^{-1} / 514.5 \times 10^{-9} \text{ m} = 5.8309 \times 10^{14} \text{ s}^{-1}$, v_{Ar} as calculated above is 912.32 m s^{-1} . Therefore the frequency is

$$f_1 = 5.8309 \times 10^{14} \text{ s}^{-1} \left(1 + \frac{913.32 \text{ m s}^{-1}}{3 \times 10^8 \text{ m s}^{-1}} \right) = 5.830922 \times 10^{14} \text{ s}^{-1}$$

The corresponding wavelength is therefore, $\lambda_1 = c/f_1 = 3 \times 10^8 \text{ m s}^{-1} / 5.830922 \times 10^{14} \text{ s}^{-1} = \mathbf{514.4984 \text{ nm}}$. In the case when the emitting source is moving away from the observer, the frequency is

$$f_2 = 5.8309 \times 10^{14} \text{ s}^{-1} \left(1 - \frac{913.32 \text{ m s}^{-1}}{3 \times 10^8 \text{ m s}^{-1}} \right) = 5.830886 \times 10^{14} \text{ s}^{-1}$$

The corresponding wavelength is therefore, $\lambda_2 = c/f_2 = 3 \times 10^8 \text{ m s}^{-1} / 5.830886 \times 10^{14} \text{ s}^{-1} = \mathbf{514.5016 \text{ nm}}$. The range of wavelengths observed by the observer is between **514.4984 nm and 514.5016 nm**. The wavelength width

$$\Delta\lambda = \lambda_2 - \lambda_1 = 514.5016 - 514.4984 \text{ nm} = \mathbf{0.0032 \text{ nm, very small.}}$$

Author's Note: The question asks for the change in the wavelength or the width in the emitted wavelengths. Four decimal places were kept in the calculations of frequency and wavelength because we are interested in these changes and the changes in the frequency and wavelength are small. It may be thought that we should similarly use higher accuracy in the velocity calculations and a more accurate c value *etc* but that's not necessary because the change in the frequency is actually $2f_o v_{\text{Ar}}/c$:

$$\begin{aligned} \Delta f &= f_o \left(1 + \frac{v_{\text{Ar}}}{c} \right) - f_o \left(1 - \frac{v_{\text{Ar}}}{c} \right) = 2f_o \frac{v_{\text{Ar}}}{c} \\ &= 2(5.8309 \times 10^{14} \text{ s}^{-1})(913.32 \text{ m s}^{-1}) / (3 \times 10^8 \text{ m s}^{-1}) = 3.55 \times 10^9 \text{ s}^{-1} = \mathbf{3.55 \text{ GHz.}} \end{aligned}$$

Author's Note to the Instructor: Some students are known to convert a range of frequencies to a range of wavelengths by taking $\Delta\lambda = c/\Delta f$, which is wrong. To convert a small range of frequencies Δf to a range of wavelengths $\Delta\lambda$, take $\lambda = c/f$ and differentiate it,

$$\frac{d\lambda}{df} = -\frac{c}{f^2}$$

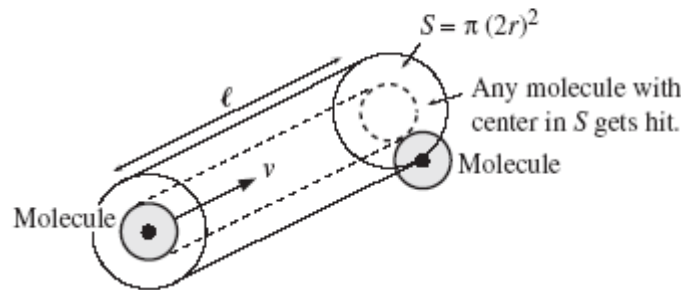
$$\therefore \Delta\lambda \approx \left| \frac{d\lambda}{df} \right| \Delta f = \frac{c}{f_o^2} \Delta f = \frac{3 \times 10^8 \text{ m s}^{-1}}{(5.8309 \times 10^{14} \text{ s}^{-1})^2} (3.55 \times 10^9 \text{ s}^{-1}) = 3.13 \times 10^{-12} \text{ m} = \mathbf{0.00313 \text{ nm}}$$

very close to the above calculation of 0.0032 nm.

***1.11 Vacuum deposition** Consider air as composed of nitrogen molecules N_2 .

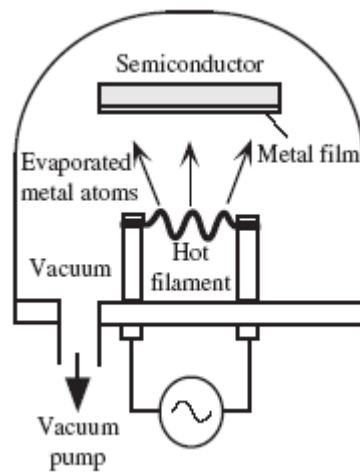
a. What is the concentration n (number of molecules per unit volume) of N_2 molecules at 1 atm and 27 °C?

- b. Estimate the mean separation between the N_2 molecules.
- c. Assume each molecule has a finite size that can be represented by a sphere of radius r . Also assume that ℓ is the **mean free path**, defined as the mean distance a molecule travels before colliding with another molecule, as illustrated in Figure 1.74a. If we consider the motion of one N_2 molecule, with all the others stationary, it is apparent that if the path of the traveling molecule crosses the cross-sectional area $S = \pi(2r)^2$, there will be a collision. Since ℓ is the mean distance between collisions, there must be at least one stationary molecule within the volume $S\ell$,



(a) A molecule moving with a velocity, travels a mean distance ℓ between collisions. Since the collision cross-sectional area is S , in the volume $S\ell$ there must be at least one molecule.

Consequently, $n(S\ell) = 1$.



(b) Vacuum deposition of metal electrodes by thermal evaporation.

Figure 1.74

as shown in Figure 1.74a. Since n is the concentration, we must have $n(S\ell) = 1$ or $\ell = 1/(\pi 4r^2 n)$. However, this must be corrected for the fact that all the molecules are in motion, which only introduces a numerical factor, so that

$$\ell \approx \frac{1}{2^{1/2} \pi 4r^2 n}$$

Assuming a radius r of 0.1 nm, calculate the mean free path of N_2 molecules between collisions at 27 °C and 1 atm.

- d. Assume that an Au film is to be deposited onto the surface of a Si chip to form metallic interconnections between various devices. The deposition process is generally carried out in a vacuum chamber and involves the condensation of Au atoms from the vapor phase onto the chip surface. In one procedure, a gold wire is wrapped around a tungsten filament, which is heated by passing a large current through the filament (analogous to the heating of the filament in a light bulb) as depicted in Figure 1.74b. The Au wire melts and wets the filament, but as the temperature of the filament increases, the gold evaporates to form a vapor. Au atoms from this vapor then condense onto the chip surface, to solidify and form the metallic connections. Suppose that the source (filament)-to-substrate (chip) distance L is 10 cm. Unless the mean free path of air molecules is much longer than L , collisions between the metal atoms and air molecules will prevent the deposition of the Au onto the chip surface. Taking the mean free path ℓ to be $100L$, what should be the pressure inside the vacuum system? (Assume the same r for Au atoms.)

Solution

Assume $T = 300$ K throughout. The radius of the nitrogen molecule (given approximately) is $r = 0.1 \times 10^{-9}$ m. Also, we know that pressure $P = 1$ atm $= 1.013 \times 10^5$ Pa.

Let N = total number of molecules, V = volume and k = Boltzmann's constant. Then,

$$PV = NkT$$

(Note: this equation can be derived from the more familiar form of $PV = \eta RT$, where η is the total number of moles, which is equal to N/N_A , and R is the gas constant, which is equal to $k \times N_A$.)

The concentration n (number of molecules per unit volume) is defined as:

$$n = N/V$$

Substituting this into the previous equation, the following equation is obtained:

$$P = nkT$$

- a. The concentration n of N_2 molecules at 1 atm and $T = 27$ °C + 273 = 300 K:

$$n = P/(kT)$$

$$\therefore n = \frac{1.013 \times 10^5 \text{ Pa}}{(1.381 \times 10^{-23} \text{ J K}^{-1})(300 \text{ K})}$$

$$\therefore n = 2.45 \times 10^{25} \text{ molecules per m}^3$$

- b. The mean separation between the N_2 molecules: We first consider a cube of a simple crystal that contains n atoms per unit volume, and suppose that each side of the cube is unit length as shown in Figure 1Q11-1. The volume $V = 1$. To each atom we can attribute *a portion of the whole volume* which for simplicity is a cube of side d . Thus, each atom is considered to occupy a volume of d^3 .

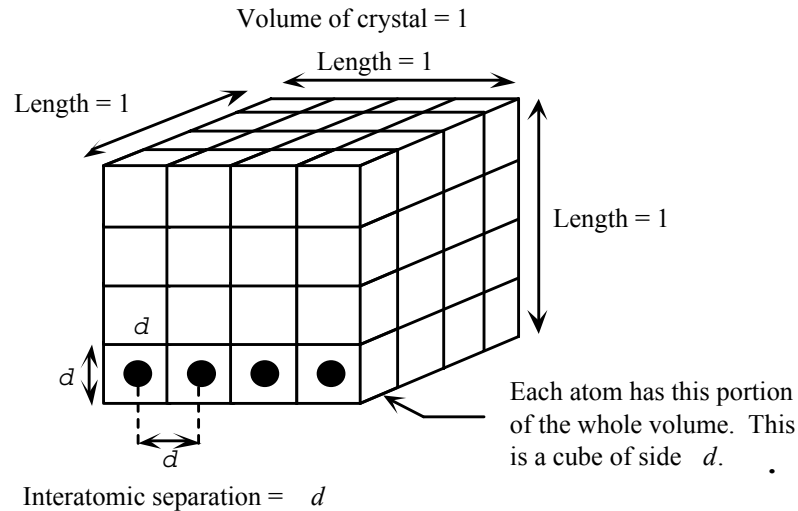


Figure 1Q11-1: Relationship between interatomic separation and the number of atoms per unit volume.

The actual or true volume of the atom does not matter. All we need to know is how much volume an atom has around it, given, all the atoms are identical and that adding all the atomic volumes must give the whole volume of the substance.

Suppose that there are n atoms in this crystal. Then n is the atomic concentration, number of atoms per unit volume. Clearly, n atoms make up the crystal so that

$$n d^3 = \text{Crystal volume} = 1$$

Remember that this is only an approximation. The separation between any two atoms is d , and hence

$$d = \frac{1}{n^{\frac{1}{3}}}$$

While $n = d^{-\frac{1}{3}}$ was derived assuming a simple crystal, it would also apply to a random distribution of atoms where d is the mean separation between the atoms. Thus,

$$d = \frac{1}{n^{\frac{1}{3}}} = \frac{1}{\sqrt[3]{(2.45 \times 10^{25} \text{ m}^{-3})}} = 3.44 \times 10^{-9} \text{ m or } 3.4 \text{ nm}$$

c. Assuming a radius, r , of 0.1 nm, what is the mean free path, ℓ , between collisions?

$$\ell = \frac{1}{\sqrt{2} \cdot 4\pi r^2 n} = \frac{1}{\sqrt{2} \cdot 4\pi (0.1 \times 10^{-9} \text{ m})^2 (2.45 \times 10^{25} \text{ m}^{-3})}$$

$$\therefore \ell = 2.30 \times 10^{-7} \text{ m or } 230 \text{ nm}$$

d. We need the new mean free path, $\ell' = 100L$, or $0.1 \text{ m} \times 100$ (L is the source-to-substrate distance)

$$\ell = 10 \text{ m}$$

The new ℓ' corresponds to a new concentration n' of nitrogen molecules.

$$\ell' = \frac{1}{\sqrt{2} \cdot 4\pi r^2 n'}$$

$$\therefore n' = \frac{1}{8} \frac{\sqrt{2}}{\pi r^2 n'} = \frac{1}{8} \frac{\sqrt{2}}{\pi (0.1 \times 10^{-9} \text{ m})^2 (10 \text{ m})} = 5.627 \times 10^{17} \text{ m}^{-3}$$

This new concentration of nitrogen molecules requires a new pressure, P' :

$$P' = n' k T = (5.627 \times 10^{17} \text{ m}^{-3}) (1.381 \times 10^{-23} \text{ J K}^{-1}) (300 \text{ K}) = \mathbf{0.00233 \text{ Pa}}$$

In atmospheres this is:

$$P' = \frac{0.00233 \text{ Pa}}{1.013 \times 10^5 \text{ Pa/atm}} = \mathbf{2.30 \times 10^{-8} \text{ atm}}$$

In units of torr this is:

$$P' = (2.30 \times 10^{-8} \text{ atm}) (760 \text{ torr/atm}) = \mathbf{1.75 \times 10^{-5} \text{ torr}}$$

There is an important assumption made, namely that the cross sectional area of the Au atom is about the same as that of N_2 so that the expression for the mean free path need not be modified to account for different sizes of Au atoms and N_2 molecules. The calculation gives a magnitude that is quite close to those used in practice, e.g. a pressure of 10^{-5} torr.

1.12 Heat capacity

- Calculate the heat capacity per mole and per gram of N_2 gas, neglecting the vibrations of the molecule. How does this compare with the experimental value of $0.743 \text{ J K}^{-1} \text{ g}^{-1}$?
- Calculate the heat capacity per mole and per gram of CO_2 gas, neglecting the vibrations of the molecule. How does this compare with the experimental value of $0.648 \text{ J K}^{-1} \text{ g}^{-1}$? Assume that CO_2 molecule is linear (O-C-O), so that it has two rotational degrees of freedom.
- Based on the Dulong-Petit rule, calculate the heat capacity per mole and per gram of solid silver. How does this compare with the experimental value of $0.235 \text{ J K}^{-1} \text{ g}^{-1}$?
- Based on the Dulong-Petit rule, calculate the heat capacity per mole and per gram of the silicon crystal. How does this compare with the experimental value of $0.71 \text{ J K}^{-1} \text{ g}^{-1}$?

Solution

a. N_2 has 5 degrees of freedom: 3 translational and 2 rotational. Its molar mass is $M_{at} = 2 \times 14.01 \text{ g/mol} = 28.02 \text{ g/mol}$.

Let C_m = heat capacity per mole, c_s = specific heat capacity (heat capacity per gram), and R = gas constant, then:

$$C_m = \frac{5}{2} R = \frac{5}{2} (8.315 \text{ J K}^{-1} \text{ mol}^{-1}) = \mathbf{20.8 \text{ J K}^{-1} \text{ mol}^{-1}}$$

$$\therefore c_s = C_m / M_{at} = (20.8 \text{ J K}^{-1} \text{ mol}^{-1}) / (28.02 \text{ g/mol}) = \mathbf{0.742 \text{ J K}^{-1} \text{ g}^{-1}}$$

This is close to the experimental value.

b. CO_2 has the linear structure $\text{O}=\text{C}=\text{O}$. Rotations about the molecular axis have negligible rotational energy as the moment of inertia about this axis is negligible. There are therefore 2 rotational degrees of freedom. In total there are 5 degrees of freedom: 3 translational and 2 rotational. Its molar mass is

$$M_{at} = 12.01 + 2 \times 16 = 44.01 \text{ g/mol}.$$

$$C_m = \frac{5}{2} R = \frac{5}{2} (8.315 \text{ J K}^{-1} \text{ mol}^{-1}) = \mathbf{20.8 \text{ J K}^{-1} \text{ mol}^{-1}}$$

$$\therefore c_s = C_m / M_{at} = (20.8 \text{ J K}^{-1} \text{ mol}^{-1}) / (44.01 \text{ g/mol}) = \mathbf{0.47 \text{ J K}^{-1} \text{ g}^{-1}}$$

This is smaller than the experimental value $0.648 \text{ J K}^{-1} \text{ g}^{-1}$. The 5 degrees of freedom assigned to the CO_2 molecule did not include molecular vibrations and “vibrations” of the atoms that include “flexing” or bond bending.

c. For solid silver, there are 6 degrees of freedom: 3 vibrational *KE* and 3 elastic *PE* terms. Its molar mass is,

$$M_{at} = 107.87 \text{ g/mol.}$$

$$C_m = \frac{6}{2} R = \frac{6}{2} (8.315 \text{ J K}^{-1} \text{ mol}^{-1}) = \mathbf{24.9 \text{ J K}^{-1} \text{ mol}^{-1}}$$

$$\therefore c_s = C_m / M_{at} = (24.9 \text{ J K}^{-1} \text{ mol}^{-1}) / (107.87 \text{ g/mol}) = \mathbf{0.231 \text{ J K}^{-1} \text{ g}^{-1}}$$

This is very close to the experimental value.

d. For a solid, heat capacity per mole is $3R$. The molar mass of Si is $M_{at} = 28.09 \text{ g/mol}$.

$$C_m = \frac{6}{2} R = \frac{6}{2} (8.315 \text{ J K}^{-1} \text{ mol}^{-1}) = \mathbf{24.9 \text{ J K}^{-1} \text{ mol}^{-1}}$$

$$\therefore c_s = C_m / M_{at} = (24.9 \text{ J K}^{-1} \text{ mol}^{-1}) / (28.09 \text{ g/mol}) = \mathbf{0.886 \text{ J K}^{-1} \text{ g}^{-1}}$$

The experimental value is substantially less and is due to the failure of classical physics. One has to consider the quantum nature of the atomic vibrations and also the distribution of vibrational energy among the atoms. The student is referred to modern physics texts (under heat capacity in the Einstein model and the Debye model of lattice vibrations).

1.13 Dulong-Petit atomic heat capacity Express the Dulong-Petit rule for the molar heat capacity as heat capacity per atom and in the units of eV K^{-1} per atom, called the **atomic heat capacity**. CsI is an ionic crystal used in optical applications that require excellent infrared transmission at very long wavelengths (up to $55 \mu\text{m}$). It has the CsCl crystal structure with one Cs^+ and one I^- ion in the unit cell. Calculate the specific heat capacity of CsI and compare it with the experimental value of $0.20 \text{ J K}^{-1} \text{ g}^{-1}$. What is your conclusion?

Solution

Molar heat capacity from Dulong-Petit rule is given by

$$C_m = \frac{dU}{dT} = 3R = 25 \text{ J K}^{-1} \text{ mol}^{-1}$$

The relation provides us with the heat capacity per mole. In one mole of a substance there are $N_A = 6.022 \times 10^{23}$ molecules or atoms. Therefore the heat capacity per atom will be

$$C_{at} = \frac{25 \text{ J K}^{-1} \text{ mol}^{-1}}{6.022 \times 10^{23} \text{ mol}^{-1}} = \mathbf{4.15 \times 10^{-23} \text{ J K}^{-1} \text{ atom}^{-1}}$$

In terms of eV K^{-1} , the heat capacity will be

$$C_{\text{at}} = \frac{4.15 \times 10^{-4} \text{ J K}^{-1} \text{ atom}^{-1}}{1.6 \times 10^{-19} \text{ J eV}^{-1}} = 2.59 \times 10^4 \text{ eV K}^{-1} \text{ atom}^{-1}$$

In the Dulong-Petit rule, C_{at} does not depend on the type of atoms. It is the same for all types of atoms in the crystal. Cs and I have $M_{\text{at}}(\text{Cs}) = 132.91 \text{ g mol}^{-1}$ and $M_{\text{at}}(\text{I}) = 126.90 \text{ g mol}^{-1}$. There are equal number of Cs^+ and I^- atoms. Consider a mass m of CsI that has 1 mole (N_A atoms) of Cs^+ and 1 mole of I^- . There will be $2N_A$ atoms in total in this sample, and the heat capacity will be $2N_A C_{\text{at}} = 2(3R)$. The Cs atoms in this sample will have a mass $M_{\text{at}}(\text{Cs})$ and I atoms will have a mass $M_{\text{at}}(\text{I})$ so that the sample mass $m = M_{\text{at}}(\text{Cs}) + M_{\text{at}}(\text{I}) = 132.91 \text{ g} + 126.90 \text{ g}$.

The specific heat capacity of CsI will be,

$$c_s = \frac{2(3R)}{m} = \frac{2(3R)}{M_{\text{Cs}} + M_{\text{I}}} \frac{2(25) \text{ J K}^{-1} \text{ mol}^{-1}}{(132.92 + 126.90) \text{ g mol}^{-1}} = 0.1924 \text{ J K}^{-1} \text{ g}^{-1}$$

Another method of calculating the specific heat capacity of CsI is by considering the unit cell. In one unit cell of CsCl type structure, there is the central atom of Cs^+ and $\frac{1}{8} \times 8 = 1$ atom of I^- . The heat capacity of the unit cell is,

$$C_{\text{cell}} = 2C_{\text{at}}$$

The specific heat capacity of CsI is then

$$c_s = \frac{\text{Heat capacity of unit cell}}{\text{Mass of unit cell}} = \frac{\text{Number of atoms in unit cell} \times \text{heat capacity per atom}}{\text{Mass of unit cell}}$$

$$\therefore c_s = \frac{2C_{\text{at}}}{M_{\text{at}}(\text{Cs}) + M_{\text{at}}(\text{I})}$$

which is the calculation above.

Conclusion: The Dulong-Petit rule applies well to CsI.

Note to the Instructor: The Dulong-petit rule is blind to the species of atoms in the solid. Take a solid that has two types of atoms A and B . If $\bar{M}$ is the mean atomic mass in the solid, *i.e.* $\bar{M} = n_A M_A + n_B M_B$ where, n_A is the atomic fraction of A , n_B the atomic fraction of B , and M_A and M_B are the atomic weights (g mol^{-1}) of A and B , then the Dulong-petit rule is

$$c_s = \frac{C_m}{\bar{M}} = \frac{25}{n_A M_A + n_B M_B} \text{ J K}^{-1} \text{ g}^{-1}$$

(See Question 1.14)

1.14 Dulong-Petit specific heat capacity of alloys and compounds

- a. Consider an alloy AB , such as solder, or a compound material such as MgO , composed of n_A , atomic fractions of A , and n_B , atomic fractions of B . (The atomic fraction of A is the same as its molar fraction.) Let M_A and M_B be the atomic weights of A and B , in g mol^{-1} . The mean atomic weight per atom in the alloy or compound is then

$$\bar{M} = n_A M_A + n_B M_B \quad \text{Average atomic weight}$$

Show that the *Dulong-Petit rule* for the specific heat capacity c_s leads to

$$c_s = \frac{C_m}{\bar{M}} = \frac{25}{n_A M_A + n_B M_B} \text{ J K}^{-1} \text{ g}^{-1} \quad \text{Specific heat capacity}$$

- b. Calculate the specific heat capacity of Pb-Sn solder assuming that its composition is 38 wt.% Pb and 62 wt.% Sn.
- c. Calculate the specific heat capacities of Pb and Sn individually as c_{sA} and c_{sB} , respectively, and then calculate the c_s for the alloy using

$$c_s = c_{sA}w_A + c_{sB}w_B \quad \text{Alloy specific heat capacity}$$

where w_A and w_B are weight fractions of A (Pb) and B (Sn) in the alloy (solder). Compare your result with part (a). What is your conclusion?

- d. ZnSe is an important optical material (used in infrared windows and lenses and high power CO₂ laser optics) and also an important II-VI semiconductor that can be used to fabricate blue-green laser diodes. Calculate the specific heat capacity of ZnSe, and compare the calculation to the experimental value of $0.345 \text{ J K}^{-1} \text{ g}^{-1}$.

Solution

a. In the Dulong-Petit rule, the molar heat capacity $C_m = 3R$ does not depend on the species (type) of atoms. Stated differently, the atomic heat capacity (heat capacity per atom) $C_{\text{at}} = 3R/N_A = 3k$, does not depend on the chemical nature of the atom. It is the same for all types of atoms in the crystal.

Consider 1 mole of a compound that has n_A moles of A and n_B moles of B. ($n_A + n_B = 1$). Let m be the mass of the compound. The atomic masses are M_A and M_B . The heat capacity will be $(n_A N_A)C_{\text{at}} + (n_B N_A)C_{\text{at}}$ or C_m . The sample mass $m = (M_A(n_A) + M_B(n_B))$ grams, since atomic masses are g mol^{-1} . The specific heat capacity will be

$$c_s = \frac{C_m}{m} = \frac{C_m}{n_A M_A + n_B M_B} = \frac{C_m}{\bar{M}} = \frac{25}{\bar{M}} \text{ J K}^{-1} \text{ g}^{-1}$$

where the mean atomic mass is defined by

$$\bar{M} = n_A M_A + n_B M_B$$

Author's Note: This is a trivial derivation that is a direct consequence of the fact that the Dulong-Petit rule does *not* depend on the species of atoms making up the crystal.

b. The molar heat capacity, i.e. heat capacity per mole, from Dulong-Petit rule is $3R$. M_{at} for Pb = 207.2 g mol^{-1} and M_{at} for Sn = 118.7 g mol^{-1} . The atomic fraction of Pb and Sn is

$$n_{\text{Pb}} = \frac{\frac{w_{\text{Pb}}}{M_{\text{Pb}}}}{\frac{w_{\text{Pb}}}{M_{\text{Pb}}} + \frac{w_{\text{Sn}}}{M_{\text{Sn}}}} = \frac{\frac{0.38}{207.2}}{\frac{0.38}{207.2} + \frac{0.62}{118.7}} = 0.2599$$

and

$$n_{\text{Sn}} = \frac{\frac{w_{\text{Sn}}}{M_{\text{Sn}}}}{\frac{w_{\text{Pb}}}{M_{\text{Pb}}} + \frac{w_{\text{Sn}}}{M_{\text{Sn}}}} = \frac{\frac{0.62}{118.7}}{\frac{0.38}{207.2} + \frac{0.62}{118.7}} = 0.7401$$

The specific heat capacity using the relation provided in section a is

$$c_s = \frac{25 \text{ J K}^{-1} \text{ mol}^{-1}}{(0.2599)(207.2 \text{ g mol}^{-1}) + (0.7401)(118.7 \text{ g mol}^{-1})} = \mathbf{0.1764 \text{ J K}^{-1} \text{ g}^{-1}}$$

Another way of calculating the specific heat capacity is as follows. The specific heat capacity of a compound consisting of *A* and *B* is

$$c_s = \frac{(\text{Heat capacity per mole})(\text{Number of moles of Sn and Pb in } M_{\text{alloy}})}{\text{Mass of the alloy } M_{\text{alloy}}}$$

$$= \frac{(3R) \left(\frac{w_{\text{Sn}} M_{\text{alloy}}}{M_{\text{Sn}}} + \frac{w_{\text{Pb}} M_{\text{alloy}}}{M_{\text{Pb}}} \right)}{M_{\text{alloy}}} = (3R) \left(\frac{w_{\text{Sn}}}{M_{\text{Sn}}} + \frac{w_{\text{Pb}}}{M_{\text{Pb}}} \right)$$

The specific heat capacity of PbSn solder with composition 38 wt.% Pb and 62 wt.% Sn is therefore

$$c_s = (25 \text{ J K}^{-1} \text{ mol}^{-1}) \left(\frac{0.62}{207.2 \text{ g mol}^{-1}} + \frac{0.38}{118.7 \text{ g mol}^{-1}} \right) = \mathbf{0.1764 \text{ J K}^{-1} \text{ g}^{-1}}$$

c. The specific heat capacity for Pb is $\frac{3R}{M_{\text{pb}}} = \frac{25 \text{ J K}^{-1} \text{ mol}^{-1}}{118.7 \text{ g mol}^{-1}} = 0.2106 \text{ J K}^{-1} \text{ g}^{-1}$

The specific heat capacity for Sn is $\frac{3R}{M_{\text{Sn}}} = \frac{25 \text{ J K}^{-1} \text{ mol}^{-1}}{207.2 \text{ g mol}^{-1}} = 0.1207 \text{ J K}^{-1} \text{ g}^{-1}$

The specific heat capacity of alloy is

$$c_s = c_{sA} w_A + c_{sB} w_B \\ = (0.2106 \text{ J K}^{-1} \text{ g}^{-1})(0.38) + (0.1207 \text{ J K}^{-1} \text{ g}^{-1})(0.62) = \mathbf{0.1764 \text{ J K}^{-1} \text{ g}^{-1}}$$

Conclusion: This is the same as the calculation in Part *b*. The Dulong-Petit rule allows the overall specific heat capacity c_s of an alloy to be readily calculated from individual specific heat capacities c_{sA} and c_{sB} weighted by their weight fractions w_A and w_B in the alloys; not their atomic fractions.

d. The atomic mass of Zn $M_{\text{at}} = 65.41 \text{ g mol}^{-1}$ and for Se $M_{\text{at}} = 78.96 \text{ g mol}^{-1}$. In the molecule ZnSe, $n_{\text{Zn}} = n_{\text{Se}} = 0.5$, so the specific heat capacity of the compound ZnSe is therefore

$$c_s = \frac{25 \text{ J mol}^{-1} \text{ K}^{-1}}{n_{\text{Zn}} M_{\text{Zn}} + n_{\text{Se}} M_{\text{Se}}} = \frac{25 \text{ J mol}^{-1} \text{ K}^{-1}}{0.5(65.41 + 78.96) \text{ g mol}^{-1}} = 0.3463 \text{ J K}^{-1} \text{ g}^{-1}.$$

The value is in close agreement with the experimental value $0.345 \text{ J K}^{-1} \text{ g}^{-1}$.

1.15 Thermal expansion

a. If λ is the thermal expansion coefficient, show that the thermal expansion coefficient for an area is 2λ . Consider an aluminum square sheet of area 1 cm^2 . If the thermal expansion coefficient of Al at room temperature (25°C) is about $24 \times 10^{-6} \text{ K}^{-1}$, at what temperature is the percentage change in the area +1%?

- b. A particular incandescent light bulb (100 W, 120 V) has a tungsten (W) filament of length 57.9 cm, and a diameter of 63.5 μm . Calculate the length of the filament at 2300 °C, the approximate operating temperature of the filament inside the bulb. The linear expansion coefficient λ of W is approximately $4.50 \times 10^{-6} \text{ K}^{-1}$ at 300 K. How would you improve your calculation?

Solution

- a. Consider a rectangular area with sides x_o and y_o . Then at temperature T_o ,

$$A_o = x_o y_o$$

and at temperature T ,

$$A = [x_o(1 + \lambda\Delta T)][y_o(1 + \lambda\Delta T)] = x_o y_o (1 + \lambda\Delta T)^2$$

that is

$$A = x_o y_o [1 + 2\lambda\Delta T + (\lambda\Delta T)^2]$$

We can now use that $A_o = x_o y_o$ and neglect the term $(\lambda\Delta T)^2$ because it is very small in comparison with the linear term $\lambda\Delta T$ ($\lambda \ll 1$) to obtain

$$A = A_o(1 + 2\lambda\Delta T) = A_o(1 + \alpha_A \Delta T)$$

So, the thermal expansion coefficient for an area is

$$\alpha_A = 2\lambda$$

The area of the aluminum sheet at any temperature is given by

$$A = A_o[1 + 2\lambda(T - T_o)]$$

where A_o is the area at the reference temperature T_o . Solving for T we obtain,

$$T = T_o + \frac{1}{2} \frac{A/A_o - 1}{\lambda} = 25^\circ\text{C} + \frac{1}{2} \frac{(1.01) - 1}{24 \times 10^{-6} \text{ }^\circ\text{C}^{-1}} = \mathbf{233.3^\circ\text{C}}.$$

- b. The linear expansion coefficient λ of tungsten (W) is approximately $4.50 \times 10^{-6} \text{ K}^{-1}$ at 300 K. The length of W at 2300 °C (2573 K) is required. Considering the linear expansion coefficient λ is temperature independent the length will be

$$\begin{aligned} L &= L_o[1 + \lambda(T - T_o)] \\ &= 57.9 \text{ cm} [1 + 4.50 \times 10^{-6} \text{ K}^{-1}(2573 \text{ K} - 300 \text{ K})] = \mathbf{58.49 \text{ cm}}. \end{aligned}$$

From Fig 1.20, we note that the thermal expansion coefficient is a function of temperature. The linear expansion coefficient λ of W at 2573 K is approximately $7.70 \times 10^{-6} \text{ K}^{-1}$. The calculated length now will be

$$\begin{aligned} L &= L_o[1 + \lambda(T - T_o)] \\ &= 57.9 \text{ cm} [1 + 7.70 \times 10^{-6} \text{ K}^{-1}(2573 - 300)] = \mathbf{58.91 \text{ cm}}. \end{aligned}$$

Author's Note: The diameter of the filament is not required in the length expansion calculations; it is given for information only, to get a feel for its dimensions. (The same filament is used in Chapter 2 in calculating the operating temperature of the filament.)

1.16 Thermal expansion of Si The expansion coefficient of silicon over the temperature range 120-1500 K is given by Okada and Tokumaru (1984) as

$$\lambda = 3.725 \times 10^{-6} [1 - e^{-5.88 \times 10^{-3}(T-124)}] + 5.548 \times 10^{-10} T \quad \text{Silicon linear expansion coefficient}$$

where λ is in K^{-1} (or $^{\circ}\text{C}^{-1}$) and T is in Kelvins.

a. By expanding the above function around 20 $^{\circ}\text{C}$ (293 K) show that,

$$\lambda = 2.5086 \times 10^{-6} + (8.663 \times 10^{-9})(T - 293) - (2.3839 \times 10^{-11})(T - 293)^2$$

Silicon linear expansion coefficient

b. The change $\delta\rho$ in the density due to a change δT in the temperature, from [Example 1.8](#), is given by

$$\delta\rho = -\rho_o \alpha_v \delta T = -3\rho_o \lambda \delta T$$

Given density of Si as 2.329 g cm^{-3} at 20 $^{\circ}\text{C}$, calculate the density at 1000 $^{\circ}\text{C}$ by using the full expression, and by using the polynomials expansion of λ . What is your conclusion?

NOTE: The exponential term is -5.88×10^{-3} NOT -3.725×10^{-3}

NOTE: The example referred to is Example 1.8 NOT Example 1.5

Solution

a. We use Taylor series for series expansion of the function about a point. Taylor series expansion of a function $f(x)$ about a point $x = a$ is given by

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \dots$$

We have to expand the thermal expansion coefficient about $a = T_0 = 293 \text{ K}$,

$$f(x) = \lambda(T) = 3.725 \times 10^{-6} [1 - e^{-5.88 \times 10^{-3}(T-124)}] + 5.548 \times 10^{-10} T$$

$$f(a) = \lambda(293 \text{ K}) = 3.725 \times 10^{-6} [1 - e^{-5.88 \times 10^{-3}(293-124)}] + 5.548 \times 10^{-10} (293) = 2.5086 \times 10^{-6} \text{ K}^{-1}.$$

$$\begin{aligned} f'(a) &= \frac{d\lambda}{dT}(293 \text{ K}) = (-3.725 \times 10^{-6})(-5.88e-3)e^{-5.88 \times 10^{-3}(293-124)} + 5.548 \times 10^{-10} \\ &= 8.6632 \times 10^{-9} \text{ K}^{-1}. \end{aligned}$$

$$f''(a) = \frac{d^2\lambda}{dT^2}(293 \text{ K}) = (2.1903 \times 10^{-8})(-5.88e-3)e^{-5.88 \times 10^{-3}(293-124)} = -4.7678 \times 10^{-11} \text{ K}^{-1}.$$

Therefore the expansion will result in

$$\lambda(T) = 2.5086 \times 10^{-6} + 8.6632 \times 10^{-9}(T - 293) + \frac{-4.7678 \times 10^{-11}}{2!}(T - 293)^2$$

$$\lambda(T) = 2.5086 \times 10^{-6} + 8.663 \times 10^{-9}(T - 293) - 2.3839 \times 10^{-11}(T - 293)^2$$

b. The expansion coefficient at 1000 $^{\circ}\text{C}$ (1273 K) will be

$$\lambda(1273 \text{ K}) = 3.725 \times 10^{-6} [1 - e^{-5.88 \times 10^{-3}(1273-124)}] + 5.548 \times 10^{-10} (1273)$$

$$\lambda(1273 \text{ K}) = 4.4269 \times 10^{-6} \text{ K}^{-1}.$$

The density of Si at 293 K is 2.329 g cm^{-3} , the density at 1000 $^{\circ}\text{C}$ (1273 K) is thus

$$\begin{aligned}\rho &= \rho_0 + \delta\rho = \rho_0 + (-\rho_0\alpha_v\delta T) = \rho_0 - 3\rho_0\lambda\delta T \\ &= 2.329 \text{ g cm}^{-3} - 3(2.329 \text{ g cm}^{-3})(4.4269 \times 10^{-6} \text{ K}^{-1})(1273 \text{ K} - 293 \text{ K}) = 2.2987 \text{ g cm}^{-3}.\end{aligned}$$

The density changes from 2.329 K to 2.2987 K which is very small change, therefore the density can be roughly assumed as constant.

1.17 Thermal expansion of GaP and GaAs

- a. GaP has the zinc blende structure. The linear expansion coefficient in GaP has been measured as follows: $\lambda = 4.65 \times 10^{-6} \text{ K}^{-1}$ at 300 K; $5.27 \times 10^{-6} \text{ K}^{-1}$ at 500 K; $5.97 \times 10^{-6} \text{ K}^{-1}$ at 800 K. Calculate the coefficients A , B and C in

$$\frac{dL}{L_0 dT} = \lambda(T) = A + B(T - T_0) + C(T - T_0)^2 + \dots$$

where $T_0 = 300 \text{ K}$. The lattice constant of GaP, a , at 27°C is 0.5451 nm . Calculate the lattice constant at 300°C .

- b. The linear expansion coefficient of GaAs over 200 K-1000 K is given by

$$\lambda = 4.25 \times 10^{-6} + (5.82 \times 10^{-9})T - (2.82 \times 10^{-12})T^2 \quad \text{GaAs linear expansion coefficient}$$

where T is in Kelvins. The lattice constant a at 300 K is 0.56533 nm . Calculate the lattice constant and the density at -40°C .

Solution

- a. Given that at

$$\begin{aligned}T = 300 \text{ K}, \quad \lambda &= 4.65 \times 10^{-6} \text{ K}^{-1} \\ T = 500 \text{ K}, \quad \lambda &= 5.27 \times 10^{-6} \text{ K}^{-1} \\ T = 800 \text{ K}, \quad \lambda &= 5.97 \times 10^{-6} \text{ K}^{-1}\end{aligned}$$

Thermal expansion coefficient is written as

$$\lambda(T) = A + B(T - T_0) + C(T - T_0)^2 + \dots$$

Substitute the value of T_0 , and λ for 300 K in the above relation neglecting higher order terms

$$4.65 \times 10^{-6} \text{ K}^{-1} = A + B(300 - 300) + C(300 - 300)^2$$

$$\therefore \quad A = 4.65 \times 10^{-6} \text{ K}^{-1}$$

At 500 K, the equation becomes

$$5.27 \times 10^{-6} \text{ K}^{-1} = 4.65 \times 10^{-6} \text{ K}^{-1} + B(500 - 300) + C(500 - 300)^2$$

$$\text{or} \quad 0.62 \times 10^{-6} \text{ K}^{-1} = 200B + 40 \times 10^3 C \quad (i)$$

At 800 K, the equation becomes

$$5.97 \times 10^{-6} = 4.65 \times 10^{-6} + B(800 - 300) + C(800 - 300)^2$$

$$\text{or} \quad 1.32 \times 10^{-6} = 500B + 250 \times 10^3 C \quad (ii)$$

Solving the two simultaneous equations (i) and (ii) we have,

$$B = 3.41 \times 10^{-9} \text{ K}^{-2}$$

$$\text{and} \quad C = -1.53 \times 10^{-12} \text{ K}^{-3}$$

The equation for $\lambda(T)$ thus becomes

$$\lambda(T) = 4.62 \times 10^{-6} \text{ K}^{-1} + 3.41 \times 10^{-9} \text{ K}^{-2} (T - 300 \text{ K}) - 1.53 \times 10^{-12} \text{ K}^{-3} (T - 300 \text{ K})^2$$

For λ at 300 °C (573 K), we substitute $T = 573 \text{ K}$ we have

$$\begin{aligned} \lambda(573 \text{ K}) &= 4.62 \times 10^{-6} \text{ K}^{-1} + 3.41 \times 10^{-9} \text{ K}^{-2} (573 \text{ K} - 300 \text{ K}) - 1.53 \times 10^{-12} \text{ K}^{-3} (573 - 300 \text{ K})^2 \\ &= 5.44 \times 10^{-6} \text{ K}^{-1}. \end{aligned}$$

The lattice constant a at $T = 573 \text{ K}$ is therefore

$$a = a_0 [1 + \lambda(T - T_0)] = 0.5451 \text{ nm} [1 + 5.44 \times 10^{-6} \text{ K}^{-1} (573 - 300)] = \mathbf{0.5459 \text{ nm}}.$$

b. The coefficient of thermal expansion at -40°C (233 K) is

$$\begin{aligned} \lambda(233 \text{ K}) &= 4.25 \times 10^{-6} + (5.82 \times 10^{-9}) (233 \text{ K}) - (2.82 \times 10^{-12}) (233 \text{ K})^2 \\ &= 5.45 \times 10^{-6} \text{ K}^{-1} \end{aligned}$$

The lattice constant at -40°C is

$$a = a_0 [1 + \lambda(T - T_0)] = 0.56533 \text{ nm} [1 + 5.45 \times 10^{-6} \text{ K}^{-1} (233 - 300)] = \mathbf{0.56512 \text{ nm}}.$$

Referring to the diamond crystal structure in Figure 1Q17-1, we can identify the following types of atoms

- 8 corner atoms labeled C,
- 6 face center atoms (labeled FC) and
- 4 inside atoms labeled 1, 2, 3, 4.

The effective number of atoms within the unit cell is:

$$(8 \text{ Corners}) \times (1/8 \text{ C-atom}) + (6 \text{ Faces}) \times (1/2 \text{ FC-atom}) + 4 \text{ atoms within the cell (1, 2, 3, 4)} = 8$$

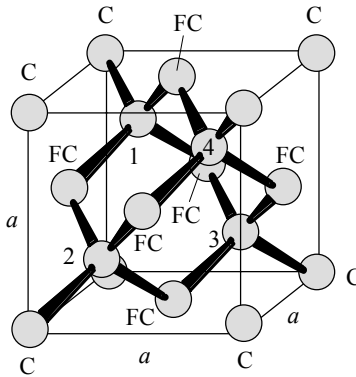


Figure 1Q17-1: The diamond crystal structure.

In the GaAs, it is apparent that there are 4 Ga and 4 As atoms in the unit cell. The concentration of Ga (or As) atoms per unit volume (n_{Ga}) is

$$n_{\text{Ga}} = \frac{4}{a^3}$$

There are $4/a^3$ Ga-As pairs per m^3 . We can calculate the mass of the Ga and As atoms from their relative atomic masses in the Periodic Table using Equation (1) with $M_{\text{at}} = M_{\text{Ga}} = 69.72 \text{ g/mol}$ for Ga and $M_{\text{at}} = M_{\text{As}} = 74.92 \text{ g/mol}$ for As. Thus,

$$\rho = \left(\frac{4}{a^3} \right) \left[\frac{(10^{-3} \text{ kg/g})(M_{\text{Ga}} + M_{\text{As}})}{N_A} \right]$$

or
$$\rho = \left[\frac{4}{(0.56512 \times 10^{-9} \text{ m})^3} \right] \left[\frac{(10^{-3} \text{ kg/g})(69.72 \text{ g/mol} + 74.92 \text{ g/mol})}{6.022 \times 10^{23} \text{ mol}^{-1}} \right]$$

i.e.
$$\rho = 5.323 \times 10^3 \text{ kg m}^{-3} \text{ or } 5.323 \text{ g cm}^{-3}$$

1.18 Electrical noise Consider an amplifier with a bandwidth B of 5 kHz, corresponding to a typical speech bandwidth. Assume the input resistance of the amplifier is 1 M Ω . What is the rms noise voltage at the input? What will happen if the bandwidth is doubled to 10 kHz? What is your conclusion?

Solution

Bandwidth: $B = 5 \times 10^3 \text{ Hz}$

Input resistance: $R_{in} = 10^6 \Omega$

Assuming room temperature: $T = 300 \text{ K}$

The noise voltage (or the rms voltage, v_{rms}) across the input is given by:

$$v_{rms} = \sqrt{4kTR_{in}B} = \sqrt{4(1.381 \times 10^{-23} \text{ J K}^{-1})(300 \text{ K})(10^6 \Omega)(5 \times 10^3 \text{ Hz})}$$

$\therefore v_{rms} = 9.10 \times 10^{-6} \text{ V or } 9.1 \mu\text{V}$

If the bandwidth is doubled: $B' = 2 \times 5 \times 10^3 = 10 \times 10^3 \text{ Hz}$

$\therefore v'_{rms} = \sqrt{4kTR_{in}B'} = 1.29 \times 10^{-5} \text{ V or } 12.9 \mu\text{V}$

The larger the bandwidth, the greater the noise voltage. 5-10 kHz is a typical speech bandwidth and the input signal into the amplifier must be much greater than $\sim 13 \mu\text{V}$ for amplification without noise added from the amplifier itself.

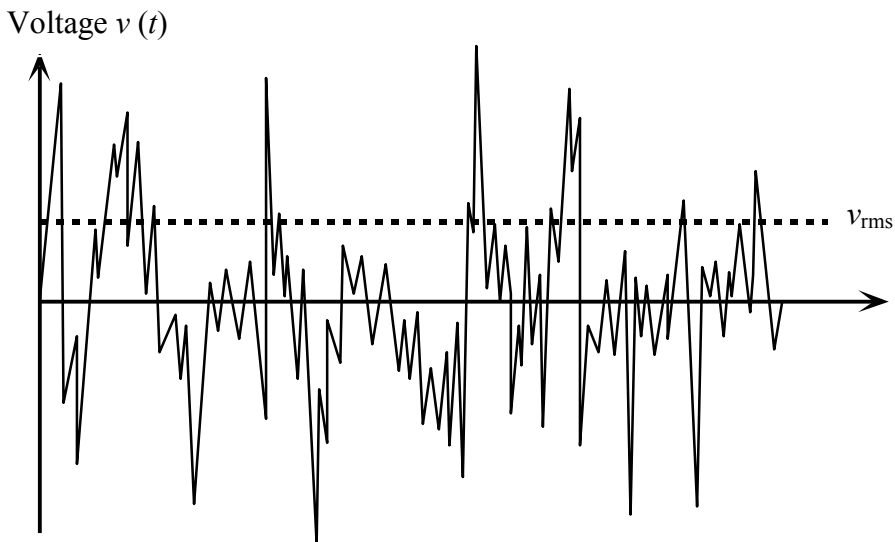


Figure 1Q18-1: Random motion of conduction electrons in a conductor results in electrical noise.

1.19 Thermal activation A certain chemical oxidation process (e.g., SiO₂) has an activation energy of 2 eV atom⁻¹.

- Consider the material exposed to pure oxygen gas at a pressure of 1 atm. at 27 °C. Estimate how many oxygen molecules per unit volume will have energies in excess of 2 eV? (Consider numerical integration of Equation 1.24)
- If the temperature is 900 °C, estimate the number of oxygen molecules with energies more than 2 eV. What happens to this concentration if the pressure is doubled?

Solution 1: Method of Estimation

The activation energy (E_A) is 2 eV per atom or

$$E_A = (2 \text{ eV/atom})(1.602 \times 10^{-19} \text{ J/eV}) = 3.204 \times 10^{-19} \text{ J/atom}$$

a. We are given pressure $P = 1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$ and temperature $T = 300 \text{ K}$. If we consider a portion of oxygen gas of volume $V = 1 \text{ m}^3$ (the unit volume), the number of molecules present in the gas (N) will be equal to the concentration of molecules in the gas (n), i.e.: $N = n$. And since we know $\eta = N/N_A$, where η is the total number of moles and N_A is Avogadro's number, we can make the following substitution into the equation $PV = \eta RT$:

$$PV = \frac{n}{N_A} RT$$

isolating n ,

$$n = \frac{N_A PV}{RT} = \frac{(6.022 \times 10^{23} \text{ mol}^{-1})(1.013 \times 10^5 \text{ Pa})(1 \text{ m}^3)}{(8.315 \text{ m}^3 \text{ Pa K}^{-1} \text{ mol}^{-1})(300 \text{ K})} \left(\frac{1}{\text{m}^3} \right)$$

$$\therefore n = 2.445 \times 10^{25} \text{ m}^{-3}$$

Therefore there are 2.445×10^{25} oxygen molecules per unit volume.

For an estimation of the concentration of molecules with energy above 2 eV, we can use the following approximation (remember to convert E_A into Joules). If n_A is the concentration of molecules with $E > E_A$, then:

$$\frac{n_A}{n} = \exp\left(-\frac{E_A}{kT}\right)$$

$$\therefore n_A = n \exp\left(-\frac{E_A}{kT}\right) = (2.445 \times 10^{25} \text{ m}^{-3}) \exp\left[-\frac{(3.204 \times 10^{-19} \text{ J})}{(1.381 \times 10^{-23} \text{ J K}^{-1})(300 \text{ K})}\right]$$

$$\therefore n_A = 6.34 \times 10^{-9} \text{ m}^{-3}$$

However, this answer is only in the right order of magnitude. For a better calculation we need to use a numerical integration of $n(E)$ from E_A to ∞ .

b. At $T = 900 \text{ °C} + 273 = 1173 \text{ K}$ and $P = 1 \text{ atm}$, the same method as above can be used to find the concentration of molecules with energy greater than E_A . After calculations, the following numbers will be obtained:

$$n = 6.254 \times 10^{24} \text{ m}^{-3}$$

$$n_A = 1.61 \times 10^{16} \text{ m}^{-3}$$

This corresponds to an increase by a factor of 10^{25} compared to a temperature of $T = 300 \text{ K}$.

Doubling the pressure doubles n and hence doubles n_A . In the oxidation of Si wafers, high pressures lead to more rapid oxidation rates and a shorter time for the oxidation process.

Solution 2: Method of Numerical Integration

To find the number of molecules with energies greater than $E_A = 2 \text{ eV}$ more accurately, numerical integration must be used. Suppose that N is the total number of molecules. Let

$$y = n_E/N$$

where n_E is the number of molecules per unit energy, so that $n_E dE$ is the number of molecules in the energy range dE .

Define a new variable x for the sake of convenience:

$$x = \frac{E}{kT}$$

$$\therefore E = kTx$$

where E is the energy of the molecules. Using partial differentiation:

$$dE = kT dx$$

The energy distribution function is given by:

$$y = \frac{2}{\pi^{\frac{1}{2}}} \left(\frac{1}{kT} \right)^{\frac{3}{2}} E^{\frac{1}{2}} \exp\left(-\frac{E}{kT}\right)$$

substitute:
$$y = 2 \frac{\left(\frac{1}{kT} \right)^{\frac{3}{2}} \exp(-x)}{\sqrt{\pi}} \sqrt{kTx}$$

simplify:
$$y = 2 \frac{\exp(-x)\sqrt{x}}{\sqrt{\pi kT}}$$

Integration of y with respect to E can be changed to that with respect to x as follows:

$$\int y dE = 2 \frac{\int \exp(-x)\sqrt{x} dE}{\sqrt{\pi kT}}$$

$$\therefore \int y dE = 2 \frac{\int \exp(-x)\sqrt{x} dx}{\sqrt{\pi}} \quad (\text{substitute } dE = kT dx)$$

We need the lower limit x_A for x corresponding to E_A

$$x_A = \frac{E_A}{kT} = \frac{3.204 \times 10^{-19} \text{ J}}{(1.381 \times 10^{-23} \text{ J K}^{-1})(300 \text{ K})} = 77.36$$

This is the activation energy E_A in terms of x .

We also need the upper limit, x_B which should be ∞ , but we will take it to be a multiple of:

$$x_B = 2 \times x_A = 154.72$$

We do this because the numerical integration is difficult with very small numbers, e.g. $\exp(-x_A) = 2.522 \times 10^{-34}$.

The fraction, F , of molecules with energies greater than E_A ($= x_A$) is:

$$F = \int_{E_A}^{\infty} y dE = \int_{x_A}^{x_B} \left(2 \frac{\exp(-x)\sqrt{x}}{\sqrt{\pi}} \right) dx = 2.519 \times 10^{-33}$$

At $T = 300 \text{ K}$, $P = 1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$, and $V = 1 \text{ m}^3$, the number of molecules per unit volume is n :

$$PV = \frac{n}{N_A} RT$$

$$\therefore n = \frac{N_A PV}{RT} = 2.445 \times 10^{25} \text{ m}^{-3}$$

The concentration of molecules with energy greater than E_A (n_A) can be found using the fraction F :

$$n_A = nF = (2.445 \times 10^{25} \text{ m}^{-3})(2.519 \times 10^{-33}) = \mathbf{6.16 \times 10^{-8} \text{ m}^{-3}}$$

If we estimate n_A by multiplying n by the Boltzmann factor as previously:

$$n_A = n \exp\left(-\frac{E_A}{kT}\right) = 6.34 \times 10^{-9} \text{ m}^{-3}$$

The estimate is out by a factor of about 10.

b. The concentration n_A of molecules with energies greater than E_A can be found at $T = 900^\circ\text{C} + 273 = 1173 \text{ K}$, using the same method as in part **a**. After calculations, the following values will be obtained:

$$F = 1.3131 \times 10^{-8}$$

$$n = 6.254 \times 10^{24} \text{ m}^{-3}$$

$$\mathbf{n_A = 8.21 \times 10^{16} \text{ m}^{-3}}$$

If we compare this value to the one obtained previously through estimation ($1.61 \times 10^{16} \text{ m}^{-3}$), we see that the estimate is out by a factor of about 5. As stated previously, doubling the pressure doubles n and hence doubles n_A . In the oxidation of Si wafers, high pressures lead to more rapid oxidation rates and a shorter time for the oxidation process.

1.20 Diffusion in Si The diffusion coefficient of boron (B) atoms in a single crystal of Si has been measured to be $1.5 \times 10^{-18} \text{ m}^2 \text{ s}^{-1}$ at 1000°C and $1.1 \times 10^{-16} \text{ m}^2 \text{ s}^{-1}$ at 1200°C .

a. What is the activation energy for the diffusion of B, in eV/atom?

b. What is the preexponential constant D_o ?

- c. What is the rms distance (in micrometers) diffused in 1 hour by the B atom in the Si crystal at 1200 °C and 1000 °C?
- d. The diffusion coefficient of B in polycrystalline Si has an activation energy of 2.4-2.5 eV/atom and $D_o = (1.5-6) \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$. What constitutes the diffusion difference between the single crystal sample and the polycrystalline sample?

Solution

Given diffusion coefficients at two temperatures:

$$T_1 = 1200 \text{ °C} + 273 = 1473 \text{ K} \quad D_1 = 1.1 \times 10^{-16} \text{ m}^2/\text{s}$$

$$T_2 = 1000 \text{ °C} + 273 = 1273 \text{ K} \quad D_2 = 1.5 \times 10^{-18} \text{ m}^2/\text{s}$$

a. The diffusion coefficients D_1 and D_2 at T_1 and T_2 respectively are given by:

$$D_1 = D_o \exp\left(-\frac{qE_A}{kT_1}\right) \quad D_2 = D_o \exp\left(-\frac{qE_A}{kT_2}\right)$$

where E_A is the activation energy in eV/atom and $q = 1.6 \times 10^{-19} \text{ J/eV}$ (conversion factor from eV to J). Since,

$$\frac{\exp(-x)}{\exp(-y)} = \exp(y-x)$$

we can take the ratio of the diffusion coefficients to express them in terms of E_A (eV):

$$\frac{D_1}{D_2} = \frac{D_o \exp\left(-\frac{qE_A}{kT_1}\right)}{D_o \exp\left(-\frac{qE_A}{kT_2}\right)} = \exp\left(\frac{qE_A}{kT_2} - \frac{qE_A}{kT_1}\right) = \exp\left(\frac{q[T_1 - T_2]E_A}{kT_1T_2}\right)$$

$$\therefore E_A = \frac{kT_1T_2 \ln\left(\frac{D_1}{D_2}\right)}{q(T_1 - T_2)} \quad (\text{in eV})$$

$$\therefore E_A = \frac{(1.381 \times 10^{-23} \text{ J K}^{-1})(1473 \text{ K})(1273 \text{ K}) \ln\left(\frac{1.1 \times 10^{-16} \text{ m}^2/\text{s}}{1.5 \times 10^{-18} \text{ m}^2/\text{s}}\right)}{(1.602 \times 10^{-19} \text{ J/eV})(1473 \text{ K} - 1273 \text{ K})}$$

$$\therefore E_A = 3.47 \text{ eV/atom}$$

b. To find D_o , use one of the equations for the diffusion coefficients:

$$D_1 = D_o \exp\left(-\frac{qE_A}{kT_1}\right)$$

$$\therefore D_o = \frac{D_1}{\exp\left(-\frac{qE_A}{kT_1}\right)} = \frac{(1.1 \times 10^{-16} \text{ m}^2/\text{s})}{\exp\left[-\frac{(1.602 \times 10^{-19} \text{ J/eV})(3.47 \text{ eV})}{(1.381 \times 10^{-23} \text{ J K}^{-1})(1473 \text{ K})}\right]}$$

$$\therefore D_0 = 8.12 \times 10^{-5} \text{ m}^2/\text{s}$$

c. Given: time (t) = (1 hr) $\times$ (3600 s/hr) = 3600 s

At 1000 °C, rms diffusion distance ($L_{1000 \text{ °C}}$) in time t is given by:

$$L_{1000 \text{ °C}} = \sqrt{2D_2t} = \sqrt{2(1.5 \times 10^{-18} \text{ m}^2/\text{s})(3600 \text{ s})}$$

$$\therefore L_{1000 \text{ °C}} = 1.04 \times 10^{-7} \text{ m or } 0.104 \text{ }\mu\text{m}$$

At 1200 °C:

$$L_{1200 \text{ °C}} = \sqrt{2D_1t} = \sqrt{2(1.1 \times 10^{-16} \text{ m}^2/\text{s})(3600 \text{ s})}$$

$$\therefore L_{1200 \text{ °C}} = 8.90 \times 10^{-7} \text{ m or } 0.89 \text{ }\mu\text{m (almost 10 times longer than at 1000 °C)}$$

d. Diffusion in polycrystalline Si would involve diffusion along grain boundaries, which is easier than diffusion in the bulk. The activation energy is smaller because it is easier for an atom to break bonds and jump to a neighboring site; there are vacancies or voids, broken bonds, and strained bonds in a grain boundary.

1.21 Diffusion in SiO₂ The diffusion coefficient of P atoms in SiO₂ has an activation energy of 2.30 eV/atom and $D_0 = 5.73 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$. What is the rms distance diffused in 1 hour by P atoms in SiO₂ at 1200 °C?

Solution

Given: Temperature: $T = 1200 \text{ °C} + 273 = 1473 \text{ K}$

Diffusion coefficient: $D_0 = 5.73 \times 10^{-9} \text{ m}^2/\text{s}$

Activation energy: $E_A = 2.30 \text{ eV/atom}$

Time: $t = (1 \text{ hr}) \times (3600 \text{ s/hr}) = 3600 \text{ s}$

Using the following equation, find the diffusion coefficient D :

$$D = D_0 \exp\left(-\frac{E_A}{kT}\right)$$

$$\therefore D = (5.73 \times 10^{-9} \text{ m}^2/\text{s}) \exp\left(-\frac{(2.30 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})}{(1.381 \times 10^{-23} \text{ J K}^{-1})(1473 \text{ K})}\right) = 7.793 \times 10^{-17} \text{ m}^2/\text{s}$$

The rms diffusion distance ($L_{1200 \text{ °C}}$) is given as:

$$L_{1200 \text{ °C}} = \sqrt{2Dt} = \sqrt{2(7.793 \times 10^{-17} \text{ m}^2/\text{s})(3600 \text{ s})}$$

$$\therefore L_{1200 \text{ °C}} = 7.49 \times 10^{-7} \text{ m or } 0.75 \text{ }\mu\text{m}$$

Comment: The P atoms at 1200 °C seem to be able to diffuse through an oxide thickness of about 0.75 μm .

1.22 BCC and FCC crystals

- a. Molybdenum has the BCC crystal structure, has a density of 10.22 g cm^{-3} and an atomic mass of 95.94 g mol^{-1} . What is the atomic concentration, lattice parameter a , and atomic radius of molybdenum?
- b. Gold has the FCC crystal structure, a density of 19.3 g cm^{-3} and an atomic mass of $196.97 \text{ g mol}^{-1}$. What is the atomic concentration, lattice parameter a , and atomic radius of gold?

Solution

a. Since molybdenum has BCC crystal structure, there are 2 atoms in the unit cell. The density is

$$\rho = \frac{\text{Mass of atoms in unit cell}}{\text{Volume of unit cell}} = \frac{(\text{Number of atoms in unit cell}) \times (\text{Mass of one atom})}{\text{Volume of unit cell}}$$

that is,
$$\rho = \frac{2 \left(\frac{M_{at}}{N_A} \right)}{a^3}$$

Solving for the lattice parameter a we receive

$$a = \sqrt[3]{\frac{2M_{at}}{\rho N_A}} = \sqrt[3]{\frac{2(95.94 \times 10^{-3} \text{ kg mol}^{-1})}{(10.22 \times 10^3 \text{ kg m}^{-3})(6.022 \times 10^{23} \text{ mol}^{-1})}} = 3.147 \times 10^{-10} \text{ m} = 0.3147 \text{ nm}$$

The Atomic concentration is 2 atoms in a cube of volume a^3 , i.e.

$$n_{at} = \frac{2}{a^3} = \frac{2}{(3.147 \times 10^{-10} \text{ m})^3} = 6.415 \times 10^{22} \text{ cm}^{-3} = 6.415 \times 10^{28} \text{ m}^{-3}$$

For a BCC cell, the lattice parameter a and the radius of the atom R are in the following relation (listed in Table 1.3):

$$R = \frac{a\sqrt{3}}{4} = \frac{(3.147 \times 10^{-10} \text{ m})\sqrt{3}}{4} = 1.363 \times 10^{-10} \text{ m} = 0.1363 \text{ nm}$$

b. Gold has the FCC crystal structure, hence, there are 4 atoms in the unit cell (as shown in Table 1.3).

The lattice parameter a is

$$a = \left[\frac{4M_{at}}{\rho N_A} \right]^{1/3} = \left[\frac{4(196.97 \times 10^{-3} \text{ kg mol}^{-1})}{(19.3 \times 10^3 \text{ kg m}^{-3})(6.022 \times 10^{23} \text{ mol}^{-1})} \right]^{1/3} = 4.077 \times 10^{-10} \text{ m} = 0.4077 \text{ nm}$$

The atomic concentration is

$$n_{at} = \frac{4}{a^3} = \frac{4}{(4.077 \times 10^{-10} \text{ m})^3} = 5.901 \times 10^{22} \text{ cm}^{-3} = 5.901 \times 10^{28} \text{ m}^{-3}$$

For an FCC cell, the lattice parameter a and the radius of the atom R are in the following relation (shown in Table 1.3):

$$R = \frac{a\sqrt{2}}{4} = \frac{(4.077 \times 10^{-10} \text{ m})\sqrt{2}}{4} = 1.442 \times 10^{-10} \text{ m} = 0.1442 \text{ nm}$$

1.23 BCC and FCC Crystals

- a. Tungsten (W) has the BCC crystal structure. The radius of the W atom is 0.1371 nm. The atomic mass of W is 183.8 amu (g mol^{-1}). Calculate the number of W atoms per unit volume and density of W.
- b. Platinum (Pt) has the FCC crystal structure. The radius of the Pt atom is 0.1386 nm. The atomic mass of Pt is 195.09 amu (g mol^{-1}). Calculate the number of Pt atoms per unit volume and density of Pt.

Solution

a. Consider a cube diagonal with two corner atoms and the central atom in contact. The length of the diagonal is $4R$. Therefore

$$a^2 + a^2 + a^2 = (4R)^2$$

or
$$a = \frac{4}{\sqrt{3}} R$$

Thus for W, $a = \frac{4}{\sqrt{3}} (0.1371 \text{ nm}) = 0.3166 \text{ nm}$

In the BCC structure, the total number of atoms = $8\left(\frac{1}{8}\right) + 1 = 2$ atoms.

The density of W is therefore

$$\begin{aligned} \rho &= \frac{\text{Mass of atoms in unit cell}}{\text{Volume of unit cell}} = \frac{2\left(\frac{M_{at}}{N_A}\right)}{a^3} = \frac{2\left(\frac{183.8 \times 10^{-3} \text{ kg mol}^{-1}}{6.022 \times 10^{23} \text{ mol}^{-1}}\right)}{(0.3166 \times 10^{-9} \text{ m})^3} \\ &= 19.23 \times 10^3 \text{ kg m}^{-3} \text{ or } 19.23 \text{ g cm}^{-3} \end{aligned}$$

The atomic concentration of W with 2 atoms per unit cell is

$$\begin{aligned} n_{at} &= \frac{\text{Number of atoms per unit cell}}{\text{Volume of unit cell}} = \frac{2}{a^3} = \frac{2}{(0.3166 \times 10^{-9} \text{ m})^3} \\ &= 6.303 \times 10^{28} \text{ m}^{-3} \text{ or } 6.303 \times 10^{22} \text{ cm}^{-3} \end{aligned}$$

b. Consider a face diagonal with corner atoms and the central atom in contact. The length of the diagonal is $4R$. Therefore

$$a^2 + a^2 = (4R)^2$$

or
$$a = \frac{4}{\sqrt{2}} R = 2\sqrt{2} R$$

Thus for Pt, $a = 2\sqrt{2} (0.1386 \text{ nm}) = 0.3920 \text{ nm}$

In the FCC structure, the total number of atoms = $8\left(\frac{1}{8}\right) + 6\left(\frac{1}{2}\right) = 4$ atoms

The density of Pt is therefore

$$\rho = \frac{\text{Mass of atoms in unit cell}}{\text{Volume of unit cell}} = \frac{4 \left(\frac{M_{at}}{N_A} \right)}{a^3} = \frac{4 \left(\frac{195.09 \times 10^{-3} \text{ kg mol}^{-1}}{6.022 \times 10^{23} \text{ mol}^{-1}} \right)}{(0.3920 \times 10^{-9} \text{ m})^3}$$

$$= 21.51 \times 10^3 \text{ kg m}^{-3} \text{ or } 21.51 \text{ g cm}^{-3}.$$

The atomic concentration of Pt with 4 atoms per unit cell is

$$n_{at} = \frac{\text{Number of atoms per unit cell}}{\text{Volume of unit cell}} = \frac{4}{a^3} = \frac{2}{(0.3920 \times 10^{-9} \text{ m})^3}$$

$$= 6.64 \times 10^{28} \text{ m}^{-3} \text{ or } 6.64 \times 10^{22} \text{ cm}^{-3}.$$

1.24 Planar and surface concentrations Niobium (Nb) has the BCC crystal with a lattice parameter $a = 0.3294 \text{ nm}$. Find the planar concentrations as the number of atoms per nm^2 of the (100), (110) and (111) planes. Which plane has the most concentration of atoms per unit area? Sometimes the number of atoms per unit area n_{surface} on the surface of a crystal is estimated by using the relation $n_{\text{surface}} = n_{\text{bulk}}^{2/3}$ where n_{bulk} is the concentration of atoms in the bulk. Compare n_{surface} values with the planar concentrations that you calculated and comment on the difference. [Note: The BCC (111) plane does not cut through the center atom and the (111) has one-sixth of an atom at each corner.]

Solution

Planar concentration (or density) is the number of atoms per unit area on a given plane in the crystal. It is the surface concentration of atoms on a given plane. To calculate the planar concentration $n_{(hkl)}$ on a given (hkl) plane, we consider a bound area A . Only atoms whose centers lie on A are involved in the calculation of $n_{(hkl)}$. For each atom, we then evaluate what portion of the atomic cross section cut by the plane (hkl) is contained within A .

For the BCC crystalline structure the planes (100), (110) and (111) are drawn in Figure 1Q24-1.

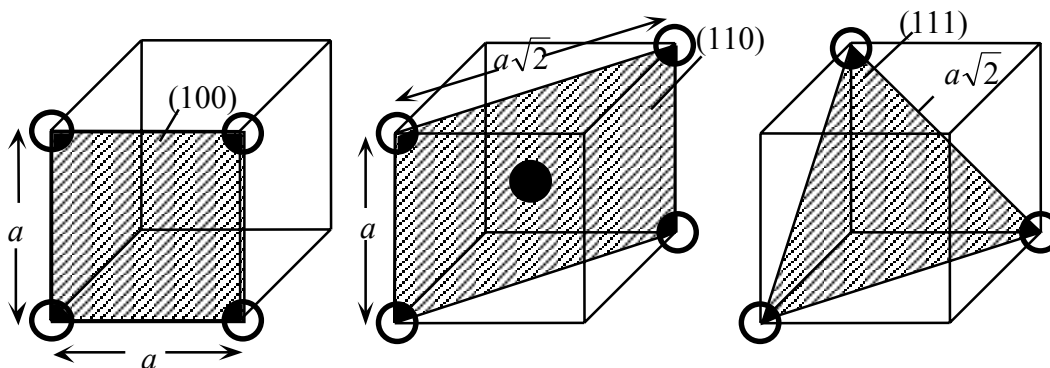


Figure 1Q24-1: (100), (110), (111) planes in the BCC crystal

Consider the (100) plane.

Number of atoms in the area $a \times a$, which is the cube face = (4 corners) $\times$ (1/4th atom at corner) = 1.

Planar concentration is

$$n_{(100)} = \frac{4\left(\frac{1}{4}\right)}{a^2} = \frac{1}{(3.294 \times 10^{-10} \text{ m})^2} = 9.216 \times 10^{18} \text{ atoms m}^{-2}$$

The most populated plane for BCC structure is (110).

Number of atoms in the area $a \times a\sqrt{2}$ defined by two face-diagonals and two cube-sides

$$= (4 \text{ corners}) \times (1/4 \text{th atom at corner}) + 1 \text{ atom at face center} = 2$$

Planar concentration is

$$n_{(110)} = \frac{4\left(\frac{1}{4}\right) + 1}{a^2 \sqrt{2}} = \frac{2}{(3.294 \times 10^{-10} \text{ m})^2 \sqrt{2}} = 1.303 \times 10^{19} \text{ atoms m}^{-2}$$

The plane (111) for the BCC structure is the one with rarest population. The area of interest is an equilateral triangle defined by face diagonals of length $a\sqrt{2}$ (see Figure 1Q24-1). The height of the triangle is $a\sqrt{\frac{3}{2}}$ so that the triangular area is $\frac{1}{2} \times a\sqrt{2} \times a\sqrt{\frac{3}{2}} = \frac{a^2 \sqrt{3}}{2}$. An atom at a corner only contributes a fraction ($60^\circ/360^\circ = 1/6$) to this area.

So, the planar concentration is

$$n_{(111)} = \frac{\frac{1}{6}(3)}{\frac{a^2 \sqrt{3}}{2}} = \frac{1}{a^2 \sqrt{3}} = \frac{1}{(3.294 \times 10^{-10} \text{ m})^2 \sqrt{3}} = 5.321 \times 10^{18} \text{ atoms m}^{-2}$$

For the BCC structure there are two atoms in unit cell and the bulk atomic concentration is

$$n_{\text{bulk}} = \frac{\text{number of atoms in unit cell}}{\text{volume of the cell}} = \frac{2}{a^3} = \frac{2}{(3.294 \times 10^{-10} \text{ m})^3} = 5.596 \times 10^{28} \text{ atoms m}^{-3}$$

and the surface concentration is

$$n_{\text{surface}} = (n_{\text{bulk}})^{\frac{2}{3}} = (5.596 \times 10^{28} \text{ m}^{-3})^{\frac{2}{3}} = 1.463 \times 10^{19} \text{ atoms m}^{-2}$$

1.25 Diamond and zinc blende Si has the diamond and GaAs has the zinc blende crystal structure. Given the lattice parameters of Si and GaAs, $a = 0.543 \text{ nm}$ and $a = 0.565 \text{ nm}$, respectively, and the atomic masses of Si, Ga, and As as 28.08, 69.73 g/mol, and 74.92, respectively, calculate the density of Si and GaAs. What is the atomic concentration (atoms per unit volume) in each crystal?

Solution

Referring to the diamond crystal structure in Figure 1Q25-1, we can identify the following types of atoms

8 corner atoms labeled C,

6 face center atoms (labeled FC) and
4 inside atoms labeled 1,2,3,4.

The effective number of atoms within the unit cell is:

$$(8 \text{ Corners}) \times (1/8 \text{ C-atom}) + (6 \text{ Faces}) \times (1/2 \text{ FC-atom}) + 4 \text{ atoms within the cell (1, 2, 3, 4)} = 8$$

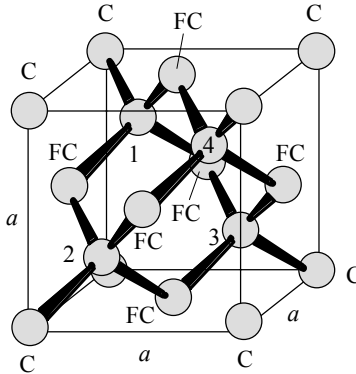


Figure 1Q25-1: The diamond crystal structure.

The lattice parameter (lengths of the sides of the unit cell) of the unit cell is a . Thus the atomic concentration in the Si crystal (n_{Si}) is

$$n_{\text{Si}} = \frac{8}{a^3} = \frac{8}{(0.543 \times 10^{-9} \text{ m})^3} = 5.0 \times 10^{28} \text{ atoms per m}^{-3}$$

If M_{at} is the atomic mass in the Periodic Table then the mass of the atom (m_{at}) in kg is

$$m_{\text{at}} = (10^{-3} \text{ kg/g}) M_{\text{at}} / N_A \quad (1)$$

where N_A is Avogadro's number. For Si, $M_{\text{at}} = M_{\text{Si}} = 28.09 \text{ g/mol}$, so then the density of Si is

$$\rho = (\text{number of atoms per unit volume}) \times (\text{mass per atom}) = n_{\text{Si}} m_{\text{at}}$$

or

$$\rho = \left(\frac{8}{a^3} \right) \left[\frac{(10^{-3} \text{ kg/g}) M_{\text{Si}}}{N_A} \right]$$

i.e.

$$\rho = \left[\frac{8}{(0.543 \times 10^{-9} \text{ m})^3} \right] \left[\frac{(10^{-3} \text{ kg/g})(28.09 \text{ g mol}^{-1})}{6.022 \times 10^{23} \text{ mol}^{-1}} \right] = 2.33 \times 10^3 \text{ kg m}^{-3} \text{ or } 2.33 \text{ g cm}^{-3}$$

In the case of GaAs, it is apparent that there are 4 Ga and 4 As atoms in the unit cell. The concentration of Ga (or As) atoms per unit volume (n_{Ga}) is

$$n_{\text{Ga}} = \frac{4}{a^3} = \frac{4}{(0.565 \times 10^{-9} \text{ m})^3} = 2.22 \times 10^{28} \text{ m}^{-3}$$

Total atomic concentration (counting both Ga and As atoms) is twice n_{Ga} .

$$n_{\text{Total}} = 2n_{\text{Ga}} = 4.44 \times 10^{28} \text{ m}^{-3}$$

There are 2.22×10^{28} Ga-As pairs per m^3 . We can calculate the mass of the Ga and As atoms from their relative atomic masses in the Periodic Table using Equation (1) with $M_{\text{at}} = M_{\text{Ga}} = 69.72 \text{ g/mol}$ for Ga and $M_{\text{at}} = M_{\text{As}} = 74.92 \text{ g/mol}$ for As. Thus,

$$\rho = \left(\frac{4}{a^3} \right) \left[\frac{(10^{-3} \text{ kg/g})(M_{\text{Ga}} + M_{\text{As}})}{N_A} \right]$$

or

$$\rho = \left[\frac{4}{(0.565 \times 10^{-9} \text{ m})^3} \right] \left[\frac{(10^{-3} \text{ kg/g})(69.72 \text{ g/mol} + 74.92 \text{ g/mol})}{6.022 \times 10^{23} \text{ mol}^{-1}} \right]$$

i.e. $\rho = 5.33 \times 10^3 \text{ kg m}^{-3}$ or 5.33 g cm^{-3}

1.26 Zinc blende, NaCl and CsCl

- InAs is a III-V semiconductor that has the zinc blende structure with a lattice parameter of 0.606 nm. Given the atomic masses of In (114.82 g mol⁻¹) and As (74.92 g mol⁻¹), find the density.
- CdO has the NaCl crystal structure with a lattice parameter of 0.4695 nm. Given the atomic masses of Cd (112.41 g mol⁻¹) and O (16.00 g mol⁻¹), find the density.
- KCl has the same crystal structure as NaCl. The lattice parameter a of KCl is 0.629 nm. The atomic masses of K and Cl are 39.10 g mol⁻¹ and 35.45 g mol⁻¹ respectively. Calculate the density of KCl.

Solution

a. For zinc blende structure there are 8 atoms per unit cell (as shown in Table 1.3). In the case of InAs, it is apparent that there are 4 In and 4 As atoms in the unit cell. The density of InAs is then

$$\begin{aligned} \rho_{\text{InAs}} &= \frac{4 \left(\frac{M_{\text{atIn}}}{N_A} \right) + 4 \left(\frac{M_{\text{atAs}}}{N_A} \right)}{a^3} = \frac{4(M_{\text{atIn}} + M_{\text{atAs}})}{N_A a^3} = \\ &= \frac{4(114.82 + 74.92) \times (10^{-3} \text{ kg mol}^{-1})}{(6.022 \times 10^{23} \text{ mol}^{-1})(0.606 \times 10^{-9} \text{ m})^3} = 5.66 \times 10^3 \text{ kg m}^{-3} = 5.66 \text{ g cm}^{-3} \end{aligned}$$

b. For NaCl crystal structure, there are 4 cations and 4 anions per unit cell. For the case of CdO we have 4 Cd atoms and 4 O atoms per unit cell and the density of CdO is

$$\begin{aligned} \rho_{\text{CdO}} &= \frac{4 \left(\frac{M_{\text{atCd}}}{N_A} \right) + 4 \left(\frac{M_{\text{atO}}}{N_A} \right)}{a^3} = \frac{4(M_{\text{atCd}} + M_{\text{atO}})}{N_A a^3} = \\ &= \frac{4(112.41 + 16.00) \times (10^{-3} \text{ kg mol}^{-1})}{(6.022 \times 10^{23} \text{ mol}^{-1})(0.4695 \times 10^{-9} \text{ m})^3} = 8.24 \times 10^3 \text{ kg m}^{-3} = 8.24 \text{ g cm}^{-3} \end{aligned}$$

c. Similarly to **b**, for the density of KCl we receive

$$\begin{aligned} \rho_{\text{KCl}} &= \frac{4 \left(\frac{M_{\text{atK}}}{N_A} \right) + 4 \left(\frac{M_{\text{atCl}}}{N_A} \right)}{a^3} = \frac{4(M_{\text{atK}} + M_{\text{atCl}})}{N_A a^3} = \\ &= \frac{4(39.1 + 35.45) \times (10^{-3} \text{ kg mol}^{-1})}{(6.022 \times 10^{23} \text{ mol}^{-1})(0.629 \times 10^{-9} \text{ m})^3} = 1.99 \times 10^3 \text{ kg m}^{-3} = 1.99 \text{ g cm}^{-3} \end{aligned}$$

1.27 Crystallographic directions and planes Consider the cubic crystal system.

- a. Show that the line $[hkl]$ is perpendicular to the (hkl) plane.
 b. Show that the spacing between adjacent (hkl) planes is given by

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

Solution

This problem assumes that students are familiar with three dimensional geometry and vector products.

Figure 1Q27-1(a) shows a typical $[hkl]$ line, labeled as **ON**, and a (hkl) plane in a cubic crystal. $\mathbf{u}_x$, $\mathbf{u}_y$ and $\mathbf{u}_z$ are the unit vectors along the x , y , z coordinates. This is a cubic lattice so we have Cartesian coordinates and $\mathbf{u}_x \cdot \mathbf{u}_x = 1$ and $\mathbf{u}_x \cdot \mathbf{u}_y = 0$ etc.

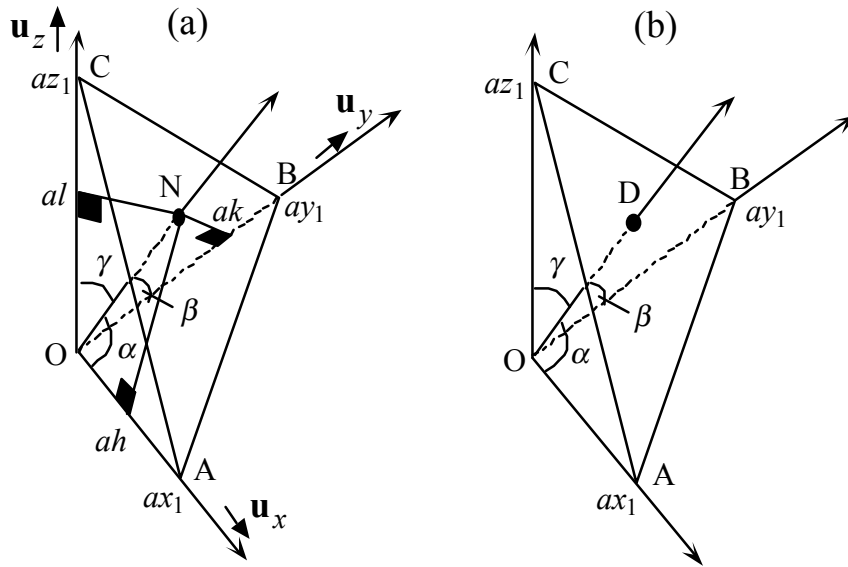


Figure 1Q27-1: Crystallographic directions and planes

a. Given a = lattice parameter, then from the definition of Miller indices ($h = 1/x_1$, $k = 1/y_1$ and $l = 1/z_1$), the plane has intercepts: $x_o = ax_1 = a/h$; $y_o = ay_1 = a/k$; $z_o = az_1 = a/l$.

The vector $\mathbf{ON} = ah\mathbf{u}_x + ak\mathbf{u}_y + al\mathbf{u}_z$

If **ON** is perpendicular to the (hkl) plane then the product of this vector with any vector in the (hkl) plane will be zero. We only have to choose 2 non-parallel vectors (such as **AB** and **BC**) in the plane and show that the dot product of these with **ON** is zero.

$$\mathbf{AB} = \mathbf{OB} - \mathbf{OA} = (a/k)\mathbf{u}_y - (a/h)\mathbf{u}_x$$

$$\mathbf{ON} \cdot \mathbf{AB} = (ah\mathbf{u}_x + ak\mathbf{u}_y + al\mathbf{u}_z) \cdot ((a/k)\mathbf{u}_y - (a/h)\mathbf{u}_x) = a^2 - a^2 = 0$$

Recall that $\mathbf{u}_x \cdot \mathbf{u}_x = \mathbf{u}_y \cdot \mathbf{u}_y = 1$ and $\mathbf{u}_x \cdot \mathbf{u}_y = \mathbf{u}_x \cdot \mathbf{u}_z = \mathbf{u}_y \cdot \mathbf{u}_z = 0$

Similarly, $\mathbf{ON} \cdot \mathbf{BC} = (ah\mathbf{u}_x + ak\mathbf{u}_y + al\mathbf{u}_z) \cdot ((a/l)\mathbf{u}_z - (a/k)\mathbf{u}_y) = 0$

Therefore **ON** or $[hkl]$ is normal to the (hkl) plane.

b. Suppose that **OD** is the normal from the plane to the origin as shown in Figure 1Q27-1(b). Shifting a plane by multiples of lattice parameters does not change the miller indices. We can therefore assume the adjacent plane passes through O. The separation between the adjacent planes is then simply the distance **OD** in Figure 1Q27-1(b).

Let α , β and γ be the angles of **OD** with the x , y and z axes. Consider the direction cosines of the line **OD**:

$$\cos\alpha = d/(ax_1) = dh/a; \quad \cos\beta = d/(ay_1) = dk/a; \quad \cos\gamma = d/(az_1) = dl/a$$

But, in 3 dimensions, $(\cos\alpha)^2 + (\cos\beta)^2 + (\cos\gamma)^2 = 1$

Thus, $(d^2h^2/a^2) + (d^2k^2/a^2) + (d^2l^2/a^2) = 1$

Rearranging, $d^2 = a^2 / [h^2 + k^2 + l^2]$

or, $d = a / [h^2 + k^2 + l^2]^{1/2}$

1.28 Si and SiO₂

- Given the Si lattice parameter $a = 0.543$ nm, calculate the number of Si atoms per unit volume, in nm^{-3} .
- Calculate the number of atoms per m^2 and per nm^2 on the (100), (110) and (111) planes in the Si crystal as shown on Figure 1.75. Which plane has the most number of atoms per unit area?
- The density of SiO₂ is 2.27 g cm^{-3} . Given that its structure is amorphous, calculate the number of molecules per unit volume, in nm^{-3} . Compare your result with (a) and comment on what happens when the surface of an Si crystal oxidizes. The atomic masses of Si and O are 28.09 and 16, respectively.

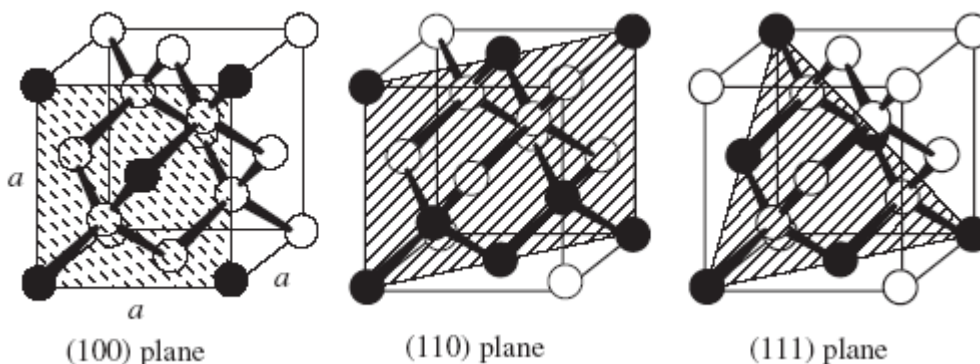


Figure 1.75: Diamond cubic crystal structure and planes. Determine what portion of a black-colored atom belongs to the plane that is hatched.

Solution

a. Si has the diamond crystal structure with 8 atoms in the unit cell, and we are given the lattice parameter $a = 0.543 \times 10^{-9} \text{ m}$ and atomic mass $M_{at} = 28.09 \times 10^{-3} \text{ kg/mol}$. The concentration of atoms per unit volume (n) in nm^{-3} is therefore:

$$n = \frac{8}{a^3} \frac{1}{(10^9 \text{ nm/m})^3} = \frac{8}{(0.543 \times 10^{-9} \text{ m})^3} \frac{1}{(10^9 \text{ nm/m})^3} = 50.0 \text{ atoms/nm}^3$$

If desired, the density ρ can be found as follows:

$$\rho = \frac{8 \frac{M_{at}}{N_A}}{a^3} = \frac{8 \frac{28.09 \times 10^{-3} \text{ kg/mol}}{6.022 \times 10^{23} \text{ mol}^{-1}}}{(0.543 \times 10^{-9} \text{ m})^3} = 2331 \text{ kg m}^{-3} \text{ or } 2.33 \text{ g cm}^{-3}$$

b. The (100) plane has 4 shared atoms at the corners and 1 unshared atom at the center. The corner atom is shared by 4 (100) type planes. Number of atoms per square nm of (100) plane area (n) is shown in Fig. 1Q28-1:

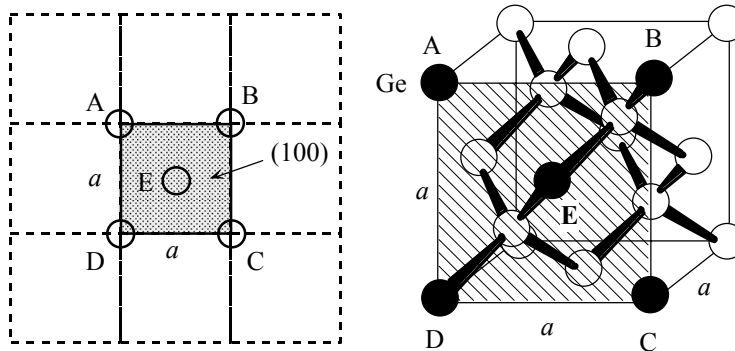


Figure 1Q28-1: The (100) plane of the diamond crystal structure.

The number of atoms per nm^2 , n_{100} , is therefore:

$$n_{100} = \frac{4\left(\frac{1}{4}\right) + 1}{a^2} \frac{1}{(10^9 \text{ nm/m})^2} = \frac{4\left(\frac{1}{4}\right) + 1}{(0.543 \times 10^{-9} \text{ m})^2} \frac{1}{(10^9 \text{ nm/m})^2}$$

$$\therefore n_{100} = 6.78 \text{ atoms/nm}^2 \text{ or } 6.78 \times 10^{18} \text{ atoms/m}^2$$

The (110) plane is shown below in Fig. 1Q28-2. There are 4 atoms at the corners and shared with neighboring planes (hence each contributing a quarter), 2 atoms on upper and lower sides shared with upper and lower planes (hence each atom contributing 1/2) and 2 atoms wholly within the plane.

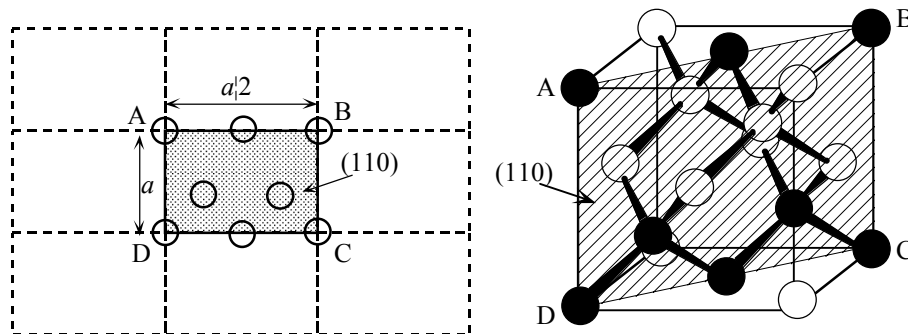


Figure 1Q28-2: The (110) plane of the diamond crystal structure.

The number of atoms per nm^2 , n_{110} , is therefore:

$$n_{110} = \frac{4\left(\frac{1}{4}\right) + 2\left(\frac{1}{2}\right) + 2\left(\frac{1}{\left(10^9 \text{ nm/m}\right)^2}\right)}{a(a\sqrt{2})}$$

$$\therefore n_{110} = \frac{4\left(\frac{1}{4}\right) + 2\left(\frac{1}{2}\right) + 2}{\left[\left(0.543 \times 10^{-9} \text{ m}\right)\left(\left(0.543 \times 10^{-9} \text{ m}\right)\sqrt{2}\right)\right]\left(\left(10^9 \text{ nm/m}\right)^2\right)}$$

$$\therefore n_{110} = 9.59 \text{ atoms/nm}^2 \text{ or } 9.59 \times 10^{18} \text{ atoms/m}^2$$

This is the most crowded plane with the most number of atoms per unit area.

The (111) plane is shown below in Fig. 1Q28-3:

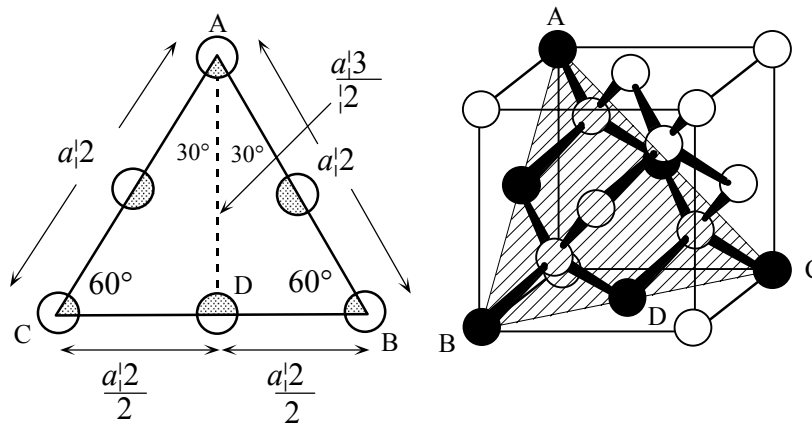


Figure 1Q28-3: The (111) plane of the diamond crystal structure

The number of atoms per nm^2 , n_{111} , is therefore:

$$n_{111} = \frac{3\left(\frac{60}{360}\right) + 3\left(\frac{1}{2}\right)}{\left[2\left(\frac{1}{2}\right)\left(a\frac{\sqrt{2}}{2}\right)\left(a\frac{\sqrt{3}}{\sqrt{2}}\right)\right]\left(\left(10^9 \text{ nm/m}\right)^2\right)}$$

$$\therefore n_{111} = \frac{3\left(\frac{60}{360}\right) + 3\left(\frac{1}{2}\right)}{\left[2\left(\frac{1}{2}\right)\left(\left(0.543 \times 10^{-9} \text{ m}\right)\frac{\sqrt{2}}{2}\right)\left(\left(0.543 \times 10^{-9} \text{ m}\right)\frac{\sqrt{3}}{\sqrt{2}}\right)\right]\left(\left(10^9 \text{ nm/m}\right)^2\right)}$$

$$\therefore n_{111} = 7.83 \text{ atoms/nm}^2 \text{ or } 7.83 \times 10^{18} \text{ atoms/m}^2$$

c. Given:

Molar mass of SiO_2 : $M_{at} = 28.09 \times 10^{-3} \text{ kg/mol} + 2 \times 16 \times 10^{-3} \text{ kg/mol} = 60.09 \times 10^{-3} \text{ kg/mol}$

Density of SiO_2 : $\rho = 2.27 \times 10^3 \text{ kg m}^{-3}$

Let n be the number of SiO_2 molecules per unit volume, then:

$$\rho = n \frac{M_{at}}{N_A}$$

$$\therefore n = \frac{N_A \rho}{M_{at}} = \frac{(6.022 \times 10^{23} \text{ mol}^{-1})(2.27 \times 10^3 \text{ kg m}^{-3})}{(60.09 \times 10^{-3} \text{ kg/mol})} = 2.27 \times 10^{28} \text{ molecules per m}^3$$

Or, converting to molecules per nm³:

$$n = \frac{2.27 \times 10^{28} \text{ molecules/m}^3}{(10^9 \text{ nm/m})^3} = \mathbf{22.7 \text{ molecules per nm}^3}$$

Oxide has less dense packing so it has a more open structure. For every 1 micron of oxide formed on the crystal surface, only about 0.5 micron of the Si crystal is consumed.

1. 29 Vacancies in metals

- The energy of formation of a vacancy in the copper crystal is about 1 eV. Calculate the concentration of vacancies at room temperature (300K) and just below the melting temperature, 1084 °C. Neglect the change in the density which is small.
- The table below shows the energies of vacancy formation in various metals with *close-packed* crystal structures and the melting temperature T_m . Plot E_v in eV vs. T_m in Kelvins, and explore if there is a correlation between a and T_m . Some materials engineers take E_v to be very roughly $10kT_m$. Do you think that they are correct? (Justify.)

	Metal								
	Al	Ag	Au	Cu	Mg	Pt	Pb	Ni	Pd
Crystal	FCC	FCC	FCC	FCC	HCP	FCC	FCC	FCC	FCC
E_v (eV)	0.70–0.76	1.0–1.1	0.90–0.98	1–1.28	0.89	1.3–1.5	0.50	1.63–1.79	1.54–1.85
T_m (°C)	660	962	1064	1085	650	1768	328	1455	1555

Solution

- Equilibrium concentration of vacancies is given by

$$n_v = N \exp\left(-\frac{E_v}{kT}\right)$$

where N is atomic concentrations. Vacancy formation energy $E_v = 1 \text{ eV} \times (1.6 \times 10^{-19} \text{ J eV}^{-1}) = 1.6 \times 10^{-19} \text{ J}$. The atomic mass of Cu is $M_{at} = 63.54 \text{ g mol}^{-1}$ and the density at 293 K is 8.96 g cm^{-3} .

Atomic concentration is calculated as

$$N = \frac{\rho N_A}{M_{at}} = \frac{(8.96 \text{ g cm}^{-3})(6.022 \times 10^{23} \text{ mol}^{-1})}{63.54 \text{ g mol}^{-1}}$$

$$= 8.49 \times 10^{22} \text{ cm}^{-3} \text{ or } 8.49 \times 10^{26} \text{ m}^{-3}$$

At $T = 300 \text{ K}$

$$n_v = (8.49 \times 10^{26} \text{ m}^{-3}) \exp\left(-\frac{1.6 \times 10^{-19} \text{ J}}{(1.3806 \times 10^{-23} \text{ J K}^{-1})(300 \text{ K})}\right)$$

$$n_v = \mathbf{1.42 \times 10^{10} \text{ m}^{-3}}.$$

This implies that 1 in 5.97×10^{10} atoms is a vacancy.

At $T = 1084^\circ\text{C}$ (1357 K)

$$n_v = (8.49 \times 10^{26} \text{ m}^{-3}) \exp\left(-\frac{1.6 \times 10^{-19} \text{ J}}{(1.3806 \times 10^{-23} \text{ J K}^{-1})(1357 \text{ K})}\right)$$

$$n_v = 1.66 \times 10^{23} \text{ m}^{-3}.$$

This implies that 1 in 5306 atoms is a vacancy.

b. The vacancy formation E_v and the melting temperature T_m are plotted as follows. The $10kT_m$ line is also plotted on the same graph.

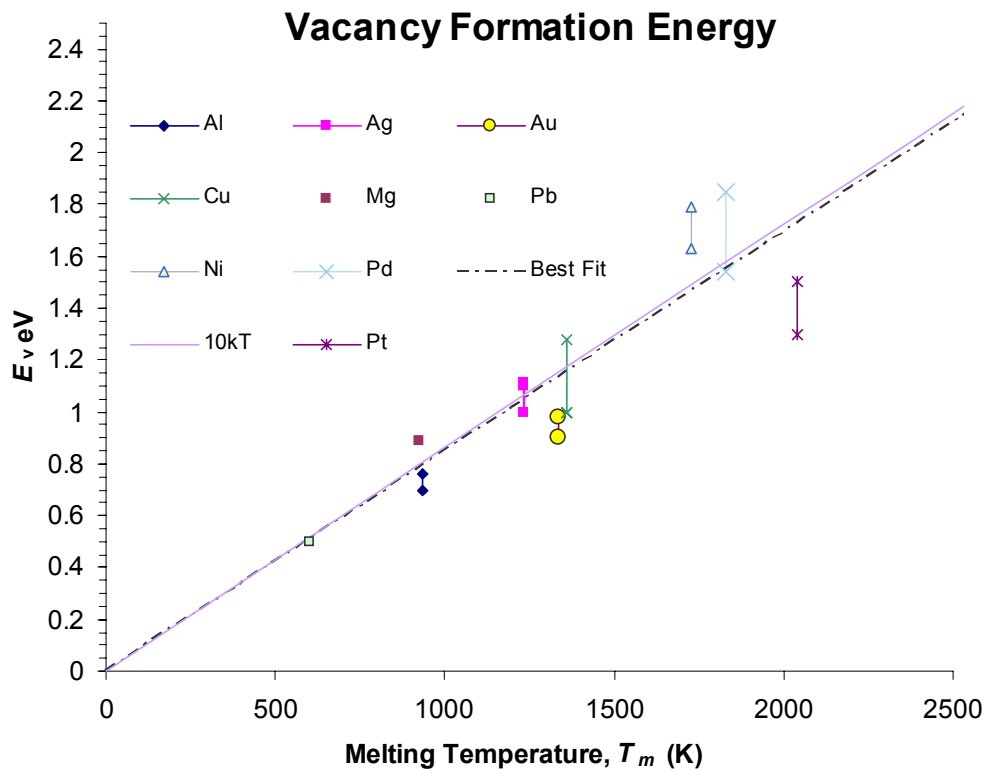


Figure 1Q29-1: Plot of vacancy formation energy versus melting temperature T_m . The $10kT_m$ line is also on plotted on the same graph.

Observing the plot we can say that vacancy formation energy can be approximated by $10kT_m$.

1.30 Vacancies in silicon

In device fabrication, Si is frequently doped by the diffusion of impurities (dopants) at high temperatures, typically 950–1200 °C. The energy of vacancy formation in the Si crystal is about 3.6 eV. What is the equilibrium concentration of vacancies in an Si crystal at 1000 °C. Neglect the change in the density with temperature which is less than 1 percent in this case.

Solution

Equilibrium concentration of vacancies is

$$n_v = N \exp\left(-\frac{E_v}{kT}\right)$$

Vacancy formation energy $E_v = 3.6 \text{ eV} \times (1.6 \times 10^{-19} \text{ J eV}^{-1}) = 5.76 \times 10^{-19} \text{ J}$. The equilibrium concentration is therefore,

$$n_v = N \exp\left(-\frac{E_v}{kT}\right)$$

For Si the atomic mass $M_{\text{at}} = 28.09 \text{ g mol}^{-1}$ and density $\rho = 2.33 \text{ g cm}^{-3}$. Therefore the atomic concentration is,

$$N = \frac{\rho N_A}{M_{\text{at}}} = \frac{(2.33 \text{ g cm}^{-3})(6.022 \times 10^{23} \text{ mol}^{-1})}{28.09 \text{ g mol}^{-1}} = 5.0 \times 10^{22} \text{ cm}^{-3} \text{ or } 5.0 \times 10^{26} \text{ m}^{-3}$$

At $T = 1000^\circ\text{C}$ (1273 K), equilibrium concentration of vacancies is,

$$n_v = (5.0 \times 10^{26} \text{ m}^{-3}) \exp\left(-\frac{1.6 \times 10^{-19} \text{ J}}{(1.3806 \times 10^{-23} \text{ J K}^{-1})(1273 \text{ K})}\right)$$

$$n_v = 5.55 \times 10^{22} \text{ m}^{-3}.$$

We neglected the change in the density from 300 K to 1273 K. Its effect on the final result is small.

1.31 Pb-Sn solder Consider the soldering of two copper components. When the solder melts, it wets both metal surfaces. If the surfaces are not clean or have an oxide layer, the molten solder cannot wet the surfaces and the soldering fails. Assume that soldering takes place at 250°C , and consider the diffusion of Sn atoms into the copper (the Sn atom is smaller than the Pb atom and hence diffuses more easily).

- The diffusion coefficient of Sn in Cu at two temperatures is $D = 1.69 \times 10^{-9} \text{ cm}^2 \text{ hr}^{-1}$ at 400°C and $D = 2.48 \times 10^{-7} \text{ cm}^2 \text{ hr}^{-1}$ at 650°C . Calculate the rms distance diffused by an Sn atom into the copper, assuming the cooling process takes 10 seconds.
- What should be the composition of the solder if it is to begin freezing at 250°C ?
- What are the components (phases) in this alloy at 200°C ? What are the compositions of the phases and their relative weights in the alloy?
- What is the microstructure of this alloy at 25°C ? What are weight fractions of the α and β phases assuming near equilibrium cooling?

Solution

a. Given information:

$$\text{Temperatures: } T_1 = 400^\circ\text{C} + 273 = 673 \text{ K} \quad T_2 = 650^\circ\text{C} + 273 = 923 \text{ K}$$

$$\text{Diffusion coefficients: } D_1 = 1.69 \times 10^{-9} \text{ cm}^2/\text{hr} = (1.69 \times 10^{-9} \text{ cm}^2/\text{hr})(0.01 \text{ m/cm})^2 / (1 \text{ hr}) \times (3600 \text{ sec/hr})$$

$$D_1 = 4.694 \times 10^{-17} \text{ m}^2/\text{s}$$

$$D_2 = 2.48 \times 10^{-7} \text{ cm}^2/\text{hr} = (2.48 \times 10^{-7} \text{ cm}^2/\text{hr})(0.01 \text{ m/cm})^2 / (1 \text{ hr}) \times (3600 \text{ sec/hr})$$

$$D_2 = 6.889 \times 10^{-15} \text{ m}^2/\text{s}$$

The diffusion coefficients D_1 and D_2 at T_1 and T_2 respectively are given by:

$$D_1 = D_o \exp\left(-\frac{qE_A}{kT_1}\right) \qquad D_2 = D_o \exp\left(-\frac{qE_A}{kT_2}\right)$$

where E_A is the activation energy in eV/atom and $q = 1.6 \times 10^{-19}$ J/eV (conversion factor from eV to J).
Since

$$\frac{\exp(-x)}{\exp(-y)} = \exp(y-x)$$

we can take the ratio of the diffusion coefficients to express them in terms of E_A (eV):

$$\frac{D_1}{D_2} = \frac{D_o \exp\left(-\frac{qE_A}{kT_1}\right)}{D_o \exp\left(-\frac{qE_A}{kT_2}\right)} = \exp\left(\frac{qE_A}{kT_2} - \frac{qE_A}{kT_1}\right) = \exp\left(\frac{q[T_1 - T_2]E_A}{kT_1T_2}\right)$$

$$\therefore E_A = \frac{kT_1T_2 \ln\left(\frac{D_1}{D_2}\right)}{q(T_1 - T_2)} \quad (\text{in eV})$$

$$\therefore E_A = \frac{(1.381 \times 10^{-23} \text{ J K}^{-1})(673 \text{ K})(923 \text{ K}) \ln\left(\frac{4.694 \times 10^{-17} \text{ m}^2/\text{s}}{6.889 \times 10^{-15} \text{ m}^2/\text{s}}\right)}{(1.602 \times 10^{-19} \text{ J/eV})(673 \text{ K} - 923 \text{ K})} \quad (\text{in eV})$$

$$\therefore E_A = 1.068 \text{ eV/atom}$$

Now the diffusion coefficient D_o can be found as follows:

$$D_1 = D_o \exp\left(-\frac{qE_A}{kT_1}\right)$$

$$\therefore D_o = \frac{D_1}{\exp\left(-\frac{qE_A}{kT_1}\right)} = \frac{4.694 \times 10^{-17} \text{ m}^2/\text{s}}{\exp\left(-\frac{(1.602 \times 10^{-19} \text{ J/eV})(1.068 \text{ eV})}{(1.381 \times 10^{-23} \text{ J K}^{-1})(673 \text{ K})}\right)} = 4.638 \times 10^{-9} \text{ m}^2/\text{s}$$

To check our value for D_o , we can substitute it back into the equation for D_2 and compare values:

$$D_2 = D_o \exp\left(-\frac{qE_A}{kT_2}\right) = (4.638 \times 10^{-9} \text{ m}^2/\text{s}) \exp\left(-\frac{(1.602 \times 10^{-19} \text{ J/eV})(1.068 \text{ eV})}{(1.381 \times 10^{-23} \text{ J K}^{-1})(923 \text{ K})}\right)$$

$$\therefore D_2 = 6.870 \times 10^{-15} \text{ m}^2/\text{s}$$

This agrees with the given value of $6.889 \times 10^{-15} \text{ m}^2/\text{s}$ for D_2 .

Now we must calculate the diffusion coefficient D_3 at $T_3 = 250^\circ\text{C} + 273 = 523 \text{ K}$ (temperature at which soldering is taking place).

$$D_3 = D_o \exp\left(-\frac{qE_A}{kT_3}\right) = (4.638 \times 10^{-9} \text{ m}^2/\text{s}) \exp\left(-\frac{(1.602 \times 10^{-19} \text{ J/eV})(1.068 \text{ eV})}{(1.381 \times 10^{-23} \text{ J K}^{-1})(523 \text{ K})}\right)$$

$$\therefore D_3 = 2.391 \times 10^{-19} \text{ m}^2/\text{s}$$

The rms distance diffused by the Sn atom in time $t = 10$ s (L_{rms}) is given by:

$$L_{rms} = \sqrt{2D_3t} = \sqrt{2(2.391 \times 10^{-19} \text{ m}^2/\text{s})(10 \text{ s})} = 2.19 \times 10^{-9} \text{ m or } 2 \text{ nm}$$

b. From Figure 1Q31-1, 250 °C cuts the liquidus line approximately at 33 wt.% Sn composition (C_o).

$$\therefore C_o = 0.33 \quad (\text{Sn})$$

c. For α -phase and liquid phase (L), the compositions as wt.% of Sn from Figure 1.69 are:

$$C_\alpha = 0.18 \quad \text{and} \quad C_L = 0.56$$

The weight fraction of α and L phases are:

$$W_\alpha = \frac{C_L - C_o}{C_L - C_\alpha} = \frac{0.56 - 0.33}{0.56 - 0.18} = 0.605 \text{ or } 60 \text{ wt.\% } \alpha\text{-phase}$$

$$W_L = \frac{C_o - C_\alpha}{C_L - C_\alpha} = \frac{0.33 - 0.18}{0.56 - 0.18} = 0.395 \text{ or } 39.5 \text{ wt.\% liquid phase}$$

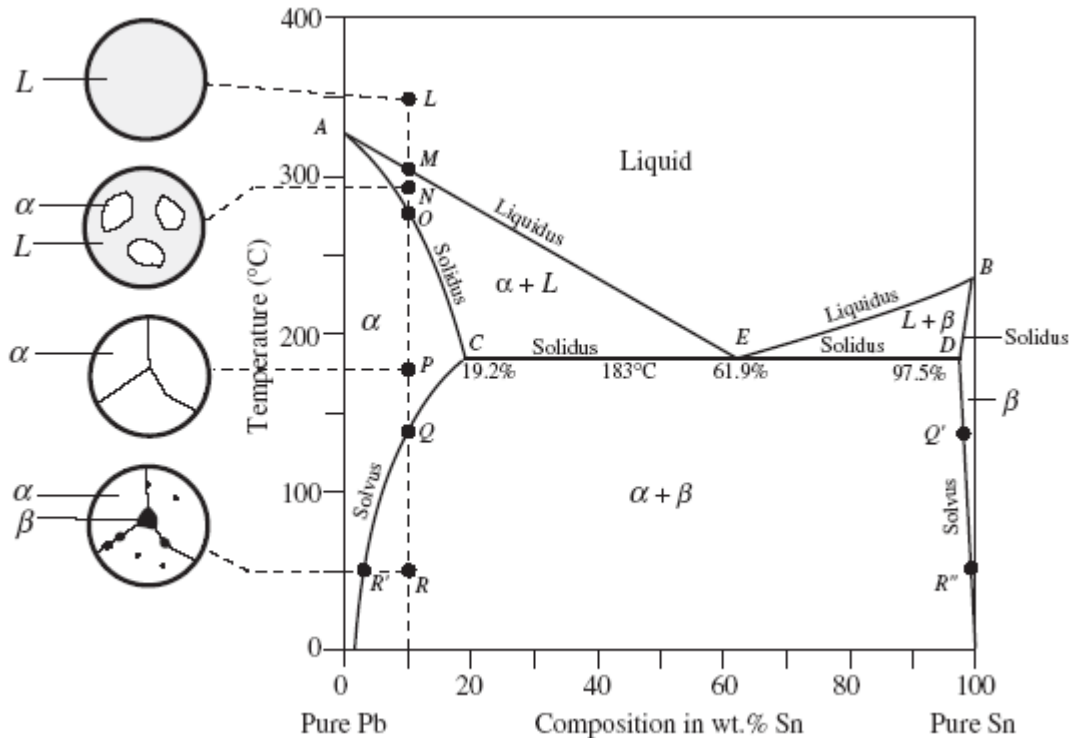


Figure 1.69: The equilibrium phase diagram of the Pb-Sn alloy.

d. The microstructure is a primary α -phase and a eutectic solid ($\alpha + \beta$) phase. There are two phases present, $\alpha + \beta$. (See Figure 1Q31-1)



Figure 1Q31-1: Microstructure of Pb-Sn at temperatures less than 183°C.

Assuming equilibrium concentrations have been reached:

$$C'_\alpha = 0.02 \quad \text{and} \quad C'_\beta = 1$$

The weight fraction of α in the whole alloy is then:

$$W'_\alpha = \frac{C'_\beta - C_o}{C'_\beta - C'_\alpha} = \frac{1 - 0.33}{1 - 0.02} = \mathbf{0.684 \text{ or } 68.4 \text{ wt.\% } \alpha\text{-phase}}$$

The weight fraction of β in the whole alloy is:

$$W'_\beta = \frac{C_o - C_\alpha}{C'_\beta - C'_\alpha} = \frac{0.33 - 0.02}{1 - 0.02} = \mathbf{0.316 \text{ or } 31.6 \text{ wt.\% } \beta\text{-phase}}$$

1.31 Pb-Sn solder Consider the 50% Pb-50% Sn solder.

- Sketch the temperature-time profile and the microstructure of the alloy at various stages as it is cooled from the melt.
- At what temperature does the solid melt?
- What is the temperature range over which the alloy is a mixture of melt and solid? What is the structure of the solid?
- Consider the solder at room temperature following cooling from 182 °C. Assume that the rate of cooling from 182 °C to room temperature is faster than the atomic diffusion rates needed to change the compositions of the α and β phases in the solid. Assuming the alloy is 1 kg, calculate the masses of the following components in the solid:
 - The primary α .
 - α in the whole alloy.
 - α in the eutectic solid.
 - β in the alloy (Where is the β -phase?)
- Calculate the specific heat of the solder given the atomic masses of Pb (207.2) and Sn (118.71).

Solution

a.

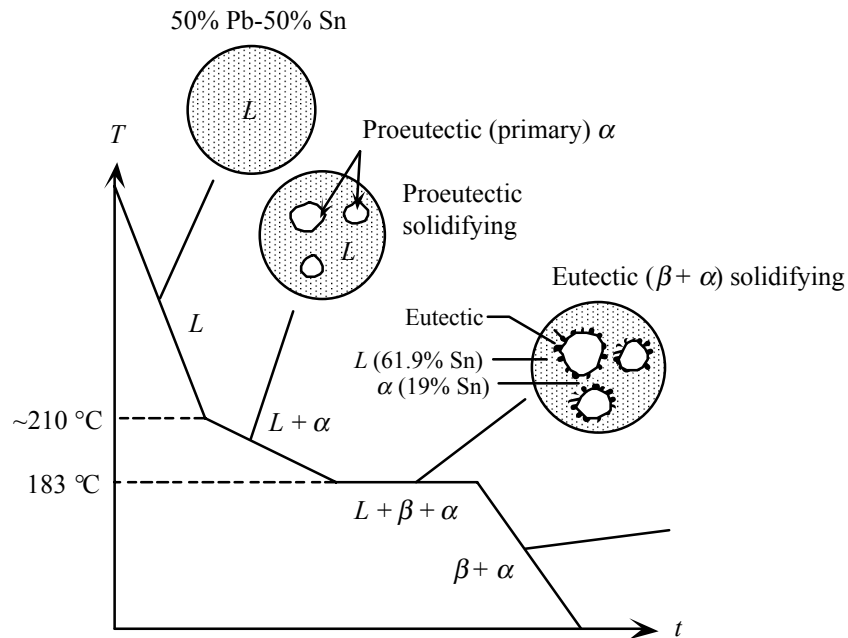


Figure 1Q31-1: Temperature - time profile and microstructure diagram of 50% Pb-50% Sn.

All compositions are in weight %.

b. When 50% Pb-50% Sn is cooled from the molten state down to room temperature, it begins to solidify at point A at about 210 °C. Therefore at 210 °C, the solid begins to melt.

c. Between 210 °C and 183 °C, the liquid has the eutectic composition and undergoes the eutectic transformation to become the eutectic solid. Below 183 °C, all the liquid has solidified, and the structure is a combination of the solid α -phase and the eutectic structure (which is composed of α and β layers).

d. At 182 °C, the composition of the proeutectic or primary α is given by the solubility limit of Sn in α , **19.2% Sn**. The primary or proeutectic α (pro- α) exists just above and below 183 °C (eutectic temperature), *i.e.* it is stable just above and below 183 °C. Thus the mass of pro- α at 182 °C is the same as at 184 °C. Applying the lever rule (at 50% Sn):

$$W_{pro-\alpha} = \frac{C_L - C_o}{C_L - C_\alpha} = \frac{61.9 - 50}{61.9 - 19.2} = \mathbf{0.279}$$

Assume that the whole alloy is 1 kg. The mass of the primary or proeutectic α is thus **27.9% of the whole alloy**; or **0.279 kg**. The mass of the total eutectic solid ($\alpha + \beta$) is thus $1 - 0.279 = \mathbf{0.721}$ or **72.1% or 0.721 kg**.

If we apply the lever rule at 182 °C at 50% Sn, we obtain the weight percentage of α in the whole solid:

$$W = \frac{C_\beta - C_o}{C_\beta - C_\alpha} = \frac{97.5 - 50}{97.5 - 19.2} = \mathbf{0.607}$$

Thus the mass of α in the whole solid is **0.607 kg**. Of this, 0.28 kg is in the primary (proeutectic) α phase. Thus $0.607 - 0.279$ or **0.328 kg** of α is in the eutectic solid.

Since the total mass of α in the solid is 0.607 kg, the remainder of the mass must be the β -phase. Thus the mass of the β -phase is $1 \text{ kg} - 0.607 \text{ kg}$ or **0.393 kg**. The β -phase is present solely in the eutectic

solid, with no primary β , because the solid is forming from a point where it consisted of the α and liquid phases only.

e. Suppose that n_A and n_B are atomic fractions of A and B in the whole alloy,

$$n_A + n_B = 1$$

Suppose that we have 1 mole of the alloy. Then it has n_A moles of A and n_B moles of B (atomic fractions also represent molar fractions in the alloy). Suppose that we consider 1 gram of the alloy. Since w_A is the weight fraction of A , w_A is also the mass of A in grams in the alloy. The number of moles of A in the alloy is then w_A/M_A where M_A is the atomic mass of A . Thus,

$$\text{Number of moles of } A = w_A/M_A.$$

$$\text{Number of moles of } B = w_B/M_B.$$

$$\text{Number of moles of the whole alloy} = w_A/M_A + w_B/M_B.$$

Molar fraction of A is the same as n_A . Thus,

$$n_A = \frac{w_A / M_A}{\frac{w_A}{M_A} + \frac{w_B}{M_B}} \quad \text{and} \quad n_B = \frac{w_B / M_B}{\frac{w_A}{M_A} + \frac{w_B}{M_B}}$$

We are given the molar masses of Pb and Sn:

$$M_{\text{Pb}} = 207.2 \text{ g/mol}$$

$$M_{\text{Sn}} = 118.71 \text{ g/mol}$$

Let n = mole (atomic) fraction and W = weight fraction. Then W_{Pb} = weight fraction of Pb in the alloy. We know this to be 0.5 (50% Pb).

The mole fractions of Pb and Sn are n_{Pb} and n_{Sn} and are given by:

$$n_{\text{Pb}} = - \frac{M_{\text{Sn}} W_{\text{Pb}}}{(M_{\text{Pb}} - M_{\text{Sn}}) W_{\text{Pb}} - M_{\text{Pb}}}$$

$$\therefore n_{\text{Pb}} = - \frac{(118.71 \text{ g/mol})(0.5)}{(207.2 \text{ g/mol} - 118.71 \text{ g/mol})(0.5) - 207.2 \text{ g/mol}} = 0.3642$$

$$n_{\text{Sn}} = 1 - n_{\text{Pb}} = 1 - 0.3642 = 0.6358$$

The mole fraction of Pb is 0.3642 and that of Sn is $1 - 0.3642 = 0.6358$.

The heat capacity of a metal is $3R$ per mole, where R is the gas constant. We want the specific heat capacity C_s (heat capacity per gram of alloy). 1 mole of the alloy has a mass M_{PbSn} :

$$M_{\text{PbSn}} = n_{\text{Pb}} M_{\text{Pb}} + n_{\text{Sn}} M_{\text{Sn}}$$

Thus the specific heat capacity C_s (*i.e.* heat capacity per gram) is:

$$c_s = \frac{3R}{M_{\text{PbSn}}} = \frac{3R}{n_{\text{Pb}} M_{\text{Pb}} + n_{\text{Sn}} M_{\text{Sn}}}$$

$$\therefore c_s = \frac{3(8.315 \text{ J K}^{-1} \text{ mol}^{-1})}{(0.3642)(207.2 \text{ g/mol}) + (0.6358)(118.71 \text{ g/mol})}$$

$$\therefore c_s = 0.165 \text{ J K}^{-1} \text{ g}^{-1}$$

"If your experiment needs statistics, you ought to have done a better experiment."

Ernest Rutherford (1871-1937)

Third Edition (© 2005 McGraw-Hill)

Chapter 2

2.1 Electrical conduction Na is a monovalent metal (BCC) with a density of 0.9712 g cm^{-3} . Its atomic mass is 22.99 g mol^{-1} . The drift mobility of electrons in Na is $53 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$.

- Consider the collection of conduction electrons in the solid. If each Na atom donates one electron to the electron sea, *estimate* the mean separation between the electrons. (Note: if n is the concentration of particles, then the particles' mean separation $d = 1/n^{1/3}$.)
- Estimate the mean separation between an electron (e^-) and a metal ion (Na^+), assuming that most of the time the electron prefers to be between two neighboring Na^+ ions. What is the approximate Coulombic interaction energy (in eV) between an electron and an Na^+ ion?
- How does this electron/metal-ion interaction energy compare with the average thermal energy per particle, according to the kinetic molecular theory of matter? Do you expect the kinetic molecular theory to be applicable to the conduction electrons in Na? If the mean electron/metal-ion interaction energy is of the same order of magnitude as the mean KE of the electrons, what is the mean speed of electrons in Na? Why should the mean kinetic energy be comparable to the mean electron/metal-ion interaction energy?
- Calculate the electrical conductivity of Na and compare this with the experimental value of $2.1 \times 10^7 \Omega^{-1} \text{ m}^{-1}$ and comment on the difference.

Solution

a. If D is the density, M_{at} is the atomic mass and N_A is Avogadro's number, then the atomic concentration n_{at} is

$$n_{at} = \frac{DN_A}{M_{at}} = \frac{(971.2 \text{ kg})(6.022 \times 10^{23} \text{ mol}^{-1})}{(22.99 \times 10^{-3} \text{ kg mol}^{-1})} = 2.544 \times 10^{28} \text{ m}^{-3}$$

which is also the electron concentration, given that each Na atom contributes 1 conduction electron.

If d is the mean separation between the electrons then d and n_{at} are related by (see Chapter 1 Solutions, Q1.11; this is only an estimate)

$$d \approx \frac{1}{n_{at}^{1/3}} = \frac{1}{(2.544 \times 10^{28} \text{ m}^{-3})^{1/3}} = 3.40 \times 10^{-10} \text{ m or } 0.34 \text{ nm}$$

b. Na is BCC with 2 atoms in the unit cell. So if a is the lattice constant (side of the cubic unit cell), the density is given by

$$D = \frac{(\text{atoms in unit cell})(\text{mass of 1 atom})}{\text{volume of unit cell}} = \frac{2\left(\frac{M_{at}}{N_A}\right)}{a^3}$$

isolate for a ,
$$a = \left[\frac{2M_{at}}{DN_A} \right]^{1/3} = \left[\frac{2(22.99 \times 10^{-3} \text{ kg mol}^{-1})}{(0.9712 \times 10^3 \text{ kg m}^{-3})(6.022 \times 10^{23} \text{ mol}^{-1})} \right]^{1/3}$$

so that $a = 4.284 \times 10^{-10} \text{ m}$ or 0.4284 nm

For the BCC structure, the radius of the metal ion R and the lattice parameter a are related by $(4R)^2 = 3a^2$, so that,

$$R = (1/4)\sqrt{(3a^2)} = 1.855 \times 10^{-10} \text{ m or } 0.1855 \text{ nm}$$

If the electron is somewhere roughly between two metal ions, then the mean electron to metal ion separation $d_{\text{electron-ion}}$ is roughly R . If $d_{\text{electron-ion}} \approx R$, the electrostatic potential energy PE between a conduction electron and one metal ion is then

$$PE = \frac{(-e)(+e)}{4\pi\epsilon_0 d_{\text{electron-ion}}} = \frac{(-1.602 \times 10^{-19} \text{ C})(+1.602 \times 10^{-19} \text{ C})}{4\pi(8.854 \times 10^{-12} \text{ F m}^{-1})(1.855 \times 10^{-10} \text{ m})} \quad (1)$$

$$\therefore \quad PE = -1.24 \times 10^{-18} \text{ J or } -7.76 \text{ eV}$$

c. This electron-ion PE is much larger than the average thermal energy expected from the kinetic theory for a collection of “free” particles, that is $E_{\text{average}} = KE_{\text{average}} = 3(kT/2) \approx 0.039 \text{ eV}$ at 300 K . In the case of Na, the electron-ion interaction is very strong so we cannot assume that the electrons are moving around freely as if in the case of free gas particles in a cylinder. If we assume that the mean KE is roughly the same order of magnitude as the mean PE ,

$$KE_{\text{average}} = \frac{1}{2} m_e u^2 \approx |PE|_{\text{average}} = -1.24 \times 10^{-18} \text{ J} \quad (2)$$

where u is the mean speed (strictly, u = root mean square velocity) and m_e is the electron mass.

$$\text{Thus,} \quad u \approx \left[\frac{2PE_{\text{average}}}{m_e} \right]^{1/2} = \left[\frac{2(1.24 \times 10^{-18} \text{ J})}{(9.109 \times 10^{-31} \text{ kg})} \right]^{1/2} \quad (3)$$

$$\text{so that} \quad u = 1.65 \times 10^6 \text{ m/s}$$

There is a theorem in classical physics called the *Virial theorem* which states that if the interactions between particles in a system obey the inverse square law (as in Coulombic interactions) then the magnitude of the mean KE is equal to the magnitude of the mean PE . The Virial Theorem states that:

$$KE_{\text{average}} = -\frac{1}{2} PE_{\text{average}}$$

Indeed, using this expression in Eqn. (2), we would find that $u = 1.05 \times 10^6 \text{ m/s}$. If the conduction electrons were moving around freely and obeying the kinetic theory, then we would expect $(1/2)m_e u^2 = (3/2)kT$ and $u = 1.1 \times 10^5 \text{ m/s}$, a much lower mean speed. Further, kinetic theory predicts that u increases as $T^{1/2}$ whereas according to Eqns. (1) and (2), u is insensitive to the temperature. The experimental linear dependence between the resistivity ρ and the absolute temperature T for most metals (non-magnetic) can only be explained by taking u = constant as implied by Eqns. (1) and (2).

d. If μ is the drift mobility of the conduction electrons and n is their concentration, then the electrical conductivity of Na is $\sigma = en\mu$. Assuming that each Na atom donates one conduction electron ($n = n_{\text{at}}$), we have

$$\sigma = en\mu = (1.602 \times 10^{-19} \text{ C})(2.544 \times 10^{28} \text{ m}^{-3})(53 \times 10^{-4} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1})$$

$$\text{i.e.} \quad \sigma = 2.16 \times 10^7 \Omega^{-1} \text{ m}^{-1}$$

which is quite close to the experimental value.

Nota Bene: If one takes the $\text{Na}^+ - \text{Na}^+$ separation $2R$ to be roughly the mean electron-electron separation then this is 0.37 nm and close to $d = 1/(n^{1/3}) = 0.34$ nm. In any event, all calculations are only approximate to highlight the main point. The interaction PE is substantial compared with the mean thermal energy and we cannot use $(3/2)kT$ for the mean KE !

2.2 Electrical conduction The resistivity of aluminum at 25 °C has been measured to be $2.72 \times 10^{-8} \Omega \text{ m}$. The thermal coefficient of resistivity of aluminum at 0 °C is $4.29 \times 10^{-3} \text{ K}^{-1}$. Aluminum has a valency of 3, a density of 2.70 g cm^{-3} , and an atomic mass of 27.

- Calculate the resistivity of aluminum at -40°C .
- What is the thermal coefficient of resistivity at -40°C ?
- Estimate the mean free time between collisions for the conduction electrons in aluminum at 25 °C, and hence estimate their drift mobility.
- If the mean speed of the conduction electrons is about $2 \times 10^6 \text{ m s}^{-1}$, calculate the mean free path and compare this with the interatomic separation in Al (Al is FCC). What should be the thickness of an Al film that is deposited on an IC chip such that its resistivity is the same as that of bulk Al?
- What is the percentage change in the power loss due to Joule heating of the aluminum wire when the temperature drops from 25 °C to -40°C ?

Solution

a. Apply the equation for temperature dependence of resistivity, $\rho(T) = \rho_o[1 + \alpha_o(T - T_o)]$. We have the temperature coefficient of resistivity, α_o , at T_o where T_o is the reference temperature. The two given reference temperatures are 0 °C or 25 °C, depending on choice. Taking $T_o = 0^\circ\text{C} + 273 = 273 \text{ K}$,

$$\rho(-40^\circ\text{C} + 273 = 233 \text{ K}) = \rho_o[1 + \alpha_o(233 \text{ K} - 273 \text{ K})]$$

$$\rho(25^\circ\text{C} + 273 = 298 \text{ K}) = \rho_o[1 + \alpha_o(298 \text{ K} - 273 \text{ K})]$$

Divide the above two equations to eliminate ρ_o ,

$$\rho(-40^\circ\text{C})/\rho(25^\circ\text{C}) = [1 + \alpha_o(-40 \text{ K})] / [1 + \alpha_o(25 \text{ K})]$$

Next, substitute the given values $\rho(25^\circ\text{C}) = 2.72 \times 10^{-8} \Omega \text{ m}$ and $\alpha_o = 4.29 \times 10^{-3} \text{ K}^{-1}$ to obtain

$$\rho(-40^\circ\text{C}) = (2.72 \times 10^{-8} \Omega \text{ m}) \frac{[1 + (4.29 \times 10^{-3} \text{ K}^{-1})(-40 \text{ K})]}{[1 + (4.29 \times 10^{-3} \text{ K}^{-1})(25 \text{ K})]} = \mathbf{2.03 \times 10^{-8} \Omega \text{ m}}$$

b. In $\rho(T) = \rho_o[1 + \alpha_o(T - T_o)]$ we have α_o at T_o where T_o is the reference temperature, for example, 0°C or 25°C depending on choice. We will choose T_o to be first at $0^\circ\text{C} = 273 \text{ K}$ and then at $-40^\circ\text{C} = 233 \text{ K}$ so that

$$\rho(-40^\circ\text{C}) = \rho(0^\circ\text{C})[1 + \alpha_o(233 \text{ K} - 273 \text{ K})]$$

and
$$\rho(0^\circ\text{C}) = \rho(-40^\circ\text{C})[1 + \alpha_{-40}(273 \text{ K} - 233 \text{ K})]$$

Multiply and simplify the two equations above to obtain

$$[1 + \alpha_o(233 \text{ K} - 273 \text{ K})][1 + \alpha_{-40}(273 \text{ K} - 233 \text{ K})] = 1$$

$$\text{or} \quad [1 - 40\alpha_o][1 + 40\alpha_{40}] = 1$$

Rearranging,

$$\alpha_{40} = (1 / [1 - 40\alpha_o] - 1)(1 / 40)$$

$$\therefore \alpha_{40} = \alpha_o / [1 - 40\alpha_o]$$

$$\text{i.e.} \quad \alpha_{40} = (4.29 \times 10^{-3} \text{ K}^{-1}) / [1 - (40 \text{ K})(4.29 \times 10^{-3} \text{ K}^{-1})] = \mathbf{5.18 \times 10^{-3} \text{ K}^{-1}}$$

Alternatively, consider, $\rho(25^\circ\text{C}) = \rho(-40^\circ\text{C})[1 + \alpha_{40}(298 \text{ K} - 233 \text{ K})]$ so that

$$\alpha_{40} = [\rho(25^\circ\text{C}) - \rho(-40^\circ\text{C})] / [\rho(-40^\circ\text{C})(65 \text{ K})]$$

$$\therefore \alpha_{40} = [2.72 \times 10^{-8} \Omega \text{ m} - 2.03 \times 10^{-8} \Omega \text{ m}] / [(2.03 \times 10^{-8} \Omega \text{ m})(65 \text{ K})]$$

$$\therefore \alpha_{40} = \mathbf{5.23 \times 10^{-3} \text{ K}^{-1}}$$

c. We know that $1/\rho = \sigma = en\mu$ where σ is the electrical conductivity, e is the electron charge, and μ is the electron drift mobility. We also know that $\mu = e\tau / m_e$, where τ is the mean free time between electron collisions and m_e is the electron mass. Therefore,

$$1/\rho = e^2 n \tau / m_e$$

$$\therefore \tau = m_e / \rho e^2 n \quad (1)$$

Here n is the number of conduction electrons per unit volume. But, from the density d and atomic mass M_{at} , atomic concentration of Al is

$$n_{Al} = \frac{N_A d}{M_{at}} = \frac{(6.022 \times 10^{23} \text{ mol}^{-1})(2700 \text{ kg/m}^3)}{(0.027 \text{ kg/mol})} = 6.022 \times 10^{28} \text{ m}^{-3}$$

$$\text{so that} \quad n = 3n_{Al} = 1.807 \times 10^{29} \text{ m}^{-3}$$

assuming that each Al atom contributes 3 "free" conduction electrons to the metal and substituting into (1),

$$\tau = \frac{m_e}{\rho e^2 n} = \frac{(9.109 \times 10^{-31} \text{ kg})}{(2.72 \times 10^{-8} \Omega \text{ m})(1.602 \times 10^{-19} \text{ C})^2 (1.807 \times 10^{29} \text{ m}^{-3})}$$

$$\therefore \tau = \mathbf{7.22 \times 10^{-15} \text{ s}}$$

(Note: If you do not convert to meters and instead use centimeters you will not get the correct answer because seconds is an SI unit.)

The relation between the drift mobility μ_d and the mean free time is given by Equation 2.5, so that

$$\mu_d = \frac{e\tau}{m_e} = \frac{(1.602 \times 10^{-19} \text{ C})(7.22 \times 10^{-15} \text{ s})}{(9.109 \times 10^{-31} \text{ kg})}$$

$$\therefore \mu_d = \mathbf{1.27 \times 10^{-3} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1} = 12.7 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}}$$

d. The mean free path is $l = u\tau$, where u is the mean speed. With $u \approx 2 \times 10^6 \text{ m s}^{-1}$ we find the mean free path:

$$l = u\tau = (2 \times 10^6 \text{ m s}^{-1})(7.22 \times 10^{-15} \text{ s}) \approx \mathbf{1.44 \times 10^{-8} \text{ m} \approx 14.4 \text{ nm}}$$

A thin film of Al must have a much greater thickness than l to show bulk behavior. Otherwise, scattering from the surfaces will increase the resistivity by virtue of Matthiessen's rule.

e. Power $P = I^2 R$ and is proportional to resistivity ρ , assuming the rms current level stays relatively constant. Then we have

$$\begin{aligned} [P(-40^\circ\text{C}) - P(25^\circ\text{C})] / P(25^\circ\text{C}) &= P(-40^\circ\text{C}) / P(25^\circ\text{C}) - 1 = \rho(-40^\circ\text{C}) / \rho(25^\circ\text{C}) - 1 \\ &= (2.03 \times 10^{-8} \Omega\text{m} / 2.72 \times 10^{-8} \Omega\text{m}) - 1 = -0.254, \text{ or } -25.4\% \end{aligned}$$

(Negative sign means a reduction in the power loss).

2.3 Conduction in gold Gold is in the same group as Cu and Ag. Assuming that each Au atom donates one conduction electron, calculate the drift mobility of the electrons in gold at 22°C . What is the mean free path of the conduction electrons if their mean speed is $1.4 \times 10^6 \text{ m s}^{-1}$? (Use ρ_0 and α_0 in Table 2.1.)

Solution

The drift mobility of electrons can be obtained by using the conductivity relation $\sigma = en\mu_d$.

Resistivity of pure gold from Table 2.1 at 0°C (273 K) is $\rho_0 = 22.8 \text{ n}\Omega\text{m}$. Resistivity at 20°C can be calculated by using Eq. 2.19.

$$\rho = \rho_0[1 + \alpha_0(T - T_0)]$$

The TCR α_0 for Au from Table 2.1 is $1/251 \text{ K}^{-1}$. Therefore the resistivity for Au at 22°C is

$$\rho(22^\circ\text{C}) = 22.8 \text{ n}\Omega\text{m} [1 + 1/251 \text{ K}^{-1}(293\text{K} - 273\text{K})] = 24.62 \text{ n}\Omega\text{m}$$

Since one Au atom donates one conduction electron, the electron concentration is

$$n = \frac{dN_A}{M_{at}}$$

where for gold d = density = 19300 kg m^{-3} , atomic mass $M_{at} = 196.67 \text{ g mol}^{-1}$. Substituting for d , N_A , and M_{at} , we have $n = 5.91 \times 10^{28} \text{ m}^{-3}$, or $5.91 \times 10^{22} \text{ cm}^{-3}$.

$$\begin{aligned} \mu_d &= \frac{\sigma}{en} = \frac{(24.62 \times 10^{-9} \Omega\text{m})^{-1}}{(1.6 \times 10^{-19} \text{ C})(5.91 \times 10^{28} \text{ m}^{-3})} \\ &= 4.26 \times 10^{-3} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1} = 42.6 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}. \end{aligned}$$

Given the mean speed of electron is $u = 1.4 \times 10^6 \text{ m s}^{-1}$, mean free path from Equation 2.10 is

$$\begin{aligned} l &= \frac{\mu_d m_e u}{e} = \frac{(4.26 \times 10^{-3} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1})(9.1 \times 10^{-31} \text{ kg})(1.4 \times 10^6 \text{ m s}^{-1})}{1.6 \times 10^{-19} \text{ C}} \\ &= 3.39 \times 10^{-8} \text{ m} = 33.9 \text{ nm}. \end{aligned}$$

2.4 Effective number of conduction electrons per atom

a. Electron drift mobility in tin (Sn) is $3.9 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$. The room temperature (20°C) resistivity of Sn is about $110 \text{ n}\Omega\text{m}$. Atomic mass M_{at} and density of Sn are $118.69 \text{ g mol}^{-1}$ and 7.30 g cm^{-3} , respectively. How many “free” electrons are donated by each Sn atom in the crystal? How does this compare with the position of Sn in Group IVB of the Periodic Table?

- b. Consider the resistivity of few selected metals from Groups I to IV in the Periodic Table in Table 2.7. Calculate the number of conduction electrons contributed per atom and compare this with the location of the element in the Periodic Table. What is your conclusion?

Table 2.7 Selection of metals from Groups I to IV in the Periodic Table

Metal	Periodic Group	Valency	Density (g cm ⁻³)	Resistivity (nΩ m)	Mobility (cm ² V ⁻¹ s ⁻¹)
Na	IA	1	0.97	42.0	53
Mg	IIA	2	1.74	44.5	17
Ag	IB	1	10.5	15.9	56
Zn	IIB	2	7.14	59.2	8
Al	IIIB	3	2.7	26.5	12
Sn	IVB	4	7.30	110	3.9
Pb	IVB	4	11.4	206	2.3

NOTE: Mobility from Hall-effect measurements.

Solution

- a. Electron concentration can be calculated from the conductivity of Sn, $\sigma = en\mu_d$.

$$n_e = \frac{\sigma}{e\mu_d} = \frac{(110 \times 10^{-9} \Omega m)^{-1}}{(1.6 \times 10^{-19} C)(3.9 \times 10^{-4} m^2 V^{-1} s^{-1})}$$

$$= 1.46 \times 10^{29} \text{ electrons m}^{-3}.$$

The atomic concentration, i.e. number of Sn atoms per unit volume is

$$n_{at} = \frac{dN_A}{M_{at}} = \frac{(7.3 \times 10^3 kg)(6.022 \times 10^{23} mol^{-1})}{(118.69 \times 10^{-3} kg mol^{-1})}$$

$$= 3.70 \times 10^{28} \text{ Sn atoms m}^{-3}.$$

Hence the number of electrons donated by each atom is $(n_e/n_{at}) = 3.94$ or **4** electrons per Sn atom. This is in good agreement with the position of the Sn in the Periodic Table (IVB) and its valency of 4.

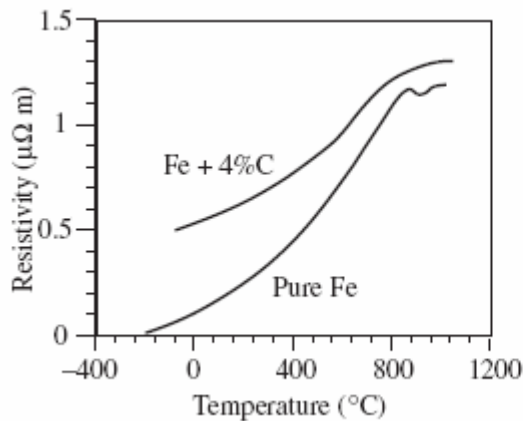
- b. Using the same method used above, the number of electrons donated by each atom of the element are calculated and tabulated as follows:

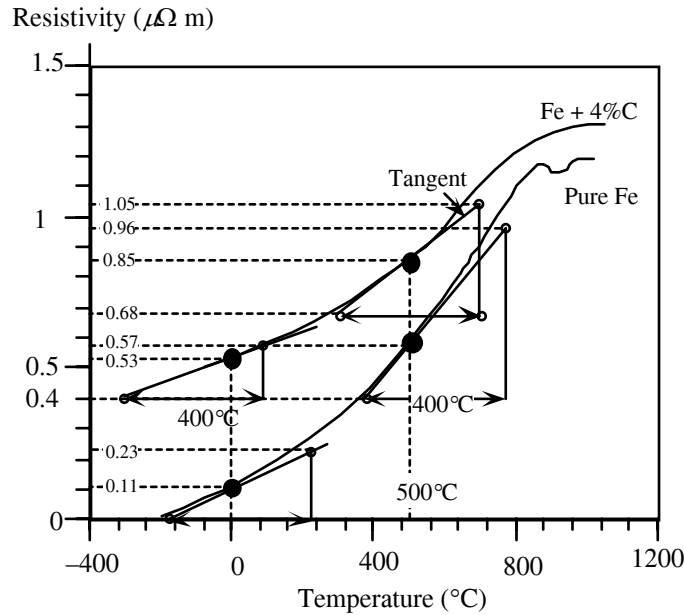
Meta l	Periodi c Group	Valency	Atomic Concentration $n_{at} \text{ (m}^{-3}\text{)}$	Electron Concentration $n_e \text{ (m}^{-3}\text{)}$	Number of electrons n_e/n_{at}	Integer (n_e/n_{at})
Na	IA	1	2.541×10^{28}	2.808×10^{28}	1.105	1
Mg	IIA	2	4.311×10^{28}	8.262×10^{28}	1.916	2
Ag	IB	1	5.862×10^{28}	7.019×10^{28}	1.197	1
Zn	IIB	2	6.575×10^{28}	1.320×10^{29}	2.007	2
Al	IIIB	3	6.026×10^{28}	1.965×10^{29}	3.262	3
Sn	IVB	4	3.703×10^{28}	1.457×10^{29}	3.934	4
Pb	IVB	4	3.313×10^{28}	1.319×10^{29}	3.981	4

Table 2Q4-1: Number of electrons donated by various elements

As evident from the above table, the calculated number of electrons donated by one atom of the element is the same as the valency of that element and the position in the periodic table.

2.5 TCR and Matthiessen's rule Determine the temperature coefficient of resistivity of pure iron and of electrotechnical steel (Fe with 4% C), which are used in various electrical machinery, at two temperatures: 0 °C and 500 °C. Comment on the similarities and differences in the resistivity versus temperature behavior shown in Figure 2.39 for the two materials.

**Figure 2.39** Resistivity versus temperature for pure iron and 4% C steel.

Solution**Figure 2Q5-1:** Resistivity versus temperature for pure iron and 4% C steel.

The temperature coefficient of resistivity α_o (TCR) is defined as follows:

$$\alpha_o = \frac{1}{\rho_o} \left(\frac{d\rho}{dT} \right)_{T_o} = \frac{\text{Slope at } T_o}{\rho_o}$$

where the slope is $d\rho/dT$ at $T = T_o$ and ρ_o is the resistivity at $T = T_o$.

To find the slope, we draw a tangent to the curve at $T = T_o$ ($T_o = 0^\circ\text{C}$ and then $T_o = 500^\circ\text{C}$) and obtain $\Delta\rho/\Delta T \approx d\rho/dT$. One convenient way is to define $\Delta T = 400^\circ\text{C}$ and find $\Delta\rho$ on the tangent line and then calculate $\Delta\rho/\Delta T$.

Iron at 0°C :

$$\text{Slope}_o \approx (0.23 \times 10^{-6} \Omega \text{ m} - 0 \Omega \text{ m}) / (400^\circ\text{C}) = 5.75 \times 10^{-10} \Omega \text{ m } ^\circ\text{C}^{-1}$$

$$\text{Since } \rho_o \approx 0.11 \times 10^{-6} \Omega \text{ m}, \alpha_o = \text{Slope}_o / \rho_o \approx 0.00523^\circ\text{C}^{-1}$$

Fe + 4% C at 0°C :

$$\text{Slope}_o \approx (0.57 \times 10^{-6} \Omega \text{ m} - 0.4 \times 10^{-6} \Omega \text{ m}) / (400^\circ\text{C}) = 4.25 \times 10^{-10} \Omega \text{ m } ^\circ\text{C}^{-1}$$

$$\text{Since } \rho_o \approx 0.53 \times 10^{-6} \Omega \text{ m}, \alpha_o = \text{Slope}_o / \rho_o \approx 0.00802^\circ\text{C}^{-1}$$

Iron at 500°C :

$$\text{Slope}_o \approx (0.96 \times 10^{-6} \Omega \text{ m} - 0.4 \times 10^{-6} \Omega \text{ m}) / (400^\circ\text{C}) = 1.40 \times 10^{-9} \Omega \text{ m } ^\circ\text{C}^{-1}$$

$$\text{Since } \rho_o \approx 0.57 \times 10^{-6} \Omega \text{ m}, \alpha_o = \text{Slope}_o / \rho_o \approx 0.00245^\circ\text{C}^{-1}$$

Fe + 4% C at 500°C :

$$\text{Slope}_o \approx (1.05 \times 10^{-6} \Omega \text{ m} - 0.68 \times 10^{-6} \Omega \text{ m}) / (400^\circ\text{C}) = 9.25 \times 10^{-10} \Omega \text{ m } ^\circ\text{C}^{-1}$$

$$\text{Since } \rho_o \approx 0.85 \times 10^{-6} \Omega \text{ m}, \alpha_o = \text{Slope}_o / \rho_o \approx 0.00109^\circ\text{C}^{-1}$$

*2.6 TCR of isomorphous alloys

- a. Show that for an isomorphous alloy $A\%-B\%$ ($B\%$ solute in $A\%$ solvent), the temperature coefficient of resistivity α_{AB} is given by

$$\alpha_{AB} \approx \frac{\alpha_A \rho_A}{\rho_{AB}}$$

where ρ_{AB} is the resistivity of the alloy (AB) and ρ_A and α_A are the resistivity and TCR of pure A . What are the assumptions behind this equation?

- b. Determine the composition of the Cu-Ni alloy that will have a TCR of $4 \times 10^{-4} \text{ K}^{-1}$, that is, a TCR that is an order of magnitude less than that of Cu. Over the composition range of interest, the resistivity of the Cu-Ni alloy can be calculated from $\rho_{\text{CuNi}} \approx \rho_{\text{Cu}} + C_{\text{eff}}X(1-X)$, where C_{eff} , the effective Nordheim coefficient, is about $1310 \text{ n}\Omega \text{ m}$.

Solution

- a. By the Nordheim rule, the resistivity of the alloy is $\rho_{\text{alloy}} = \rho_o + CX(1-X)$. We can find the TCR of the alloy from its definition

$$\alpha_{\text{alloy}} = \frac{1}{\rho_{\text{alloy}}} \frac{d\rho_{\text{alloy}}}{dT} = \frac{1}{\rho_{\text{alloy}}} \frac{d}{dT} [\rho_o + CX(1-X)]$$

To obtain the desired equation, **we must assume that C is temperature independent** (*i.e.* the increase in the resistivity depends on the lattice distortion induced by the impurity) so that $d[CX(1-X)]/dT = 0$, enabling us to substitute for $d\rho_o/dT$ using the definition of the TCR: $\alpha_o = (d\rho_o/dT)/\rho_o$. Substituting into the above equation:

$$\alpha_{\text{alloy}} = \frac{1}{\rho_{\text{alloy}}} \frac{d\rho_o}{dT} = \frac{1}{\rho_{\text{alloy}}} \alpha_o \rho_o$$

$$\text{i.e.} \quad \alpha_{\text{alloy}} \rho_{\text{alloy}} = \alpha_o \rho_o \quad \text{or} \quad \alpha_{AB} \rho_{AB} = \alpha_A \rho_A$$

Remember that all values for the alloy and pure substance must all be taken at the same temperature, or the equation is invalid.

- b. Assume room temperature $T = 293 \text{ K}$. Using values for copper from Table 2.1 in Equation 2.19, $\rho_{\text{Cu}} = 17.1 \text{ n}\Omega \text{ m}$ and $\alpha_{\text{Cu}} = 4.0 \times 10^{-3} \text{ K}^{-1}$, and from Table 2.3 the Nordheim coefficient of Ni dissolved in Cu is $C = 1570 \text{ n}\Omega \text{ m}$. We want to find the composition of the alloy such that $\alpha_{\text{CuNi}} = 4 \times 10^{-4} \text{ K}^{-1}$. Then,

$$\alpha_{\text{alloy}} = \frac{\alpha_{\text{Cu}} \rho_{\text{Cu}}}{\alpha_{\text{alloy}}} = \frac{(0.0040 \text{ K}^{-1})(17.1 \text{ n}\Omega \text{ m})}{0.0004 \text{ K}^{-1}} = 171.0 \text{ n}\Omega \text{ m}$$

Using Nordheim's rule:

$$\rho_{\text{alloy}} = \rho_{\text{Cu}} + CX(1-X)$$

$$\text{i.e.} \quad 171.0 \text{ n}\Omega \text{ m} = 17.1 \text{ n}\Omega \text{ m} + (1570 \text{ n}\Omega \text{ m})X(1-X)$$

$$\therefore \quad X^2 - X + 0.0879 = 0$$

solving the quadratic, we find $X = 0.11$.

Thus the composition is **89% Cu-11% Ni**. However, this value is in *atomic percent* as the Nordheim coefficient is in atomic percent. Note that as Cu and Ni are very close in the Periodic Table this would also be the weight percentage. Note: the quadratic will produce another value, namely $X = 0.86$. However, using this number to obtain a composition of 11% Cu-89% Ni is incorrect because the values we used in calculations corresponded to a solution of Ni dissolved in Cu, not vice-versa (*i.e.* Ni was taken to be the impurity).

2.7 Resistivity of isomorphous alloys and Nordheim's rule What are the maximum atomic and weight percentages of Cu that can be added to Au without exceeding a resistivity that is twice that of pure gold? What are the maximum atomic and weight percentages of Au that can be added to pure Cu without exceeding twice the resistivity of pure copper? (Alloys are normally prepared by mixing the elements in weight.)

Solution

From combined Matthiessen and Nordheim rule,

$$\rho_{\text{Alloy}} = \rho_{\text{Au}} + \rho_{\text{Cu}},$$

with $\rho_{\text{Cu}} = CX(1-X)$. In order to keep the resistivity of the alloy less than twice of pure gold, the resistivity of solute (Cu), should be less than resistivity of pure gold, *i.e.* $\rho_{\text{Cu}} = CX(1-X) < \rho_{\text{Au}}$. From Table 2.3, Nordheim coefficient for Cu in Au at 20°C is, $C = 450 \text{ n}\Omega \text{ m}$. Resistivity of Au at 20°C, using $\alpha_0 = 1/251 \text{ K}^{-1}$, is

$$\rho = \rho_0[1 + \alpha_0(T - T_0)] = 22.8 \text{ n}\Omega \text{ m} \left[1 + \frac{1}{251} \text{ K}^{-1}(293 \text{ K} - 273 \text{ K})\right] = 24.62 \text{ n}\Omega \text{ m}$$

Therefore the condition for solute (Cu) atomic fraction is $CX(1-X) < 24.62 \text{ n}\Omega \text{ m}$.

$$X(1-X) < (24.62 \text{ n}\Omega \text{ m}) / (450 \text{ n}\Omega \text{ m}) = 0.0547$$

$$X^2 - X - 0.0547 < 0$$

Solving the above equation, we have $X < 0.0581$ or 5.81%. Therefore the atomic fraction of Cu should be less than 0.0581 in order to keep the overall resistivity of the alloy less than twice the resistivity of pure Au. The weight fraction for Cu for this atomic fraction can be calculated from

$$\begin{aligned} w_{\text{Cu}} &= \frac{XM_{\text{Cu}}}{XM_{\text{Cu}} + (1-X)M_{\text{Au}}} = \frac{(0.0581)(63.54 \text{ g mol}^{-1})}{(0.0581)(63.54 \text{ g mol}^{-1}) + (1-0.0581)(196.67 \text{ g mol}^{-1})} \\ &= 0.01956 \text{ or } 1.956\%. \end{aligned}$$

Now, we discuss the case of Au in Cu, *i.e.* Au as solute in Cu alloy. Resistivity of Cu at 0°C is 15.7 nΩ m. Therefore the resistivity of Cu at 20°C is

$$\rho = \rho_0[1 + \alpha_0(T - T_0)] = 15.7 \text{ n}\Omega \text{ m} \left[1 + \frac{1}{232} \text{ K}^{-1}(293 \text{ K} - 273 \text{ K})\right] = 17.05 \text{ n}\Omega \text{ m}$$

Therefore the condition for solute (Au) atomic fraction is $CX(1-X) < 17.03 \text{ n}\Omega \text{ m}$. Nordheim coefficient for Au in Cu at 20°C is, $C = 5500 \text{ n}\Omega \text{ m}$.

$$X(1-X) < (17.03 \text{ n}\Omega \text{ m}) / (5500 \text{ n}\Omega \text{ m}) = 3.10 \times 10^{-3}.$$

$$X^2 - X - 3.10 \times 10^{-3} < 0$$

Solving the inequality we get the condition, $X < 3.106 \times 10^{-3}$, required to keep the resistivity of alloy less than twice of pure Cu. The weight fraction for Cu for this atomic fraction can be calculated from

$$w_{Au} = \frac{XM_{Au}}{XM_{Au} + (1-X)M_{Cu}} = \frac{(3.106 \times 10^{-3})(196.67 \text{ g mol}^{-1})}{(3.106 \times 10^{-3})(196.67 \text{ g mol}^{-1}) + (1 - 3.106 \times 10^{-3})(63.54 \text{ g mol}^{-1})}$$

$$= 9.55 \times 10^{-3} \text{ or } 0.955\%.$$

2.8 Nordheim's rule and brass Brass is a Cu–Zn alloy. Table 2.8 shows some typical resistivity values for various Cu–Zn compositions in which the alloy is a solid solution (up to 30% Zn).

- Plot ρ versus $X(1-X)$. From the slope of the best-fit line find the mean (effective) Nordheim coefficient C for Zn dissolved in Cu over this compositional range.
- Since X is the atomic fraction of Zn in brass, for each atom in the alloy, there are X Zn atoms and $(1-X)$ Cu atoms. The conduction electrons consist of each Zn donating two electrons and each copper donating one electron. Thus, there are $2(X) + 1(1-X) = 1 + X$ conduction electrons per atom. Since the conductivity is proportional to the electron concentration, the combined Nordheim-Matthiessens rule must be scaled up by $(1 + X)$.

$$\rho_{\text{brass}} = \frac{\rho_0 + CX(1-X)}{(1+X)}$$

Plot the data in Table 2.8 as $\rho(1+X)$ versus $X(1-X)$. From the best-fit line find C and ρ_0 . What is your conclusion? (Compare the correlation coefficients of the best-fit lines in your two plots).

NOTE: More rigorously, $\rho_{\text{brass}} = \rho_{\text{matrix}} + C_{\text{eff}} X(1-X)$, in which ρ_{matrix} is the resistivity of the perfect matrix. Accounting for the extra electrons, $\rho_{\text{matrix}} \approx \rho_0/(1+X)$, where ρ_0 is the pure metal matrix resistivity and C_{eff} is the Nordheim coefficient at the composition of interest, given by $C_{\text{eff}} \approx C/(1+X)^{2/3}$. (It is assumed that the atomic concentration does not change significantly.) As always, there are also other theories; part *b* is more than sufficient for most practical purposes.

Table 2.8 Cu–Zn brass alloys

Zn at. % in Cu–Zn	0	0.34	0.5	0.93	3.06	4.65	9.66	15.6	19.59	29.39
Resistivity nΩ m	17	18.1	18.84	20.7	26.8	29.9	39.1	49.0	54.8	63.5

SOURCE: H. A. Fairbank, Phys. Rev., 66, 274, 1944.

Solution

a. We know the resistivity to be $\rho_{\text{alloy}} = \rho_o + CX(1-X)$. We plot ρ_{alloy} versus $X(1-X)$. We have a best-fit straight line of the form $y = mx + b$, where m is the slope of the line. The slope is C_{eff} , the Nordheim coefficient.

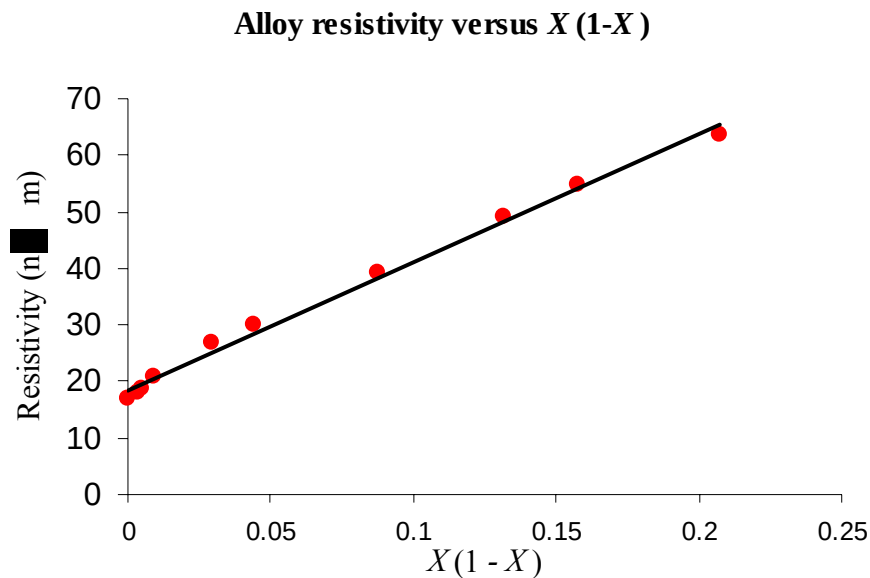


Figure 2Q8-1: Plot of alloy resistivity against $X(1-X)$

The equation of the line is $y = 225.76x + 18.523$. The slope m of the best-fit line is 225.76 nΩ m, which is the effective Nordheim coefficient C_{eff} for the compositional range of Zn provided.

b.

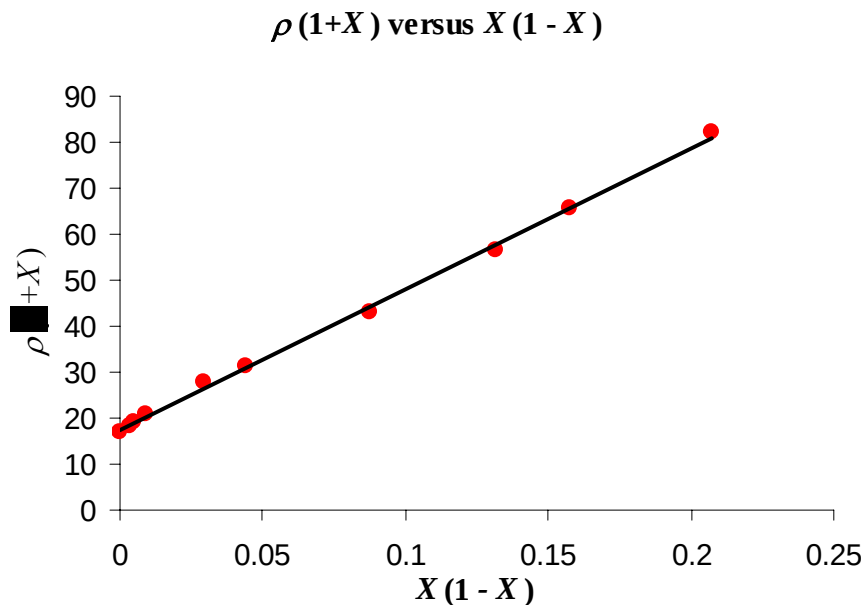


Figure 2Q8-2: Plot of $\rho(1+X)$ against $X(X-1)$

The slope of the best-fit line is 306.67. As given in the question, the modified combined Nordheim–Matthiessens rule must be scaled up by $(1 + X)$,

$$\rho_{\text{brass}} = \frac{\rho_0 + CX(1 - X)}{(1 + X)}$$

or
$$\rho_{\text{brass}}(1 + X) = \rho_0 + CX(1 - X)$$

The above equation is of the straight line form $y = mx + b$, where m is the slope of the line. Therefore from the equation of the line $y = 306.67x + 17.4$, we have the effective Nordheim coefficient is $C_{\text{eff}} = 306.67 \text{ n}\Omega \text{ m}$ and ρ_0 is $17.40 \text{ n}\Omega \text{ m}$.

If we calculate the resistivity using the values obtained above in the combined Nordheim-Mattheisen rule we obtain the following table,

Zn at.% X	Experimental Resistivity ($\text{n}\Omega \text{ m}$)	Case I	Case II
		Resistivity ($\text{n}\Omega \text{ m}$) $C_{\text{eff}} = 225.76 \text{ n}\Omega \text{ m}$	Resistivity ($\text{n}\Omega \text{ m}$) $C_{\text{eff}} = 306.67 \text{ n}\Omega \text{ m}$ Scaled by $(1+X)$
0	17	17.00	17.00
0.34	18.1	17.76	17.98
0.5	18.84	18.12	18.43
0.93	20.7	19.08	19.64
3.06	26.8	23.70	25.32
4.65	29.9	27.01	29.24
9.66	39.1	36.70	39.91
15.6	49	46.72	49.63
19.59	54.8	52.56	54.61
29.39	63.5	63.85	62.32

Table 2Q8-1: C_{eff} values calculated by fitting line to experimental data and by taking into account the effect of extra valence electron

For case I, the resistivity is calculated using effective Nordheim coefficient (C_{eff}) and for the second case the combined Nordheim–Matthiessens rule is scaled up by $(1 + X)$. It is observed that the values obtained by the later method is closer to the experimental results verifying the method of scaling taking in to consideration the number of electrons donated by the solute atoms.

2.9 Resistivity of solid solution metal alloys: testing Nordheim’s rule Nordheim’s rule accounts for the increase in the resistivity from the scattering of electrons from the random distribution of impurity (solute) atoms in the host (solvent) crystal. It can nonetheless be quite useful in approximately predicting the resistivity at one composition of a solid solution metal alloy, given the

value at another composition. Table 2.9 lists some solid solution metal alloys and gives the resistivity ρ at one composition X and asks for a prediction ρ' based on Nordheim's rule at another composition X' . Fill in the table for ρ' and compare the predicted values with the experimental values, and comment.

Table 2.9 Resistivities of some solid solution metal alloys

	Alloy							
	Ag-Au	Au-Ag	Cu-Pd	Ag-Pd	Au-Pd	Pd-Pt	Pt-Pd	Cu-Ni
X (at.%)	8.8% Au	8.77% Ag	6.2% Pd	10.1% Pd	8.88% Pd	7.66% Pt	7.1% Pd	2.16% Ni
ρ_0 (n Ω m)	16.2	22.7	17	16.2	22.7	108	105.8	17
ρ at X (n Ω m)	44.2	54.1	70.8	59.8	54.1	188.2	146.8	50
C_{eff}								
X'	15.4% Au	24.4% Ag	13% Pd	15.2% Pd	17.1% Pd	15.5% Pt	13.8% Pd	23.4% Ni
ρ' at X' (n Ω m)								
ρ' at X' (n Ω m)	66.3	107.2	121.6	83.8	82.2	244	181	300
Experimental								

NOTE: First symbol (e.g., Ag in AgAu) is the matrix (solvent) and the second (Au) is the added solute. X is in at.%, converted from traditional weight percentages reported with alloys. C_{eff} is the effective Nordheim coefficient in $\rho = \rho_0 + C_{\text{eff}}X(1 - X)$.

Solution

Combined Matthiessen and Nordheim rule is

$$\rho_{\text{alloy}} = \rho_0 + C_{\text{eff}}X(1 - X)$$

therefore, from the above equation effective Nordheim coefficient C_{eff} is

$$C_{\text{eff}} = \frac{\rho_{\text{alloy}} - \rho_0}{X(1 - X)}$$

Ag-Au:

For this alloy, it is given that for $X = 8.8\%$ Au, $\rho = 44.2$ n Ω m, with $\rho_0 = 16.2$ n Ω m, the effective Nordheim coefficient C_{eff} is

$$C_{\text{eff}} = \frac{(44.2 - 16.2) \text{ n}\Omega \text{ m}}{0.088 \times (1 - 0.088)} = 348.89 \text{ n}\Omega \text{ m}$$

Now, for $X' = 15.4\%$ Au, the resistivity of the alloy will be

$$\rho' = 16.2 \text{ n}\Omega \text{ m} + (348.88 \text{ n}\Omega \text{ m})(0.154)(1 - 0.154) = 61.65 \text{ n}\Omega \text{ m}$$

Similarly, the effective Nordheim coefficient C_{eff} and the resistivities of the alloys at X' are calculated for the various alloys and tabulated as follows,

	Alloy							
	Ag-Au	Au-Ag	Cu-Pd	Ag-Pd	Au-Pd	Pd-Pt	Pt-Pd	Cu-Ni
X (at.%)	8.8% Au	8.77% Ag	6.2% Pd	10.1% Pd	8.88% Pd	7.66% Pt	7.1% Pd	2.16% Ni

ρ_0 (n Ω m)	16.2	22.7	17	16.2	22.7	108	105.8	17
ρ at X (n Ω m)	44.2	54.1	70.8	59.8	54.1	188.2	146.8	50
C_{eff}	348.88	392.46	925.10	480.18	388.06	1133.85	621.60	1561.51
X'	15.4% Au	24.4% Ag	13% Pd	15.2% Pd	17.1% Pd	15.5% Pt	13.8% Pd	23.4% Ni
ρ' at X' (n Ω m)	61.65	95.09	121.63	78.09	77.71	256.51	179.74	296.89
ρ' at X' (n Ω m) Experimental	66.3	107.2	121.6	83.8	82.2	244	181	300
Percentage Difference	7.01% less	11.29% less	0.02% more	6.81% less	5.46% less	4.88% more	0.69% less	1.04% less

Table 2Q9-1: Resistivities of solid solution metal alloys

Comment: From the above table, the best case has a 0.02% difference and the worst case has a 7% difference. It is clear that the Nordheim rule is very useful in predicting the approximate resistivity of a solid solution at one composition from the resistivity at a known composition.

***2.10 TCR and alloy resistivity** Table 2.10 shows the resistivity and TCR (α) of Cu–Ni alloys. Plot TCR versus $1/\rho$, and obtain the best-fit line. What is your conclusion? Consider the Matthiessen rule, and explain why the plot should be a straight line. What is the relationship between ρ_{Cu} , α_{Cu} , ρ_{CuNi} , and α_{CuNi} ? Can this be generalized?

Table 2.10 Cu–Ni alloys, resistivity, and TCR

	Ni wt.% in Cu–Ni				
	0	2	6	11	20
Resistivity (n Ω m)	17	50	100	150	300
TCR (ppm $^{\circ}\text{C}^{-1}$)	4270	1350	550	430	160

NOTE: ppm-parts per million, i.e., 10^{-6} .

Solution

The plot of temperature coefficient of resistivity TCR (α) versus $1/\rho$, is as follows

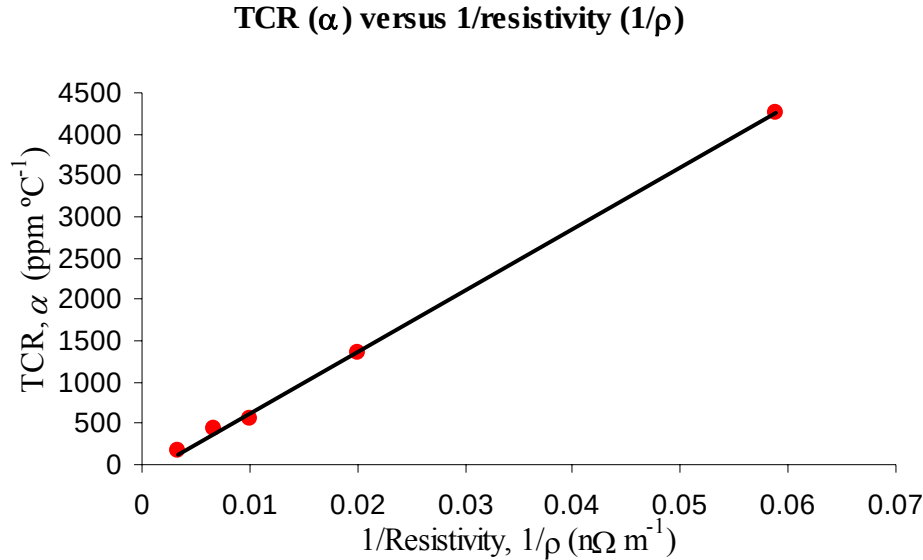


Figure 2Q10-1: TCR (α) versus reciprocal of resistivity $1/\rho$

From Matthiessen's rules, we have

$$\rho_{\text{alloy}} = \rho_{\text{matrix}} + \rho_I = \rho_o + \rho_I$$

where ρ_o is the resistivity of the matrix, determined by scattering of electrons by thermal vibrations of crystal atoms and ρ_I is the resistivity due to scattering of electrons from the impurities. Obviously, ρ_o is a function of temperature, but ρ_I shows very little temperature dependence. From the definition of temperature coefficient of resistivity,

$$\alpha_o = \frac{1}{\rho_o} \left(\frac{\delta \rho_o}{\delta T} \right) \quad \text{or} \quad \frac{\delta \rho_o}{\delta T} = \alpha_o \rho_o$$

and

$$\alpha_{\text{alloy}} = \frac{1}{\rho_{\text{alloy}}} \left(\frac{\delta \rho_{\text{alloy}}}{\delta T} \right) = \frac{\delta(\rho_o + \rho_I)}{\rho_{\text{alloy}} \delta T} \approx \frac{\delta \rho_o}{\rho_{\text{alloy}} \delta T} = \frac{\alpha_o \rho_o}{\rho_{\text{alloy}}} \propto \frac{1}{\rho_{\text{alloy}}}$$

Clearly the TCR of the alloy is inversely proportional to the resistivity of the alloy. The higher the resistivity, the smaller the TCR, which is evident from the plot.

2.11 Electrical and thermal conductivity of In Electron drift mobility in indium has been measured to be $6 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$. The room temperature (27°C) resistivity of In is $8.37 \times 10^{-8} \Omega\text{m}$, and its atomic mass and density are 114.82 amu or g mol^{-1} and 7.31 g cm^{-3} , respectively.

- Based on the resistivity value, determine how many free electrons are donated by each In atom in the crystal. How does this compare with the position of In in the Periodic Table (Group IIIB)?
- If the mean speed of conduction electrons in In is $1.74 \times 10^8 \text{ cm s}^{-1}$, what is the mean free path?
- Calculate the thermal conductivity of In. How does this compare with the experimental value of $81.6 \text{ W m}^{-1} \text{ K}^{-1}$?

Solution

a. From $\sigma = en\mu_d$ (σ is the conductivity of the metal, e is the electron charge, and μ_d is the electron drift mobility) we can calculate the concentration of conduction electrons (n):

$$n = \frac{\sigma}{e\mu_d} = \frac{(8.37 \times 10^{-8} \Omega \text{ m})^{-1}}{(1.602 \times 10^{-19} \text{ C})(6 \times 10^{-4} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1})}$$

i.e. $n = 1.243 \times 10^{29} \text{ m}^{-3}$

Atomic concentration n_{at} is

$$n_{\text{at}} = \frac{dN_A}{M_{\text{at}}} = \frac{(7.31 \times 10^3 \text{ kg m}^{-3})(6.022 \times 10^{23} \text{ mol}^{-1})}{(114.82 \times 10^{-3} \text{ kg mol}^{-1})}$$

i.e. $n_{\text{at}} = 3.834 \times 10^{28} \text{ m}^{-3}$

Effective number of conduction electrons donated per In atom (n_{eff}) is:

$$n_{\text{eff}} = n / n_{\text{at}} = (1.243 \times 10^{29} \text{ m}^{-3}) / (3.834 \times 10^{28} \text{ m}^{-3}) = 3.24$$

Conclusion: There are therefore about three electrons per atom donated to the conduction-electron sea in the metal. This is in good agreement with the position of the In element in the Periodic Table (III) and its valency of 3.

b. If τ is the mean scattering time of the conduction electrons, then from $\mu_d = e\tau/m_e$ (m_e = electron mass) we have:

$$\tau = \frac{\mu_d m_e}{e} = \frac{(6 \times 10^{-4} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1})(9.109 \times 10^{-31} \text{ kg})}{(1.602 \times 10^{-19} \text{ C})} = 3.412 \times 10^{-15} \text{ s}$$

Taking the mean speed $u \approx 1.74 \times 10^6 \text{ m s}^{-1}$, the mean free path (l) is given by

$$l = u\tau = (1.74 \times 10^6 \text{ m s}^{-1})(3.412 \times 10^{-15} \text{ s}) = 5.94 \times 10^{-9} \text{ m or } 5.94 \text{ nm}$$

c. From the Wiedemann-Franz-Lorenz law, thermal conductivity is given as:

$$\kappa = \sigma TC_{\text{WFL}} = (8.37 \times 10^{-8} \Omega \text{ m})^{-1}(20^\circ \text{C} + 273 \text{ K})(2.44 \times 10^{-8} \text{ W } \Omega \text{ K}^{-2})$$

i.e. $\kappa = 85.4 \text{ W m}^{-1} \text{ K}^{-1}$

This value reasonably agrees with the experimental value.

2.12 Electrical and thermal conductivity of Ag The electron drift mobility in silver has been measured to be $56 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ at 27°C . The atomic mass and density of Ag are given as 107.87 amu or g mol^{-1} and 10.50 g cm^{-3} , respectively.

a. Assuming that each Ag atom contributes one conduction electron, calculate the resistivity of Ag at 27°C . Compare this value with the measured value of $1.6 \times 10^{-8} \Omega \text{ m}$ at the same temperature and suggest reasons for the difference.

b. Calculate the thermal conductivity of silver at 27°C and at 0°C .

Solution

a. Atomic concentration n_{at} is

$$n_{\text{at}} = \frac{dN_A}{M_{\text{at}}} = \frac{(10.50 \times 10^3 \text{ kg m}^{-3})(6.022 \times 10^{23} \text{ mol}^{-1})}{(107.87 \times 10^{-3} \text{ kg mol}^{-1})} = 5.862 \times 10^{28} \text{ m}^{-3}$$

If we assume there is one conduction electron per Ag atom, the concentration of conduction electrons (n) is $5.862 \times 10^{28} \text{ m}^{-3}$, and the conductivity is therefore:

$$\sigma = en\mu_d = (1.602 \times 10^{-19} \text{ C})(5.862 \times 10^{28} \text{ m}^{-3})(56 \times 10^{-4} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}) = 5.259 \times 10^7 \Omega^{-1} \text{ m}^{-1}$$

and the resistivity, $\rho = 1/\sigma = 19.0 \text{ n}\Omega \text{ m}$

The experimental value of ρ is $16 \text{ n}\Omega \text{ m}$. We assumed that exactly 1 "free" electron per Ag atom contributes to conduction. This is not necessarily true. We need to use energy bands to describe conduction more accurately and this is addressed in Chapter 4.

b. From the Wiedemann-Franz-Lorenz law at 27°C ,

$$\kappa = \sigma T C_{\text{WFL}} = (5.259 \times 10^7 \Omega^{-1} \text{ m}^{-1})(27 + 273 \text{ K})(2.44 \times 10^{-8} \text{ W } \Omega \text{ K}^{-2})$$

i.e. $\kappa = 385 \text{ W m}^{-1} \text{ K}^{-1}$

For pure metals such as Ag this is nearly independent of temperature (same at 0°C).

2.13 Mixture rules

A 70% Cu - 30% Zn brass electrical component has been made of powdered metal and contains 15 vol. % porosity. Assume that the pores are dispersed randomly. Given that the resistivity of 70% Cu-30% Zn brass is $62 \text{ n}\Omega \text{ m}$, calculate the effective resistivity of the brass component using the simple conductivity mixture rule, Equation 2.26 and the Reynolds and Hough rule.

Solution

The component has 15% air pores. Apply the empirical mixture rule in Equation 2.26. The fraction of volume with air pores is $\chi = 0.15$. Then,

$$\rho_{\text{eff}} = \rho \frac{(1 + \frac{1}{2}\chi)}{(1 - \chi)} = 62 \text{ n}\Omega \text{ m} \frac{(1 + 0.5 \times 0.15)}{(1 - 0.15)} = 78.41 \text{ n}\Omega \text{ m}$$

Reynolds and Hough rule is given by Equation 2.28 as

$$\frac{\sigma - \sigma_{\text{alloy}}}{\sigma + 2\sigma_{\text{alloy}}} = \chi \frac{\sigma_{\text{air}} - \sigma_{\text{alloy}}}{\sigma_{\text{air}} + 2\sigma_{\text{alloy}}}$$

For the given case $\sigma_{\text{air}} = 0$, $\sigma_{\text{alloy}} = (62 \text{ n}\Omega \text{ m})^{-1}$. Substituting the conductivity values in the RHS of the equation we have

$$\chi \frac{\sigma_{\text{air}} - \sigma_{\text{alloy}}}{\sigma_{\text{air}} + 2\sigma_{\text{alloy}}} = (0.15) \left(\frac{0 - (62 \text{ n}\Omega \text{ m})^{-1}}{0 + 2(62 \text{ n}\Omega \text{ m})^{-1}} \right) = -0.075.$$

Solving for effective conductivity, we have $\sigma = 1.2753 \times 10^7 \Omega^{-1} \text{ m}^{-1}$

$$\therefore \rho_{\text{eff}} = 78.41 \times 10^{-9} \Omega \text{m or } 78.41 \text{ n}\Omega \text{ m.}$$

Hence the values obtained are the same. Equation 2.26 is in fact the simplified version of Reynolds and Hough rule for the case when the resistivity of dispersed phase is considerably larger than the continuous phase.

2.14 Mixture rules

- A certain carbon electrode used in electrical arcing applications is 47 percent porous. Given that the resistivity of graphite (in polycrystalline form) at room temperature is about $9.1 \mu\Omega \text{ m}$, estimate the effective resistivity of the carbon electrode using the appropriate Reynolds and Hough rule and the simple conductivity mixture rule. Compare your estimates with the measured value of $18 \mu\Omega \text{ m}$ and comment on the differences.
- Silver particles are dispersed in a graphite paste to increase the effective conductivity of the paste. If the volume fraction of dispersed silver is 30 percent, what is the effective conductivity of this paste?

Solution

- The effective conductivity of mixture can be estimated using Reynolds and Hough rule that states

$$\frac{\sigma - \sigma_c}{\sigma + 2\sigma_c} = \chi \frac{\sigma_d - \sigma_c}{\sigma_d + 2\sigma_c}$$

If the conductivity of the dispersed medium is very small compared to the continuous phase, as in the given case conductivity of pores is extremely small compared to polycrystalline carbon, i.e. $\sigma_c \gg \sigma_d$. Equation 2.26 is the simplified version of Reynolds and Hough rule.

The volume fraction of air pores is $\chi = 0.47$ and the conductivity of graphite is $\rho_c = 9.1 \mu\Omega \text{ m}$, therefore

$$\rho_{\text{eff}} = \rho_c \frac{(1 + \frac{1}{2} \chi_d)}{(1 - \chi_d)} = (9.1 \mu\Omega \text{ m}) \frac{(1 + 0.5 \times 0.47)}{(1 - 0.47)} = 21.21 \mu\Omega \text{ m}$$

Conductivity mixture rule is based on the assumption that the two phases α and β are parallel to each other and the effective conductivity from Equation 2.25 is

$$\sigma_{\text{eff}} = \chi_\alpha \sigma_\alpha + \chi_\beta \sigma_\beta$$

For the given situation $\chi_{\text{air}} = 0.47$, $\chi_{\text{graphite}} = (1 - 0.47)$, $\sigma_{\text{air}} = 0$, $\sigma_{\text{graphite}} = (9.1 \mu\Omega \text{ m})^{-1}$, therefore the effective resistivity using the conductivity mixture rule is

$$\frac{1}{\rho_{\text{eff}}} = \frac{(1 - 0.47)}{9.1 \mu\Omega \text{ m}} + 0$$

$$\rho_{\text{eff}} = 17.17 \mu\Omega \text{ m}$$

- If the dispersed phase has higher conductivity than the continuous phase, the Reynolds and Hough rule is reduced to Equation 2.27. From Table 2.1, resistivity for silver at 273 K is $14.6 \mu\Omega \text{ m}$. Using $\alpha_0 = 1/244 \text{ K}^{-1}$, the resistivity at room temperature (20°C) can be estimated as

$$\rho = \rho_0 [1 + \alpha_0 (T - T_0)] = (14.6 \text{ n}\Omega \text{ m}) \left[1 + \frac{1}{244} \text{ K}^{-1} (293 \text{ K} - 273 \text{ K}) \right] = 15.79 \text{ n}\Omega \text{ m}$$

Since $\rho_d < 0.1\rho_c$, we can use Equation 2.27. Volume fraction of dispersed silver is 30%, $\chi_d = 0.3$. The effective resistivity is

$$\rho_{eff} = \rho_c \frac{(1 - \chi_d)}{(1 + 2\chi_d)} = (9.1 \mu\Omega \text{ m}) \frac{(1 - 0.3)}{(1 - 2 \times 0.3)} = \mathbf{3.98 \mu\Omega \text{ m}}.$$

If we use the Reynolds and Hough rule to calculate the effective resistivity, we obtain **3.99 $\mu\Omega \text{ m}$** , which is reasonably close to the value calculated by using Equation 2.27.

2.15 Ag–Ni alloys (contact materials) and the mixture rules Silver alloys, particularly Ag alloys with the precious metals Pt, Pd, Ni, and Au, are extensively used as contact materials in various switches. Alloying Ag with other metals generally increases the hardness, wear resistance, and corrosion resistance at the expense of electrical and thermal conductivity. For example, Ag–Ni alloys are widely used as contact materials in switches in domestic appliances, control and selector switches, circuit breakers, and automotive switches up to several hundred amperes of current. Table 2.11 shows the resistivities of four Ag–Ni alloys used in make-and-break as well as disconnect contacts with current ratings up to ~100 A.

- Ag–Ni is a two-phase alloy, a mixture of Ag-rich and Ni-rich phases. Using an appropriate mixture rule, predict the resistivity of the alloy and compare with the measured values in Table 2.11. Explain the difference between the predicted and experimental values.
- Compare the resistivity of Ag–10% Ni with that of Ag–10% Pd in Table 2.9. The resistivity of the Ag–Pd alloy is almost a factor of **3 greater**. Ag–Pd is an isomorphous solid solution, whereas Ag–Ni is a two-phase mixture. Explain the difference in the resistivities of Ag–Ni and Ag–Pd.

Table 2.11 Resistivity of Ag–Ni contact alloys for switches

	Ni % in Ag–Ni					
	0	10	15	20	30	100
$\rho(\text{n}\Omega \text{ m})$	16.9	20.9	23.6	25	31.1	71.4
$d(\text{g cm}^{-3})$	10.5	10.3	9.76	9.4	9.47	8.9
Hardness	30	50	55	60	65	80
VHN						

NOTE: Compositions are in wt.%. Ag–10% Ni means 90% Ag–10% Ni. Vickers hardness number (VHN) is a measure of the hardness or strength of the alloy, and d is density.

Solution

- The Ni contents are given in wt.%. For volume fraction we use the relation

$$\chi_{Ni} = \frac{w_{Ni}d}{d_{Ni}}$$

where w_{Ni} is the weight fraction of Ni, d_{Ni} is the density of Ni and, d is the density of the alloy mixture. For example, for Ni-30% wt. the volume fraction of Ni in alloy will be

$$\chi_{Ni} = \frac{(0.3)(9.47 \times 10^3 \text{ kgm}^{-3})}{(8.9 \times 10^3 \text{ kgm}^{-3})} = 0.32$$

First we use Reynolds and Hough rule for mixture of dispersed phases to calculate the effective resistivity of the alloy. From Equation 2.28 we have

$$\frac{\sigma - \sigma_{Ag}}{\sigma + 2\sigma_{Ag}} = \chi_{Ni} \frac{\sigma_{Ni} - \sigma_{Ag}}{\sigma_{Ni} + 2\sigma_{Ag}}$$

Solving for Ni-30% wt., the R.H.S. of the above equation will be

$$= (0.32) \cdot \frac{(71.4 \text{ n}\Omega \text{ m})^{-1} - (16.9 \text{ n}\Omega \text{ m})^{-1}}{(71.4 \text{ n}\Omega \text{ m})^{-1} + 2(16.9 \text{ n}\Omega \text{ m})^{-1}} = -0.1089 (\text{n}\Omega \text{ m})^{-1}$$

Substitute the calculated value in the Reynolds and Hough rule, the effective resistivity of the alloy is $\rho = 23.96 \text{ n}\Omega \text{ m}$. Similarly the resistivity of alloy with other Ni contents is calculated and is tabulated below.

Ni % in Ag-Ni	d (g cm ⁻³)	χ_{Ni}	ρ_{eff} (n Ω m)	Experimental (n Ω m)
10	10.3	0.12	19.07	20.9
15	9.76	0.16	20.11	23.6
20	9.4	0.21	21.17	25
30	9.47	0.32	23.96	31.1

Table 2Q15-1 Resistivity of Ag-Ni contact alloys for switches

Now we use resistivity-mixture rule and conductivity-mixture rule to calculate the effective resistivity and compare the results. The *resistivity-mixture rule* or the *series rule of mixtures* is defined in Equation 2.24 as

$$\rho_{eff} = \chi_{\alpha} \rho_{\alpha} + \chi_{\beta} \rho_{\beta}$$

and conductivity-mixture rule is given by the Equation 2.25

$$\sigma_{eff} = \sigma_{\alpha} \rho_{\alpha} + \sigma_{\beta} \rho_{\beta}$$

The values obtained using these rules are as follows,

Ni % in Ag-Ni	χ_{Ni}	ρ_{eff} (n Ω m)			
		Resistivity-Mixture Rule	Conductivity-Mixture Rule	Reynolds & Hough Rule	Experimental
10.00	0.12	23.21	18.54	19.07	20.90
15.00	0.16	25.86	19.33	20.11	23.60
20.00	0.21	28.41	20.15	21.17	25.00
30.00	0.32	34.30	22.34	23.96	31.10

Table 2Q15-2: Resistivity of Ag-Ni contact alloys for switches calculated using different mixture rules

b. 90%Ag-10% Ni, the solid is a mixture, and has two phases with an overall $\rho = 20.9 \text{ n}\Omega \text{ m}$. 90%Ag-10% Pd, the solid is a solid solution with $\rho = 59.8 \text{ n}\Omega \text{ m}$, the value is roughly 3 times greater. The resistivity of a mixture is normally much lower than the resistivity of a *similar* solid solution. In a solid solution, the added impurities scatter electrons and increase the resistivity. In a mixture, each phase is almost like a "pure" metal, and the overall resistivity is simply an appropriate "averaging" or combination of the two resistivities.

12.16 Ag–W alloys (contact materials) and the mixture rule Silver–tungsten alloys are frequently used in heavy-duty switching applications (e.g., current-carrying contacts and oil circuit breakers) and in arcing tips. Ag–W is a two-phase alloy, a mixture of Ag-rich and W-rich phases. The measured resistivity and density for various Ag–W compositions are summarized in Table 2.12.

- Plot the resistivity and density of the Ag–W alloy against the W content (wt. %)
- Show that the density of the mixture, d , is given by

$$d^{-1} = w_{\alpha} d_{\alpha}^{-1} + w_{\beta} d_{\beta}^{-1}$$

where w_{α} is the weight fraction of phase α , w_{β} is the weight fraction of phase β , d_{α} is the density of phase α , and d_{β} is the density of phase β .

- Show that the resistivity mixture rule is

$$\rho = \rho_{\alpha} \frac{dw_{\alpha}}{d_{\alpha}} + \rho_{\beta} \frac{dw_{\beta}}{d_{\beta}}$$

where ρ is the resistivity of the alloy (mixture), d is the density of the alloy (mixture), and subscripts α and β refer to phases α and β , respectively.

- Calculate the density d and the resistivity ρ of the mixture for various values of W content (in wt. %) and plot the calculated values in the same graph as the experimental values. What is your conclusion?

Table 2.12 Dependence of resistivity in Ag–W alloy on composition as a function of wt.% W

	W(wt.%)													
	0	10	15	20	30	40	65	70	75	80	85	90	100	
ρ (n Ω m)	16.2	18.6	19.7	20.9	22.7	27.6	35.5	38.3	40	46	47.9	53.9	55.6	
d (g cm ⁻³)	10.5	10.75	10.95	11.3	12	12.35	14.485	15.02	15.325	16.18	16.6	17.25	19.1	

NOTE: ρ = resistivity and d = density.

Solution

- The plot of density versus W weight fraction is as follows.

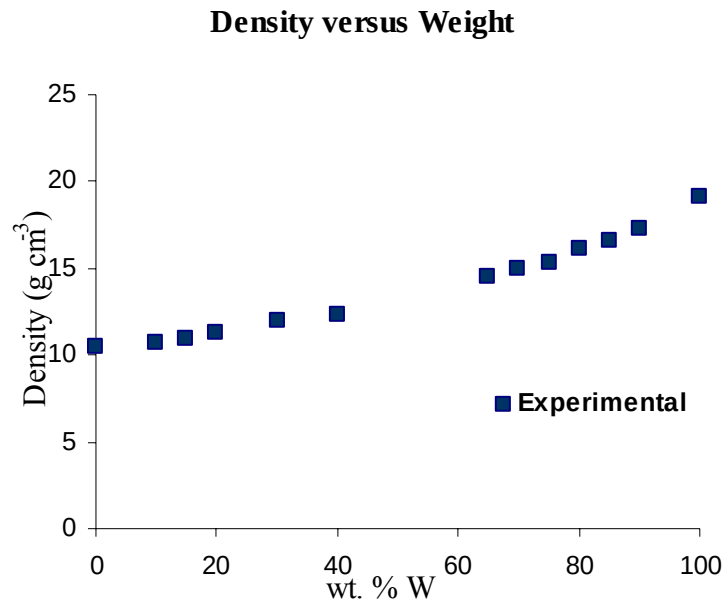


Figure 2Q16-1: Experimental density for various compositions of W and the plot of resistivity versus W wt% is,

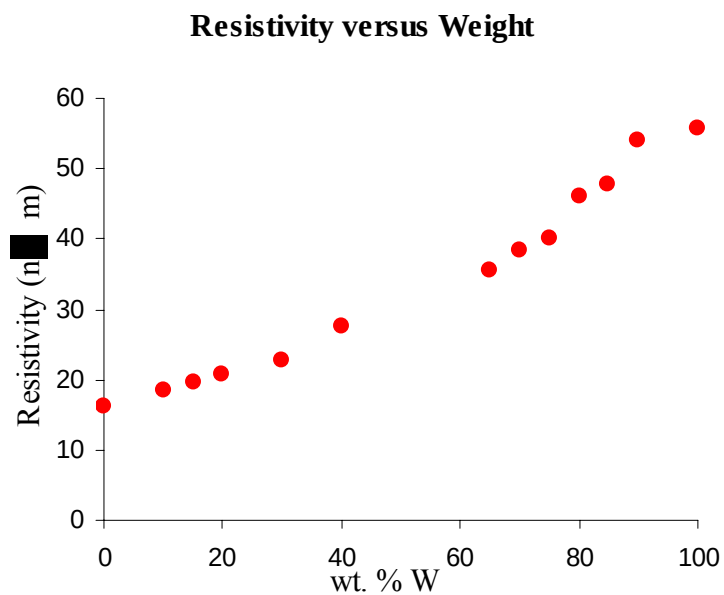


Figure 2Q16-2: Experimental resistivity for various compositions of W

b. The given mixture consists of two phases α , and β . Assume that the total mass of the alloy is M_{mixture} . If w_α and w_β are the weight fractions of α , and β phases, then their respective masses in the mixture are

$$M_\alpha = w_\alpha M_{\text{mixture}}$$

$$M_\beta = w_\beta M_{\text{mixture}}$$

The densities of the phases α , and β , are d_α and d_β , therefore the volume occupied by these phases can be calculated using the definition of density. i.e. density = mass / volume, we have

$$V_\alpha = \frac{\text{mass of } \alpha}{\text{density of } \alpha} = \frac{M_\alpha}{d_\alpha} = \frac{w_\alpha M_{\text{mixture}}}{d_\alpha}$$

$$V_\beta = \frac{\text{mass of } \beta}{\text{density of } \beta} = \frac{M_\beta}{d_\beta} = \frac{w_\beta M_{\text{mixture}}}{d_\beta}$$

The total volume of the alloy mixture is

$$V_{\text{mixture}} = V_\alpha + V_\beta = \frac{w_\alpha M_{\text{mixture}}}{d_\alpha} + \frac{w_\beta M_{\text{mixture}}}{d_\beta}$$

The density of the mixture is therefore,

$$d = \frac{M_{\text{mixture}}}{V_{\text{mixture}}} = \frac{M_{\text{mixture}}}{\frac{w_\alpha M_{\text{mixture}}}{d_\alpha} + \frac{w_\beta M_{\text{mixture}}}{d_\beta}} = \frac{M_{\text{mixture}}}{M_{\text{mixture}} \left(\frac{w_\alpha}{d_\alpha} + \frac{w_\beta}{d_\beta} \right)} = \frac{1}{\left(\frac{w_\alpha}{d_\alpha} + \frac{w_\beta}{d_\beta} \right)}$$

or

$$\frac{1}{d} = \frac{w_\alpha}{d_\alpha} + \frac{w_\beta}{d_\beta}$$

c. The resistivity-mixture rule or the series rule of mixtures is defined in Equation 2.24 as

$$\rho_{\text{eff}} = \chi_\alpha \rho_\alpha + \chi_\beta \rho_\beta$$

where χ_α and χ_β are the volume fractions of phase α and β respectively. (For detailed derivation of this rule please see Example 2.13.) Volume fractions of the two phases are,

$$\chi_\alpha = \frac{V_\alpha}{V_{\text{mixture}}} \text{ and } \chi_\beta = \frac{V_\beta}{V_{\text{mixture}}}$$

From part a of this problem, the volume of the phases α and β in the mixture are

$$V_\alpha = \frac{w_\alpha M_{\text{mixture}}}{d_\alpha} \text{ and } V_\beta = \frac{w_\beta M_{\text{mixture}}}{d_\beta}$$

and the volume of the mixture is $V = \frac{M_{\text{mixture}}}{d}$, therefore the volume fraction of the two contents is

$$\chi_\alpha = \frac{V_\alpha}{V} = \frac{\frac{w_\alpha M_{\text{mixture}}}{d_\alpha}}{\frac{M_{\text{mixture}}}{d}} = \frac{w_\alpha d}{d_\alpha} \text{ and } \chi_\beta = \frac{V_\beta}{V} = \frac{\frac{w_\beta M_{\text{mixture}}}{d_\beta}}{\frac{M_{\text{mixture}}}{d}} = \frac{w_\beta d}{d_\beta}$$

Substituting the volume fraction χ_α and χ_β , the resistivity mixture rule is

$$\rho_{\text{eff}} = \frac{w_\alpha d}{d_\alpha} \rho_\alpha + \frac{w_\beta d}{d_\beta} \rho_\beta$$

d. We calculate the density and the resistivity using the relations proved in parts (b) and (c). As an example, for 30% W wt. content, the density and resistivity are,

$$d^{-1} = w_{Ag} d_{Ag}^{-1} + w_W d_W^{-1} = \frac{(1-0.3)}{10.5 \text{ g cm}^{-3}} + \frac{(0.3)}{19.1 \text{ g cm}^{-3}}$$

$$d = 12.14 \text{ g cm}^{-3}.$$

Using resistivity-mixture rule, we have

$$\begin{aligned} \rho &= \rho_{Ag} \frac{d \cdot w_{Ag}}{d_{Ag}} + \rho_W \frac{d \cdot w_W}{d_W} = (16.2 \text{ n}\Omega \text{ m}) \frac{(12.14 \text{ g cm}^{-3})(1-0.3)}{(10.5 \text{ g cm}^{-3})} + (55.6 \text{ n}\Omega \text{ m}) \frac{(12.14 \text{ g cm}^{-3})(0.3)}{(19.1 \text{ g cm}^{-3})} \\ &= 23.71 \text{ n}\Omega \text{ m} \end{aligned}$$

The experimental resistivity as given in the table is 22.7 nΩ m. Similarly, the densities and resistivities for the given W contents are calculated and listed in the Table 2Q16-1.

W wt. %	Volume Fraction χ_W	Experimental density (g cm ⁻³)	Calculated density (g cm ⁻³)	Resistivity Mixture Rule (nΩ m)	Conductivity Mixture Rule (nΩ m)	Reynolds & Hough Rule (nΩ m)	Experimental Resistivity (nΩ m)
10	0.06	10.75	11.00	18.47	16.87	17.08	18.6
15	0.09	10.95	11.26	19.68	17.25	17.57	19.7
20	0.12	11.3	11.54	20.96	17.68	18.12	20.9
30	0.19	12	12.14	23.71	18.70	19.41	22.7
40	0.26	12.35	12.81	26.77	19.84	20.83	27.6
65	0.49	14.485	14.84	36.10	24.90	26.86	35.5
70	0.55	15.02	15.33	38.34	26.56	28.74	38.3
75	0.60	15.325	15.85	40.73	28.24	30.61	40
80	0.68	16.18	16.41	43.28	31.17	33.74	46
85	0.74	16.6	17.01	46.03	34.00	36.65	47.9
90	0.81	17.25	17.65	48.98	38.21	40.77	53.9

Table 2Q16-1: Resistivity of Ag–W alloy on composition as a function of wt.% W calculated using different mixture rules

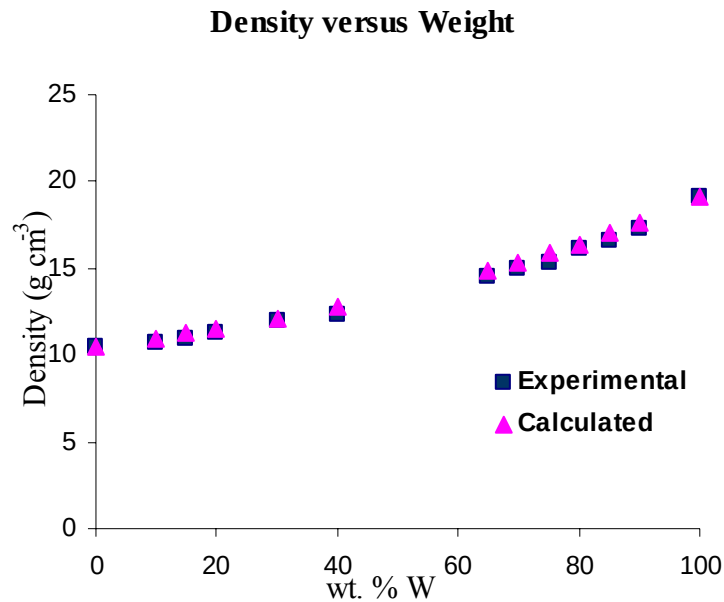


Figure 2Q16-3: Calculated and experimental density for various compositions of W

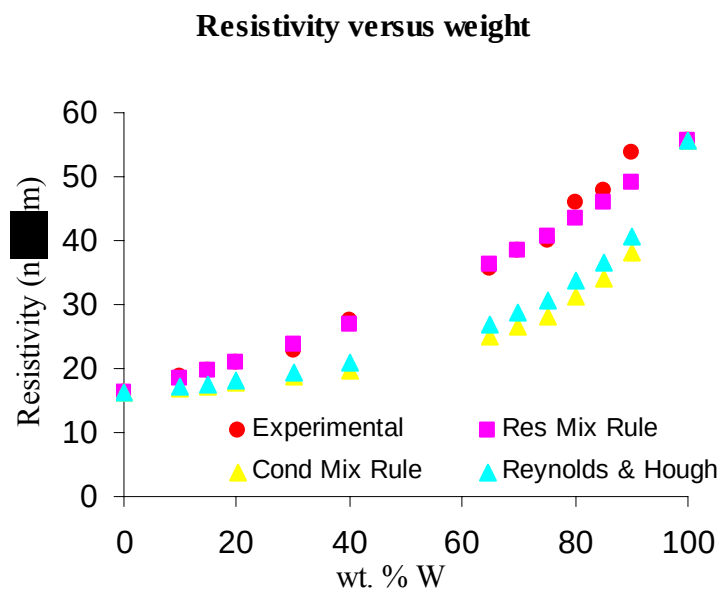


Figure 2Q16-4: Calculated and experimental resistivities for various compositions of W

Author's Comment: The data were collected from a variety of sources (various handbooks and papers) and combined into a single table. The data are not simply from a single source. Hence the experimental values show some scatter. Given the scatter, the resistivity mixture model is in good agreement with the experimental data.

2.17 Thermal conduction Consider brass alloys with an X atomic fraction of Zn. Since Zn addition increases the number of conduction electrons, we have to scale the final alloy resistivity calculated from the simple Matthiessen-Nordheim rule in Equation 2.22 down by a factor $(1+X)$ (see

Question 2.8) so that the resistivity of the alloy is $\rho \approx [\rho_o + CX(1 - X)]/(1 + X)$ in which $C = 300 \text{ n}\Omega \text{ m}$ and $\rho_o = \rho_{\text{Cu}} = 17 \text{ n}\Omega \text{ m}$.

a. An 80 at. % Cu–20 at. % Zn brass disk of 40 mm diameter and 5 mm thickness is used to conduct heat from a heat source to a heat sink.

(1) Calculate the thermal resistance of the brass disk.

(2) If the disk is conducting heat at a rate of 100 W, calculate the temperature drop along the disk.

b. What should be the composition of brass if the temperature drop across the disk is to be halved?

Solution

a.

(1) Assume $T = 20^\circ\text{C} = 293 \text{ K}$. Apply Equation 2.22 to find the resistivity of the brass in the disk with $\rho_{\text{Cu}} = 17.1 \text{ n}\Omega \text{ m}$ and $X_{\text{Zn}} = 0.20$:

$$\rho_{\text{brass}} = \rho_{\text{Cu}} + C_{\text{Zn in Cu}} X_{\text{Zn}} (1 - X_{\text{Zn}})$$

$$i.e. \quad \rho_{\text{brass}} = 17.1 \text{ n}\Omega \text{ m} + (300 \text{ n}\Omega \text{ m})(0.20)(1 - 0.20)$$

$$\therefore \quad \rho_{\text{brass}} = 65.1 \text{ n}\Omega \text{ m}$$

We know that the thermal conductivity is given by $\kappa/\sigma_{\text{brass}} = C_{\text{FWL}}T$ where σ_{brass} is the conductivity of the disk, C_{FWL} is the Lorenz number and T is the temperature. This equation can also be written as $\kappa\rho_{\text{brass}} = C_{\text{FWL}}T$ so that $\kappa = C_{\text{FWL}}T/\rho$. Applying this equation,

$$\kappa(20^\circ\text{C}) = (2.44 \times 10^{-8} \text{ W } \Omega \text{ K}^{-2})(293 \text{ K}) / (6.51 \times 10^{-8} \Omega \text{ m})$$

$$\therefore \quad \kappa(20^\circ\text{C}) = 109.8 \text{ W K}^{-1} \text{ m}^{-1}$$

The thermal resistance is $\theta = L/(\kappa A)$, where L is the thickness of the disk and A is the cross-sectional area of the disk.

$$\theta = L/(\kappa A) = (5 \times 10^{-3} \text{ m}) / [(109.8 \text{ W K}^{-1} \text{ m}^{-1})(\pi)(2 \times 10^{-2} \text{ m})^2] = \mathbf{0.0362 \text{ K W}^{-1}}$$

(2) From $dQ/dt = A\kappa\Delta T/\Delta x = \Delta T/\theta$ (Δx can be taken to be the same as L), and $dQ/dt = P$ (power conducted), we can substitute to obtain:

$$\Delta T = P\theta = (100 \text{ W})(3.62 \times 10^{-2} \text{ K W}^{-1}) = \mathbf{3.62 \text{ K or } 3.62^\circ\text{C}}$$

Note: Change in temperature is the same in either Kelvins or degrees Celsius, *i.e.* $\Delta T = T_1 - T_2 = (T_1 + 273) - (T_2 + 273)$.

b. Since $\Delta T = P\theta$, to get half ΔT , we need half θ or double κ or double σ or half ρ . We thus need $1/2\rho_{\text{brass}}$ or $1/2(65.1 \text{ n}\Omega \text{ m})$ which can be attained if the brass composition is X_{new} so that

$$\rho_{\text{new}} = \rho_{\text{Cu}} + C_{\text{Zn in Cu}} X_{\text{new}} (1 - X_{\text{new}})$$

$$i.e. \quad 1/2(65.1 \text{ n}\Omega \text{ m}) = 17 \text{ n}\Omega \text{ m} + (300 \text{ n}\Omega \text{ m})X_{\text{new}}(1 - X_{\text{new}})$$

Solving this quadratic equation we get $X_{\text{new}} = 0.0545$, or 5.5% Zn. Thus we need **94.5% Cu-5.5% Zn** brass.

2.18 Thermal resistance Consider a thin insulating disc made of mica to electrically insulate a semiconductor device from a conducting heat sink. Mica has $\kappa = 0.75 \text{ W m}^{-1} \text{ K}^{-1}$. The disk thickness is 0.1 mm, and the diameter is 10 mm. What is the thermal resistance of the disk? What is the temperature drop across the disk if the heat current through it is 25W?

Solution

The thermal resistance of the mica disk can be calculated directly from Equation 2.40.

$$\theta = \frac{L}{\kappa A} = \frac{4L}{\kappa \pi d^2} = \frac{4(1 \times 10^{-3} \text{ m})}{\pi(0.75 \text{ W m}^{-1} \text{ K}^{-1})(1 \times 10^{-2} \text{ m})^2} = \mathbf{1.698 \text{ K W}^{-1}}$$

The temperature drop across the disk according to Equation 2.36 (in the textbook) is

$$\Delta T = Q'\theta = \mathbf{42.4 \text{ }^\circ\text{C}}$$

***2.19 Thermal resistance** Consider a coaxial cable operating under steady state conditions when the current flow through the inner conductor generates Joule heat at a rate $P = I^2 R$. The heat generated per second by the core conductor flows through the dielectric; $Q' = I^2 R$. The inner conductor reaches a temperature T_i whereas the outer conductor is at T_o . Show that the thermal resistance θ of the hollow cylindrical insulation for heat flow in the radial direction is

$$\theta = \frac{(T_i - T_o)}{Q'} = \frac{\ln\left(\frac{b}{a}\right)}{2\pi\kappa L} \quad \text{Thermal resistance of hollow cylinder}$$

where a is the inside (core conductor) radius, b is the outside radius (outer conductor), κ is the thermal conductivity of the insulation, and L is the cable length. Consider a coaxial cable that has a copper core conductor and polyethylene (PE) dielectric with the following properties: Core conductor resistivity $\rho = 19 \text{ n}\Omega \text{ m}$, core radius, $a = 4 \text{ mm}$, dielectric thickness, $b - a = 3.5 \text{ mm}$, dielectric thermal conductivity $\kappa = 0.3 \text{ W m}^{-1} \text{ K}^{-1}$. The outside temperature T_o is $25 \text{ }^\circ\text{C}$. The cable is carrying a current of 500 A. What is the temperature of the inner conductor?

Solution

Consider a thin cylindrical shell of thickness dr as shown in Figure 2Q12-1. The temperature difference across dr is dT . The surface area of this shell is $2\pi rL$. Thus, from Fourier's law,

$$Q' = -(2\pi rL)\kappa \frac{dT}{dr}$$

which we can integrate with respect to r from $r = a$ where $T = T_i$ to $r = b$ where $T = T_o$,

$$Q' \int_a^b \frac{dr}{r} = -2\pi L \kappa \int_{T_i}^{T_o} dT$$

i.e.

$$Q' = (T_i - T_o) \frac{2\pi\kappa L}{\ln\left(\frac{b}{a}\right)}$$

Thus the thermal resistance of the hollow cylindrical insulation is

$$\theta = \frac{(T_i - T_o)}{Q'} = \frac{\ln\left(\frac{b}{a}\right)}{2\pi\kappa L}$$

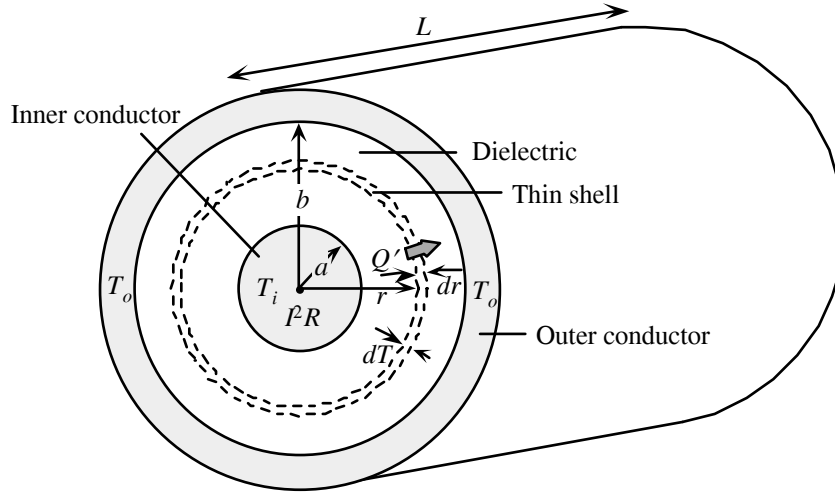


Figure 2Q12-1: Thermal resistance of a hollow cylindrical shell. Consider an infinitesimally thin cylindrical shell of radius r and thickness dr in the dielectric and concentrically around the inner conductor. The surface area is $2\pi rL$.

The actual length of the conductor does not affect the calculations as long as the length is sufficiently long such that there is no heat transfer along the length; heat flows radially from the inner to the outer conductor. We consider a portion of length L of a very long cable and we set $L = 1$ m so that the calculations are per unit length. The joule heating per unit second (power) generated by the current I through the core conductor is

$$Q' = I^2 \frac{\rho L}{\pi a^2} = (500^2) \frac{(19 \times 10^{-9})(1)}{\pi (4 \times 10^{-3})^2} = \mathbf{94.5 \text{ W}}$$

The thermal resistance of the insulation is,

$$\theta = \frac{\ln\left(\frac{b}{a}\right)}{2\pi\kappa L} = \frac{\ln\left(\frac{(4 + 3.5) \times 10^{-3}}{4 \times 10^{-3}}\right)}{2\pi(0.3)(1)} = \mathbf{0.33 \text{ } ^\circ\text{C/W}}$$

Thus, the temperature difference ΔT due to Q' flowing through θ is,

$$\Delta T = Q' \theta = (94.5 \text{ W})(0.33 \text{ } ^\circ\text{C/W}) = \mathbf{31.2 \text{ } ^\circ\text{C}}.$$

The inner temperature is therefore,

$$T_i = T_o + \Delta T = 25 + 31.2 = \mathbf{56.2 \text{ } ^\circ\text{C}}.$$

Note that for simplicity we assumed that the inner conductor resistivity ρ and thermal conductivity κ are constant (do not change with temperature).

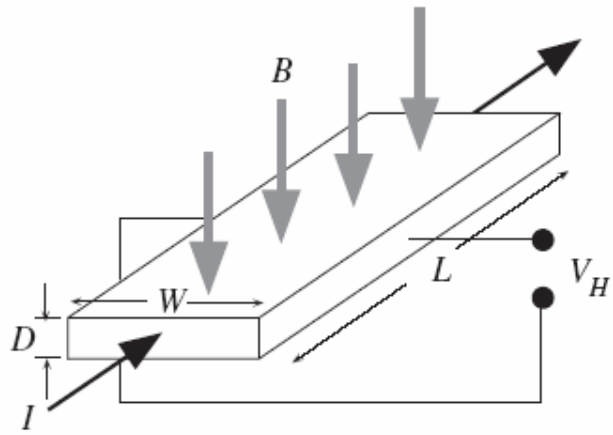
2.20 The Hall effect Consider a rectangular sample, a metal or an n -type semiconductor, with a length L , width W , and thickness D . A current I is passed along L , perpendicular to the cross-sectional area WD . The face $W \times L$ is exposed to a magnetic field density B . A voltmeter is connected across the width, as shown in Figure 2.40, to read the Hall voltage V_H .

a. Show that the Hall voltage recorded by the voltmeter is

$$V_H = \frac{IB}{Dn} \quad \text{Hall voltage}$$

b. Consider a 1-micron-thick strip of gold layer on an insulating substrate that is a candidate for a Hall probe sensor. If the current through the film is maintained at constant 100 mA, what is the magnetic field that can be recorded per μV of Hall voltage?

Figure 2.40 Hall effect in a rectangular material with length L , width W , and thickness D . The voltmeter is across the width W .



Solution

a. The Hall coefficient, R_H , is related to the electron concentration, n , by $R_H = -1 / (en)$, and is defined by $R_H = E_y / (JB)$, where E_y is the electric field in the y -direction, J is the current density and B is the magnetic field. Equating these two equations:

$$-\frac{1}{en} = \frac{E_y}{JB}$$

$$\therefore E_y = -\frac{JB}{en}$$

This electric field is in the opposite direction of the Hall field (E_H) and therefore:

$$E_H = -E_y = \frac{JB}{en} \quad (1)$$

The current density perpendicular (going through) the plane $W \times D$ (width by depth) is:

$$J = \frac{I}{WD}$$

$$\therefore W = \frac{I}{JD} \quad (2)$$

The Hall voltage (V_H) across W is:

$$V_H = WE_H$$

If we substitute expressions (1) and (2) into this equation, the following will be obtained:

$$V_H = \frac{IB}{Den}$$

Note: this expression only depends on the thickness and not on the length of the sample.

In general, the Hall voltage will depend on the specimen shape. In the elementary treatment here, the current flow lines were assumed to be nearly parallel from one end to the other end of the sample. In an irregularly shaped sample, one has to consider the current flow lines. However, if the specimen thickness is uniform, it is then possible to carry out meaningful Hall effect measurements using the van der Pauw technique as discussed in advanced textbooks.

b. We are given the depth of the film $D = 1$ micron $= 1 \mu\text{m}$ and the current through the film $I = 100$ mA $= 0.1$ A. The Hall voltage can be taken to be $V_H = 1 \mu\text{V}$, since we are looking for the magnetic field B per μV of Hall voltage. To be able to use the equation for Hall voltage in part (a), we must find the electron concentration of gold. Appendix B in the textbook contains values for gold's atomic mass ($M_{\text{at}} = 196.97 \text{ g mol}^{-1}$) and density ($d = 19.3 \text{ g/cm}^3 = 19300 \text{ kg/m}^3$). Since gold has a valency of 1 electron, the concentration of free electrons is equal to the concentration of Au atoms.

$$\therefore n = \frac{dN_A}{M_{\text{at}}} = \frac{(19300 \text{ kg m}^{-3})(6.022 \times 10^{23} \text{ mol}^{-1})}{(196.97 \times 10^{-3} \text{ kg mol}^{-1})} = 5.901 \times 10^{28} \text{ m}^{-3}$$

Now the magnetic field B can be found using the equation for Hall voltage:

$$\therefore V_H = \frac{IB}{Den}$$

$$\therefore B = \frac{V_H Den}{I} = \frac{(1 \times 10^{-6} \text{ V})(1 \times 10^{-6} \text{ m})(1.602 \times 10^{-19} \text{ C})(5.901 \times 10^{28} \text{ m}^{-3})}{(0.1 \text{ A})}$$

$$\therefore \mathbf{B = 0.0945 \text{ T}}$$

As a side note, the power (P) dissipated in the film could be found very easily. Using the value for resistivity of Au at $T = 273 \text{ K}$, $\rho = 22.8 \text{ n}\Omega \text{ m}$, the resistance of the film is:

$$R = \frac{\rho L}{A} = \frac{\rho L}{WD} = \frac{(22.8 \times 10^{-9} \Omega \text{ m})(0.001 \text{ m})}{(0.0001 \text{ m})(1 \times 10^{-6} \text{ m})} = 0.228 \Omega$$

The power dissipated is then:

$$P = I^2 R = (0.1 \text{ A})^2 (0.228 \Omega) = 0.00228 \text{ W}$$

2.21 The strain gauge A **strain gauge** is a transducer attached to a body to measure its fractional elongation $\Delta L/L$ under an applied load (force) F . The gauge is a grid of many folded runs of a thin, resistive wire glued to a flexible backing, as depicted in Figure 2.41. The gauge is attached to the body under test such that the resistive wire length is parallel to the strain.

a. Assume that the elongation does not change the resistivity and show that the change in the resistance ΔR is related to the strain, $\varepsilon = \Delta L/L$ by

$$\Delta R \approx R(1+2\nu)\varepsilon$$

Strain gauge equation

where ν is the Poisson ratio, which is defined by

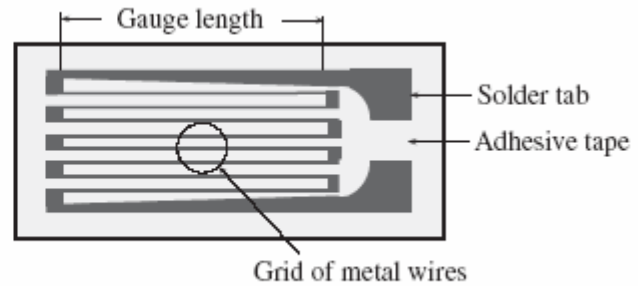
$$\nu = -\frac{\text{Transverse strain}}{\text{Longitudinal strain}} = -\frac{\varepsilon_t}{\varepsilon_l} \quad \text{Poisson ratio}$$

where ε_l is the strain along the applied load, that is, $\varepsilon_l = \Delta L/L = \varepsilon$, and ε_t is the strain in the transverse direction, that is, $\varepsilon_t = \Delta D/D$, where D is the diameter (thickness) of the wire.

- b. Explain why a nichrome wire would be a better choice than copper for the strain gauge (consider the TCR).

Figure 2.41 The strain gauge consists of a long, thin wire folded several times along its length to form a grid as shown and embedded in a self-adhesive tape.

The ends of the wire are attached to terminals (solder pads) for external connections. The tape is stuck on the component for which the strain is to be measured.



- c. How do temperature changes affect the response of the gauge? Consider the effect of temperature on ρ . Also consider the differential expansion of the specimen with respect to the gauge wire such that even if there is no applied load, there is still strain, which is determined by the differential expansion coefficient, $\lambda_{\text{specimen}} - \lambda_{\text{gauge}}$, where λ is the thermal coefficient of linear expansion: $L = L_0[1 + \lambda(T - T_0)]$, where T_0 is the reference temperature.
- d. The gauge factor for a transducer is defined as the fractional change in the measured property $\Delta R/R$ per unit input signal (ε). What is the gauge factor for a metal-wire strain gauge, given that for most metals, $\nu \approx \frac{1}{3}$?
- e. Consider a strain gauge that consists of a nichrome wire of resistivity $1 \mu\Omega \text{ m}$, a total length of 1 m, and a diameter of $25 \mu\text{m}$. What is ΔR for a strain of 10^{-3} ? Assume that $\nu \approx \frac{1}{3}$.
- f. What will ΔR be if constantan wire with resistivity of $500 \text{ n}\Omega \text{ m}$ is used?

Solution

- a. Consider the resistance R of the gauge wire,

$$R = \frac{\rho L}{\pi \left(\frac{D}{2} \right)^2} \quad (1)$$

where L and D are the length and diameter of the wire, respectively. Suppose that the applied load changes L and D by δL and δD which change R by δR . The total derivative of a function R of two variables L and D can be found by taking partial differentials (like those used for error calculations in physics labs). Assuming that ρ is constant,

$$\delta R = \left(\frac{\partial R}{\partial L} \right) \delta L + \left(\frac{\partial R}{\partial D} \right) \delta D = \left(\frac{\rho}{\frac{\pi}{4} D^2} \right) \delta L - 2 \left(\frac{\rho L}{\frac{\pi}{4} D^3} \right) \delta D$$

so that the fractional change is

$$\frac{\delta R}{R} = \frac{\left(\frac{\rho}{\frac{\pi}{4} D^2} \right)}{\left(\frac{\rho L}{\frac{\pi}{4} D^2} \right)} \delta L - 2 \frac{\left(\frac{\rho L}{\frac{\pi}{4} D^3} \right)}{\left(\frac{\rho L}{\frac{\pi}{4} D^2} \right)} \delta D = \frac{\delta L}{L} - 2 \frac{\delta D}{D} \quad (2)$$

We can now use the definitions of longitudinal and transverse strain,

$$\frac{\delta L}{L} = \varepsilon_l \quad \text{and} \quad \frac{\delta D}{D} = \varepsilon_t = -\nu \varepsilon_l$$

in the expression for $\delta R/R$ in Eqn. (2) to obtain,

$$\frac{\delta R}{R} = \varepsilon_l - 2(-\nu \varepsilon_l) = (1 + 2\nu) \varepsilon \quad (3)$$

where $\varepsilon = \varepsilon_l$.

b. The change in R was attributed to changes in L and D due to their extension by the applied load. There are two reasons why a nichrome wire will be a better choice. First is that nichrome has a higher resistivity ρ which means that its resistance R will be higher than that of a similar size copper wire and hence nichrome wire will exhibit a greater change in R (δR is easier to measure). Secondly nichrome has a very small TCR which means that ρ and hence R does not change significantly with temperature. If we were to use a Cu wire, any small change in the temperature will most likely overwhelm any change due to strain.

c. Consider a change δR in R due to a change δT in the temperature. We can differentiate R with respect to T by considering that ρ , L and D depend on T . It is not difficult to show (See Question 2.15) that if α is the temperature coefficient of resistivity and λ is the linear expansion coefficient, then

$$\left(\frac{1}{R} \right) \frac{dR}{dT} = \alpha - \lambda \quad (4)$$

Typically $\lambda \approx 2 \times 10^{-5} \text{ K}^{-1}$, and for pure metals $\alpha \approx 1/273 \text{ K}^{-1}$ or $3.6 \times 10^{-3} \text{ K}^{-1}$. A 1°C fluctuation in the temperature will result in $\delta R/R = 3.6 \times 10^{-3}$ which is about the same as $\delta R/R$ from a strain of $\varepsilon = 2 \times 10^{-3}$ at a constant temperature. It is clear that temperature fluctuations would not allow sensible strain measurements if we were to use a pure metal wire. To reduce the temperature fluctuation effects, we need $\alpha \approx \lambda$ which means we have to use a metal alloy such as a nichrome wire or a constantan wire.

Even if we make $\alpha - \lambda = 0$, the temperature change still produces a change in the resistance because the metal wire and specimen expand by different amounts and this creates a strain and hence a change in the resistance. Suppose that λ_{gauge} and $\lambda_{\text{specimen}}$ are the linear expansion coefficients of the

gauge wire and the specimen, then the differential expansion will be $\lambda_{\text{specimen}} - \lambda_{\text{gauge}}$ and this can only be zero if $\lambda_{\text{gauge}} = \lambda_{\text{specimen}}$.

d. The gauge factor is defined as

$$\text{GF} = \frac{\text{Fractional change in gauge property}}{\text{Input signal}} = \frac{\delta R / R}{\varepsilon} = \frac{(1 + 2\nu)\varepsilon}{\varepsilon}$$

i.e. **GF** = $1 + 2\nu = 1 + 2(1/3) = \mathbf{1.67}$

which applies to metals. ($\nu \approx \frac{1}{3}$)

e. For nichrome wire, given that $\rho = 10^{-6} \Omega \text{ m}$, $L = 1 \text{ m}$, and $D = 25 \times 10^{-6} \text{ m}$,

we have,
$$R = \frac{\rho L}{\pi \left(\frac{D}{2}\right)^2} = \frac{(10^{-6} \Omega \text{ m})(1 \text{ m})}{\pi \left(\frac{25 \times 10^{-6} \text{ m}}{2}\right)^2} = 2037 \Omega$$

For a strain of $\varepsilon = 10^{-3}$,

$$\delta R = R(1 + 2\nu)\varepsilon = (2037 \Omega)[1 + 2(1/3)](10^{-3}) = \mathbf{3.40 \Omega}$$

This can be easily measured with an ohm-meter, though instrumentation engineers would normally use the gauge in an electrical bridge circuit. Note that $\delta R/R = (3.4 \Omega)/(2037 \Omega) = 0.0017$, or 0.17 %.

f. For a constantan wire, given that $\rho = 5 \times 10^{-7} \Omega \text{ m}$, $L = 1 \text{ m}$, and $D = 25 \times 10^{-6} \text{ m}$,

we have,
$$R = \frac{(5 \times 10^{-7} \Omega \text{ m})(1 \text{ m})}{\pi \left(\frac{25 \times 10^{-6} \text{ m}}{2}\right)^2} = 1019 \Omega$$

so that
$$\delta R = R(1 + 2\nu)\varepsilon = (1019 \Omega)[1 + 2(1/3)](10^{-3}) = \mathbf{1.70 \Omega}$$

2.22 Thermal coefficients of expansion and resistivity

a. Consider a thin metal wire of length L and diameter D . Its resistance is $R = \rho L/A$, where $A = \pi D^2/4$. By considering the temperature dependence of L , A , and ρ individually, show that

$$\frac{1}{R} \frac{dR}{dT} = \alpha_o - \lambda_o$$

where α_o is the temperature coefficient of resistivity (TCR), and λ_o is the temperature coefficient of linear expansion (thermal expansion coefficient or expansivity), that is,

$$\lambda_o = L_o^{-1} \left(\frac{dL}{dT} \right)_{T=T_o} \quad \text{or} \quad \lambda_o = D_o^{-1} \left(\frac{dD}{dT} \right)_{T=T_o}$$

Note: Consider differentiating $R = \rho L/[(\pi D^2)/4]$ with respect to T with each parameter, ρ , L , and D , having a temperature dependence.

Given that typically, for most pure metals, $\alpha_o \approx 1/273 \text{ K}^{-1}$ and $\lambda_o \approx 2 \times 10^{-5} \text{ K}^{-1}$, confirm that the temperature dependence of ρ controls R , rather than the temperature dependence of the geometry. Is it necessary to modify the given equation for a wire with a noncircular cross section?

- b. Is it possible to design a resistor from a suitable alloy such that its temperature dependence is almost nil? Consider the TCR of an alloy of two metals A and B , for which $\alpha_{AB} \approx \alpha_A \rho_A / \rho_{AB}$.

Solution

- a. Consider the resistance R of the wire,

$$R = \frac{\rho L}{\pi \left(\frac{D}{2} \right)^2} \quad (1)$$

Consider a change δR in R due to a change δT in the temperature. We can differentiate R with respect to T by considering that ρ , L , and D depend on T ,

$$\frac{dR}{dT} = \frac{d}{dT} \left(\frac{\rho L}{\frac{\pi}{4} D^2} \right) = \left(\frac{L}{\frac{\pi}{4} D^2} \right) \frac{d\rho}{dT} + \left(\frac{\rho}{\frac{\pi}{4} D^2} \right) \frac{dL}{dT} - 2 \left(\frac{\rho L}{\frac{\pi}{4} D^3} \right) \frac{dD}{dT}$$

divide by R ,
$$\left(\frac{1}{R} \right) \frac{dR}{dT} = \left(\frac{1}{\rho} \right) \frac{d\rho}{dT} + \left(\frac{1}{L} \right) \frac{dL}{dT} - 2 \left(\frac{1}{D} \right) \frac{dD}{dT} \quad (2)$$

At $T = T_o$ we have,

$$\left(\frac{1}{R} \right) \frac{dR}{dT} = \left(\frac{1}{\rho_o} \right) \frac{d\rho}{dT} + \left(\frac{1}{L_o} \right) \frac{dL}{dT} - 2 \left(\frac{1}{D_o} \right) \frac{dD}{dT} \quad (3)$$

where the derivatives are at $T = T_o$. Recall that the temperature coefficient of resistivity, TCR (α_o), and the linear expansion coefficient λ_o are defined as follows,

$$\alpha_o = \left(\frac{1}{\rho_o} \right) \frac{d\rho}{dT} \quad \text{and} \quad \lambda_o = \left(\frac{1}{L_o} \right) \frac{dL}{dT} = \left(\frac{1}{D_o} \right) \frac{dD}{dT} \quad (4)$$

which reduces Eqn. (3) to

$$\left(\frac{1}{R} \right) \frac{dR}{dT} = \alpha_o - \lambda_o \quad (5)$$

Typically $\lambda_o \approx 2 \times 10^{-5} \text{ K}^{-1}$, and for pure metals $\alpha_o \approx 1/273 \text{ K}^{-1}$ or $3.6 \times 10^{-3} \text{ K}^{-1}$. Thus,

$$\left(\frac{1}{R} \right) \frac{dR}{dT} = 3.6 \times 10^{-3} \text{ K}^{-1} - 2 \times 10^{-5} \text{ K}^{-1} \approx 3.6 \times 10^{-3} \text{ K}^{-1} \approx \alpha_o$$

Since α_o is much larger than λ_o , it dominates the change in R .

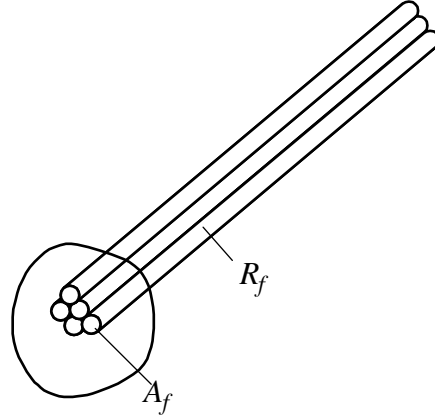


Figure 2Q22-1: Wire with non-circular cross section.

There is no need to modify Eqn. (5) for a non-circular cross sectional area. You can derive the same expression for a rectangular or an elliptic cross section, or, indeed, any arbitrary cross section. One can consider the wire to be made up of N thin fibers each of *circular* cross section A_f . Imagine holding a bunch of these in your hand and then sliding them into any cross section you like as in the Figure 2Q22-1. In all cases $A = \Sigma A_f$. However, since the fibers are in parallel, the total resistance is given by $R^{-1} = \Sigma R_f^{-1} = N R_f^{-1}$. Thus $R = R_f / N$. For each fiber, $\delta R_f = R_f(\alpha_o - \lambda_o)\delta T$ as we have derived above. Then,

$$\delta R = (\delta R_f)/N = [R_f(\alpha_o - \lambda_o)\delta T]/N = R(\alpha_o - \lambda_o)\delta T$$

[There are a few assumptions such as the resistivity is homogeneous and the cross section does not change along the wire!]

b. For pure metals, $\alpha_o \gg \lambda_o$. Alloying metal A with metal B reduces the TCR (temperature coefficient of resistivity) α_A of metal A to $\alpha_{AB} = \alpha_A(\rho_A/\rho_{AB})$. The $\alpha_o - \lambda_o$ can be brought to zero by using a suitable composition alloy for which $\alpha_o = \lambda_o$. Since α_o strictly depends (however slightly) on T , the condition $\alpha_o - \lambda_o$ can only be exactly true at one temperature or approximately true in a small region around that temperature. In practice this temperature range may be sufficient to cover a typical application range in which, for most practical purposes, $\alpha_o - \lambda_o$ is negligible.

2.23 Temperature of a light bulb

- a.* Consider a 100 W, 120 V incandescent bulb (lamp). The tungsten filament has a length of 0.579 m and a diameter of 63.5 μm . Its resistivity at room temperature is 56 n Ω m. Given that the resistivity of the filament can be represented as

$$\rho = \rho_0 \left[\frac{T}{T_0} \right]^n \quad \text{Resistivity of W}$$

where T is the temperature in K, ρ_0 is the resistance of the filament at T_0 K, and $n = 1.2$, estimate the temperature of the bulb when it is operated at the rated voltage, that is, directly from the mains outlet. Note that the bulb dissipates 100 W at 120 V.

- b.* Suppose that the electrical power dissipated in the tungsten wire is totally radiated from the surface of the filament. The radiated power at the absolute temperature T can be described by Stefan's Law

$$P_{\text{radiated}} = \epsilon \sigma_s A (T^4 - T_o^4) \quad \text{Radiated power}$$

where σ_s is Stefan's constant ($5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$), ϵ is the emissivity of the surface (0.35 for tungsten), A is the surface area of the tungsten filament, and T_o is the room temperature (293 K). Obviously, for $T > T_o$, $P_{\text{radiated}} = \epsilon \sigma_s A T^4$.

Assuming that all of the electrical power is radiated from the surface, estimate the temperature of the filament and compare it with your answer in part (a).

- c. If the melting temperature of W is 3407°C , what is the voltage that guarantees that the light bulb will blow?

Solution

- a. First, find the current through the bulb at 100 W and 120 V.

$$P = VI$$

$$\therefore I = P/V = (100 \text{ W})/(120 \text{ V}) = 0.8333 \text{ A}$$

From Ohm's law the resistance of the bulb can be found:

$$R = V/I = (120 \text{ V})/(0.8333 \text{ A}) = 144.0 \Omega$$

The values for length of the filament ($L = 0.579 \text{ m}$) and diameter of the filament ($D = 63.5 \mu\text{m}$) at operating temperature are given. Using these values we can find the resistivity of the filament when the bulb is on (ρ_1).

$$R = \frac{\rho_1 L}{\frac{\pi}{4} D^2}$$

$$\therefore \rho_1 = \frac{R \frac{\pi}{4} D^2}{L} = \frac{(144.0 \Omega) \frac{\pi}{4} (63.5 \times 10^{-6} \text{ m})^2}{(0.579 \text{ m})} = 7.876 \times 10^{-7} \Omega \text{ m}$$

Now the bulb's operating temperature (T_1) can be found using our obtained values in the equation for resistivity of W (assuming room temperature $T_o = 293 \text{ K}$ and given $n = 1.2$):

$$\rho = \rho_0 \left[\frac{T}{T_o} \right]^n$$

$$\text{isolate: } T_1 = T_o \left[\frac{\rho_1}{\rho_0} \right]^{\frac{1}{n}} = (293 \text{ K}) \left[\frac{7.876 \times 10^{-7} \Omega \text{ m}}{56 \times 10^{-9} \Omega \text{ m}} \right]^{\frac{1}{1.2}} = \mathbf{2652 \text{ K}}$$

- b. First we need the surface area A of the Tungsten filament. Since it is cylindrical in shape:

$$A = L(\pi D) = (0.579 \text{ m})(\pi)(63.5 \times 10^{-6} \text{ m}) = 0.0001155 \text{ m}^2$$

Now the temperature of the filament T_1 can be found by isolating it in Stefan's equation and substituting in the given values for emissivity ($\epsilon = 0.35$), Stefan's constant ($\sigma_s = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$) and room temperature ($T_o = 293 \text{ K}$).

$$P = \epsilon \sigma_s A (T_1^4 - T_o^4)$$

$$\therefore T_1 = \left(\frac{P}{\epsilon \sigma_s A} + T_0^4 \right)^{\frac{1}{4}}$$

$$\therefore T_1 = \left[\frac{100 \text{ W}}{(0.35)(5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-1})(0.0001155 \text{ m}^2)} + (293 \text{ K})^4 \right]^{\frac{1}{4}}$$

$$\therefore T_1 = 2570 \text{ K}$$

Note: We can even ignore T_0 to get the same temperature since $T_0 \ll T_1$:

$$P = \epsilon \sigma_s A T_1^4$$

$$\therefore T_1 = \left(\frac{P}{\epsilon \sigma_s A} \right)^{\frac{1}{4}} = \left[\frac{100 \text{ W}}{(0.35)(5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-1})(0.0001155 \text{ m}^2)} \right]^{\frac{1}{4}}$$

$$\therefore T_1 = 2570 \text{ K}$$

These values are fairly close to the answer obtained in part (a).

c. Let V be the voltage and R be the resistance when the filament is at temperature T_m . We are given the temperature $T_m = 3407^\circ\text{C} + 273 = 3680 \text{ K}$. Since we know the following:

$$R = \frac{L}{\frac{\pi}{4} D^2} \rho \quad \text{and} \quad \rho = \rho_0 \left[\frac{T_m}{T_0} \right]^n$$

We can make a substitution for ρ and use the values given for the light bulb filament to find the resistance of the filament at temperature T_m .

$$R = \frac{L}{\frac{\pi}{4} D^2} \rho_0 \left[\frac{T_m}{T_0} \right]^n = \frac{(0.579 \text{ m})}{\frac{\pi}{4} (63.5 \times 10^{-6} \text{ m})^2} (56 \times 10^{-9} \Omega \text{ m}) \left[\frac{3680 \text{ K}}{293 \text{ K}} \right]^{1.2}$$

$$\therefore R = 213.3 \Omega$$

Assuming that all electrical power is radiated from the surface of the bulb, we can use Stefan's law and substitute in V^2/R for power P :

$$\frac{V^2}{R} = \epsilon \sigma_s A (T_1^4 - T_0^4)$$

$$\therefore V^2 = R \epsilon \sigma_s A (T_m^4 - T_0^4)$$

$$\therefore V^2 = (213.3 \Omega)(0.35)(5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4})(0.0001155 \text{ m}^2)[(3680 \text{ K})^4 - (293 \text{ K})^4]$$

$$\therefore V = 299 \text{ V}$$

2.24 Einstein relation and ionic conductivity In the case of ionic conduction, ions have to jump from one interstice to the neighboring one. This process involves overcoming a potential energy barrier just like atomic diffusion, and drift and diffusion are related. The drift mobility μ of ions is

proportional to the diffusion coefficient D because drift is limited by the atomic diffusion process. The **Einstein relation** relates the two by

$$\frac{D}{\mu} = \frac{kT}{e} \quad \text{Einstein relation}$$

The diffusion coefficient of the Na^+ ion in sodium silicate ($\text{Na}_2\text{O-SiO}_2$) glasses at 400°C is $3.4 \times 10^{-9} \text{ cm}^2 \text{ s}^{-1}$. The density of such glasses is approximately 2.4 g cm^{-3} . Calculate the ionic conductivity and resistivity of (17.5 mol% Na_2O)(82.5 mol% SiO_2) sodium silicate glass at 400°C and compare your result with the experimental value of about $10^4 \Omega \text{ cm}$ for the resistivity.

Solution

We can apply the conductivity expression $\sigma = q_i n_i \mu_i$, where q_i is the charge of the ion, Na^+ , so that it is $+e$, n_i is the concentration of the Na^+ ions in the glass and μ_i is their mobility. We can think of the glass as built from $[(\text{Na}_2\text{O})_{0.175}(\text{SiO}_2)_{0.825}]$ units. The atomic masses of Na, O and Si are 23, 16 and 28.1 g mol^{-1} respectively. The atomic mass of one unit is

$$M_{at} = 0.175[2 \times (23) + 16] + 0.825[28.1 + 2 \times (16)] = \mathbf{60.43 \text{ g mol}^{-1}} \text{ of } [(\text{Na}_2\text{O})_{0.175}(\text{SiO}_2)_{0.825}]$$

The number of $[(\text{Na}_2\text{O})_{0.175}(\text{SiO}_2)_{0.825}]$ units per unit volume can be found from the density d by

$$n = \frac{d N_A}{M_{at}} = \frac{(2.4 \text{ g cm}^{-3})(6.02 \times 10^{23} \text{ mol}^{-1})}{(60.43 \text{ g mol}^{-1})} = \mathbf{2.39 \times 10^{22} \text{ units cm}^{-3}}$$

The concentration of Na^+ ions is equal to the concentration of Na atoms and then

$$\begin{aligned} n_i &= n_{Na} = (\text{atomic fraction of Na in the unit}) \times n = \\ &= \left[\frac{0.175 \times (2)}{0.175(2+1) + 0.825(1+2)} \right] (2.39 \times 10^{22} \text{ cm}^{-3}) = \mathbf{2.79 \times 10^{21} \text{ cm}^{-3}} \end{aligned}$$

The mobility of sodium ions can be calculated from Einstein relation

$$\mu_i = \frac{eD}{kT} = \frac{(1.602 \times 10^{-19} \text{ C})(3.4 \times 10^{-9} \text{ cm}^2 \text{ s}^{-1})}{(1.38 \times 10^{-23} \text{ J K}^{-1})(400 + 273) \text{ K}} = \mathbf{6.04 \times 10^{-8} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}}$$

Thus the conductivity of the glass is

$$\begin{aligned} \sigma &= e n_i \mu_i = (1.602 \times 10^{-19} \text{ C})(2.79 \times 10^{21} \text{ cm}^{-3})(6.04 \times 10^{-8} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}) \\ &= \mathbf{2.7 \times 10^{-5} \Omega^{-1} \text{ cm}^{-1}} \end{aligned}$$

and its resistivity is

$$\rho = \frac{1}{\sigma} = \frac{1}{(2.7 \times 10^{-5} \Omega^{-1} \text{ cm}^{-1})} = \mathbf{3.8 \times 10^4 \Omega \text{ cm}}$$

2.25 Skin effect

- a. What is the skin depth for a copper wire carrying a current at 60 Hz? The resistivity of copper at 27°C is $17 \text{ n}\Omega \text{ m}$. Its relative permeability is $\mu_r \approx 1$. Is there any sense in using a conductor for power transmission with a diameter of more than 2 cm?

- b. What is the skin depth for an iron wire carrying a current at 60 Hz? The resistivity of iron at 27 °C is 97 nΩ m. Assume that its relative permeability is $\mu_r \approx 700$. How does this compare with the copper wire? Discuss why copper is preferred over iron for power transmission even though the iron is nearly 100 times cheaper than copper.

Solution

a. The conductivity is $1/\rho$. The relative permeability (μ_r) for copper is 1, thus $\mu_{\text{Cu}} = \mu_o$. The angular frequency is $\omega = 2\pi f = 2\pi(60 \text{ Hz})$. Using these values in the equation for skin depth (δ):

$$\delta = \frac{1}{\sqrt{\frac{1}{2} \omega_o \frac{1}{\rho} \mu_{\text{Cu}}}} = \frac{1}{\sqrt{\frac{1}{2} (2\pi 60 \text{ s}^{-1}) \frac{(4\pi \times 10^{-7} \text{ H m}^{-1})}{(17 \times 10^{-9} \text{ } \Omega \text{ m})}}}$$

$$\therefore \quad \delta = \mathbf{0.00847 \text{ m or } 8.47 \text{ mm}}$$

This is the depth of current flow. If the radius of wire is 10 mm or more, no current flows through the core region and it is wasted. There is no point in using wire much thicker than a radius of 10 mm (diameter of 20 mm).

b. The conductivity is $1/\rho$. The relative permeability (μ_r) for Iron is 700, thus $\mu_{\text{Fe}} = 700\mu_o$. The angular frequency is $\omega = 2\pi f = 2\pi(60 \text{ Hz})$. Using these values in the equation for skin depth (δ):

$$\delta_{\text{Fe}} = \frac{1}{\sqrt{\frac{1}{2} \omega_o \frac{1}{\rho} \mu_{\text{Fe}}}} = \frac{1}{\sqrt{\frac{1}{2} (2\pi 60 \text{ s}^{-1}) \frac{(700)(4\pi \times 10^{-7} \text{ H m}^{-1})}{(97 \times 10^{-9} \text{ } \Omega \text{ m})}}}$$

$$\therefore \quad \delta = \mathbf{0.000765 \text{ m or } 0.765 \text{ mm}}$$

Thus the skin depth is 0.765 mm, about 11 times less than that for copper.

To calculate the resistance we need the cross sectional area for conduction. The material cross sectional area is πr^2 where r is the radius of the wire. But the current flow is within depth δ . We deduct the area of the core, $\pi(r - \delta)^2$, from the overall area, πr^2 , to obtain the cross sectional area for conduction.

Comparison of Cu and Fe based on solid core wires:

The resistance per unit length of the solid core Fe wire (R_{Fe}) is:

$$R_{\text{Fe}} = \frac{\rho_{\text{Fe}}}{A_{\text{Fe}}} = \frac{\rho_{\text{Fe}}}{\pi r_{\text{Fe}}^2 - \pi(r_{\text{Fe}} - \delta_{\text{Fe}})^2} = \frac{\rho_{\text{Fe}}}{2\pi r_{\text{Fe}} \delta_{\text{Fe}} - \pi \delta_{\text{Fe}}^2}$$

The resistance per unit length of solid core Cu wire is:

$$R_{\text{Cu}} = \frac{\rho_{\text{Cu}}}{A_{\text{Cu}}} = \frac{\rho_{\text{Cu}}}{\pi r_{\text{Cu}}^2 - \pi(r_{\text{Cu}} - \delta_{\text{Cu}})^2} = \frac{\rho_{\text{Cu}}}{2\pi r_{\text{Cu}} \delta_{\text{Cu}} - \pi \delta_{\text{Cu}}^2}$$

If we equate these two resistances, we can make a comparison between Fe and Cu:

$$R_{\text{Fe}} = R_{\text{Cu}}$$

$$\therefore \quad \frac{\rho_{\text{Fe}}}{2\pi r_{\text{Fe}} \delta_{\text{Fe}} - \pi \delta_{\text{Fe}}^2} = \frac{\rho_{\text{Cu}}}{2\pi r_{\text{Cu}} \delta_{\text{Cu}} - \pi \delta_{\text{Cu}}^2}$$

$$\therefore 2\pi\rho_{\text{Cu}}r_{\text{Fe}}\delta_{\text{Fe}} = \rho_{\text{Fe}}(2\pi r_{\text{Cu}}\delta_{\text{Cu}} - \pi\delta_{\text{Cu}}^2) \quad (\text{Neglect the } \delta_{\text{Fe}}^2 \text{ term which is small})$$

$$\therefore r_{\text{Fe}} = \frac{\rho_{\text{Fe}}(2\pi r_{\text{Cu}}\delta_{\text{Cu}} - \pi\delta_{\text{Cu}}^2)}{2\pi\rho_{\text{Cu}}\delta_{\text{Fe}}}$$

We can assume a value for r_{Cu} for calculation purposes, $r_{\text{Cu}} = 10$ mm. The resistivity ρ_{Cu} is given as 17 n Ω m and the skin depth of Cu is known to be $\delta_{\text{Cu}} = 8.47$ mm. The resistivity of Fe is given as $\rho_{\text{Fe}} = 97$ n Ω m and its skin depth was just calculated to be $\delta_{\text{Fe}} = 0.765$ mm. We can substitute these values into the above equation to determine the radius of Fe wire that would be equivalent to 10 mm diameter Cu wire.

$$r_{\text{Fe}} = \frac{(97 \times 10^{-9} \Omega \text{ m})[2\pi(0.010 \text{ m})(0.00847 \text{ m}) - \pi(0.00847 \text{ m})^2]}{2\pi(17 \times 10^{-9} \Omega \text{ m})(0.000765 \text{ m})}$$

$$\therefore r_{\text{Fe}} = \mathbf{0.364 \text{ m}}$$

Now compare the volume (V) of Fe per unit length to the volume of Cu per unit length:

$$\frac{V_{\text{Fe}}}{V_{\text{Cu}}} = \frac{\pi r_{\text{Fe}}^2 (1 \text{ m})}{\pi r_{\text{Cu}}^2 (1 \text{ m})} = \frac{(0.364 \text{ m})^2}{(0.010 \text{ m})^2} = \mathbf{1325}$$

Even though Fe costs 100 times less than Cu, we need about 1300 times the volume of Cu if Fe is used. The cost disadvantage is 13 times in addition to weight disadvantage.

ADDENDUM JANUARY 2001 (A Discussion by George Belev)

Comparison of Cu and Fe wires: any shape and any number:

To determine if it is worthwhile to use iron rather than copper, we must compare the amount of iron needed to perform the equivalent task of some amount of copper (*i.e.* have the same resistance). Let us first assume that by choosing a proper shape for the conductors we can *eliminate* the influence of the skin effect on conduction.

The resistance per unit length of the Fe wire (R_{Fe}) is:

$$R_{\text{Fe}} = \frac{\rho_{\text{Fe}}}{A_{\text{Fe}}}$$

The resistance per unit length of Cu wire is

$$R_{\text{Cu}} = \frac{\rho_{\text{Cu}}}{A_{\text{Cu}}}$$

If we equate these two resistances, we can make a direct comparison between Fe and Cu:

$$R_{\text{Fe}} = R_{\text{Cu}}$$

$$\therefore A_{\text{Fe}} = \frac{\rho_{\text{Fe}}}{\rho_{\text{Cu}}} A_{\text{Cu}}$$

and the volumes of iron and copper per unit length will be in the same ratio

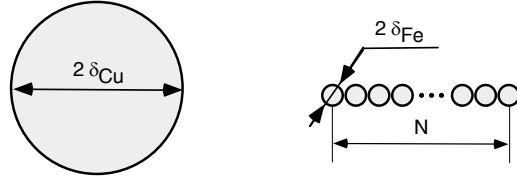
$$V_{\text{Fe}} = \frac{\rho_{\text{Fe}}}{\rho_{\text{Cu}}} V_{\text{Cu}}$$

Let us compare the masses of Fe and Cu needed per unit length

$$\frac{M_{Fe}}{M_{Cu}} = \frac{V_{Fe} D_{Fe}}{V_{Cu} D_{Cu}} = \frac{\rho_{Fe}}{\rho_{Cu}} \frac{D_{Fe}}{D_{Cu}} = \frac{97 \text{ n}\Omega\text{m}}{17 \text{ n}\Omega\text{m}} \cdot \frac{7.86 \text{ gcm}^{-3}}{8.96 \text{ gcm}^{-3}} \approx 5$$

Since Fe costs 100 times less than Cu, if we use iron conductors, we will reduce the cost for wire by $100/5 = 20$ times. So it seems that the use of Fe will have great economic advantage if we can find a reasonable way to eliminate the influence of the skin effect on conduction.

There is no sense in making the conductor with a radius bigger than the skin depth, so let us consider a single copper conductor with radius δ_{Cu} , and using N iron conductors each with radius δ_{Cu} as shown bellow:



As we have calculated above both conductors will have equal resistance per unit length if

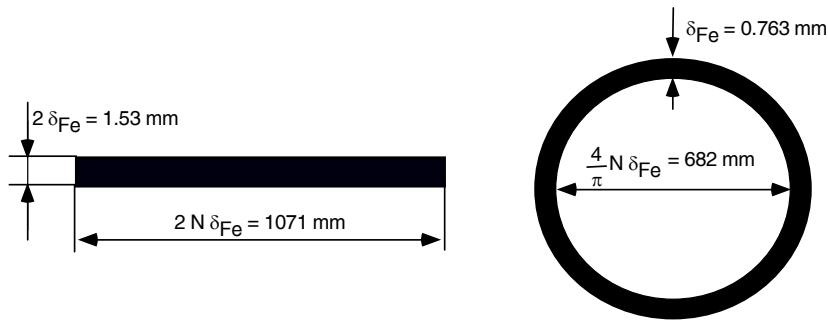
$$\frac{A_{Fe}}{A_{Cu}} = \frac{N\pi\delta_{Fe}^2}{\pi\delta_{Cu}^2} = \frac{\rho_{Fe}}{\rho_{Cu}}$$

and we can calculate the number N of Fe wires which should run in parallel

$$N = \frac{\rho_{Fe}}{\rho_{Cu}} \frac{\delta_{Cu}^2}{\delta_{Fe}^2} \approx 700$$

Thus, copper as a single conductor has 700 times better performance than a single iron conductor.

It is not necessary to manufacture 700 Fe wires and run them in parallel. The iron conductor can be produced more conveniently in the shapes shown bellow and it will be cheaper than the Cu conductor.



But, it will be of impractical size; it will have poor mechanical properties and will be 5 times heavier compared with the single Cu wire. A power grid based on Cu conductors will be much cheaper and much smaller than the one based on Fe conductors.

2.26 Thin films

- Consider a polycrystalline copper film that has $R = 0.40$. What is the approximate mean grain size d in terms of the mean free path λ in the bulk that would lead to the polycrystalline Cu film having a resistivity that is $1.5\rho_{\text{bulk}}$. If the mean free path in the crystal is about 40 nm at room temperature, what is d ?

- b. What is the thickness D of a copper film in terms of λ in which surface scattering increases the film resistivity to $1.2\rho_{\text{bulk}}$ if the specular scattering fraction p is 0.5?
- c. Consider the data of Lim *et al.* (2003) presented in Table 2.13. Show that the excess resistivity, *i.e.* resistivity above that of bulk Cu, is roughly proportional to the reciprocal film thickness.

Table 2.13 Resistivity ρ_{film} of a copper film as a function of thickness D .

D (nm)	8.61	17.2	34.4	51.9	69	85.8	102.6	120.3	173.2	224.3
ρ_{film} (n Ω m)	121.8	75.3	46.1	38.5	32.1	25.2	22.0	20.5	19.9	18.8

Solution

a. Mayadas-Shatzke formula estimates the equivalent resistivity of the resistivity of polycrystalline sample as

$$\frac{\rho}{\rho_{\text{crystal}}} \approx 1 + 1.33\beta$$

where,
$$\beta = \frac{\lambda}{d} \left(\frac{R}{1-R} \right).$$

Using the probability of reflection at a grain boundary $R = 0.40$, in the above formula we have

$$\frac{1.5\rho_{\text{crystal}}}{\rho_{\text{crystal}}} \approx 1 + 1.33 \frac{\lambda}{d} \left(\frac{0.4}{1-0.4} \right)$$

$$1.5 = 1 + 0.89 \frac{\lambda}{d} \quad \text{or} \quad 1.5 - 1 = 0.89 \frac{\lambda}{d}$$

$$d = 1.77 \lambda$$

If the mean free path in the crystal is $\lambda = 40$ nm, then the mean grain size is $d = 1.77(40 \text{ nm}) = \mathbf{70.8 \text{ nm}}$.

b. Surface scattering resistivity is given by Equation 2.60 as

$$\frac{\rho}{\rho_{\text{bulk}}} \approx 1 + \frac{3\lambda}{8D}(1-p) \quad \frac{D}{\lambda} > 0.3$$

Using $p = 0.5$, we have

$$1.2 \frac{\rho_{\text{bulk}}}{\rho_{\text{bulk}}} \approx 1 + \frac{\lambda}{D} \cdot \frac{3}{8} \cdot (0.5)$$

$$1.2 \approx 1 + 0.1875 \frac{\lambda}{D}$$

Simplifying we have $D \approx \mathbf{0.9375 \lambda}$.

c. The data in Table 2.13 are plotted as resistivity vs. $1/D$, where it can be seen that ρ depends on $1/D$.

Author's Note: The actual data, as it turns out, cannot be simply interpreted in terms of the simple surface scattering resistivity formula, which is the reason that it is not directly applied to the data. (Its application generates a negative p !) Although surface scattering is certainly dominant, as apparent from

ρ vs. $1/D$ behavior, the original paper has also indicated some grain boundary scattering contribution as well. (J.W. Lim, K. Mimura and M. Isshiki, *Applied Surface Science*, **217**, 95-99, 2003.)

Author's Note to the Instructor: Most successful models that account for the observed resistivity of a thin film as a function of thickness invariably involve combining surface scattering not only with grain boundary scattering, but also including other factors, such as the effect of surface roughness on the scattering mechanism. (A good example is H. D. Liu et al, *Thin Solid Films*, **384**, 151-156, 2001.)

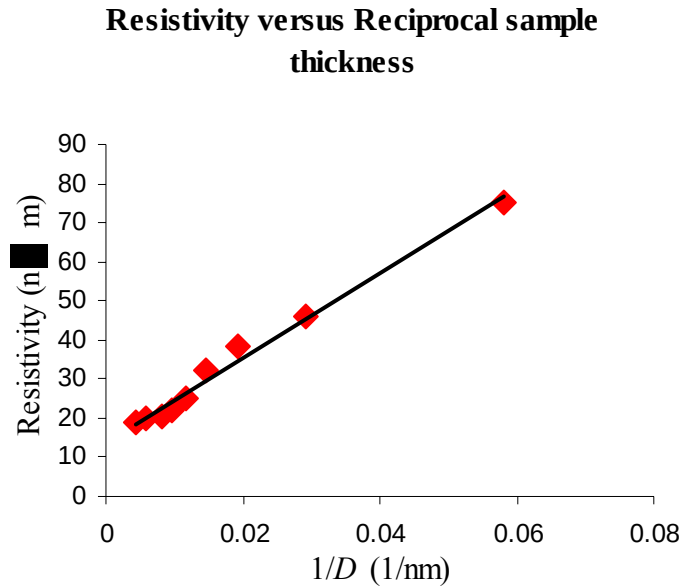


Figure 2Q26-1: Resistivity versus reciprocal sample thickness $1/D$

From the plot we see a straight line for plot of film resistivity and sample thickness reciprocal.

2.27 Interconnects Consider a high-transistor-density CMOS chip in which the interconnects are copper with a pitch P of 500 nm, interconnect thickness T of 400 nm, aspect ratio 1.4, and $H = X$. The dielectric is FSG with $\epsilon_r = 3.6$. Consider two cases, $L = 1$ mm and $L = 10$ mm, and calculate the overall effective interconnect capacitance C_{eff} and the RC delay time. Suppose that Al, which is normally Al with about 4 wt.% Cu in the microelectronics industry with a resistivity $31 \text{ n}\Omega \text{ m}$, is used as the interconnect. What is the corresponding RC delay time?

Solution

The effective capacitance in multilevel interconnect structure is given by Equation 2.61

$$C_{\text{eff}} = 2\epsilon_0\epsilon_r L \left(\frac{T}{X} + \frac{W}{H} \right)$$

Aspect ratio is defined as $A_R = \frac{T}{W}$,

or
$$W = \frac{T}{A_R} = \frac{400 \text{ nm}}{1.4} = 258.71 \text{ nm}.$$

$$X = P - W = 500 \text{ nm} - 258.71 \text{ nm} = 241.29 \text{ nm}.$$

Therefore the effective capacitance of the interconnect is

For $L = 1$ mm

$$C_{\text{eff}} = 2 \cdot (8.8542 \times 10^{-12} \text{ Fm}^{-1}) \cdot (3.6) \cdot (1 \times 10^{-3} \text{ m}) \left(\frac{400 \text{ nm}}{214.29 \text{ nm}} + \frac{258.29 \text{ nm}}{214.29 \text{ nm}} \right)$$

$$= \mathbf{0.196 \times 10^{-12} \text{ F or } 0.196 \text{ pF.}}$$

For $L = 10$ mm

$$C_{\text{eff}} = 2 \cdot (8.8542 \times 10^{-12} \text{ Fm}^{-1}) \cdot (3.6) \cdot (10 \times 10^{-3} \text{ m}) \left(\frac{400 \text{ nm}}{214.29 \text{ nm}} + \frac{258.29 \text{ nm}}{214.29 \text{ nm}} \right)$$

$$= 1.96 \times 10^{-12} \text{ F or } 1.96 \text{ pF}$$

RC time constant of interconnect is given by

$$RC = 2\varepsilon_0\varepsilon_r\rho \left(\frac{L^2}{TW} \right) \left(\frac{T}{X} + \frac{W}{H} \right)$$

Therefore for Cu, the RC delay time for $L = 1$ mm is

$$RC = 2 \cdot (8.8542 \times 10^{-12} \text{ Fm}^{-1}) \cdot (3.6) \cdot (17 \text{ n}\Omega \text{ m}) \cdot \frac{(1 \times 10^{-3} \text{ m})^2}{(400 \text{ nm})(258.71 \text{ nm})} \cdot \left(\frac{400 \text{ nm}}{214.29 \text{ nm}} + \frac{258.29 \text{ nm}}{214.29 \text{ nm}} \right)$$

$$= 3.217 \times 10^{-11} \text{ s or } 32.17 \text{ ps}$$

and the RC delay time for $L = 10$ mm is

$$RC = 2 \cdot (8.8542 \times 10^{-12} \text{ Fm}^{-1}) \cdot (3.6) \cdot (17 \text{ n}\Omega \text{ m}) \cdot \frac{(10 \times 10^{-3} \text{ m})^2}{(400 \text{ nm})(258.71 \text{ nm})} \cdot \left(\frac{400 \text{ nm}}{214.29 \text{ nm}} + \frac{258.29 \text{ nm}}{214.29 \text{ nm}} \right)$$

$$= 3.217 \times 10^{-9} \text{ s or } 32.17 \text{ ns}$$

Given that for Al $\rho = 31 \text{ n}\Omega \text{ m}$, the RC time constant for $L = 1$ mm is

$$RC = 2 \cdot (8.8542 \times 10^{-12} \text{ Fm}^{-1}) \cdot (3.6) \cdot (31 \text{ n}\Omega \text{ m}) \cdot \frac{(1 \times 10^{-3} \text{ m})^2}{(400 \text{ nm})(258.71 \text{ nm})} \cdot \left(\frac{400 \text{ nm}}{214.29 \text{ nm}} + \frac{258.29 \text{ nm}}{214.29 \text{ nm}} \right)$$

$$= 5.867 \times 10^{-11} \text{ s or } 58.67 \text{ ps}$$

For $L = 10$ mm

$$RC = 2 \cdot (8.8542 \times 10^{-12} \text{ Fm}^{-1}) \cdot (3.6) \cdot (31 \text{ n}\Omega \text{ m}) \cdot \frac{(10 \times 10^{-3} \text{ m})^2}{(400 \text{ nm})(258.71 \text{ nm})} \cdot \left(\frac{400 \text{ nm}}{214.29 \text{ nm}} + \frac{258.29 \text{ nm}}{214.29 \text{ nm}} \right)$$

$$= 5.867 \times 10^{-9} \text{ s or } 5.867 \text{ ns}$$

***2.28 Thin 50 nm interconnects** Equation 2.60 is for conduction in a thin film of thickness D and assumes scattering from two surfaces, which yields an additional resist-

ivity $\rho_2 = \rho_{\text{bulk}} \frac{3}{8} (\lambda / D) (1 - p)$. An interconnect line in an IC is not quite a thin film and has four surfaces (interfaces), because the thickness T of the conductor is comparable to the width W . If we assume $T =$

W , we can very roughly take $\rho_4 \approx \rho_2 + \rho_2 \approx \rho_{\text{bulk}} \frac{3}{4} (\lambda/D)(1-p)$ in which $D = T$. (The exact expression is more complicated, but the latter will suffice for this problem.) In addition there will be a contribution from grain boundary scattering, (Equation 2.57a). For simplicity assume $T \approx W \approx X \approx H \approx 60$ nm, $\lambda = 40$ nm, $p = 0.5$ and $\varepsilon_r = 3.6$. If the mean grain size d is roughly 40 nm and $R = 0.4$, estimate the resistivity of the interconnect and hence the RC delay for a 1 mm interconnect.

Solution

In thin films in addition to the bulk resistivity, we take into account the resistivity due to scattering from the surfaces and due to grain boundary effects. As described in the problem statement, the resistivity due to surface scattering from four surfaces in an interconnect can roughly taken into account as

$$\rho_{\text{surface}} = \rho_{\text{bulk}} \frac{3}{4} \frac{\lambda}{D} (1-p)$$

In addition to the surface scattering, there will be resistivity due to grain boundary scattering as well, which is given by the Equations 2.57a and 2.57b as

$$\rho_{\text{grain}} = \rho_{\text{bulk}} \cdot 1.33 \cdot \frac{\lambda}{d} \left(\frac{R}{1-R} \right)$$

The effective resistivity of the interconnect will therefore be the sum of these resistivities

$$\rho = \rho_{\text{bulk}} + \rho_{\text{surface}} + \rho_{\text{grain}} = \rho_{\text{bulk}} \left[1 + \frac{3}{4} \frac{\lambda}{D} (1-p) + 1.33 \frac{\lambda}{d} \left(\frac{R}{1-R} \right) \right]$$

Considering the bulk resistivity of Cu as 17 nΩ m, we have

$$\rho = 17 \text{ n}\Omega \text{ m} \left[1 + \frac{3}{4} \frac{(40 \text{ nm})}{(60 \text{ nm})} (1-0.5) + 1.33 \cdot \frac{(40 \text{ nm})}{(40 \text{ nm})} \left(\frac{0.4}{1-0.4} \right) \right] = \mathbf{36.32 \text{ n}\Omega \text{ m}}$$

$$RC = 2\varepsilon_0\varepsilon_r\rho \left(\frac{L^2}{TW} \right) \left(\frac{T}{X} + \frac{W}{H} \right)$$

Therefore for Cu, the RC delay time for $L = 1$ mm is

$$\begin{aligned} RC &= 2 \cdot (8.8542 \times 10^{-12} \text{ F m}^{-1}) \cdot (3.6) \cdot (17 \text{ n}\Omega \text{ m}) \cdot \frac{(1 \times 10^{-3} \text{ m})^2}{(60 \text{ nm})(60 \text{ nm})} \cdot \left(\frac{60 \text{ nm}}{60 \text{ nm}} + \frac{60 \text{ nm}}{60 \text{ nm}} \right) \\ &= 1.29 \times 10^{-9} \text{ s or } 1.29 \text{ ns.} \end{aligned}$$

2.29 TCR of thin films Consider Matthiessen's rule applied to a thin film. Show that, very approximately, the product of the thermal coefficient of resistivity (TCR) α_{film} and the resistivity ρ_{film} is equivalent to the product of the bulk TCR and resistivity:

$$\alpha_{\text{film}} \rho_{\text{film}} \approx \alpha_{\text{bulk}} \rho_{\text{bulk}}$$

Solution

Considering thin films, depending on a number of factors there will be additional resistivities resulting from surface scattering and grain boundary scattering. If we consider that these additional resistivities contribute a factor K , then the effective resistivity of the film in Matthiessen's rule form will be

$$\rho_{\text{film}} = \rho_{\text{bulk}} (1 + K)$$

We very roughly assume that the additional resistivities are independent of the change in temperature. From the definition of temperature coefficient of resistivity α we have

$$\alpha_{\text{film}} = \frac{1}{\rho_{\text{film}}} \left[\frac{d\rho_{\text{film}}}{dT} \right]_{T=T_0}$$

Substitute the relation for ρ_{film} , we have

$$\alpha_{\text{film}} = \frac{1}{\rho_{\text{film}}} \cdot \frac{d}{dT} (\rho_{\text{bulk}} (1 + K))$$

Since we have assumed the additional resistivity is independent of T

$$\alpha_{\text{film}} \approx \frac{1}{\rho_{\text{film}}} \cdot \left[\frac{d\rho_{\text{bulk}}}{dT} \right]$$

From $\alpha_{\text{bulk}} = \frac{1}{\rho_{\text{bulk}}} \frac{d\rho_{\text{bulk}}}{dT} \Big|_{T=T_0}$ we have $\frac{d\rho_{\text{bulk}}}{dT} = \alpha_{\text{bulk}} \rho_{\text{bulk}}$. Substituting this in the above equation we have

$$\alpha_{\text{film}} \approx \frac{1}{\rho_{\text{film}}} \cdot \alpha_{\text{bulk}} \rho_{\text{bulk}}$$

or $\alpha_{\text{film}} \rho_{\text{film}} \approx \alpha_{\text{bulk}} \rho_{\text{bulk}}$

2.30 Electromigration Although electromigration-induced failure in Cu metallization is less severe than in Al metallization, it can still lead to interconnect failure depending on current densities and the operating temperature. In a set of experiments carried out on electroplated Cu metallization lines, failure of the Cu interconnects have been examined under accelerated tests (at elevated temperatures). The mean lifetime t_{50} (time for 50 percent of the lines to break) have been measured as a function of current density J and temperature T at a given current density. The results are summarized in Table 2.14.

- Plot semi-logarithmically t_{50} versus $1/T$ (T in Kelvins) for the first three interconnects. Al(Cu) and Cu ($1.3 \times 0.7 \mu\text{m}^2$) have single activation energies E_A . Calculate E_A for these interconnects. Cu ($1.3 \times 0.7 \mu\text{m}^2$) exhibits different activation energies for the high- and low-temperature regions. Estimate these E_A .
- Plot on a log-log plot t_{50} versus J at 370°C . Show that at low J , $n \approx 1.1$ and at high J , $n \approx 1.8$.

Table 2.14 Results of electromigration failure experiments on various Al and Cu interconnects

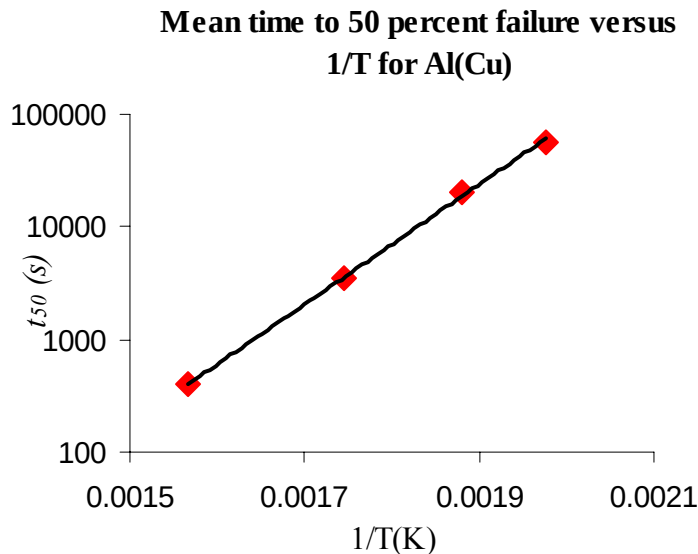
Al(Cu) [$J = 25 \text{ mA}/\mu\text{m}^2$, $A = 0.35 \times 0.2 (\mu\text{m})^2$]		Cu [$J = 25 \text{ mA}/\mu\text{m}^2$, $A = 0.24 \times 0.28 (\mu\text{m})^2$]		Cu [$J = 25 \text{ mA}/\mu\text{m}^2$, $A = 1.3 \times 0.7 (\mu\text{m})^2$]		Cu ($T = 370^\circ\text{C}$)	
$T (^\circ\text{C})$	$t_{50} (\text{hr})$	$T (^\circ\text{C})$	$t_{50} (\text{hr})$	$T (^\circ\text{C})$	$t_{50} (\text{hr})$	$J \text{ mA } \mu\text{m}^{-2}$	$t_{50} (\text{hr})$
365	0.11	397	2.87	395	40.3	3.54	131.5
300	0.98	354	12.8	360	196	11.7	25.2
259	5.73	315	70.53	314	825	24.8	14.9
233	15.7	269	180	285	2098	49.2	4.28
		232	899			74.1	2.29
						140	0.69

Solution

a. The mean time to 50 percent failure is calculated using Black's equation give by Equation 2.64 as

$$t_{MTF} = A_B J^{-n} \exp\left(\frac{E_A}{kT}\right)$$

The t_{50} versus $1/T$ plot using linear scale we obtain an exponential curve. Now, if we use logarithmic scale for t_{50} , we will have a linear plot.

**Figure 2Q30-1:** Mean time to 50 percent failure versus $1/T$ for Al(Cu)

Analyzing Black's equation, we observe that the power of the exponential curve we be equal to

$$\frac{E_A}{k} = 12770 \text{ K or } E_A = (12770 \text{ K}) \cdot (8.617 \times 10^{-5} \text{ eV K}^{-1}) = 1.10 \text{ eV.}$$

Similarly, plot for Cu ($A = 0.24 \times 0.28 (\mu\text{m})^2$) is

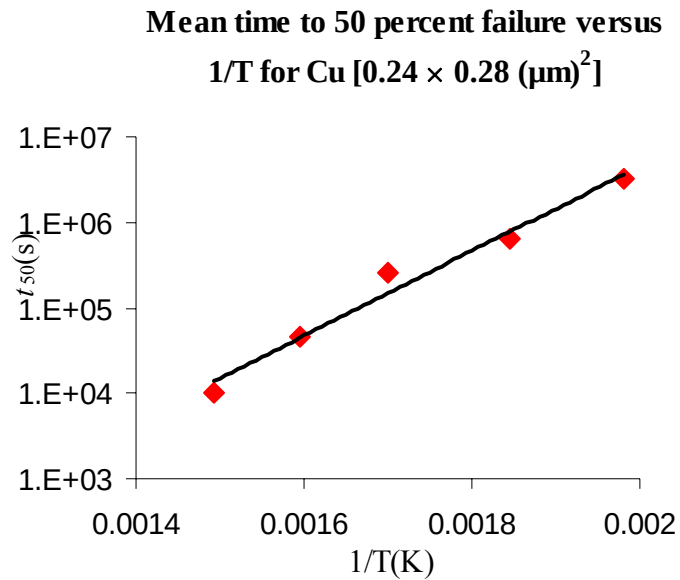


Figure 2Q30-2: Mean time to 50 percent failure versus 1/T for Cu [0.24 × 0.28 (μm)²]

From the plot we have $\frac{E_A}{k} = 11419 \text{ K}$ or $E_A = (11419 \text{ K}) \cdot (8.617 \times 10^{-5} \text{ eV K}^{-1}) = 0.9840 \text{ eV}$.

Comment: In both cases the activation energy E_A is about 1 eV, close to the activation energy for vacancy formation. See Question 1.29. Vacancies assist atomic electromigration.

Plot for Cu ($A = 1.3 \times 0.70 (\mu\text{m})^2$) is,

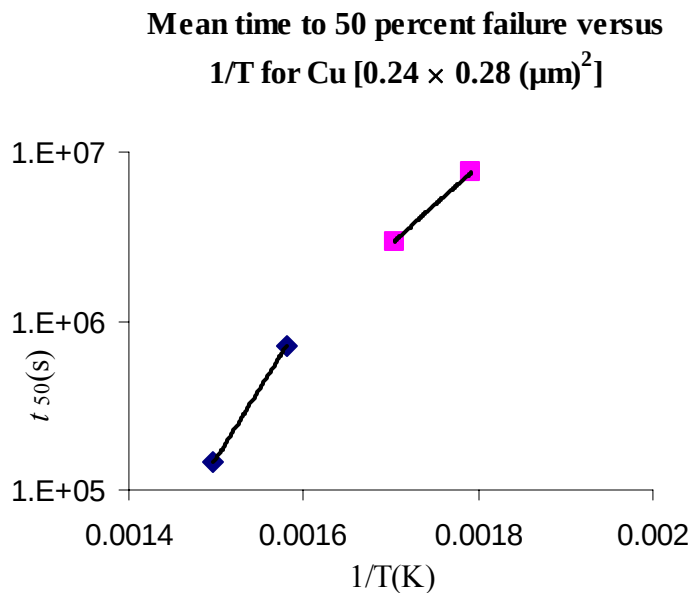


Figure 2Q30-3: Mean time to 50 percent failure versus 1/T for Cu [0.24 × 0.28 (μm)²]

The activation energies at high and low temperatures for this case are different. From the plot,
At low temperatures:

$$\frac{E_A}{k} = 10542 \text{ or } E_A = (10542 \text{ K}) \cdot (8.617 \times 10^{-5} \text{ eV K}^{-1}) = 0.9084 \text{ eV}.$$

At high temperatures:

$$\frac{E_A}{k} = 19110 \text{ or } E_A = (19110 \text{ K}) \cdot (8.617 \times 10^{-5} \text{ eV K}^{-1}) = 1.647 \text{ eV.}$$

b. The plot of current density J mA/ μm^2 against mean time to 50% failure t_{50} is as follows,

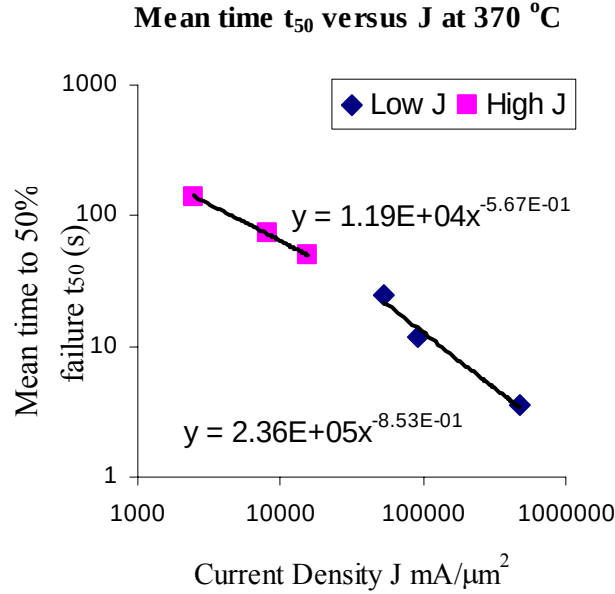


Figure 2Q30-3: Log-Log plot of t_{50} versus J at 370 °C

The equation for the best fit line are mentioned next to the plot. It is evident from the equations that at low temperatures the for J - n the n value is roughly $n \approx 1/1.1$ and at high temperatures $n \approx 1/1.8$.

Taking natural logarithm $\ln$ of both sides of the above relation

$$\ln(t_{MTF}) = \ln\left(A_B J^{-n} \exp\left(\frac{E_A}{kT}\right)\right)$$

$$\ln(t_{MTF}) = \ln A_B + \ln J^{-n} + \frac{E_A}{kT}$$

Let $C = \ln A_B + \ln J^n$, then we have

$$\ln(t_{MTF}) = C + \frac{E_A}{k} \cdot \frac{1}{T}$$

"I think there is a world market for maybe five computers"

Thomas Watson (Chairman of IBM, 1943)

Third Edition (© 2001 McGraw-Hill)

Chapter 3

Note: The first printing has a few odd typos, which are indicated in blue below. These will be corrected in the reprint.

3.1 Photons and photon flux

- Consider a 1 kW AM radio transmitter at 700 kHz. Calculate the number of photons emitted from the antenna per second.
- The average intensity of sunlight on Earth's surface is about 1 kW m^{-2} . The maximum intensity is at a wavelength around 800 nm. Assuming that all the photons have an 800 nm wavelength, calculate the number of photons arriving on Earth's surface per unit time per unit area. What is the magnitude of the electric field in the sunlight?
- Suppose that a solar cell device can convert each sunlight photon into an electron, which can then give rise to an external current. What is the maximum current that can be supplied per unit area (m^2) of this solar cell device?

Solution

- a.** Given: power $P = 1000 \text{ W}$ and frequency $\nu = 700 \times 10^3 \text{ s}^{-1}$.

Then the photon energy is

$$E_{ph} = h\nu = (6.626 \times 10^{-34} \text{ J s})(700 \times 10^3 \text{ s}^{-1})$$

$$\therefore E_{ph} = 4.638 \times 10^{-28} \text{ J/photon or } 2.895 \times 10^{-9} \text{ eV/photon}$$

The number of photons emitted from the antenna per unit time (N_{ph}) is therefore:

$$N_{ph} = \frac{P}{E_{ph}} = \frac{1000 \text{ W}}{4.638 \times 10^{-28} \text{ J}} = \mathbf{2.16 \times 10^{30} \text{ photons per second}}$$

- b.** Average intensity $I_{\text{average}} = 1000 \text{ W/m}^2$ and maximum wavelength $\lambda_{\text{max}} = 800 \text{ nm}$. The photon energy at λ_{max} is,

$$E_{ph} = hc/\lambda_{\text{max}} = (6.626 \times 10^{-34} \text{ J s})(3.0 \times 10^8 \text{ m s}^{-1})/(800 \times 10^{-9} \text{ m})$$

$$\therefore E_{ph} = 2.485 \times 10^{-19} \text{ J or } 1.551 \text{ eV}$$

The photon flux Γ_{ph} is the number of photons arriving per unit time per unit area,

$$\Gamma_{ph} = \frac{I_{\text{average}}}{E_{ph}} = \frac{1000 \text{ W m}^{-2}}{2.485 \times 10^{-19} \text{ J}} = \mathbf{4.02 \times 10^{21} \text{ photons m}^{-2} \text{ s}^{-1}}$$

For the electric field we use classical physics. $I_{\text{average}} = (1/2)c\epsilon_0 E^2$, so that

$$E = \sqrt{\frac{2I_{\text{average}}}{c\epsilon_o}} = \sqrt{\frac{2(1000 \text{ W m}^{-2})}{(3.0 \times 10^8 \text{ m s}^{-1})(8.854 \times 10^{-12} \text{ F m}^{-1})}} = 868 \text{ V m}^{-1}$$

c. If each photon gives rise to one electron then the current density J is:

$$J = e\Gamma_{ph} = (1.602 \times 10^{-19} \text{ C})(4.02 \times 10^{21} \text{ m}^{-2} \text{ s}^{-1}) = 644 \text{ A m}^{-2}$$

3.2 Yellow, cyan, magenta, and white *Three primary colors*, red, green, and blue (RGB), can be added together in various proportions to generate any color on various displays and light emitting devices in what is known as the *additive theory of color*. For example, yellow can be generated from adding red and green, cyan from blue and green, and magenta from red and blue.

- a. A device engineer wants to use three light emitting diodes (LEDs) to generate various colors in an LED-based color display that is still in the research stage. His three LEDs have wavelengths of 660 nm for red, 563 nm for green, and 450 nm for blue. He simply wishes to generate the yellow and cyan by mixing equal optical powers from these LEDs; *optical power*, or *radiant power*, is defined as the radiation energy emitted per unit time. What are the numbers of red and blue photons needed (to the nearest integer) to generate yellow and cyan, respectively, for every 100 green photons?
- b. An equi-energy white light is generated by mixing red, green, and blue light in equal optical powers. Suppose that the wavelengths are 700 nm for red, 546 nm for green, and 436 nm for blue (which is one set of possible standard primary colors). Suppose that the optical power in each primary color is 0.1 W. Calculate the *total photon flux* (photons per second) needed from each primary color.
- c. There are bright white LEDs on the market that generate the white light by mixing yellow (a combination of red and green) with blue emissions. The inexpensive types use a single blue LED to generate a strong blue radiation, some of which is absorbed by a phosphor in front of the LED which then emits yellow light. The yellow and the blue passing through the phosphor mix and make up the white light. In one type of white LED, the blue and yellow wavelengths are 450 nm and 564 nm, respectively. White light can be generated by setting the optical (radiative) power ratio of yellow to blue light emerging from the LED to be about 1.74. What is the ratio of the number of blue to yellow photons needed? (Sometimes the mix is not perfect and the white LED light tends to have a noticeable slight blue tint.) If the total optical power output from the white LED is 100 mW, calculate the blue and yellow total photon fluxes (photons per second).

Solution

a. Radiation energy can be written as, $P = Nh\nu/t$ or $P = Nhc/\lambda t$, where N is the total number of photons.

To generate **yellow** from red and green photons with same radiant power,

$$P_r = P_g \text{ or } N_r = (\lambda_r/\lambda_g) \times N_g = (660 \text{ nm}/563 \text{ nm}) \times 100 \text{ photons}$$

$$\therefore \quad N_r = 117 \text{ photons}$$

To generate **cyan** from blue and green photons with same radiant power,

$$P_b = P_g \text{ or } N_b = (\lambda_b/\lambda_g) \times N_g = (450 \text{ nm}/563 \text{ nm}) \times 100 \text{ photons}$$

$$\therefore N_b = 80 \text{ photons}$$

b. Optical power, $P = \Phi \times hc/\lambda$ or $\Phi = P\lambda/hc$ where Φ is the total photon flux.

$$\Phi_r = P\lambda_r/hc = (0.1 \text{ W})(700 \times 10^{-9} \text{ m})/(6.63 \times 10^{-34} \text{ J s})(3 \times 10^8 \text{ m s}^{-1})$$

$$\therefore \Phi_r = 3.52 \times 10^{17} \text{ photons s}^{-1}$$

$$\Phi_g = P\lambda_g/hc = (0.1 \text{ W})(546 \times 10^{-9} \text{ m})/(6.63 \times 10^{-34} \text{ J s})(3 \times 10^8 \text{ m s}^{-1})$$

$$\therefore \Phi_r = 2.75 \times 10^{17} \text{ photons s}^{-1}$$

$$\Phi_b = P\lambda_b/hc = (0.1 \text{ W})(436 \times 10^{-9} \text{ m})/(6.63 \times 10^{-34} \text{ J s})(3 \times 10^8 \text{ m s}^{-1})$$

$$\therefore \Phi_r = 2.2 \times 10^{17} \text{ photons s}^{-1}$$

c. Optical power ratio, $P_y/P_b = (N_y hc/\lambda_y)/(N_b hc/\lambda_b)$ or $P_y/P_b = N_y \lambda_b/N_b \lambda_y$

$$N_b/N_y = (\lambda_b/\lambda_y) \times (P_b/P_y) = (450 \text{ nm})/(564 \text{ nm}) \times (1/1.74)$$

$$\therefore N_b/N_y = 0.46$$

$$P_b = 100 \text{ mW} \times 1/(1+1.74) = 36.5 \text{ mW}$$

$$P_y = 100 \text{ mW} - 36.5 \text{ mW} = 63.5 \text{ mW}$$

$$\Phi_b = P_b \lambda_b/hc = (36.5 \times 10^{-3} \text{ W})(450 \times 10^{-9} \text{ m})/(6.63 \times 10^{-34} \text{ J s})(3 \times 10^8 \text{ m s}^{-1})$$

$$\therefore \Phi_b = 8.26 \times 10^{16} \text{ photons s}^{-1}$$

$$\Phi_y = P_y \lambda_y/hc = (63.5 \times 10^{-3} \text{ W})(564 \times 10^{-9} \text{ m})/(6.63 \times 10^{-34} \text{ J s})(3 \times 10^8 \text{ m s}^{-1})$$

$$\therefore \Phi_y = 1.8 \times 10^{17} \text{ photons s}^{-1}$$

3.3 Brightness of laser pointers The brightness of a light source depends not only on the radiation (optical) power emitted by the source but also on its wavelength because the human eye perceives each wavelength with a different efficiency. The visual “brightness” of a source as observed by an average daylight-adapted eye is proportional to the radiation power emitted, called the *radiant flux* Φ_e , and the efficiency of the eye to detect the spectrum of the emitted radiation. While the eye can see a red color source, it cannot see an infrared source and the brightness of the infrared source would be zero. The **luminous flux** Φ_v is a measure of *brightness*, in lumens (lm), and is defined by

$$\Phi_v = \Phi_e \times (683 \text{ lm W}^{-1}) \times \eta_{\text{eye}}$$

where Φ_e is the radiant flux or the radiation power emitted (in watts) and $\eta_{\text{eye}} = \eta_{\text{eye}}(\lambda)$ is the *relative luminous efficiency* (or the relative sensitivity) of an average light-adapted eye which depends on the

wavelength; η_{eye} is a Gaussian looking function with a peak of unity at 555 nm. (See Figure 3.46 for η_{eye} vs. λ) One lumen of luminous flux, or brightness, is obtained from a 1.58 mW light source emitting at a single wavelength of 555 nm (green). A typical 60 W incandescent lamp provides roughly 900 lm. When we buy a light bulb, we are buying lumens. Consider one 5 mW red 650 nm laser pointer, and another weaker 2 mW green 532 nm laser: $\eta_{\text{eye}}(650 \text{ nm}) = 0.11$ and $\eta_{\text{eye}}(532 \text{ nm}) = 0.86$. Find the luminous flux (brightness) of each laser pointer. Which is brighter? Calculate the number of photons emitted per unit time, the total *photon flux*, by each laser.

Solution

For 650 nm laser pointer,

$$\Phi_v(650 \text{ nm}) = \Phi_e \times (633 \text{ lm W}^{-1}) \times \eta_{\text{eye}}(650 \text{ nm}) = (5 \times 10^{-3} \text{ W}) \times (633 \text{ lm W}^{-1}) \times 0.11$$

$$\therefore \Phi_v(650 \text{ nm}) = 0.35 \text{ lm}$$

For 532 nm laser pointer,

$$\Phi_v(532 \text{ nm}) = \Phi_e \times (633 \text{ lm W}^{-1}) \times \eta_{\text{eye}}(532 \text{ nm}) = (2 \times 10^{-3} \text{ W}) \times (633 \text{ lm W}^{-1}) \times 0.86$$

$$\therefore \Phi_v(532 \text{ nm}) = 1.09 \text{ lm}$$

So the weaker 2 mW 532 nm laser pointer is brighter.

$$\Phi_{650 \text{ nm}} = P\lambda/hc = (5 \times 10^{-3} \text{ W})(650 \times 10^{-9} \text{ m}) / (6.63 \times 10^{-34} \text{ J s})(3 \times 10^8 \text{ m s}^{-1})$$

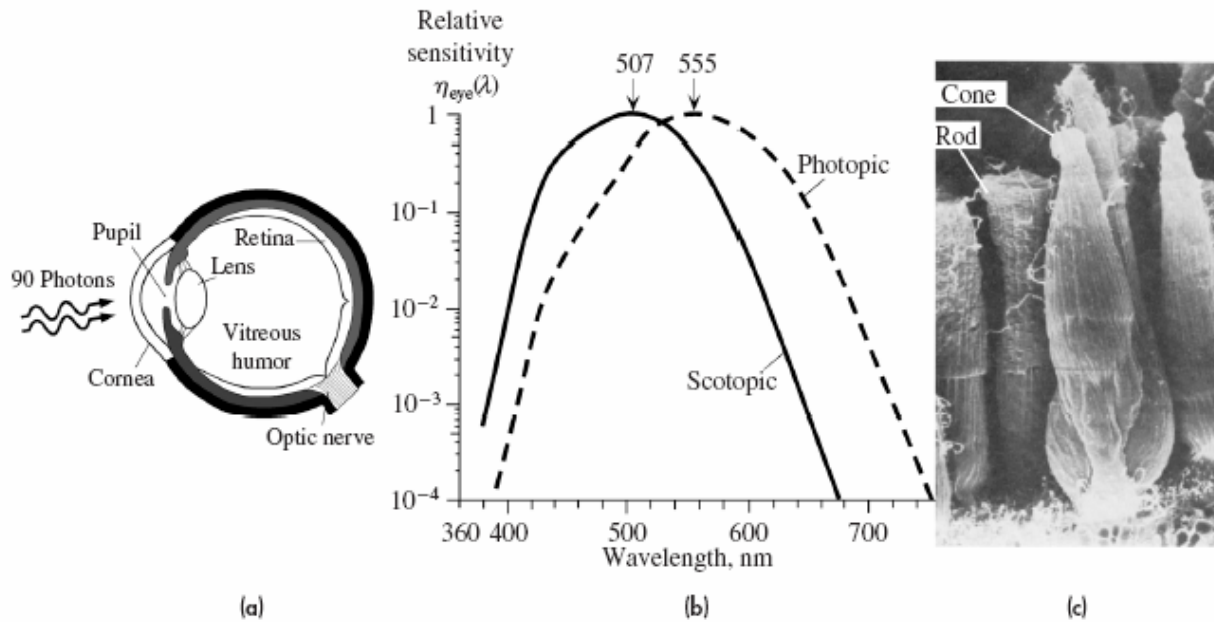
$$\therefore \Phi_{650 \text{ nm}} = 1.63 \times 10^{16} \text{ photons s}^{-1}$$

$$\Phi_{532 \text{ nm}} = P\lambda/hc = (2 \times 10^{-3} \text{ W})(532 \times 10^{-9} \text{ m}) / (6.63 \times 10^{-34} \text{ J s})(3 \times 10^8 \text{ m s}^{-1})$$

$$\therefore \Phi_{532 \text{ nm}} = 5.35 \times 10^{15} \text{ photons s}^{-1}$$

3.4 Human eye Photons passing through the pupil are focused by the lens onto the retina of the eye and are detected by two types of photosensitive cells, called *rods* and *cones*, as visualized in Figure 3.46. Rods are highly sensitive photoreceptors with a peak response at a wavelength of about 507 nm (green-cyan). They do not register color and are responsible for our vision under dimmed light conditions, termed **scotopic vision**. Cones are responsible for our color perception and daytime vision, called **photopic vision**. These three types of cone photoreceptors are sensitive to blue, green, and red at wavelengths, respectively, of 430 nm, 535 nm, and 575 nm. All three cones have an overall peak response at 555 nm (green), which represents the peak response of an average daylight-adapted eye or in our photopic vision.

- Calculate the photon energy (in eV) for the peak responsivity for each of the photoreceptors in the eye (one rod and three cones).
- Various experiments (the most well known being by Hecht et al., *J. Opt. Soc. America*, **38**, 196, 1942) have tested the threshold sensitivity of the dark-adapted eye and have estimated that visual

**Figure 3.46**

(a) The retina in the eye has photoreceptors that can sense the incident photons on them and hence provide necessary visual perception signals. It has been estimated that for minimum visual perception there must be roughly 90 photons falling on the cornea of the eye.

(b) The wavelength dependence of the relative efficiency $\eta_{eye}(\lambda)$ of the eye is different for daylight vision, or *photopic* vision (involves mainly cones), and for vision under dimmed light, or *scotopic* vision, which represents the dark-adapted eye, and involves rods.

(c) SEM photo of rods and cones in the retina.

| SOURCE: Dr. Frank Werblin, University of California, Berkeley.

perception requires a minimum of roughly 90 photons to be incident onto the cornea in front of the eye's pupil and within 1/10 second. Taking 90 incident photons every 100 ms as the threshold sensitivity, calculate the total photon flux (photons per second), total energy in eV (within 100 ms), and the optical power that is needed for threshold visual perception.

- c. Not all photons incident on the eye make it to the actual photoreceptors in the retina. It has been estimated that only 1 in 10 photons arriving at the eye's cornea actually make it to rod photoreceptors, due to various reflections and absorptions in the eye and other loss mechanisms. Thus, only nine photons make it to photoreceptors on the retina. It is estimated that the nine test photons fall randomly onto a circular area of about 0.0025 mm^2 . What is the estimated threshold intensity for visual perception? If there are $150,000 \text{ rods mm}^{-2}$ in this area of the eye, estimate the number of rods in this test spot. If there are a large number of rods, more than 100 in this spot, then it is likely that no single rod receives more than one photon since the nine photons arrive randomly. Thus, a rod must be able to *sense* a single photon, but it takes nine excited rods, somehow summed up by the visual system, to generate the visual sensation. Do you agree with the latter conclusion?
- d. It is estimated that at least 200,000 photons per second must be incident on the eye to generate a color sensation by exciting the cones. Assuming that this occurs at the peak sensitivity at 555 nm, and that as in part (b) only about 10 percent of the photons make it to the retina, estimate the threshold optical power stimulating the cones in the retina.

Solution

a. The photon energy for peak responsivity can be found as follows (firstly for rod):

$$E_{ph}(\text{rod}) = hc/\lambda = (6.63 \times 10^{-34} \text{ J s})(3.0 \times 10^8 \text{ m/s})/(507 \times 10^{-9} \text{ m})$$

$$\therefore E_{ph}(\text{rod}) = 3.92 \times 10^{-19} \text{ J or } 2.45 \text{ eV}$$

$$E_{ph}(\text{cones}) = hc/\lambda = (6.63 \times 10^{-34} \text{ J s})(3.0 \times 10^8 \text{ m/s})/(555 \times 10^{-9} \text{ m})$$

$$\therefore E_{ph}(\text{cones}) = 3.58 \times 10^{-19} \text{ J or } 2.24 \text{ eV}$$

b. The total photon flux,

$$\Phi = N/t = 90 \text{ photons} / 0.1 \text{ s} = \mathbf{900 \text{ photons s}^{-1}}$$

$$\text{Total energy, } E_{total} = N \times hc/\lambda = 90 \times (6.63 \times 10^{-34} \text{ J s})(3.0 \times 10^8 \text{ m/s})/(507 \times 10^{-9} \text{ m})$$

$$\therefore E_{total} = 3.53 \times 10^{-17} \text{ J or } 221 \text{ eV}$$

$$\text{The optical power, } P = E_{total}/t = 3.53 \times 10^{-17} \text{ J}/0.1 \text{ s} = \mathbf{3.53 \times 10^{-16} \text{ W}}$$

Author's Note: We are interested in the best response of the eye is dark adapted, which means at a wavelength of 507 nm, where the relative efficiency is unity.

c. Intensity,

$$I_{th} = \frac{N}{At} \times \frac{hc}{\lambda} = \frac{90}{(0.0025 \times 10^{-6} \text{ m}^2)(0.1 \text{ s})} \times \frac{(6.63 \times 10^{-34} \text{ Js})(3 \times 10^8 \text{ ms}^{-1})}{507 \times 10^{-9} \text{ m}}$$

$$\therefore I_{th} = \mathbf{1.41 \times 10^{-7} \text{ J s}^{-1} \text{ m}^{-2} \text{ or } 1.41 \times 10^{-7} \text{ W m}^{-2}}$$

$$\text{Total number of rods in test spot} = 150\,000 \text{ rods mm}^{-2} \times 0.0025 \text{ mm}^2 = \mathbf{375 \text{ rods}}$$

Author's Note: We are interested in the best response of the eye is dark adapted, which means at a wavelength of 507 nm, where the relative efficiency is unity.

d. Stimulating optical power can be calculated as

$$P_{st} = (0.1)(2 \times 10^5 \text{ s}^{-1}) \times (6.63 \times 10^{-34} \text{ J s})(3.0 \times 10^8 \text{ m/s})/(555 \times 10^{-9} \text{ m})$$

$$\therefore P_{st} = \mathbf{7.17 \times 10^{-15} \text{ J s}^{-1} \text{ or } 7.17 \times 10^{-15} \text{ W}}$$

3.5 X-ray photons In *chest radiology*, a patient's chest is exposed to X-rays, and the X-rays passing through the patient are recorded on a photographic film to generate an X-ray image of the chest for medical diagnosis. The average wavelength of X-rays in chest radiology is about 0.2 Å (0.02 nm). Numerous measurements indicate that the patient, on average, is exposed to total radiation energy per unit area of roughly 0.1 μJ cm⁻² for one chest X-ray image. Find the photon energy used in chest radiology, and the average number of photons incident on the patient per unit area (per cm²).

Solution

Photon energy,

$$E_{ph} = hc/\lambda = (6.63 \times 10^{-34} \text{ J s})(3.0 \times 10^8 \text{ m/s})/(0.02 \times 10^{-9} \text{ m})$$

$$\therefore E_{ph} = \mathbf{9.945 \times 10^{-15} \text{ J or } 62.15 \text{ keV}}$$

Photons per unit,

$$\Phi = P\lambda/hc = (0.1 \times 10^{-6} \text{ J cm}^{-2}) (0.02 \times 10^{-9} \text{ m}) / (6.63 \times 10^{-34} \text{ J s}) (3 \times 10^8 \text{ ms}^{-1})$$

$$\therefore \Phi = 1 \times 10^7 \text{ photons cm}^{-2}$$

***3.6 X-rays, exposure, and roentgens** X-rays are widely used in many applications such as medical imaging, security scans, X-ray diffraction studies of crystals, and for examining defects such as cracks in objects and structures. X-rays are highly energetic photons that can easily penetrate and pass **through** various objects. Different materials attenuate X-rays differently, so when X-rays are passed through an object, the emerging X-rays can be recorded on a photographic film, or be captured by a modern flat panel X-ray image detector, to generate an X-ray image of the interior of the object; this is called **radiography**. X-rays also cause ionization in a medium and hence are known as *ionization radiation*. The amount of exposure (denoted by X) to X-rays, ionizing radiation, is measured in terms of the ionizing effects of the X-ray photons. One **roentgen** (1 R) is defined as an X-ray exposure that ionizes 1 cm³ of air to generate 0.33 nC of charge in this volume at standard temperature and pressure (STP). When a body is exposed to X-rays, it will receive a certain amount of radiation energy per unit area, called **energy fluence** Ψ_E , that is, so many joules per cm², that depends on the exposure X . If X in roentgens is the exposure, then the energy fluence is given by

$$\Psi_E = \left[\frac{8.73 \times 10^{-6}}{\mu_{\text{en,air}} / \rho_{\text{air}}} \right] X \text{ J cm}^{-2} \quad [3.58]$$

where Ψ_E is in J cm⁻², and $\mu_{\text{en,air}}/\rho_{\text{air}}$ is the *mass energy absorption coefficient* of air in cm²g⁻¹ at the photon energy E_{ph} of interest; the $\mu_{\text{en,air}}/\rho_{\text{air}}$ values are listed in radiological tables. For example, for 1 R of exposure, $X = 1$, $E_{\text{ph}} = 20 \text{ keV}$, and $\mu_{\text{en,air}}/\rho_{\text{air}} = 0.539 \text{ cm}^2 \text{ g}^{-1}$. Equation 3.58 gives $\Psi_E = 1.62 \times 10^{-5} \text{ J cm}^{-2}$ incident on the object.

- In mammography (X-ray imaging of the breasts for breast cancer), the average photon energy is about 20 keV, and the X-ray mean exposure is 12 mR. At $E_{\text{ph}} = 20 \text{ keV}$, $\mu_{\text{en,air}}/\rho_{\text{air}} = 0.539 \text{ cm}^2 \text{ g}^{-1}$. Find the mean energy incident per unit area in $\mu\text{J cm}^{-2}$, and the mean number of X-ray photons incident per unit area (photons cm⁻²), called **photon fluence** Φ .
- In chest radiography, the average photon energy is about 60 keV, and the X-ray mean exposure is 300 μR . At $E_{\text{ph}} = 60 \text{ keV}$, $\mu_{\text{en,air}}/\rho_{\text{air}} = 0.0304 \text{ cm}^2 \text{ g}^{-1}$. Find the mean energy incident per unit area in $\mu\text{J cm}^{-2}$, and the mean number of X-ray photons incident per unit area.
- A modern flat panel *X-ray image detector* is a large area image sensor that has numerous arrays of tiny pixels (millions) all tiled together to make one large continuous image sensor. Each pixel is an independent X-ray detector and converts the X-rays it receives to an electrical signal. Each tiny detector is responsible for capturing a small pixel of the whole image. (Typically, the image resolution is determined by the detector pixel size.) Each pixel in a particular experimental chest radiology X-ray sensor is $150 \mu\text{m} \times 150 \mu\text{m}$. If the mean exposure is 300 μR , what is the number of photons received by each pixel detector? If each pixel is required to have at least 10 photons for an acceptable signal-to-noise ratio, what is the minimum exposure required in μR ?

Solution

a. Mammography:

Energy fluence,

$$\Psi_E = \left[\frac{8.73 \times 10^{-6}}{0.539} \right] 12 \times 10^{-3} \text{ J cm}^{-2}$$

$$\therefore \Psi_E = 1.94 \times 10^{-7} \text{ J cm}^{-2} \text{ or } 0.194 \text{ } \mu\text{J cm}^{-2}$$

Photon fluence,

$$\Phi = \Psi_E / E_{ph} = (1.94 \times 10^{-7} \text{ J cm}^{-2}) / (20 \times 10^3 \times 1.6 \times 10^{-19} \text{ J})$$

$$\therefore \Phi = 6.06 \times 10^7 \text{ photons cm}^{-2}$$

b. Chest radiography:

Energy fluence,

$$\Psi_E = \left[\frac{8.73 \times 10^{-6}}{0.0304} \right] 300 \times 10^{-6} \text{ J cm}^{-2}$$

$$\therefore \Psi_E = 8.62 \times 10^{-8} \text{ J cm}^{-2} \text{ or } 0.0862 \text{ } \mu\text{J cm}^{-2}$$

Photon fluence,

$$\Phi = \Psi_E / E_{ph} = (8.62 \times 10^{-8} \text{ J cm}^{-2}) / (60 \times 10^3 \times 1.6 \times 10^{-19} \text{ J})$$

$$\therefore \Phi = 8.99 \times 10^6 \text{ photons cm}^{-2}$$

c. Total number of photons received by each pixel,

$$= \Phi \times A = (8.99 \times 10^6 \text{ photons cm}^{-2}) (150 \times 10^{-4} \text{ cm} \times 150 \times 10^{-4} \text{ cm}).$$

$$= 2023 \text{ photons}$$

For 10 photons in each pixel,

$$\Phi_E = 10 \text{ photons} / (150 \times 10^{-4} \text{ cm} \times 150 \times 10^{-4} \text{ cm}) = 4.44 \times 10^4 \text{ photons cm}^{-2}$$

$$\begin{aligned} \Psi_E &= \Phi_E E_{ph} = (4.44 \times 10^4 \text{ photons cm}^{-2}) (60 \times 10^3 \times 1.6 \times 10^{-19} \text{ J/photon}) \\ &= 4.27 \times 10^{-10} \text{ J cm}^{-2} \end{aligned}$$

$$\text{Exposure, } X = (\Psi_E \times \mu_{en,air} / \rho_{air}) / 8.73 \times 10^{-6} \text{ R} = (4.27 \times 10^{-10}) (0.0304) / (8.73 \times 10^{-6})$$

$$\therefore X = 1.49 \times 10^{-6} \text{ R or } 1.49 \text{ } \mu\text{R}$$

3.7 Photoelectric effect A photoelectric experiment indicates that violet light of wavelength 420 nm is the longest wavelength radiation that can cause photoemission of electrons from a particular multialkali photocathode surface.

- What is the work function of the photocathode surface, in eV?
- If a UV radiation of wavelength 300 nm is incident upon the photocathode surface, what will be the maximum kinetic energy of the photoemitted electrons, in eV?
- Given that the UV light of wavelength 300 nm has an intensity of 20 mW/cm², if the emitted electrons are collected by applying a positive bias to the opposite electrode, what will be the photoelectric current density in mA cm⁻²?

Solution

a. We are given $\lambda_{\max} = 420 \text{ nm}$. The work function is then:

$$\Phi = h\nu_o = hc/\lambda_{\max} = (6.626 \times 10^{-34} \text{ J s})(3.0 \times 10^8 \text{ m s}^{-1})/(420 \times 10^{-9} \text{ m})$$

$$\therefore \Phi = 4.733 \times 10^{-19} \text{ J or } 2.96 \text{ eV}$$

b. Given $\lambda = 300 \text{ nm}$, the photon energy is then:

$$E_{\text{ph}} = h\nu = hc/\lambda = (6.626 \times 10^{-34} \text{ J s})(3.0 \times 10^8 \text{ m s}^{-1})/(300 \times 10^{-9} \text{ m})$$

$$\therefore E_{\text{ph}} = 6.626 \times 10^{-19} \text{ J} = 4.14 \text{ eV}$$

The kinetic energy KE of the emitted electron can then be found:

$$KE = \Phi - E_{\text{ph}} = 4.14 \text{ eV} - 2.96 \text{ eV} = 1.18 \text{ eV}$$

c. The photon flux Γ_{ph} is the number of photons arriving per unit time per unit area. If I_{light} is the light intensity (light energy flowing through unit area per unit time) then,

$$\Gamma_{\text{ph}} = \frac{I_{\text{light}}}{E_{\text{ph}}}$$

Suppose that each photon creates a single electron, then

$$J = \text{Charge flowing per unit area per unit time} = \text{Charge} \times \text{Photon Flux}$$

$$\therefore J = e\Gamma_{\text{ph}} = \frac{eI_{\text{light}}}{E_{\text{ph}}} = \frac{(1.602 \times 10^{-19} \text{ C})(200 \text{ W m}^{-2})}{(6.626 \times 10^{-19} \text{ J})} = 48.4 \text{ A m}^{-2} = 4.84 \text{ mA cm}^{-2}$$

3.8 Photoelectric effect and quantum efficiency Cesium metal is to be used as the photocathode material in a photoemissive electron tube because electrons are relatively easily removed from a cesium surface. The work function of a clean cesium surface is 1.9 eV.

- What is the longest wavelength of radiation which can result in photoemission?
- If blue radiation of wavelength 450 nm is incident onto the Cs photocathode, what will be the kinetic energy of the photoemitted electrons in eV? What should be the voltage required on the opposite electrode to extinguish the external photocurrent?
- Quantum efficiency (QE)** of a photocathode is defined by,

$$\text{Quantum efficiency} = \frac{\text{Number of photoemitted electrons}}{\text{Number of incident photons}} \quad [3.59]$$

QE is 100% if each incident photon ejects one electron. Suppose that blue light of wavelength 450 nm with an intensity of 30 mW cm^{-2} is incident on a Cs photocathode that is a circular disk of diameter 6 mm. If the emitted electrons are collected by applying a positive bias voltage to the anode, and the photocathode has a QE of 25%, what will be the photocurrent?

Solution

a. The longest wavelength will be $\Phi = h\nu_o = hc/\lambda_{\max}$

$$\lambda_{\max} = \frac{hc}{\Phi} = \frac{(6.626 \times 10^{-34} \text{ J s})(3 \times 10^8 \text{ m s}^{-1})}{(1.9 \times 1.602 \times 10^{-19} \text{ J})} = 653.9 \times 10^{-7} \text{ m} = 653.9 \text{ nm}$$

b. The energy of the photon at 450 nm is

$$E_{ph} = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34} \text{ J s})(3 \times 10^8 \text{ m s}^{-1})}{(450 \times 10^{-9} \text{ m})} = 4.417 \times 10^{-19} \text{ J} = 2.76 \text{ eV}$$

The excess energy over Φ goes to kinetic energy of the photoelectron. Thus the electron kinetic energy is

$$KE = E_{ph} - \Phi = (2.76 \text{ eV}) - (1.9 \text{ eV}) = 0.86 \text{ eV}$$

and the stopping voltage for this wavelength will be **-0.86 V**.

c. The number of photons arriving per unit area per unit time at the photocathode is

$$\Gamma_{ph} = \frac{I}{E_{ph}} = \frac{(30 \times 10^{-3} \times 10^4 \text{ J s}^{-1} \text{ m}^2)}{(4.417 \times 10^{-19} \text{ J})} = 6.792 \times 10^{20} \text{ s}^{-1} \text{ m}^{-2}$$

The current density is then simply

$$J = e\Gamma_{ph}QE = (1.602 \times 10^{-19} \text{ C})(6.792 \times 10^{20} \text{ s}^{-1} \text{ m}^{-2})(0.25) = 27.2 \text{ A m}^{-2}$$

The photoelectric current is then

$$I = AJ = \frac{\pi d^2}{4} J = \frac{\pi (6 \times 10^{-3} \text{ m})^2}{4} (27.2 \text{ A m}^{-2}) = 7.691 \times 10^{-4} \text{ A} = 0.769 \text{ mA}$$

3.9 Photoelectric effect A multi-alkali metal alloy is to be used as the photocathode material in a photoemissive electron tube. The work function of the metal is 1.6 eV, and the photocathode area is 0.5 cm². Suppose that blue light of wavelength 420 nm with an intensity of 50 mW cm⁻² is incident on the photo- cathode.

- If the photoemitted electrons are collected by applying a positive bias to the anode, what will be the photoelectric current density assuming that the quantum efficiency η is 15 percent? *Quantum efficiency* as a percentage is the number of photoemitted electrons per 100 absorbed photons and is defined in Equation 3.59. What is the kinetic energy of a photoemitted electron at 420 nm?
- What should be the voltage and its polarity to extinguish the current?
- What should be the intensity of an incident red light beam of wavelength 600 nm that would give the same photocurrent if the quantum efficiency is 5 percent at this wavelength? (Normally the quantum efficiency depends on the wavelength.)

Solution

a. The number of photons arriving per unit area per unit time at the photocathode is

$$\Gamma_{ph} = \frac{I\lambda}{hc} = \frac{(50 \times 10^{-3} \text{ J s}^{-1} \text{ cm}^2)(420 \times 10^{-9} \text{ m})}{(6.63 \times 10^{-34} \text{ J s})(3 \times 10^8 \text{ m s}^{-1})} = 1.06 \times 10^{17} \text{ s}^{-1} \text{ cm}^{-2}$$

The photoelectric current density,

$$J = e\Gamma_{\text{ph}}QE = (1.6 \times 10^{-19} \text{ C})(1.06 \times 10^{21} \text{ s}^{-1} \text{ m}^{-2})(0.15) = 2.54 \times 10^{-3} \text{ A cm}^{-2}$$

The excess energy of the photon over Φ goes to kinetic energy of the photoelectron. Thus the electron kinetic energy is

$$KE = E_{\text{ph}} - \Phi = \frac{hc}{e\lambda} - \Phi = \left(\frac{6.63 \times 10^{-34} \text{ Js} \times 3 \times 10^8 \text{ ms}^{-1}}{1.6 \times 10^{-19} \text{ J} \times 420 \times 10^{-9} \text{ m}} \right) - (1.6 \text{ eV}) = 1.36 \text{ eV}$$

b. The voltage to extinguish the current is - **1.36 V**.

c.
$$J = e\Gamma_{\text{ph}}QE = e \frac{I\lambda}{hc} QE$$

or
$$I = \frac{J \times hc}{e\lambda \times QE} = \frac{2.54 \times 10^{-3} \text{ Acm}^{-2} \times 6.63 \times 10^{-34} \text{ Js} \times 3 \times 10^8 \text{ ms}^{-1}}{1.6 \times 10^{-19} \text{ C} \times 600 \times 10^{-9} \text{ m} \times 0.05} = 0.105 \text{ Js}^{-1} \text{ cm}^{-2}$$

or
$$I = 0.105 \text{ Wcm}^{-2}.$$

3.10 Planck's law and photon energy distribution of radiation

NOTE: First printing has a typo in Equation 3.60. The correct version is reproduced below.

Planck's law, stated in Equation 3.9, provides the spectral distribution of the black body radiation intensity in terms of wavelength through I_λ , intensity per unit wavelength. Suppose that we wish to find the distribution in terms of frequency ν or photon energy $h\nu$. Frequency $\nu = c/\lambda$ and the wavelength range λ to $\lambda + d\lambda$ corresponds to a frequency range ν to $\nu + d\nu$. ($d\lambda$ and $d\nu$ have opposite signs since ν increases as λ decreases.) The intensity $I_\lambda d\lambda$ in λ to $\lambda + d\lambda$ must be the same as the intensity in ν to $\nu + d\nu$, which we can write as $I_\nu d\nu$ where I_ν is the radiation intensity per unit frequency. Thus,

$$I_\nu = I_\lambda \left| \frac{d\lambda}{d\nu} \right|$$

The magnitude sign is needed because $\lambda = c/\nu$ results in a negative $d\lambda/d\nu$, and I_ν must be positive by definition. We can simply substitute $\lambda = c/\nu$ for λ in I_λ and obtain I_λ as a function of ν , and then find $|d\lambda/d\nu|$ to find I_ν from the preceding expression.

a. Show that

$$I_\nu = \frac{2\pi(h\nu)^3}{c^2 h^2 [\exp(h\nu/kT) - 1]} \quad 3.60$$

Equation 3.60 is written to highlight that it is a function of the *photon energy* $h\nu$, which is in joules in Equation 3.60 but can be converted to eV by dividing by $1.6 \times 10^{-19} \text{ J eV}^{-1}$.

b. If we integrate I_ν over all photon energies (numerically on a calculator or a computer from 0 to say 6 eV), we would obtain the total intensity at a temperature T . Find the total intensity I_T emitted at $T = 2600 \text{ K}$ (a typical incandescent light bulb filament temperature) and at 6000 K

(roughly representing the sun's spectrum). Plot $y = I_\nu/I_T$ versus the photon energy in eV. What are the photon energies for the peaks in the distributions? Calculate the corresponding wavelength for each using $\lambda = c/\nu$ and then compare these wavelengths with those predicted by Wien's law, $\lambda_{\max}T \approx 2.89 \times 10^{-3} \text{ m K}$.

Solution

a. $\lambda = c/\nu$

or $|d\lambda/d\nu| = c/\nu^2$

$$I_\nu = I_\lambda \left| \frac{d\lambda}{d\nu} \right| = \frac{2\pi hc^2}{\lambda^5 \left[\exp\left(\frac{hc}{\lambda kT}\right) - 1 \right]} \left| \frac{d\lambda}{d\nu} \right| = \frac{2\pi hc^2}{\left(\frac{c}{\nu}\right)^5 \left[\exp\left(\frac{h\nu}{kT}\right) - 1 \right]} \times \frac{c}{\nu^2}$$

or
$$I_\nu = \frac{2\pi(h\nu)^3}{c^2 h^2 \left[\exp\left(\frac{h\nu}{kT}\right) - 1 \right]}$$

b. By integrating numerically using any math software, the total intensity

$$I_\lambda(2600 \text{ K}) = 1.07 \times 10^{-7} \text{ J m}^{-2} = 1.07 \times 10^{-8} \text{ W s}^{-1} \text{ m}^{-2}$$

and

$$I_\lambda(6000 \text{ K}) = 3.03 \times 10^{-7} \text{ J m}^{-2} = 3.03 \times 10^{-7} \text{ W s}^{-1} \text{ m}^{-2}$$

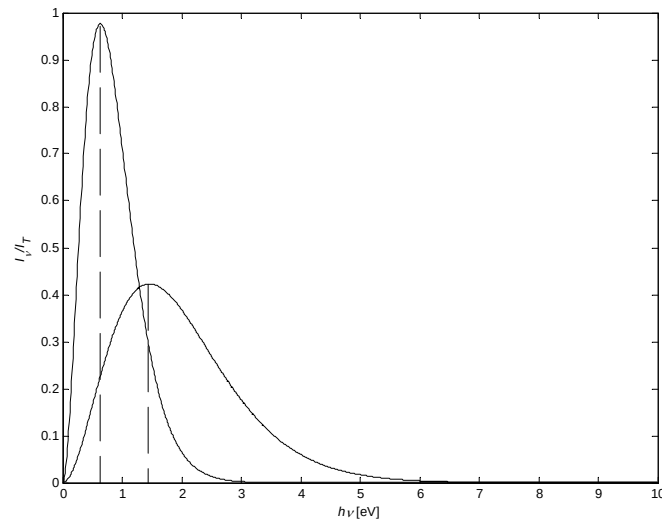


Figure 3Q10-1: Plot of I_ν/I_T versus photon energy.

At 2600 K:

From the Fig. 3Q10-1, at the peak in intensity, we get

$$h\nu = 0.63 \text{ eV or } hc/\lambda = 0.63 \text{ eV}$$

or $\lambda_{2600} = (4.135 \times 10^{-15} \text{ eV s})(3 \times 10^8 \text{ ms}^{-1})/(0.63 \text{ eV}) = \mathbf{1.97 \times 10^{-6} \text{ m or } 1.97 \mu\text{m}}$

At 6000 K:

From the Fig. 3Q10-1, at the peak in intensity, we get

$$h\nu = 1.46 \text{ eV or } hc/\lambda = 1.46 \text{ eV}$$

or $\lambda_{6000} = (4.135 \times 10^{-15} \text{ eV s})(3 \times 10^8 \text{ ms}^{-1}) / (0.63 \text{ eV}) = \mathbf{8.5 \times 10^{-7} \text{ m or 850 nm}}$

Using Wien's law,

$$\lambda_{\max} T = 2.89 \times 10^{-3} \text{ m K}$$

$$\lambda_{\max} = (2.89 \times 10^{-3} \text{ m K} / 2600 \text{ K}) = 1.11 \times 10^{-6} \text{ m or } \mathbf{1.11 \mu\text{m}} \text{ at 2600 K}$$

and $\lambda_{\max} = (2.89 \times 10^{-3} \text{ m K} / 6000 \text{ K}) = 4.8 \times 10^{-7} \text{ m or } \mathbf{480 \text{ nm}}$ at 6000 K

Temperature (T)	$\lambda_{\max}$ (Planck's law)	$\lambda_{\max}$ (Wien's law)
2600	1.97 μm	1.11 μm
6000	850 nm	480 nm

Table 3Q10-1: Comparison of wavelengths obtained using Planck's law and Wien's law.

3.11 Wien's law The maximum in the intensity distribution of black body radiation depends on the temperature. Substitute $x = hc/(\lambda kT)$ in Planck's law and plot I_λ versus x and find $\lambda_{\max}$ which corresponds to the peak of the distribution, and hence derive Wien's law. Find the peak intensity wavelength $\lambda_{\max}$ for a 40 W light bulb given that its filament operates at roughly 2400 °C.

Solution

$$x = hc / \lambda kT$$

$$I_\lambda = \frac{2\pi hc^2}{\lambda^5 \left[\exp\left(\frac{hc}{\lambda kT}\right) - 1 \right]} = \left(\frac{kT}{hc}\right)^5 2\pi hc^2 \frac{x^5}{\exp(x) - 1} = C \frac{x^5}{\exp(x) - 1}$$

Here C is a constant and it does not change the shape of the I_λ versus x plot. Taking $C = 1$, the plot of I_λ versus x is shown in the figure below.

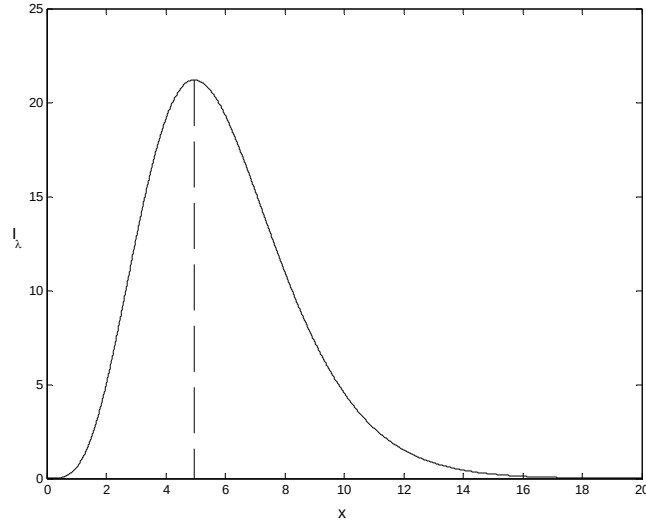


Figure 3Q11-1: Plot of I_λ versus x .

For peak intensity, $x = 5$

$$\therefore hc/\lambda_{\max}kT = 5$$

$$\therefore (6.626 \times 10^{-34} \text{ Js})(3 \times 10^8 \text{ ms}^{-1}) / (1.38 \times 10^{-23} \text{ JK}^{-1})(\lambda_{\max} T) = 5$$

$$\therefore \lambda_{\max} T = 2.88 \times 10^{-3} \text{ m K}$$

$$\lambda_{\max} = 2.88 \times 10^{-3} \text{ m K} / T = 2.88 \times 10^{-3} \text{ m K} / 2673 \text{ K} = \mathbf{1.07 \times 10^{-6} \text{ m} = 1.07 \mu\text{m}}$$

3.12 Diffraction by X-rays and an electron beam Diffraction studies on a polycrystalline Al sample using X-rays gives the smallest diffraction angle (2θ) of 29.5° corresponding to diffraction from the (111) planes. The lattice parameter a of Al (FCC), is 0.405 nm. If we wish to obtain the same diffraction pattern (same angle) using an electron beam, what should be the voltage needed to accelerate the electron beam? Note that the interplanar separation d for planes (h, k, l) and the lattice parameter a for cubic crystals are related by

$$d = a / [h^2 + k^2 + l^2]^{1/2}$$

Solution

The plane indices are given as $h = 1$, $k = 1$ and $l = 1$, and the lattice parameter is given as $a = 0.405$ nm. Therefore,

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}} = \frac{0.405 \text{ nm}}{\sqrt{1^2 + 1^2 + 1^2}} = 0.2338 \text{ nm}$$

From the Bragg condition with $\theta = 29.5^\circ / 2 = 14.75^\circ$ we have,

$$\lambda = 2d \sin(\theta) = 2(0.2338 \text{ nm}) \sin(14.75^\circ) = 0.1191 \text{ nm}$$

The voltage V required to accelerate the electron is (see Example 3.4),

$$V = \left(\frac{1.226 \text{ nm}}{\lambda} \right)^2 = \left(\frac{1.226 \text{ nm}}{0.1191 \text{ nm}} \right)^2 = \mathbf{106 \text{ V}}$$

3.13 Heisenberg's uncertainty principle Show that if the uncertainty in the position of a particle is on the order of its de Broglie wavelength, then the uncertainty in its momentum is about the same as the momentum value itself.

Solution

The de Broglie wavelength is

$$\lambda = \frac{h}{p}$$

where p is the momentum. From Heisenberg's uncertainty principle we have,

$$\Delta x \Delta p \approx \hbar$$

If we take the uncertainty in the position to be of the order of the wavelength, $\Delta x \sim \lambda$, then

$$\Delta p \approx \frac{\hbar}{\Delta x} \approx \left(\frac{1}{2\pi} \right) \frac{h}{\lambda} = \left(\frac{1}{2\pi} \right) p \sim p$$

so that the uncertainty in the momentum will be of the same order as the momentum itself.

3.14 Heisenberg's uncertainty principle An excited electron in an Na atom emits radiation at a wavelength 589 nm and returns to the ground state. If the mean time for the transition is about 20 ns, calculate the inherent width in the emission line. What is the length of the photon emitted?

Solution

The emitted wavelength is $\lambda = 589 \text{ nm}$ and the transition time is $\Delta t = 20 \text{ ns}$. This time corresponds to an uncertainty in the energy that is ΔE . Using the uncertainty principle we can calculate ΔE ,

$$\Delta t \Delta E \approx \hbar$$

$$\therefore \Delta E \approx \frac{\hbar}{\Delta t} = \frac{\frac{1}{2\pi} (6.626 \times 10^{-34} \text{ J s})}{20 \times 10^{-9} \text{ s}} = 5.273 \times 10^{-27} \text{ J or } 3.29 \times 10^{-8} \text{ eV}$$

The corresponding uncertainty in the emitted frequency $\Delta \nu$ is

$$\Delta \nu = \frac{\Delta E}{h} = \frac{(5.273 \times 10^{-27} \text{ J})}{(6.626 \times 10^{-34} \text{ J s})} = 7.958 \times 10^6 \text{ s}^{-1}$$

To find the corresponding spread in wavelength and hence the line width $\Delta \lambda$, we can differentiate $\lambda = c/\nu$.

$$\frac{d\lambda}{d\nu} = -\frac{c}{\nu^2} = -\frac{\lambda^2}{c}$$

Thus,
$$\Delta\lambda = \frac{\lambda^2}{c} \Delta\nu = \frac{(589 \times 10^{-9} \text{ m})^2}{3.0 \times 10^8 \text{ m/s}} (7.958 \times 10^6 \text{ s}^{-1}) = 9.20 \times 10^{-15} \text{ m or } 9.20 \text{ fm}$$

The length of the photon l is

$$l = (\text{light velocity}) \times (\text{emission duration}) = (3.0 \times 10^8 \text{ m/s})(20 \times 10^{-9} \text{ s}) = 6.0 \text{ m}$$

3.15 Tunneling

- a. Consider the phenomenon of tunneling through a potential energy barrier of height V_o and width a , as shown in Figure 3.16. What is the probability that the electron will be reflected? Given the transmission coefficient T , can you find the reflection coefficient R ? What happens to R as a or V_o or both become very large?
- b. For a wide barrier ($a\alpha \gg 1$), show that T_o can at most be 4 and that $T_o = 4$ when $E = \frac{1}{2} V_o$.

Solution

a. The relative reflection probability or reflection coefficient R is given as the ratio of the square of the amplitude of the reflected wave to that of the incident wave, which is:

$$R = \frac{A_2^2}{A_1^2}$$

Also, R can be found from the transmission coefficient T by the equation $R = 1 - T$, as stated in Equation 3.33. From Equation 3.29, T is given as:

$$T = \frac{1}{1 + D(\sinh[a\alpha])^2}$$

where a is the width of the potential energy barrier, α is the rate of decay, and D is given by:

$$D = \frac{V_o^2}{4E(V_o - E)}$$

To determine the behavior of R as a or V_o or both become very large, we can use the equation $R = 1 - T$ to express R in terms of a and D (remember D is a function of V_o).

$$R = 1 - T = 1 - \frac{1}{1 + D(\sinh[a\alpha])^2}$$

$\therefore$
$$R = \frac{D(\sinh[a\alpha])^2}{1 + D(\sinh[a\alpha])^2}$$

We know that $\sinh(\infty) = \infty$, and also that $1 / \infty = 0$. Therefore, as V_o becomes large, so does D , which leads to $T = 0$ and $R = 1$, meaning total reflection occurs. If a becomes large then $\sinh(\infty) = \infty$ and $T = 0$, making $R = 1$ for total reflection.

b. We need to find the maximum value of T_o . Since T_o depends on the energy E , we can differentiate it with respect to E , set the result to 0 and isolate E .

$$T_o = \frac{16E(V_o - E)}{V_o^2} \quad (\text{See Equation 3.32})$$

$$\therefore \frac{dT_o}{dE} = 16 \left(\frac{-2E + V_o}{V_o^2} \right) = 0$$

$$\therefore E = \frac{1}{2} V_o$$

Thus T_o is maximum when $E = V_o / 2$. If this expression for energy is substituted back into the equation for T_o to find its maximum value (T_o'):

$$T_o' = \frac{16E(V_o - E)}{V_o^2} = \frac{16 \left(\frac{1}{2} V_o \right) \left(V_o - \frac{1}{2} V_o \right)}{V_o^2} = \frac{16V_o^2}{4V_o^2} = 4$$

3.16 Electron impact excitation

- A projectile electron of kinetic energy 12.2 eV collides with a hydrogen atom in a gas discharge tube. Find the n -th energy level to which the electron in the hydrogen atom gets excited.
- Calculate the possible wavelengths of radiation (in nm) that will be emitted from the excited H atom in part (a) as the electron returns to its ground state. Which one of these wavelengths will be in the visible spectrum?
- In neon street lighting tubes, gaseous discharge in the Ne tube involves electrons accelerated by the electric field impacting Ne atoms and exciting some of them to the $2p^5 3p^1$ states, as shown in Figure 3.42. What is the wavelength of emission? Can the Ne atom fall from the $2p^5 3p^1$ state to the ground state by spontaneous emission?

Solution

a. The energy of the electron in the hydrogenic atom at the n th level is given by the equation:

$$E_n = \frac{-13.6 \text{ eV}}{n^2} Z^2$$

where Z is the atomic number, in this case 1 for hydrogen. Thus,

$$E_n = -\frac{13.6 \text{ eV}}{n^2}$$

The projectile electron, at best, will pass all its KE to the electron in the H-atom and excite it to the n -th level,

$$KE = E_n - E_{\text{ground}}$$

The ground energy of hydrogen's electron is -13.6 eV. Thus,

$$KE = \frac{-13.6 \text{ eV}}{n^2} + 13.6 \text{ eV}$$

$$\therefore n = \sqrt{\frac{13.6 \text{ eV}}{13.6 \text{ eV} - KE}}$$

Substituting in the given value for KE of 12.2 eV:

$$n = \sqrt{\frac{13.6 \text{ eV}}{13.6 \text{ eV} - 12.2 \text{ eV}}} = \mathbf{3.12}$$

Therefore the electron is at the 3rd energy level in the hydrogen atom (since n must be an integer). The amount of energy absorbed by hydrogen's electron in the excitation is:

$$\Delta E = (-13.6 \text{ eV}) / 3^2 - E_{\text{ground}} = 12.09 \text{ eV}.$$

b. The electron can return down to the ground level either from $n = 3$ to $n = 1$, or from $n = 3$ to $n = 2$ then to $n = 1$. For the first transition ($n = 3$ to $n = 1$):

$$\Delta E_{31} = \frac{-13.6 \text{ eV}}{3^2} - \left(\frac{-13.6 \text{ eV}}{1^2} \right) = 12.09 \text{ eV}$$

This value can be put into the following equation to find the emitted wavelength:

$$\Delta E_{31} = h\nu = h \frac{c}{\lambda_{31}}$$

$$\therefore \lambda_{31} = h \frac{c}{\Delta E_{31}} = (6.626 \times 10^{-34} \text{ J s}) \frac{3.0 \times 10^8 \text{ m/s}}{(12.09 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})}$$

$$\therefore \lambda_{31} = \mathbf{1.03 \times 10^{-7} \text{ m} \text{ or } 103 \text{ nm}}$$

This wavelength is not in the visible spectrum. For the $n = 3$ to $n = 2$ to $n = 1$ transition, two different wavelengths of light will be released, one in the first transition from 3 to 2 and the other in the second.

$$\Delta E_{32} = \frac{-13.6 \text{ eV}}{3^2} - \left(\frac{-13.6 \text{ eV}}{2^2} \right) = 1.889 \text{ eV}$$

$$\therefore \lambda_{32} = h \frac{c}{\Delta E_{32}} = (6.626 \times 10^{-34} \text{ J s}) \frac{3.0 \times 10^8 \text{ m/s}}{(1.889 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})}$$

$$\therefore \lambda_{32} = \mathbf{6.57 \times 10^{-7} \text{ m} \text{ or } 657 \text{ nm}}$$

This wavelength is visible (red). For the 2 to 1 transition:

$$\Delta E_{21} = \frac{-13.6 \text{ eV}}{2^2} - \left(\frac{-13.6 \text{ eV}}{1^2} \right) = 10.2 \text{ eV}$$

$$\therefore \lambda_{21} = h \frac{c}{\Delta E_{21}} = (6.626 \times 10^{-34} \text{ J s}) \frac{3.0 \times 10^8 \text{ m/s}}{(10.2 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})}$$

$$\therefore \lambda_{21} = 1.22 \times 10^{-7} \text{ m or } 122 \text{ nm}$$

This wavelength is not visible.

c. The wavelength is given in Figure 3.42, as 600 nm approximately which is in the red. The Ne atom must change its orbital quantum number ℓ in a transition involving a photon (emission or absorption). The transition ($2p^5 3p^1$) to ($2p^6$) does not change ℓ and hence it is NOT allowed in a photon-emission transition.

3.17 Line spectra of hydrogenic atoms Spectra of hydrogen-like atoms are classified in terms of electron transitions to a common lower energy level.

- All transitions from energy levels $n = 2, 3, \dots$ to $n = 1$ (the *K* shell) are labeled *K* lines and constitute the **Lyman series**. The spectral line corresponding to the smallest energy difference ($n = 2$ to $n = 1$) is labeled the K_α line, next is labeled K_β , and so on. The transition from $n = \infty$ to $n = 1$ has the largest energy difference and defines the greatest photon energy (shortest wavelength) in the *K* series; hence it is called the absorption edge K_{ae} . What is the range of wavelengths for the *K* lines? What is K_{ae} ? Where are these lines with respect to the visible spectrum?
- All transitions from energy levels $n = 3, 4, \dots$ to $n = 2$ (*L* shell) are labeled *L* lines and constitute the **Balmer series**. What is the range of wavelengths for the *L* lines (*i.e.*, L_α and L_{ae})? Are these in the visible range?
- All transitions from energy levels $n = 4, 5, \dots$ to $n = 3$ (*M* shell) are labeled *M* lines and constitute the **Paschen series**. What is the range of wavelengths for the *M* lines? Are these in the visible range?
- How would you expect the spectral lines to depend on the atomic number Z ?

Solution

Consider the expression for the energy change for a transition from $n = n_2$ to $n = n_1$,

$$\Delta E = -Z^2 E_I \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right) \quad (1)$$

where $E_I = 13.6 \text{ eV}$ is the ionization energy from the ground state ($n = 1$) and Z is the atomic number; for hydrogen $Z = 1$.

The absorption wavelength is given by $\lambda = hc / \Delta E$.

- a.** For the Lyman series (*K*-shell), $n_1 = 1$. For K_α , $n_2 = 2$, which gives $\Delta E = 10.2 \text{ eV}$ from Eqn. (1) and

$$\lambda(K_\alpha) = \frac{hc}{\Delta E} = \frac{(6.626 \times 10^{-34} \text{ J s})(3.0 \times 10^8 \text{ m s}^{-1})}{(10.2 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})} = 122 \text{ nm}$$

For K_{ae} , $n_2 = \infty$ and $\Delta E = 13.6 \text{ eV}$ and

$$\lambda(K_{\alpha e}) = \frac{hc}{\Delta E} = \frac{(6.626 \times 10^{-34} \text{ J s})(3.0 \times 10^8 \text{ m s}^{-1})}{(13.6 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})} = \mathbf{91.2 \text{ nm}}$$

The wavelengths range from **91.2 nm to 122 nm; much shorter than the visible spectrum.**

b. For the Balmer series (L -shell), $n_1 = 2$. For L_{α} , $n_2 = 3$, which gives $\Delta E = 1.889 \text{ eV}$ and

$$\lambda(L_{\alpha}) = \frac{hc}{\Delta E} = \frac{(6.626 \times 10^{-34} \text{ J s})(3.0 \times 10^8 \text{ m s}^{-1})}{(1.889 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})} = \mathbf{657 \text{ nm}}$$

For $L_{\alpha e}$, $n_2 = \infty$ and $\Delta E = 3.4 \text{ eV}$ and

$$\lambda(L_{\alpha e}) = \frac{hc}{\Delta E} = \frac{(6.626 \times 10^{-34} \text{ J s})(3.0 \times 10^8 \text{ m s}^{-1})}{(3.4 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})} = \mathbf{365 \text{ nm}}$$

The wavelengths range from **365 nm to 657 nm. Part of this is in the visible spectrum (blue and green).**

c. For the Paschen series (M -shell), $n_1 = 3$. For M_{α} , $n_2 = 4$, which gives $\Delta E = 0.6611 \text{ eV}$ and

$$\lambda(M_{\alpha}) = \frac{hc}{\Delta E} = \frac{(6.626 \times 10^{-34} \text{ J s})(3.0 \times 10^8 \text{ m s}^{-1})}{(0.6611 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})} = \mathbf{1877 \text{ nm}}$$

For $M_{\alpha e}$, $n_2 = \infty$ and $\Delta E = 1.511 \text{ eV}$ and

$$\lambda(M_{\alpha e}) = \frac{hc}{\Delta E} = \frac{(6.626 \times 10^{-34} \text{ J s})(3.0 \times 10^8 \text{ m s}^{-1})}{(1.511 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})} = \mathbf{821 \text{ nm}}$$

The wavelengths range from **821 nm to 1877 nm, which is in the infrared region.**

d. The transition energy depends on Z^2 , and therefore the emitted photon wavelength of the spectral lines depends inversely on Z^2 .

3.18 Ionization energy and effective Z

- Consider the singly ionized Li ion, Li^+ , which has lost its $2s$ electron. If the energy required to ionize one of the $1s$ electrons in Li^+ is 18.9 eV , calculate the effective nuclear charge seen by a $1s$ electron, that is, $Z_{\text{effective}}$ in the hydrogenic atom ionization energy expression in Equation 3.45; $E_{l,n} = (Z_{\text{effective}}/n)^2 (13.6 \text{ eV})$.
- The B atom has a total of five electrons, two in the $1s$ orbital, two in the $2s$, and one in the $2p$. The experimental ionization energy of B is 8.30 eV . Calculate $Z_{\text{effective}}$.
- The experimental ionization energy of Na is 3.49 eV . Calculate the effective nuclear charge seen by the $3s$ valence electron.
- The chemical tables typically list the first, second, and third ionization energies E_1 , E_2 , E_3 , respectively, and so on. Consider Al. E_1 represents the energy required to remove the first electron from neutral Al; E_2 , the second electron from Al^+ ; E_3 , the third electron from Al^{2+} to generate Al^{3+} . For Al, experimentally, $E_1 = 6.0 \text{ eV}$, $E_2 = 18.8 \text{ eV}$, and $E_3 = 28.4 \text{ eV}$. For each case find the $Z_{\text{effective}}$ seen by the electron that is removed

Solution

a. For 1s electron, $n = 1$, hence from Eq. 3.45

$$18.9 \text{ eV} = Z_{\text{effective}}^2 (13.6 \text{ eV})/(1)^2$$

or $Z_{\text{effective}} = 1.18$

b. For 2p electron, $n = 2$, hence from Eq. 3.45

$$8.30 \text{ eV} = Z_{\text{effective}}^2 (13.6 \text{ eV})/(2)^2$$

or $Z_{\text{effective}} = 1.56$

c. Na has 11 electron with two electrons in 1s orbital, two electrons in 2s orbital, six electrons in 2p orbital and one electron in 3s orbital. For 3s electron, $n = 3$, hence from Eq. 3.45

$$3.49 \text{ eV} = Z_{\text{effective}}^2 (13.6 \text{ eV})/(3)^2$$

or $Z_{\text{effective}} = 1.52$

d. Al has 13 electrons with two electrons in 1s orbital, two electrons in 2s orbital, six electrons in 2p orbital and two electrons in 3s orbital and one electron in the 3p orbital.

For first ionization energy E_1 that means removal of the 3p electron, $n = 3$

$$\text{So, } E_1 = Z_{\text{effective}}^2 (13.6 \text{ eV})/(3)^2 \text{ or } 6.0 \text{ eV} = Z_{\text{effective}}^2 (13.6 \text{ eV})/(3)^2$$

or $Z_{\text{effective}} = 1.99$

For second ionization energy E_2 that means removal of one of the 3s electron, $n = 3$

$$\text{So, } E_2 = Z_{\text{effective}}^2 (13.6 \text{ eV})/(3)^2 \text{ or } 18.8 \text{ eV} = Z_{\text{effective}}^2 (13.6 \text{ eV})/(3)^2$$

or $Z_{\text{effective}} = 3.53$

For third ionization energy E_3 that means removal of the second 3s electron, $n = 3$

$$\text{So, } E_2 = Z_{\text{effective}}^2 (13.6 \text{ eV})/(3)^2 \text{ or } 28.4 \text{ eV} = Z_{\text{effective}}^2 (13.6 \text{ eV})/(3)^2$$

or $Z_{\text{effective}} = 4.34$

3.19 Atomic and ionic radii The maximum in the radial probability distribution of an electron in a hydrogen-like atom is given by Equation 3.44, that is, $r_{\text{max}} = (n^2 a_0)/Z$, for $l = n - 1$. The average distance $\bar{r}$ of an electron from the nucleus can be calculated by using the definition of an average and the probability distribution function $P_{n,l}(r)$, that is,

$$\bar{r} = \int_0^\infty r P_{n,l}(r) dr = \frac{a_0 n^2}{Z} \left[\frac{3}{2} - \frac{l(l+1)}{2n^2} \right]$$

in which the right-hand side represents the result of the integration (which has been done by physicists). Calculate r_{max} and $\bar{r}$ for the 2p valence electron in the B atom. Which value is closer to the radius of the B atom, 0.085 nm, given in the Period Table? Consider only the outermost electrons, and calculate r_{average} for Li, Li^+ , Be^{2+} , and B, and compare with the experimental values of the atomic or ionic sizes: 0.15 nm for Li, 0.070 nm for Li^+ , 0.035 nm for Be^{2+} , and 0.085 nm for B.

Solution**B:**

For **hydrogen-like** B atom

$Z = 1$, $n = 2$, hence from Eq. 3.44,

$$r_{\max} = (2^2)(0.0529 \text{ nm})/1 = \mathbf{0.21 \text{ nm}}$$

For average distance,

$Z = 5$, $n = 2$ and $l = 1$; hence

$$\bar{r} = \frac{(0.0529 \text{ nm})(2^2)}{5} \left[\frac{3}{2} - \frac{1(2)}{2(2^2)} \right] = 0.0529 \text{ nm}$$

So average distance is close to the radius (0.085 nm) of the B atom

Li:

$Z = 3$, $n = 2$ and $l = 1$; hence

$$\bar{r} = \frac{(0.0529 \text{ nm})(2^2)}{3} \left[\frac{3}{2} - \frac{1(2)}{2(2^2)} \right] = 0.088 \text{ nm}$$

Li⁺:

$Z = 3$, $n = 1$ and $l = 0$; hence

$$\bar{r} = \frac{(0.0529 \text{ nm})(1^2)}{3} \left[\frac{3}{2} - \frac{0(1)}{2(2^2)} \right] = 0.026 \text{ nm}$$

Be²⁺:

$Z = 4$, $n = 1$ and $l = 0$; hence

$$\bar{r} = \frac{(0.0529 \text{ nm})(1^2)}{4} \left[\frac{3}{2} - \frac{0(1)}{2(2^2)} \right] = 0.0198 \text{ nm}$$

Atom/Ion	r_{average} (theoretical)	Experimental ionic sizes
Li	0.088 nm	0.15 nm
Li ⁺	0.026 nm	0.070 nm
Be ²⁺	0.0198 nm	0.035 nm
B	0.0529 nm	0.085 nm

Table 3Q19-1: Comparison of r_{average} (theoretical) with the experimental values of the ionic sizes

***3.20 X-rays and the Moseley relation** Xrays are photons with wavelengths in the range 0.01 nm-10 nm, with typical energies in the range 100 eV to 100 keV. When an electron transition occurs in an atom from the L to the K shell, the emitted radiation is generally in the X-ray spectrum.

For all atoms with atomic number $Z > 2$, the K shell is full. Suppose that one of the electrons in the K shell has been knocked out by an energetic projectile electron impacting the atom (the projectile electron would have been accelerated by a large voltage difference). The resulting vacancy in the K shell can then be filled by an electron in the L shell transiting down and emitting a photon. The emission resulting from the L to K shell transition is labeled the K_α line. The table shows the K_α line data obtained for various materials.

	Material								
	Mg	Al	S	Ca	Cr	Fe	Cu	Rb	W
Z	12	13	16	20	24	26	29	37	74
K_α line (nm)	0.987	0.834	0.537	0.335	0.229	0.194	0.154	0.093	0.021

- a. If ν is the frequency of emission, plot $\nu^{1/2}$ against the atomic number Z of the element.
- b. H.G.Moseley, while still a graduate student of E. Rutherford in 1913, found the empirical relationship

$$\nu^{1/2} = B(Z - C) \quad \text{Moseley relation}$$

where B and C are constants. What are B and C from the plot? Can you give a simple explanation as to why K_α absorption should follow this relationship?

Solution

- a. If $\lambda(K_\alpha)$ is the wavelength of the K_α emission, then the corresponding frequency is

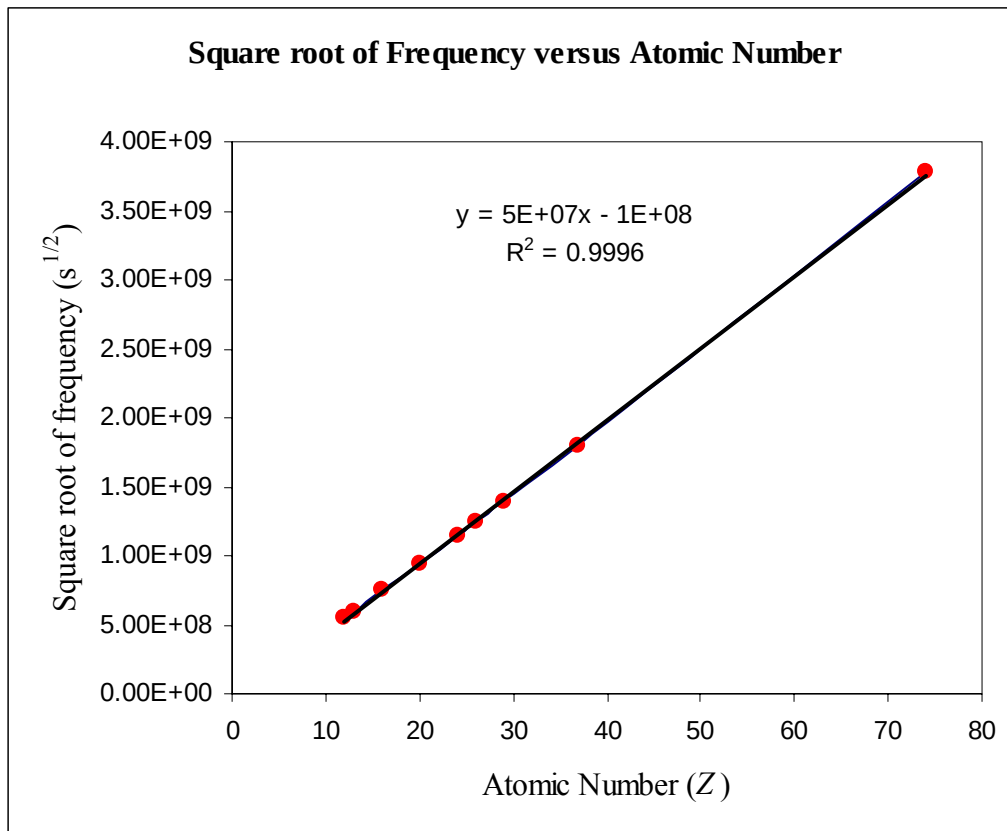
$$\nu = c / \lambda(K_\alpha)$$

For example, for Mg, $\lambda(K_\alpha) = 0.987$ nm and

$$\nu = c / \lambda(K_\alpha) = (3.0 \times 10^8 \text{ m s}^{-1}) / (0.987 \times 10^{-9} \text{ m}) = 3.04 \times 10^{17} \text{ s}^{-1}$$

We can then calculate ν for each metal and plot $\sqrt{\nu}$ vs. Z (atomic number) as in the figure below. The result is a straight line indicating that

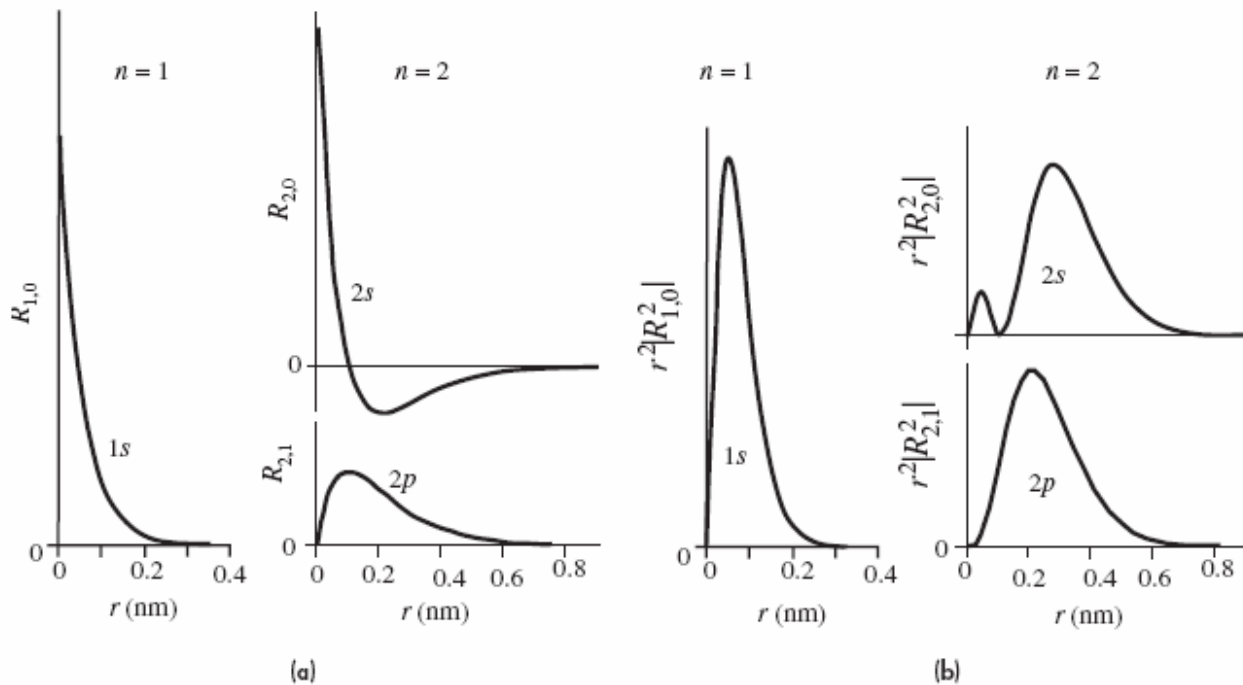
$$(\nu)^{1/2} = (5.210 \times 10^7 \text{ s}^{-1/2})Z - 9.626 \times 10^7 \text{ s}^{-1/2}$$

**Figure 3Q20-1:** Plot of square root of frequency versus atomic number

Metal	Mg	Al	S	Ca	Cr	Fe	Cu	Rb	W
Z	12	13	16	20	24	26	29	37	74
$\nu(K_\alpha) \times 10^{17} \text{ s}^{-1}$	3.04	3.60	5.59	8.96	13.1	15.5	19.5	32.3	143

Table 3Q20-1: Summarized values for frequency of K_α emissions.

b.

**Figure 3.21**(a) Radial wavefunctions of the electron in a hydrogenic atom for various n and ℓ values.(b) $r^2|R_{n,\ell}^2|$ gives the radial probability density. Vertical axis scales are linear in arbitrary units.

The first energy level, given by Equation 3.43a, depends on Z^2 . This is for a hydrogenic atom *i.e.*, only 1 electron in the atom but a nuclear charge of $+Ze$. In the present case, the transiting electron in the L shell sees an effective charge of $+Ze - 1e$ because there is 1 electron in the K -shell shielding the nucleus. Therefore the photon energy should be proportional to $(Z - 1)^2$. However, some of the L -shell electrons also spend time near the nucleus (see Figure 3.21) and therefore provide some additional shielding. This means the photon energy should be proportional to $(Z - C)^2$ where $C > 1$ due to the additional shielding effect. It seems that the additional shielding is not negligible.

The best fit line is $(\nu)^{1/2} = (5.210 \times 10^7 \text{ s}^{-1/2})Z - 9.626 \times 10^7 \text{ s}^{-1/2}$

$$\therefore B = 5.210 \times 10^7 \text{ s}^{-1/2}$$

$$\therefore C = (9.626 \times 10^7 \text{ s}^{-1/2})/B = 1.85$$

3.21 The He atom Suppose that for the He atom, zero energy is taken to be the two electrons stationary at infinity (and infinitely apart) from the nucleus (He^{++}). Estimate the energy (in eV) of the electrons in the He atom by neglecting the electron-electron repulsion, that is, neglecting the potential energy due to the mutual Coulombic repulsion between the electrons. How does this compare with the experimental value of -79 eV? How strong is the electron-electron repulsion energy?

Solution

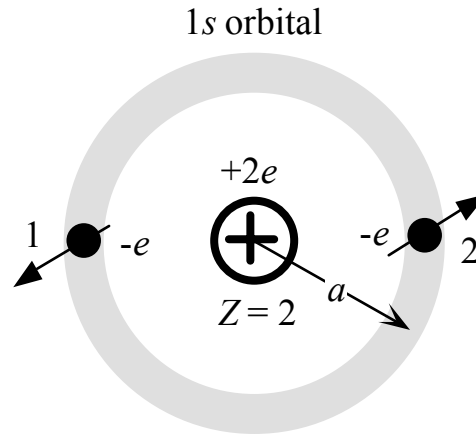


Figure 3Q21-1: Basic diagram of the He atom.

In He there are 2 electrons and the nucleus has a positive charge of $+2e$. Both electrons occupy the same orbital and hence have the same spatial distribution (same n, l, m_l). The two electrons, however, have opposite spins, *i.e.* different spin quantum numbers (Pauli exclusion principle). Due to their Coulombic repulsion, we envisage the two electrons as doing their best to avoid each other as shown. Let a be the distance to the maximum of the probability distribution. Then, $a = a_o / Z$ where a_o is the Bohr radius. In the hydrogen atom ($Z = 1$), a is simply the Bohr radius (a_o).

Energy in eV of an electron in a hydrogenic atom is given by:

$$E_n = -\frac{13.6 \text{ eV}}{n^2} Z^2$$

where n is the principal quantum number and $n = 1$ corresponds to the ground energy state. This energy is with respect to the electron and nucleus infinitely separated. It also assumes that there is also no other electron so that there is no electron-electron repulsion energy.

Electron 1 sees a net charge of $+2e$ in the nucleus so that $Z = 2$. Obviously the electron is in the ground state ($1s$), thus $n = 1$.

$$E_1 = -\frac{13.6 \text{ eV}}{1^2} (2)^2 = -54.4 \text{ eV}$$

Similarly in the absence of electron 1, electron 2 will have the energy:

$$E_2 = -\frac{13.6 \text{ eV}}{1^2} (2)^2 = -54.4 \text{ eV}$$

Thus neglecting electron-electron interactions, the total energy of the electrons in the He atom is:

$$E_{\text{total}} = E_1 + E_2 = -108.8 \text{ eV}$$

As we would expect, this value is negative, corresponding to an attractive force (between a positive and negative charge). However, when this value is compared to the experimentally obtained value of -79 eV , it is apparent that ignoring the electron-electron repulsion gives a substantially different number. The experimental value is different because it includes the interaction energy between the two electrons. Using the value for E_{total} and the experimental, the interaction energy can be found:

$$E_{\text{interaction}} = E_{\text{experiment}} - E_{\text{total}} = 29.8 \text{ eV}$$

As expected this is a positive quantity corresponding to repulsion (two negative charges).

As a footnote to demonstrate the power of intuition, let us estimate this repulsion energy differently (“back of an envelope” type calculation).

Both electrons occupy the same orbital and thus have the time averaged same spatial distribution except that due to electron-electron repulsion they will be avoiding each other. They will correlate their motions to avoid each other as follows: At one instant when one is at the extreme right the other will be at the extreme left and vice versa later on. The maximum probability radius for the 1s state for $Z = 2$ is given as:

$$a = a_0 / Z = (5.29 \times 10^{-11} \text{ m}) / (2) = 2.645 \times 10^{-11} \text{ m}$$

And the potential energy is then:

$$PE = \frac{e^2}{4\pi\epsilon_0(2a)} \left(\frac{1}{q} \right)$$

$$\therefore PE = \frac{(1.602 \times 10^{-19} \text{ C})^2}{4\pi(8.854 \times 10^{-12} \text{ F/m})(2 \times 2.645 \times 10^{-11} \text{ m})} \left(\frac{1}{1.602 \times 10^{-19} \text{ J/eV}} \right)$$

$$\therefore PE = 27.2 \text{ eV}$$

This is very close to what we found above (29.8 eV).

3.22 Excitation energy of He In the HeNe laser, an energetic electron is accelerated by the applied field impacts and excites the He from its ground state, $1s^2$, to an excited state He^* , $1s^1 2s^1$, which has one of the electrons in the 2s orbital. The ground energy of the He atom is -79 eV with respect to both electrons isolated at infinity, which defines the zero energy. Consider the $1s^1 2s^1$ state. If we neglect the electron–electron interactions, we can calculate the energy of the 1s and 2s electrons using the energy for a hydrogenic atom, $E_n = -(Z^2/n^2)(13.6 \text{ eV})$. We can then add the electron–electron interaction energy by assuming that the 1s and 2s electrons are effectively separated by $3a_0$, which is the difference, $4a_0 - 1a_0$, between the 1s and 2s Bohr radii. Calculate the overall energy of He^* and hence the excitation energy from He to He^* . The experimental value is about 20.6 eV.

Solution

For the 1s electron, $Z = 2$, $n = 1$; hence

$$E_{1s} = -(2^2)(13.6 \text{ eV})/(1^2) = -54.4 \text{ eV}$$

For the 2s electron, $Z = 1$, $n = 2$; hence

$$E_{2s} = -(1^2)(13.6 \text{ eV})/(2^2) = -3.4 \text{ eV}$$

And electron-electron interaction energy,

$$E_{\text{int}} = \frac{e^2}{4\pi\epsilon_0 3a_0} = \left(\frac{e^2}{4\pi\epsilon_0 a_0} \right) \frac{1}{3} = \frac{27.2 \text{ eV}}{3} = 9.07 \text{ eV}$$

$$\therefore E_{\text{He}^*} = E_{1s} + E_{2s} + E_{\text{int}} = -54.4 \text{ eV} - 3.4 \text{ eV} + 9.07 \text{ eV} = -48.73 \text{ eV}$$

$$\therefore \text{Excitation energy} = E_{\text{He}^*} - E_{\text{He}} = (48.73 - 79) \text{ eV} = 30.27 \text{ eV}$$

3.23 Electron affinity The fluorine atom has the electronic configuration $[\text{He}]2s^2 2p^5$. The F atom can actually capture an electron to become a F^- ion, and release energy, which is listed as its *electron affinity*, 328 kJ mol^{-1} . We will assume that the two $1s$ electrons in the closed K shell (very close to the nucleus) and the two electrons in the $2s$ orbitals will shield four positive charges and thereby expose $+9e - 4e = +5e$ for the $2p$ orbital. Suppose that we try to calculate the energy of the F^- ion by simply assuming that the additional electron is attracted by an effective positive charge, $+e(5 - Z_{2p})$ or $+eZ_{\text{effective}}$, where Z_{2p} is the overall shielding effect of the five electrons in the $2p$ orbital, so that the tenth electron we have added sees an effective charge of $+eZ_{\text{effective}}$. Calculate Z_{2p} and $Z_{\text{effective}}$. The F atom does not enjoy losing an electron. The ionization energy of the F atom is 1681 kJ mol^{-1} . What is the $Z_{\text{effective}}$ that is experienced by a $2p$ electron? (Note: $1 \text{ kJ mol}^{-1} = 0.01036 \text{ eV/atom}$.)

Solution

For the $2p$ orbital, $n = 2$.

$$E_I = (Z_{\text{effective}}/n)^2 (13.6 \text{ eV})$$

$$\text{or} \quad (328 \text{ kJ mol}^{-1})(0.01036 \text{ eV/atom/kJ mol}^{-1}) = (Z_{\text{effective}}/2)^2 (13.6 \text{ eV})$$

$$\text{or} \quad Z_{\text{effective}} = 1$$

$$\text{now,} \quad 5 - Z_{2p} = 1 \text{ or } Z_{2p} = 4$$

In case of ionization,

$$E_I = (Z_{\text{effective}}/n)^2 (13.6 \text{ eV})$$

$$\text{or} \quad (1681 \text{ kJ mol}^{-1})(0.01036 \text{ eV/atom/kJ mol}^{-1}) = (Z_{\text{effective}}/2)^2 (13.6 \text{ eV})$$

$$\text{or} \quad Z_{\text{effective}} = 2.26$$

***3.24 Electron spin resonance (ESR)** It is customary to write the spin magnetic moment of an electron as

$$\mu_{\text{spin}} = -\frac{ge}{2m_e} S$$

where S is the spin angular momentum, and g is a numerical factor, called the **g factor**, which is 2 for a free electron. Consider the interaction of an electron's spin with an external magnetic field. Show that the additional potential energy E_{BS} is given by

$$E_{BS} = \beta g m_s B$$

where $\beta = e\hbar/2m_e$ is called the **Bohr magneton**. Frequently **electron spin resonance** is used to examine various defects and impurities in semiconductors. A defect such as a dangling bond, for example, will have a single unpaired electron in an orbital and thus will possess a spin magnetic moment. A strong magnetic field is applied to the specimen to split the energy level E_1 of the unpaired spin to two levels $E_1 - E_{BS}$ and $E_1 + E_{BS}$, separated by ΔE_{BS} . The electron occupies the lower level $E_1 - E_{BS}$. Electromagnetic waves (usually in the microwave range) of known frequency ν , and

hence of known photon energy $h\nu$, are passed through the specimen. The magnetic field B is varied until the EM waves are absorbed by the specimen, which corresponds to the excitation of the electron at each defect from $E_1 - E_{BS}$ to $E_1 + E_{BS}$, that is, $h\nu = \Delta E_{BS}$ at a certain field B . This maximum absorption condition is called **electron spin resonance**, as the electron's spin is made to resonate with the EM wave. If $B = 2$ T, calculate the frequency of the EM waves needed for ESR, taking $g = 2$. Note: For many molecules, and impurities and defects in crystals, g is not exactly 2, because the electron is in a different environment in each case. The experimentally measured value of g can be used to characterize molecules, impurities, and defects.

Solution

The potential energy E_{BS} due to spin magnetic moment, μ_{spin} and B interacting is given by

$$E_{BS} = -\mu_{spin} B \cos \theta$$

$$\therefore E_{BS} = \frac{ge}{2m_e} SB \cos \theta = \frac{ge}{2m_e} B(S \cos \theta)$$

where θ is the angle between μ_{spin} and B .

By definition S_z is the component of S along z axis (along B), and is quantized so that $S_z = S \cos \theta = m_s \hbar$

$$\text{So, } E_{BS} = \frac{ge}{2m_e} B m_s \hbar$$

$$\therefore E_{BS} = \frac{e\hbar}{2m_e} g m_s B$$

$$\therefore E_{BS} = \beta g m_s B$$

There are two values corresponding $m_s = -1/2$ and $m_s = +1/2$, which differ by an amount $\Delta E_{BS} = \beta g B$. Taking $\Delta E_{BS} = h\nu$, we get

$$h\nu = \Delta E_{BS} = \beta g B.$$

$$\therefore (6.626 \times 10^{-34} \text{ Js}) \times \nu = (9.27 \times 10^{-24} \text{ J T}^{-1})(2)(2\text{T})$$

$$\text{or } \nu = 5.6 \times 10^{10} \text{ Hz or 56 GHz}$$

3.25 Spin-orbit coupling

NOTE: First printing, 2 in the denominator of the second equation should be 4.

An electron in an atom will experience an internal magnetic field B_{int} because, from the electron's reference frame, it is the positive nucleus that is orbiting the electron. The electron will "see" the nucleus, take as charge $+e$, circling around it, which is equivalent to a current $I = +ef$ where f is the electron's frequency of rotation around the nucleus. The current I generates the internal magnetic field B_{int} at the electron. From electromagnetism texts, B_{int} is given by

$$B_{\text{int}} = \frac{\mu_o I}{2r}$$

where r is the radius of the electron's orbit and μ_o is the absolute permeability. Show that

$$B_{\text{int}} = \frac{\mu_o e}{4\pi m_e r^3} L$$

Consider the hydrogen atom with $Z = 1$, $2p$ orbital, $n = 2$, $l = 1$, and take $r \approx n^2 a_o$. Calculate B_{int} .

The electron's spin magnetic moment μ_{spin} will couple with this internal field, which means that the electron will now possess a magnetic potential energy E_{SL} that is due to the coupling of the *spin* with the *orbital motion*, called spin-orbit coupling. E_{SL} will be either negative or positive, with only two values, depending on whether the electron's spin magnetic moment is along or opposite $\mathbf{B}_{\text{int}}$. Take z along B_{int} so that $E_{SL} = -B_{\text{int}}\mu_{\text{spin},z}$ where $\mu_{\text{spin},z}$ is μ_{spin} along z , and then show that the energy E_2 of the $2p$ orbital splits into two closely separated levels whose separation is

$$\Delta E_{SL} = \left(\frac{e\eta}{m_e} \right) B_{\text{int}}$$

Calculate ΔE_{SL} in eV and compare it with $E_2(n = 2)$ and the separation $\Delta E = E_2 - E_1$. (The exact calculation of E_{SL} is much more complicated, but the calculated value here is sufficiently close to be useful.) What is the effect of E_{SL} on the observed emission spectrum from the H-atom transition from $2p$ to $1s$? What is the separation of the two wavelengths? The observation is called **fine structure splitting**.

Solution

Orbital angular momentum, $L = m_e \omega r^2 = 2\pi f m_e r^2$

$$\therefore f = \frac{L}{2\pi m_e r^2}$$

$$\text{Now, } B_{\text{int}} = \frac{\mu_o I}{2r}$$

$$\therefore B_{\text{int}} = \frac{\mu_o}{2\pi r} e f$$

$$\therefore B_{\text{int}} = \frac{\mu_o}{2r} e \frac{L}{2\pi m_e r^2}$$

$$\therefore B_{\text{int}} = \frac{\mu_o e}{4\pi m_e r^3} L$$

Hydrogen atom with $Z = 1$, and for the $2p$ orbital $n = 2$ and $\ell = 1$, $r = n^2 a_o = 4a_o$.

The angular momentum, $L = \eta[\lambda(\lambda+1)]^{1/2}$ in which $\lambda = 1$

$$B_{\text{int}} = \frac{\mu_o e}{4\pi m_e r^3} L = \frac{(4\pi \times 10^{-7} \text{ WbA}^{-1}\text{m}^{-1})(1.6 \times 10^{-19} \text{ C})}{(4\pi)(9.1 \times 10^{-31} \text{ kg})(4 \times 0.0529 \times 10^{-9} \text{ m})^3} \left[(1.0546 \times 10^{-34} \text{ Js}) \sqrt{(1)(1+1)} \right]$$

$$\therefore B_{\text{int}} = 30.7 \text{ Wbm}^{-2} \text{ or about } 31 \text{ T}$$

Consider the spin magnetic moment

$$\mu_{\text{spin},z} = -\frac{e}{m_e} S_z = -\frac{e}{m_e} (m_s \hbar) = -\frac{e\hbar}{m_e} m_s$$

The PE is

$$E_{SL} = -\mu_{\text{spin},z} B_{\text{int}} = -\left(-\frac{e\hbar}{m_e} m_s\right) B_{\text{int}} = \frac{e\hbar}{m_e} m_s B_{\text{int}}$$

which has only two values, corresponding to $m_s = -1/2$ and $+1/2$. The difference between the two values is

$$\begin{aligned} \Delta E_{SL} &= \frac{e\hbar}{m_e} B_{\text{int}} = [(1.6 \times 10^{-19} \text{ C})(1.0546 \times 10^{-34} \text{ J s}) / (9.1 \times 10^{-31} \text{ kg})] (30.7 \text{ T}) \\ &= 5.69 \times 10^{-22} \text{ J or } 3.55 \times 10^{-3} \text{ eV} \end{aligned}$$

For hydrogen atom,

$$E_2 = -(1^2)(13.6 \text{ eV}) / (2^2) = -3.4 \text{ eV}$$

$$E_1 = -(1^2)(13.6 \text{ eV}) / (1^2) = -13.6 \text{ eV}$$

$$\therefore \Delta E_{12} = E_2 - E_1 = 10.2 \text{ eV}$$

Therefore ΔE_{SL} is much less than E_2 or ΔE_{12}

Transmission will be two wavelengths separated by ΔE_{SL} .

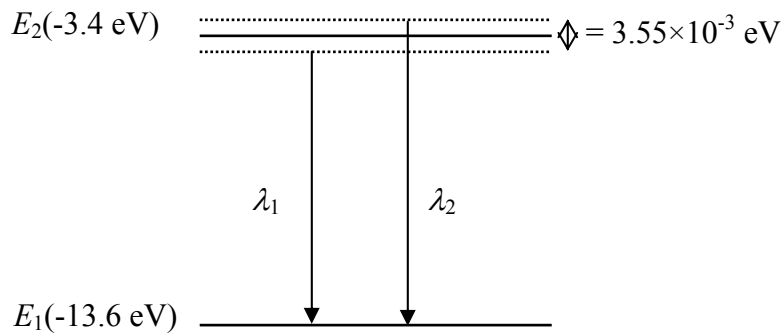


Figure 3Q25-1: Energy levels of hydrogen atom with spin orbital coupling.

E_2 splits into two levels: E_2' and E_2''

$$E_2' = E_2 - (1/2)\Delta E_{SL} \text{ and } E_2'' = E_2 + (1/2)\Delta E_{SL}$$

Thus there are two emission wavelengths,

$$\frac{hc}{\lambda_1} = E_2' - E_1 = E_2 - \frac{1}{2}\Delta E_{SL} - E_1 = \Delta E_{12} - \frac{1}{2}\Delta E_{SL}$$

$$\frac{(6.626 \times 10^{-34} \text{ J s})(3 \times 10^8 \text{ m s}^{-1})}{\lambda_1} = [10.2 \text{ eV} - (1/2)(3.55 \times 10^{-3} \text{ eV})](1.602 \times 10^{-19} \text{ J/eV})$$

$$\lambda_1 = 121.574 \text{ nm}$$

Similarly,

$$\frac{hc}{\lambda_2} = E_2'' - E_1 = E_2 + \frac{1}{2} \Delta E_{SL} - E_1 = \Delta E_{12} + \frac{1}{2} E_{SL}$$

$$\frac{(6.626 \times 10^{-34} \text{ J s})(3 \times 10^8 \text{ m s}^{-1})}{\lambda_1} = [10.2 \text{ eV} + (1/2)(3.55 \times 10^{-3} \text{ eV})](1.602 \times 10^{-19} \text{ J/eV})$$

$$\lambda_2 = 121.532 \text{ nm}$$

The difference is

$$\Delta\lambda = \lambda_1 - \lambda_2 = 0.042 \text{ nm, very small}$$

Note, we can also use calculus to obtain the wavelength difference. Given $hc / \lambda = \Delta E$,

$$\therefore \lambda = \frac{hc}{\Delta E}$$

$$\therefore \frac{d\lambda}{d\Delta E} = -\frac{hc}{\Delta E^2} \quad \text{or} \quad |\delta\lambda| \approx \frac{hc}{\Delta E^2} |\delta\Delta E|$$

$$\text{or} \quad |\delta\lambda| \approx \frac{hc}{\Delta E^2} |\Delta E_{SL}|$$

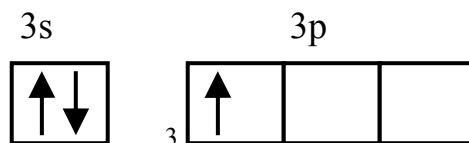
$$|\delta\lambda| \approx \frac{(6.626 \times 10^{-34} \text{ J s})(3 \times 10^8 \text{ m s}^{-1})}{[(10.2 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})]^2} [(3.55 \times 10^{-3} \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})]$$

$$= 4.23 \times 10^{-11} \text{ m or } \mathbf{0.042 \text{ nm}}$$

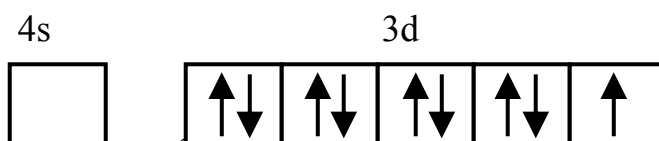
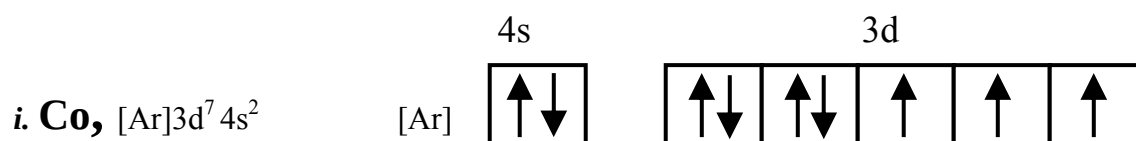
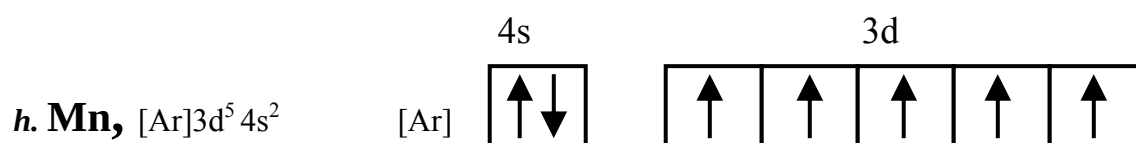
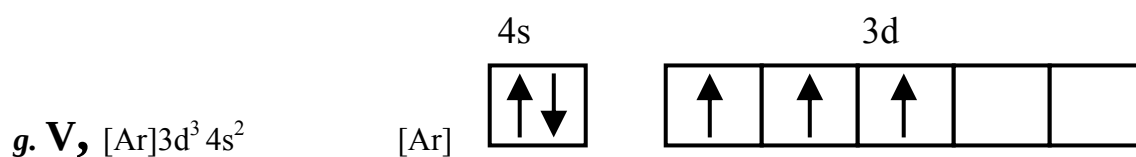
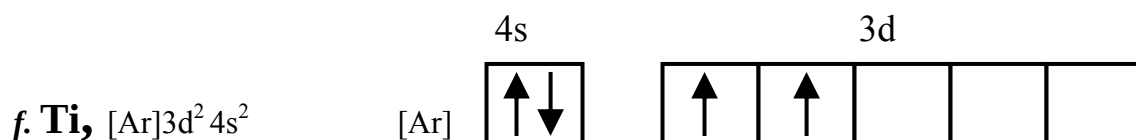
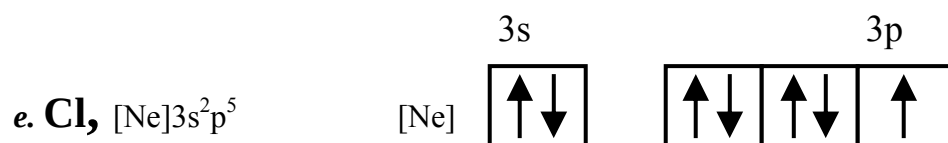
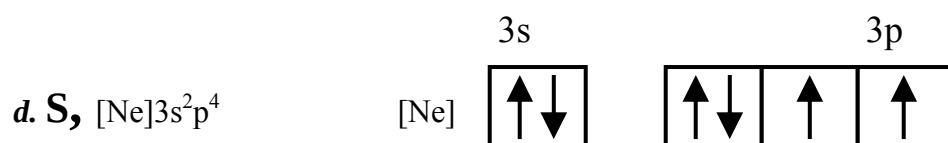
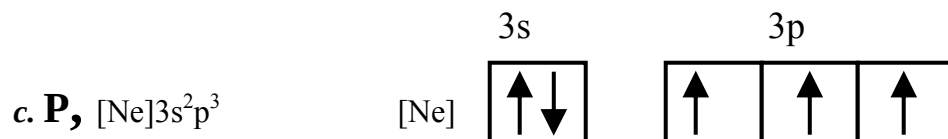
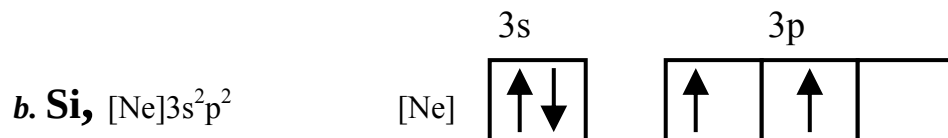
3.26 Hund's rule For each of the following atoms and ions, sketch the electronic structure, using a box for an orbital wavefunction and an arrow (up or down) for an electron:

- | | |
|---|---|
| a. Aluminum, [Ne]3s ² p ¹ | f. Titanium, [Ar]3d ² 4s ² |
| b. Silicon, [Ne]3s ² p ² | g. Vanadium, [Ar]3d ³ 4s ² |
| c. Phosphorus, [Ne]3s ² p ³ | h. Manganese, [Ar]3d ⁵ 4s ² |
| d. Sulfur, [Ne]3s ² p ⁴ | i. Cobalt, [Ar]3d ⁷ 4s ² |
| e. Chlorine, [Ne]3s ² p ⁵ | j. Cu ²⁺ , [Ar]3d ⁹ 4s ⁰ |

Solution



a. **Al**, [Ne]3s²p¹ [Ne]



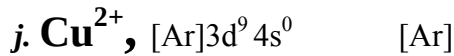


Figure 3Q26-1: Electronic structures of various atoms.

3.27 Hund's rule The carbon atom has the electronic structure $2s^2 2p^2$ in its ground state. The ground state and various possible excited states of C are shown in Figure 3.47. The following energies are known for the states *a* to *e* in Figure 3.47, not in any particular order: 0, 7.3 eV, 4.1 eV, 7.9 eV, and 1.2 eV. Using reasonable arguments match these energies to the states *a* to *e*. Use Hund's rule to establish the ground state with 0 eV. If you have to flip a spin to go from the ground to another configuration, that would cost energy. If you have to move an electron from a lower *s* to *p* or from *p* to a higher *s*, that would cost a lot of energy. Two electrons in the same orbital (obviously with paired electrons) would have substantial Coulombic repulsion energy.

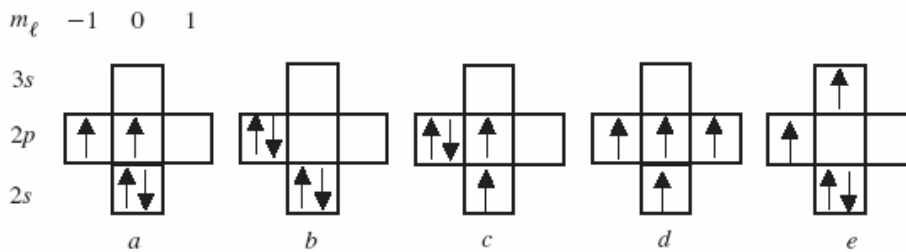
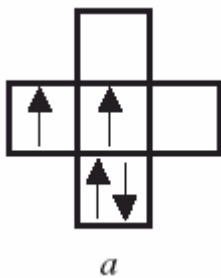
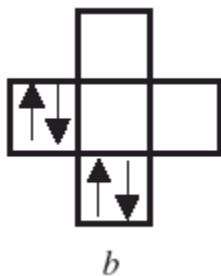


Figure 3.47
Some possible states of the carbon atom, not in any particular order.

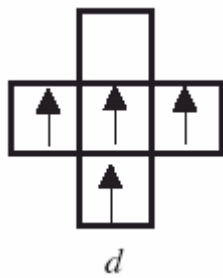
Solution



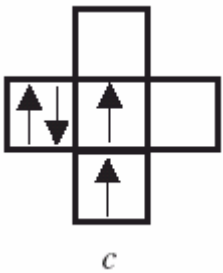
→ 0 eV



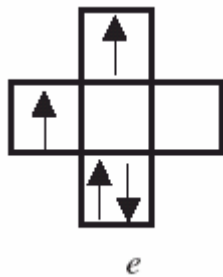
→ 1.2 eV



—————→ 4.1 eV



—————→ 7.3 eV



—————→ 7.9 eV

3.28 The HeNe Laser A particular HeNe laser operating at 632.8 nm has a tube that is 40 cm long. The operating gas temperature is about 130 °C.

- a. Calculate the Doppler-broadened linewidth $\Delta\lambda$ in the output spectrum.
- b. What are the n values that satisfy the resonant cavity condition? How many modes are therefore allowed?
- c. Calculate the frequency separation and the wavelength separation of the laser modes. How do these change as the tube warms up during operation? Taking the linear expansion coefficient to be 10^{-6} K^{-1} , estimate the change in the mode frequency separation.

Solution

Operating wavelength = $\lambda_o = 632.8 \times 10^{-9} \text{ m}$

Length of the tube = $L = 0.4 \text{ m}$

Operating temperature = $T = 403 \text{ K}$

Room temperature = 293 K (20 °C)

a. $M_{at} = 20.2 \times 10^{-3} \text{ kg mol}^{-1}$. The mass of the Ne atom is M_{at}/N_A . The effective velocity v_x of the Ne atoms along the x -direction can be found from $(1/2)Mv_x^2 = (1/2)kT$,

$$v_x = \sqrt{\frac{kT}{M}} = \sqrt{\frac{kT}{M_{at}/N_A}} = \sqrt{\frac{(1.381 \times 10^{-23} \text{ J K}^{-1})(403 \text{ K})}{(20.18 \times 10^{-3} \text{ kg mol}^{-1})/(6.022 \times 10^{23} \text{ mol}^{-1})}}$$

$$\therefore v_x = 407.5 \text{ m s}^{-1}$$

The wavelength $\lambda_o = 632.8 \text{ nm}$ corresponds to a frequency ν_o :

$$\nu_o = \frac{c}{\lambda_o} = \frac{(3.0 \times 10^8 \text{ m s}^{-1})}{(632.8 \times 10^{-9} \text{ m})} = 4.741 \times 10^{14} \text{ s}^{-1}$$

The width in the frequency spectrum, $\Delta\nu$, is given by (see Example 3.25):

$$\Delta\nu = \frac{2\nu_o v_x}{c} = \frac{2(4.741 \times 10^{14} \text{ s}^{-1})(407.5 \text{ m s}^{-1})}{(3.0 \times 10^8 \text{ m s}^{-1})} = 1.288 \times 10^9 \text{ s}^{-1}$$

From Example 3.25

$$\Delta\lambda = \frac{\lambda_o}{\nu_o} \Delta\nu = \frac{(632.8 \times 10^{-9} \text{ m})}{(4.741 \times 10^{14} \text{ s}^{-1})} (1.288 \times 10^9 \text{ s}^{-1})$$

$$\therefore \Delta\lambda = 1.72 \times 10^{-12} \text{ m or } 0.00172 \text{ nm}$$

The velocity v_x we used is the root mean square (rms) velocity along one dimension (the laser tube axis) as given by the kinetic theory, *i.e.* $(1/2)Mv_x^2 = (1/2)kT$. Intuitively we would therefore expect the linewidth $\Delta\nu$ or $\Delta\lambda$ to correspond to the width between the rms points in the Gaussian output spectrum.

b. Let n be the mode number corresponding to λ_o , then

$$n = \frac{L}{\frac{1}{2}\lambda_o} = \frac{(40 \times 10^{-2} \text{ m})}{\frac{1}{2}(632.8 \times 10^{-9} \text{ m})} = 1264222.5; \text{ a very large number.}$$

However n must be an integer so 1264222 and 126223 would be two possible modes that are within the spectrum.

The n_1 and n_2 values corresponding to the shortest λ_1 and longest λ_2 wavelengths are

$$n_1 = \frac{L}{\frac{1}{2}(\lambda_o - \frac{1}{2}\Delta\lambda)} = \frac{(40 \times 10^{-2} \text{ m})}{\frac{1}{2}(632.8 \times 10^{-9} \text{ m} - \frac{1}{2}1.7195 \times 10^{-12} \text{ m})} = 1264224.2$$

$$n_2 = \frac{L}{\frac{1}{2}(\lambda_o + \frac{1}{2}\Delta\lambda)} = \frac{(40 \times 10^{-2} \text{ m})}{\frac{1}{2}(632.8 \times 10^{-9} \text{ m} + \frac{1}{2}1.7195 \times 10^{-12} \text{ m})} = 1264220.8$$

Thus n values are: 1264221, 1264222, 1264223, 1264224. There are about four modes.

Alternatively we can find the number of modes as follows. Changes in n and λ are related by the above equation; $n = 2L/\lambda$. We can relate the change in n to the change in λ by differentiating $n = 2L/\lambda$. Then over the width $\Delta\lambda$ of the spectrum, n changes by Δn :

$$\frac{dn}{d\lambda} = -2L \frac{1}{\lambda^2}$$

Substitute: $|\Delta n| = 2 \left(\frac{n\lambda}{2} \right) \frac{1}{\lambda^2} |\Delta \lambda|$

$$\therefore |\Delta n| = \frac{n}{\lambda} |\Delta \lambda| = \frac{(1.264 \times 10^6)}{(632.8 \times 10^{-9} \text{ m})} (1.72 \times 10^{-12} \text{ m}) = 3.44$$

Since Δn must be an integer $\Delta n = 3$.

The number of modes, however, is $\Delta n + 1$ or 4 (there are 4 numbers between 13 and 16 inclusively although $16 - 13 = 3$).

Note: These 4 modes are within the central region, *i.e.* between the rms points, of the Gaussian output spectrum, that is within $\Delta \lambda$. If we wish to find *all* the modes, we can estimate this by considering the full “extent” of the spectrum. Suppose that the full extent of the spectrum is roughly $2\Delta \lambda$. Then there would be ~ 8 modes. The situation is more complicated by the fact that the exact number of modes depends on how the emission spectrum and the cavity modes coincide. Further, we need to find the net optical gain that is the optical gain minus cavity losses not just the optical gain curve. All these fine points are far beyond the scope of this text (see S. O. Kasap, *Applied Optoelectronics: Principles and Practices*, Prentice Hall, 1999).

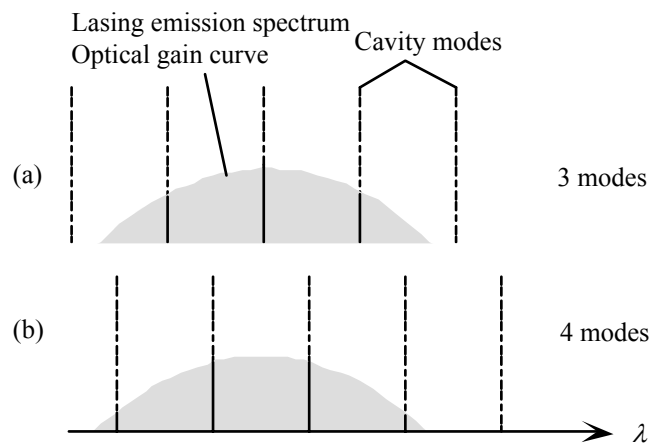


Figure 3Q28-1: Number of laser modes depends on how the cavity modes intersect the optical gain curve (lasing emission spectrum). In (a) there are 3 modes and in (b) there are 4 modes. The width of the emission spectrum here is roughly between rms (root mean square) points of the intensity vs. wavelength behavior.

c. The separation $\Delta \nu$ of two neighboring modes can be found by finding the change in ν corresponding to a change of 1 in n , that is $\Delta n = 1$. Substitute $\lambda = c/\nu$ in $n(\lambda/2) = L$ to find $\nu = cn/(2L)$. Differentiate ν with respect to n to find

$$\Delta \nu = \frac{c}{2L} \Delta n = \frac{(3.0 \times 10^8 \text{ m s}^{-1})}{2(0.40 \text{ m})} (1) = 3.75 \times 10^8 \text{ s}^{-1} \text{ or } 375 \text{ MHz}$$

When the tube warms up, L expands and $\Delta \nu$ decreases. The new length L' of the tube at 130°C is given by

$$L' = L[1 + \alpha(T - T_o)] = (0.4 \text{ m})[1 + (10^{-6} \text{ }^\circ\text{C}^{-1})(130 - 20)^\circ\text{C}] = 0.400044 \text{ m}$$

where α is the linear expansion coefficient, T_o is room temperature assumed to be 20°C . The new frequency separation is

$$\Delta\nu' = \frac{c}{2L'} \Delta n = \frac{(3.0 \times 10^8 \text{ m s}^{-1})}{2(0.400044 \text{ m})} (1) = 3.74959 \times 10^8 \text{ s}^{-1}$$

The change in the frequency separation is

$$\Delta\nu' - \Delta\nu = -4.10 \times 10^4 \text{ s}^{-1} \text{ or } 41.0 \text{ kHz (decrease)}$$

3.29 Er³⁺-doped fiber amplifier When the Er³⁺ ion in the Er³⁺-doped fiber amplifier (EDFA) is pumped with 980 nm of radiation, the Er³⁺ ions absorb energy from the pump signal and become excited to E_3 (Figure 3.44). Later the Er³⁺ ions at E_2 are stimulated to add energy (coherent photons) to the signal at 1550 nm. What is the wasted energy (in eV) from the pump to the signal at each photon amplification step? (This energy is lost as heat in the glass medium.) An Er-doped fiber amplifier is 10 m long, and the **radius of the core is 5 μm** . The Er concentration in the core is 10^{18} cm^{-3} . The nominal power gain of the amplifier is 100 (or 20 dB). The pump wavelength is 980 nm, and the signal wavelength is 1550 nm. If the output power from the amplifier is 100 mW and assuming the signal and pump are confined to the core, what is the minimum intensity of the pump signal? How much power is wasted in this EDFA? (The pump must provide enough photons to pump the Er³⁺ ions needed to generate the additional output photons over that of input photons. The concentration of Er³⁺ ions in the fiber is given for information only.)

Solution

$$\text{Wasted energy} = E_3 - E_2 = hc/\lambda_{\text{pump}} - hc/\lambda_{\text{signal}}$$

$$= (4.13 \times 10^{-15} \text{ eV})(3 \times 10^8 \text{ ms}^{-1}) / (980 \times 10^{-9} \text{ m}) - (4.13 \times 10^{-15} \text{ eV})(3 \times 10^8 \text{ ms}^{-1}) / (1550 \times 10^{-9} \text{ m})$$

$$= 1.27 \text{ eV} - 0.8 \text{ eV} = 0.47 \text{ eV}$$

$$P_{\text{out}} = 100 \text{ mW.}$$

$$\text{Now, } P_{\text{out}}/P_{\text{in}} = 100$$

$$\therefore P_{\text{in}} = 1 \text{ mW}$$

$$\therefore \text{Output power from the pumped signal} = P_{\text{out}} - P_{\text{in}} = 99 \text{ mW}$$

$$\text{Wasted power} = (0.47)/(0.8) \times 99 \text{ mW} = \mathbf{58.16 \text{ mW}}$$

$$\text{Total pumped signal} = 99 \text{ mW} + 58.16 \text{ mW} = 157.16 \text{ mW}$$

$$\therefore \text{Intensity of the pump signal} = \text{Total pump signal/core area}$$

$$= (0.15716 \text{ W})/(\pi r^2) = (0.15716 \text{ W})/(\pi 5^2 \times 10^{-8} \text{ cm}^2) = \mathbf{0.2 \times 10^6 \text{ W cm}^{-2}}$$

"It was not unusual in theoretical physics to spend a lot of time in speculative notion that turns out to be wrong. I do it all the time. Having a lot of crazy ideas is the secret of my success. Some of them turned out to be right!"

Sheldon L. Glashow (Higgins Professor of Physics at Harvard University; Nobel Laureate, 1979)

Third Edition (© 2001 McGraw-Hill)

Chapter 4

Note: The first printing has a few odd typos, which are indicated in blue below. These will be corrected in the reprint.

4.1 Phase of an atomic orbital

- What is the functional form of a 1s wavefunction, $\psi(r)$? Sketch schematically the atomic wavefunction $\psi_{1s}(r)$ as a function of distance from the nucleus.
- What is the total wavefunction $\Psi_{1s}(r, t)$?
- What is meant by two wavefunctions $\Psi_{1s}(A)$ and $\Psi_{1s}(B)$ that are out of phase?
- Sketch schematically the two wavefunctions $\Psi_{1s}(A)$ and $\Psi_{1s}(B)$ at one instant.

Solution

- $\psi_{1s}(r)$ decays exponentially as $\exp(-r/a_0)$, where a_0 is the Bohr radius (Table 3.2).

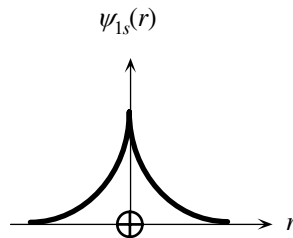


Figure 4Q1-1: Atomic wavefunction as a function of distance from the nucleus.

- $\Psi_{1s}(r, t) = \psi_{1s}(r) \times \exp(-jEt/\hbar) = \psi_{1s}(r) \times \exp(-j\omega t)$ where $\omega = E/\hbar$ is an angular frequency (Section 3.2.2). The total wavefunction is harmonic in time. Recall that $\exp(j\theta) = \cos\theta + j\sin\theta$ so that $\exp(j\omega t) = \cos\omega t + j\sin\omega t$.
- Two sine waves of the same frequency will have a certain phase difference which represents the time delay between the time oscillations of the two waves. Two waves will be in phase if their maxima coincide and out of phase if the maximum of one coincides with the minimum of the other. Two 1s wavefunctions, $\Psi_{1s}(A)$ and $\Psi_{1s}(B)$ will have the same frequency. Like two sine waves of the same frequency, the waves can be in phase or out of phase. When they are out of phase, if $\Psi_{1s}(A) = 1$ then $\Psi_{1s}(B) = -1$ and vice versa.

d.

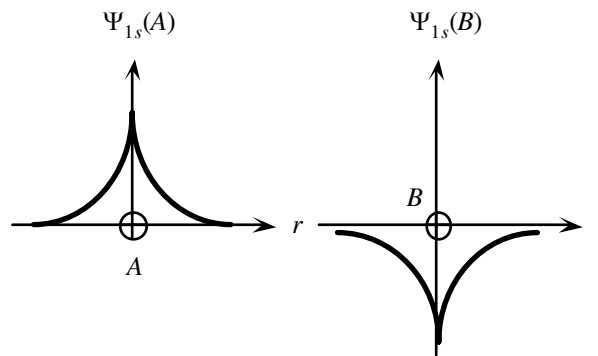


Figure 4Q1-2: Two $1s$ wavefunctions which are out of phase (sketches at the same instant)

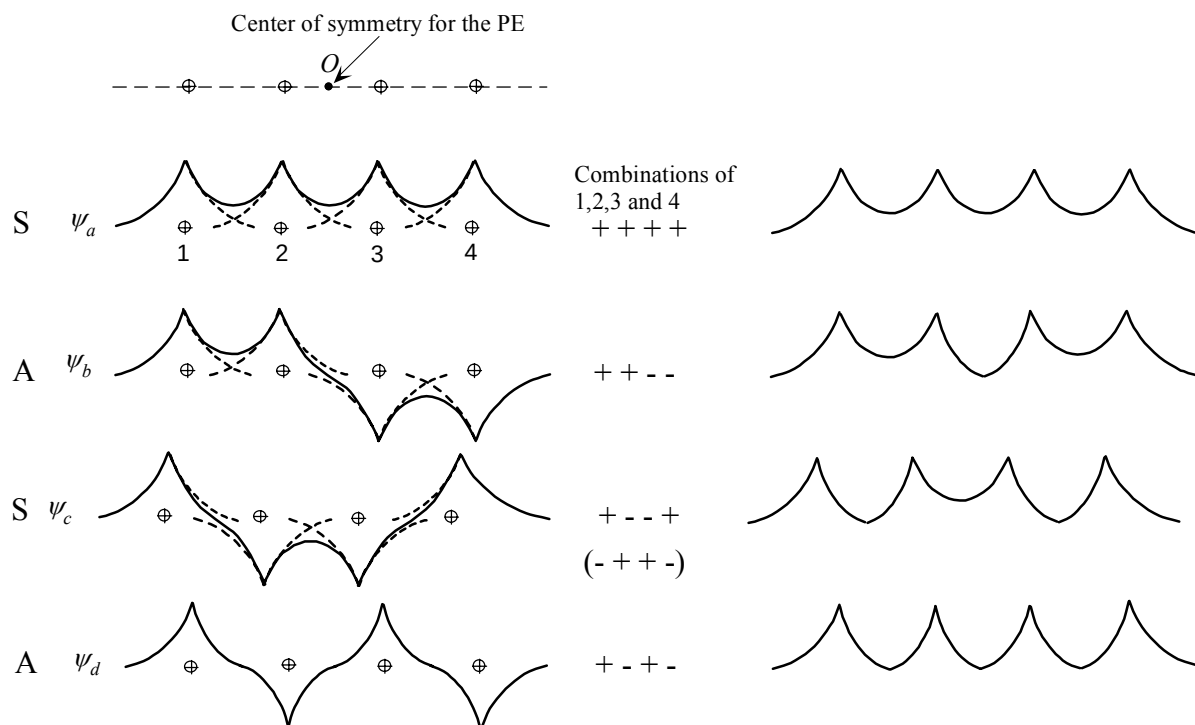
4.2 Molecular orbitals and atomic orbitals Consider a linear chain of four identical atoms representing a hypothetical molecule. Suppose that each atomic wavefunction is $1s$ wavefunction. This system of identical atoms has a center of symmetry C with respect to the center of the molecule (midway between the second and the third atom), and all molecular wavefunctions must be either symmetric or antisymmetric about C .

- Using LCAO principle, sketch the possible molecular orbitals.
- Sketch the probability distribution $|\psi|^2$
- If more nodes in the wavefunction lead to greater energies, order the energies of the molecular orbitals.

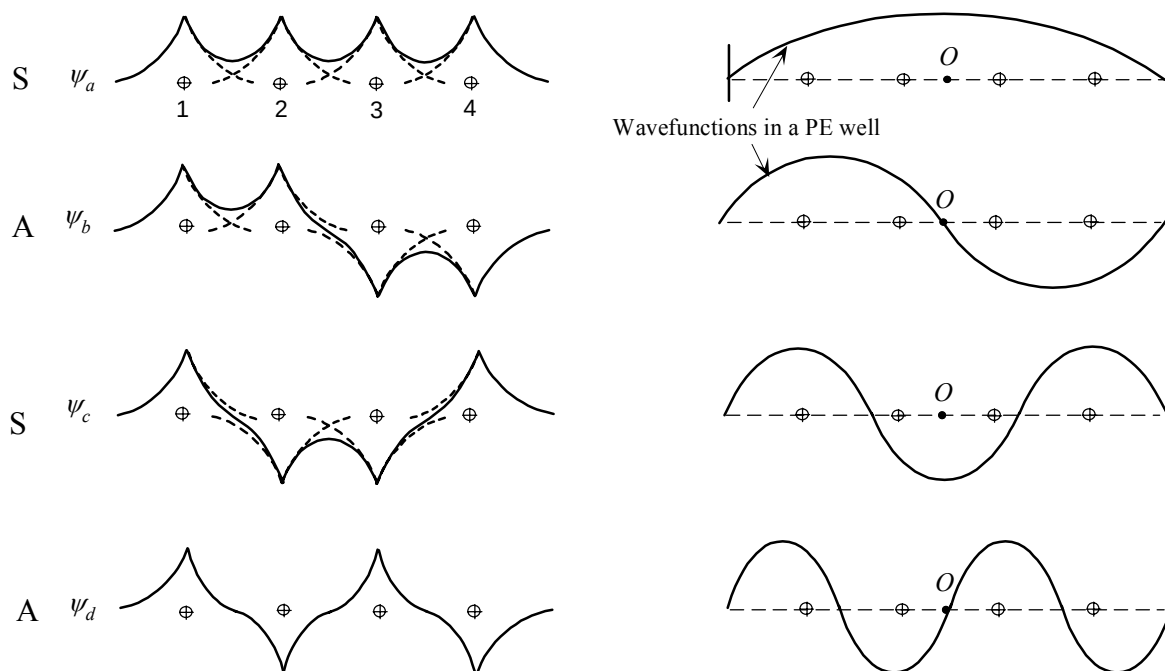
Note: The electron wavefunctions, and the related probability distributions, in a simple potential energy well that are shown in Figure 3.15 can be used as a rough *guide* towards finding the appropriate molecular wavefunctions in the four-atom symmetric molecule. For example, if we were to smooth the electron potential energy in the four-atom molecule into a constant potential energy, that is, generate a potential energy well, we should be able to modify or distort, without flipping, the molecular orbitals to somewhat resemble ψ_1 to ψ_4 sketched in Figure 3.15. Consider also that the number of nodes increases from none for ψ_1 to three for ψ_4 in Figure 3.15.

Author's Note to the Instructor: This has to changed to four atoms to simplify the problem.

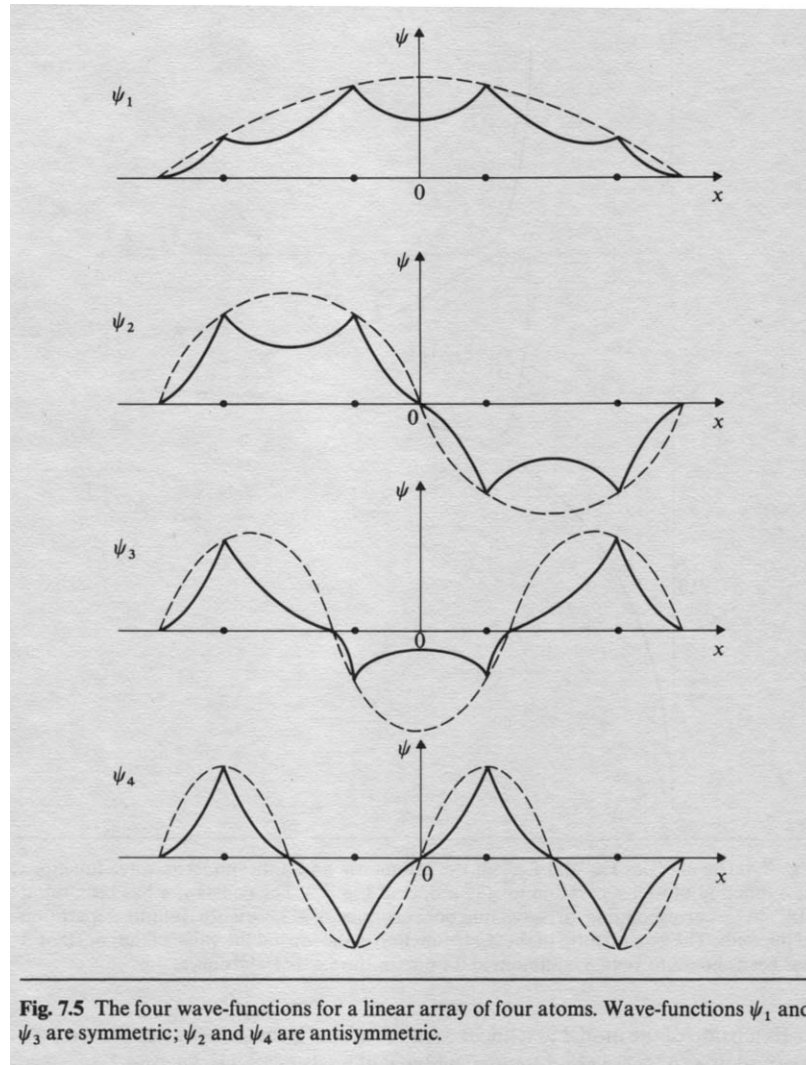
Solution



Left: Molecular orbitals based on LCAO for a linear array of 4 atoms. S is symmetric and A is antisymmetric. Right: Probability distributions. All sketches are rough schematic illustrations. Energy increases with the number of nodes. In the above figures, energy increases downwards from ψ_a to ψ_d . Lowest for the top MO (molecular orbital) and highest for the bottom MO.



Comparison of molecular orbitals based on LCAO for a linear array of 4 atoms and the first four wavefunctions in a potential energy well. Notice the resemblance.



Author's Note: Generally, the actual molecular orbitals are not a simple linear combination but may be a linear combination in which each individual atomic wavefunction is scaled or weighted by a certain amount. The similarity between the molecular orbitals and the wavefunctions in a PE well is more apparent in the above figure from waves, *Atoms and Solids*, D.A. Davies, Longman (England), 1978, Figure 7.5, p214

4.3 Diamond and Tin Germanium, silicon, and diamond have the same crystal structure, that of diamond. Bonding in each case involves sp^3 hybridization. The bonding energy decreases as we go from C to Si to Ge, as noted in Table 4.7.

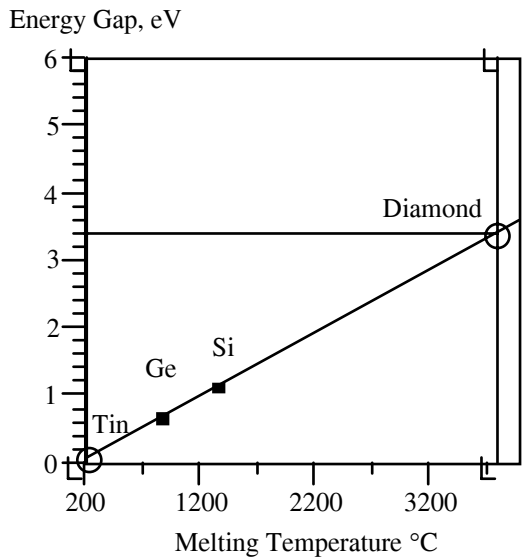
- a. What would you expect for the band gap of diamond? How does it compare with the experimental value of 5.5 eV?
- b. Tin has a tetragonal crystal structure, which makes it different than its group members, diamond, silicon, and germanium.
 1. Is it a metal or a semiconductor?
 2. What experiments do you think would expose its semiconductor properties?

Table 4.7

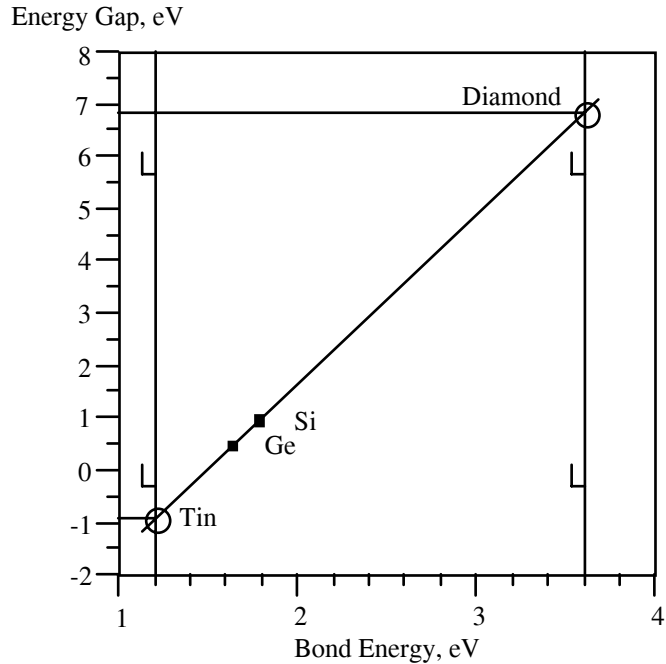
Property	Diamond	Silicon	Germanium	Tin
Melting temperature, °C	3800	1417	937	232
Covalent radius, nm	0.077	0.117	0.122	0.146
Bond energy, eV	3.60	1.84	1.7	1.2
First ionization energy, eV	11.26	8.15	7.88	7.33
Bandgap, eV	?	1.12	0.67	?

Solution

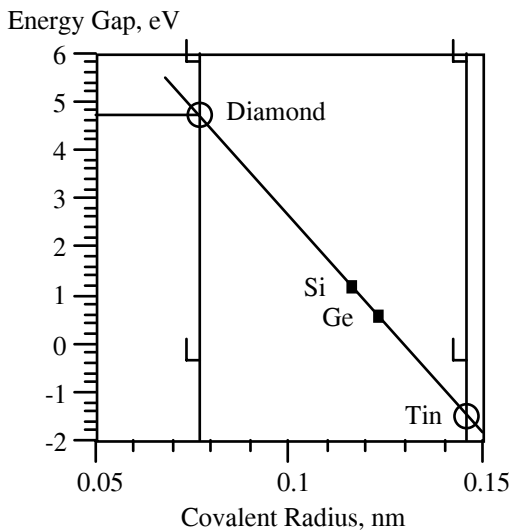
Given the properties in Table 4.7, we have the following plots:



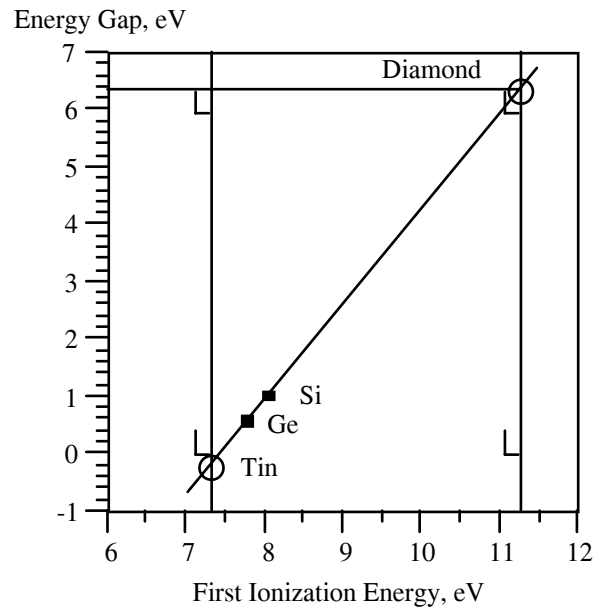
(a) Energy gap versus melting temperature. From the plot, it seems that diamond has an $E_g = 3.4$ eV and Sn has $E_g \approx 0$ or is a metal



(b) Energy gap versus bond energy. From the plot, it seems that diamond has an $E_g = 6.8$ eV and Sn has $E_g \approx -0.9$ or is a metal



(c) Energy gap versus covalent radius. From the plot, it seems that diamond has an $E_g = 4.7$ eV and Sn has $E_g \approx -1.5$ eV or is a metal.



(d) Energy gap versus first ionization energy. From the plot, it seems that diamond has an $E_g = 6.3$ eV and Sn has $E_g \approx -0.3$ eV or is a metal.

Figure 4Q3-1

Each is a plot of the band gap E_g (or energy gap) vs. some property. The straight line in each is drawn to pass through Si and Ge. Diamond and tin points are then located on this straight line at the intersections with the vertical lines representing the corresponding properties of diamond and tin.

a. Diamond has an E_g greater than Si and Ge. Averaging the four E_g for diamond we find 5.3 eV which is close to the experimental value of 5.5 eV.

b.

(1) All the four properties indicate that tin has $E_g \leq 0$ or is a metal.

(2) Tin's semiconductivity can be tested by examining its electrical conductivity and optical absorption (see Chapter 5). For example, for metals the conductivity should NOT be thermally activated over a wide temperature range, whereas for semiconductors there will be an Arrhenius temperature dependence over at least some temperature range. Further, semiconductors have an absorption edge that corresponds to $h\nu > E_g$ (Chapter 5).

4.4 Compound III-V semiconductors Indium as an element is a metal. It has a valency of III. Sb as an element is a metal and has a valency of V. InSb is a semiconductor, with each atom bonding to four neighbors, just like in silicon. Explain how this is possible and why InSb is a semiconductor and not a metal alloy. (Consider the electronic structure and sp^3 hybridization for each atom.)

Solution

The one s and three p orbitals hybridize to form 4 ψ_{hyb} orbitals. In Sb there are 5 valence electrons. One ψ_{hyb} has two paired electrons and 3 ψ_{hyb} has 1 electron as shown in Figure 4Q4-1. In In there are 3 electrons so one ψ_{hyb} is empty. This empty ψ_{hyb} of In can overlap the full ψ_{hyb} of Sb. The overlapped orbital, the bonding orbital, has two paired electrons. This is a bond between In and Sb even though the electrons come from Sb (this type of bonding is called *dative bonding*). It is a bond because the electrons in the overlapped orbital are shared by both Sb and In. The other 3 ψ_{hyb} of Sb can overlap 3 ψ_{hyb} of neighboring In to form "normal bonds". Repeating this in three dimensions generates the InSb crystal where each atom bonds to four neighboring atoms as shown. As all the bonding orbitals are full, the valence band formed from these orbitals is also full. The crystal structure is reminiscent of that of Si, as all the valence electrons are in bonds. Since it is similar to Si, InSb is a semiconductor.

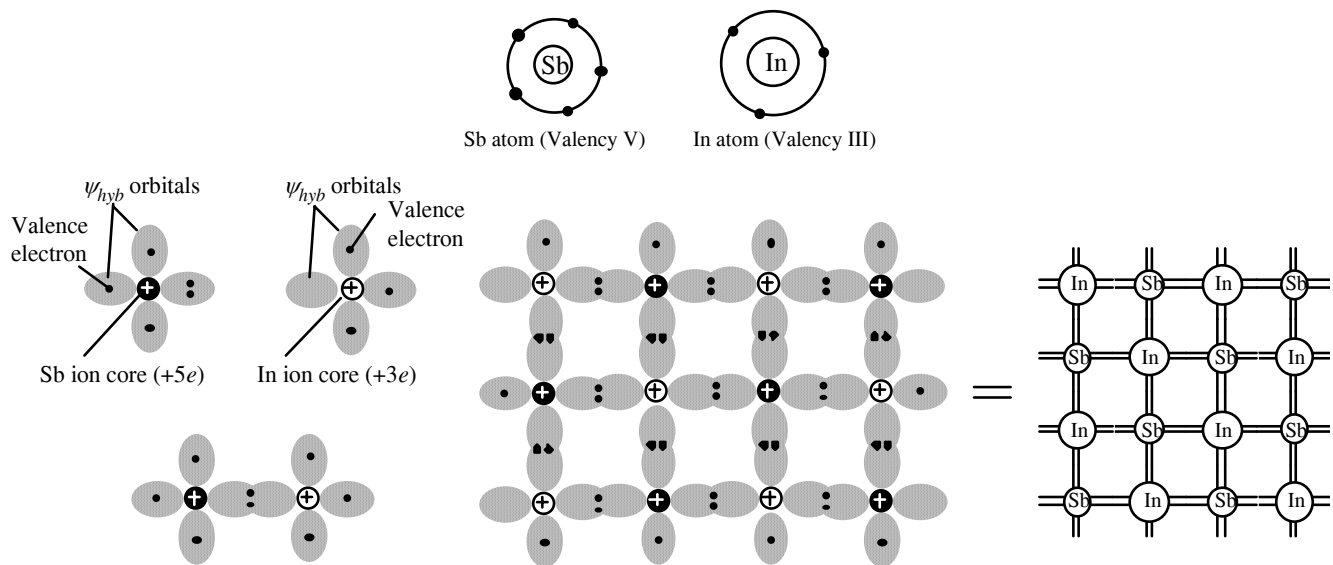


Figure 4Q4-1: Bonding structure of InSb.

4.5 Compound II-VI semiconductors CdTe is a semiconductor, with each atom bonding to four neighbors, just like in silicon. In terms of covalent bonding and the positions of Cd and Te in the Periodic Table, explain how this is possible. Would you expect the bonding in CdTe to have more ionic character than that in III-V semiconductors?

Solution

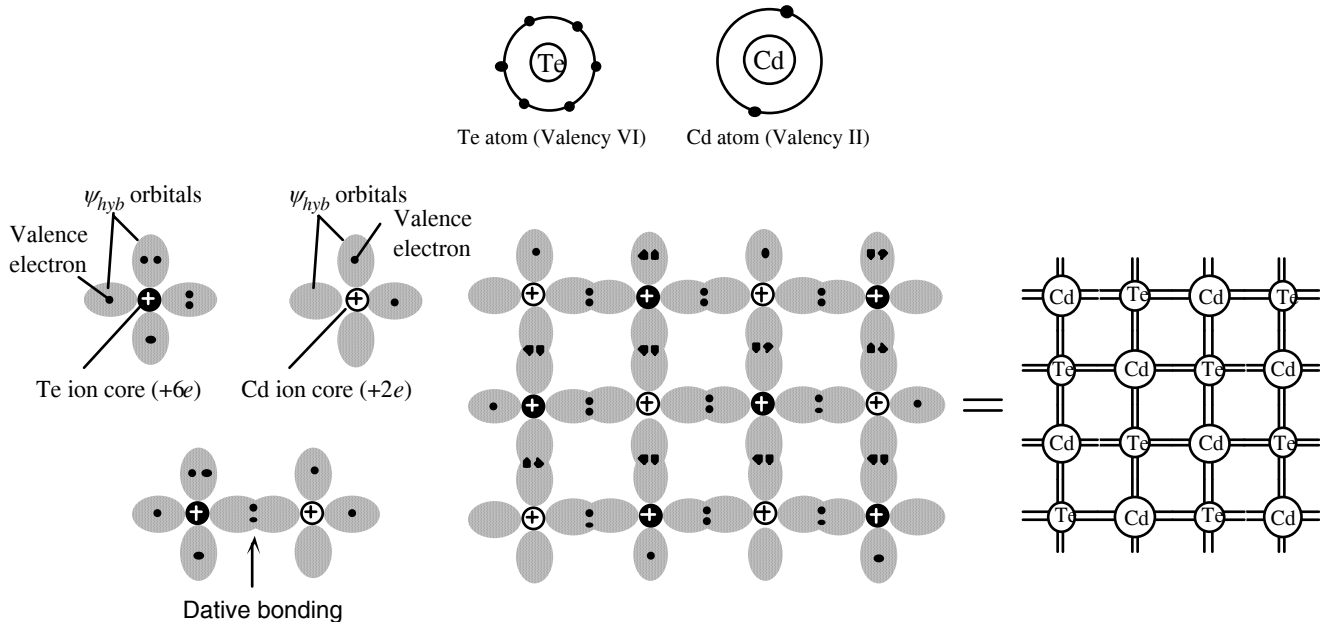


Figure 4Q5-1: Bonding structure of CdTe.

In CdTe one would expect a mixture of covalent and ionic bonding. Transferring 2 Cd electrons to Te would generate Te^{2-} and Cd^{2+} which then bond ionically. In covalent bonding we expect hybridization of s and p orbitals. The one s and three p orbitals hybridize to form 4 ψ_{hyb} orbitals. In Te there are 6 valence electrons. Two ψ_{hyb} have two paired electrons each and two ψ_{hyb} have 1 electron each as shown. In Cd there are 2 electrons so two ψ_{hyb} are empty. An empty ψ_{hyb} of Cd can overlap a full ψ_{hyb} of Te. The overlapped orbital, the bonding orbital, then has two paired electrons. This is a bond between Cd and Te even though the electrons come from Te (this type of bonding is called *dative bonding*). It is a bond because the electrons in the overlapped orbital are shared by both Te and Cd. The other full ψ_{hyb} of Te can similarly overlap another empty ψ_{hyb} of a different neighboring Cd to form another dative bond. The half occupied orbitals (two on Te and two on Cd) overlap and form "normal bonds". Repeating two dative bonds and two normal bonds in three dimensions generates the CdTe crystal where each atom bonds to four neighboring atoms as shown. Since all the bonding orbitals are full, the valence band formed from these orbitals is also full. Strictly the bonding is neither fully covalent nor fully ionic but a mixture.

***4.6 Density of states for a two-dimensional electron gas** Consider a two-dimensional electron gas in which the electrons are restricted to move freely within a square area a^2 in the xy plane. Following the procedure in Section 4.5, show that the density of states $g(E)$ is constant (independent of energy).

Solution

For a two dimensional electron gas confined within a square region of sides a we have:

$$E = \frac{h^2}{8m_e a^2} (n_1^2 + n_2^2)$$

Only positive n_1 and n_2 are allowed. Each n_1 and n_2 combination is an orbital state. Define a new variable n as:

$$n^2 = n_1^2 + n_2^2$$

substitute:
$$E = \frac{h^2}{8m_e a^2} n^2$$

Let us consider how many states there are with energies less than E' . E' corresponds to $n \leq n'$.

$$E' = \frac{h^2}{8m_e a^2} n'^2$$

$$\therefore n' = \sqrt{\frac{8a^2 m_e E'}{h^2}}$$

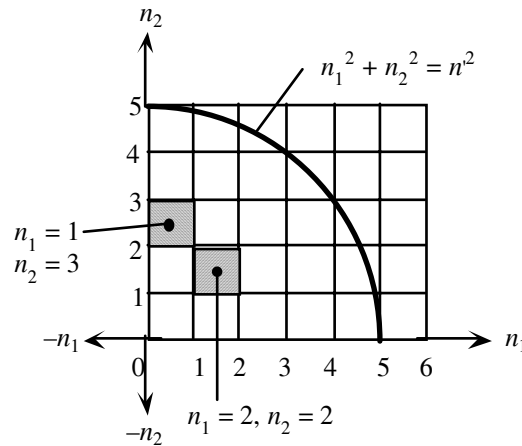


Figure 4Q6-1: Each state, or electron wavefunction in the crystal, can be represented by a box at n_1, n_2 .

Consider Figure 4Q6-1. All states within the quarter arc defined by n' have $E < E'$. The area of this quarter arc is the total number of orbital states. The total number of states, S , including spin is twice as many,

$$S = 2 \left(\frac{1}{4} \pi n'^2 \right) = 2 \left(\frac{1}{4} \pi \left[\sqrt{\frac{8a^2 m_e E'}{h^2}} \right]^2 \right)$$

$$\therefore S = \frac{4\pi a^2 m_e E'}{h^2}$$

The density of states g is defined as the number of states per unit area per unit energy.

$$\therefore g = \frac{1}{a^2} \frac{dS}{dE'} = \frac{1}{a^2} \frac{4\pi a^2 m_e}{h^2} = \frac{4\pi m_e}{h^2}$$

Thus, for a two dimensional gas, the density of states is constant.

4.7 Fermi energy of Cu The Fermi energy of electrons in copper at room temperature is 7.0 eV. The electron drift mobility in copper, from Hall effect measurements, is $33 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$.

- What is the speed v_F of conduction electrons with energies around E_F in copper? By how many times is this larger than the average thermal speed v_{thermal} of electrons, if they behaved like an ideal gas (Maxwell-Boltzmann statistics)? Why is v_F much larger than v_{thermal} ?
- What is the De Broglie wavelength of these electrons? Will the electrons get diffracted by the lattice planes in copper, given that interplanar separation in Cu = $2.09 \text{ \AA}$? (Solution guide: Diffraction of waves occurs when $2d\sin\theta = \lambda$, which is the Bragg condition. Find the relationship between λ and d that results in $\sin\theta > 1$ and hence no diffraction.)
- Calculate the mean free path of electrons at E_F and comment.

Solution

a. The Fermi speed v_F is given by:

$$E_F = \frac{1}{2} m_e v_F^2$$

$$\therefore v_F = \sqrt{2 \frac{E_F}{m_e}} = \sqrt{2 \frac{(7 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})}{(9.109 \times 10^{-31} \text{ kg})}}$$

$$\therefore v_F = 1.57 \times 10^6 \text{ m/s}$$

Maxwell - Boltzmann statistics predicts an effective velocity (rms velocity) which is called “thermal velocity” given by (assume room temperature $T = 20 \text{ }^\circ\text{C} = 293 \text{ K}$):

$$\frac{1}{2} m_e v_{\text{thermal}}^2 = \frac{3}{2} kT$$

$$\therefore v_{\text{thermal}} = \sqrt{\frac{3Tk}{m_e}} = \sqrt{\frac{3(293 \text{ K})(1.381 \times 10^{-23} \text{ J/K})}{(9.109 \times 10^{-31} \text{ kg})}}$$

$$\therefore v_{\text{thermal}} = 1.15 \times 10^5 \text{ m/s}$$

Comparing the two values:

$$\text{Ratio} = v_F/v_{\text{thermal}} = 13.7$$

v_F is about 14 times greater than v_{thermal} . This is because v_{thermal} assumes that electrons do not interact and obey Maxwell-Boltzmann statistics ($E_{av} = 3/2 kT$). However, in a metal there are many conduction electrons. They interact with the metal ions and obey the Pauli exclusion principle, i.e. Fermi-Dirac statistics. They extend to higher energies to avoid each other and thereby fulfill the Pauli exclusion principle.

b. The De Broglie wavelength is $\lambda = h/p$ where $p = m_e v_F$ is the momentum of the electrons.

$$\lambda = \frac{h}{m_e v_F} = \frac{6.626 \times 10^{-34} \text{ J s}}{(9.109 \times 10^{-31} \text{ kg})(1.57 \times 10^6 \text{ m/s})}$$

$$\therefore \lambda = 4.63 \times 10^{-10} \text{ m or } 4.63 \text{ \AA}$$

The interplanar separation, d , is given as $2.09 \text{ \AA}$. The diffraction condition is:

$$\lambda = 2d \sin \theta$$

$$\therefore \sin \theta = \frac{1}{2} \frac{\lambda}{d} = \frac{1}{2} \frac{(4.63 \text{ \AA})}{(2.09 \text{ \AA})} = 1.11$$

Since this is greater than 1, and $\sin \theta$ cannot be greater than 1, the electrons will not be diffracted.

c. The drift mobility is related to the mean scattering time τ by:

$$\tau = \frac{\mu m_e}{e} = \frac{(33 \times 10^{-4} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1})(9.109 \times 10^{-31} \text{ kg})}{1.602 \times 10^{-19} \text{ C}} = 1.876 \times 10^{-14} \text{ s}$$

The mean free path, l_F , of electrons with speed, v_F is:

$$l_F = v_F \tau = (1.57 \times 10^6 \text{ m/s})(1.876 \times 10^{-14} \text{ s}) = 2.95 \times 10^{-8} \text{ m or } 295 \text{ \AA}$$

The mean free path of those electrons with effective speeds v_e (close to mean speed) can be found as follows (E_F has little change with temperature, therefore $E_F \approx E_{FO}$):

$$\frac{1}{2} m_e v_e^2 = \frac{3}{5} E_{FO} = \frac{3}{5} E_F$$

$$\therefore v_e = \sqrt{\frac{6 E_F}{5 m_e}} = \sqrt{\frac{6 (7.0 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})}{(9.109 \times 10^{-31} \text{ kg})}} = 1.215 \times 10^6 \text{ m/s}$$

$$\therefore l_e = v_e \tau = (1.215 \times 10^6 \text{ m/s})(1.876 \times 10^{-14} \text{ s}) = 2.28 \times 10^{-8} \text{ m or } 228 \text{ \AA}$$

4.8 Free electron model, Fermi energy, and density of states Na and Au both are valency I metals; that is, each atom donates one electron to the sea of conduction electrons. Calculate the Fermi energy (in eV) of each at 300 K and 0 K. Calculate the mean speed of all the conduction electrons and also the speed of electrons at E_F for each metal. Calculate the density of states as states per eV cm^{-3} at the Fermi energy and also at the center of the band, to be taken at $(E_F + \Phi)/2$. (See Table 4.1 for Φ)

Solution

Since Na and Au are valency I metal, their electron concentrations, n are then the atomic concentrations multiplied by the group number, or:

$$n_{\text{Na}} = (\text{Valency}) \frac{N_A d_{\text{Na}}}{M_{\text{Na}}} = (1) \frac{(6.022 \times 10^{23} \text{ mol}^{-1})(968 \text{ kgm}^{-3})}{23 \times 10^{-3} \text{ kg/mol}} = 2.53 \times 10^{28} \text{ m}^{-3}$$

$$n_{\text{Au}} = (\text{Valency}) \frac{N_A d_{\text{Au}}}{M_{\text{Au}}} = (1) \frac{(6.022 \times 10^{23} \text{ mol}^{-1})(19300 \text{ kgm}^{-3})}{197 \times 10^{-3} \text{ kg/mol}} = 5.9 \times 10^{28} \text{ m}^{-3}$$

At 0 K,

$$E_{F0}(\text{Na}) = \frac{h^2}{8m_e} \left(\frac{3n}{\pi} \right)^{\frac{2}{3}} = \frac{(6.626 \times 10^{-34} \text{ Js})^2}{8 \times 9.1 \times 10^{-31} \text{ Kg}} \left(\frac{3 \times 2.53 \times 10^{28} \text{ m}^{-3}}{\pi} \right)^{2/3}$$

$$= 5.04 \times 10^{-19} \text{ J or } 3.15 \text{ eV}$$

$$E_{F0}(\text{Au}) = \frac{h^2}{8m_e} \left(\frac{3n}{\pi} \right)^{\frac{2}{3}} = \frac{(6.626 \times 10^{-34} \text{ Js})^2}{8 \times 9.1 \times 10^{-31} \text{ Kg}} \left(\frac{3 \times 5.9 \times 10^{28} \text{ m}^{-3}}{\pi} \right)^{2/3}$$

$$= 8.863 \times 10^{-19} \text{ J or } 5.54 \text{ eV}$$

At 300 K,

$$E_F(\text{Na}) = E_{F0}(\text{Na}) \left[1 - \frac{\pi^2}{12} \left(\frac{kT}{E_{F0}(\text{Na})} \right)^2 \right] = 3.15 \text{ eV} \left[1 - \frac{\pi^2}{12} \left(\frac{0.02585 \text{ eV}}{3.15 \text{ eV}} \right)^2 \right]$$

$$\therefore E_F(\text{Na}) = 3.15 \text{ eV}$$

$$E_F(\text{Au}) = E_{F0}(\text{Au}) \left[1 - \frac{\pi^2}{12} \left(\frac{kT}{E_{F0}(\text{Au})} \right)^2 \right] = 5.54 \text{ eV} \left[1 - \frac{\pi^2}{12} \left(\frac{0.02585 \text{ eV}}{5.54 \text{ eV}} \right)^2 \right]$$

$$\therefore E_F(\text{Au}) = 5.54 \text{ eV}$$

Mean speed of conduction electrons:

$$\frac{1}{2} m_e v_e^2 = E_{av} = \frac{3}{5} E_0$$

$$\therefore v_e = (6E_{F0} / 5m_e)^{1/2}$$

$$\therefore v_e(\text{Na}) = (6E_{F0}(\text{Na})/5m_e)^{1/2} = ((6 \times 3.15 \times 1.6 \times 10^{-19} \text{ J}) / (5 \times 9.1 \times 10^{-31} \text{ kg}))^{1/2}$$

$$\therefore v_e(\text{Na}) = 8.15 \times 10^5 \text{ ms}^{-1}$$

and $v_e(\text{Au}) = (6E_{F0}(\text{Au})/5m_e)^{1/2} = ((6 \times 5.54 \times 1.6 \times 10^{-19} \text{ J}) / (5 \times 9.1 \times 10^{-31} \text{ kg}))^{1/2}$

$$\therefore v_e(\text{Au}) = 1.08 \times 10^6 \text{ ms}^{-1}$$

Speed at E_F :

$$\frac{1}{2} m_e v_e^2 = E_{F0}$$

$$\therefore v_e = (2E_{F0} / m_e)^{1/2}$$

$$\therefore v_e(\text{Na}) = (2E_{F0}(\text{Na})/m_e)^{1/2} = ((2 \times 3.15 \times 1.6 \times 10^{-19} \text{ J}) / (9.1 \times 10^{-31} \text{ kg}))^{1/2}$$

$$\therefore v_e(\text{Na}) = 1.05 \times 10^6 \text{ ms}^{-1}$$

and $v_e(\text{Au}) = (2E_{F0}(\text{Au})/m_e)^{1/2} = ((2 \times 5.54 \times 1.6 \times 10^{-19} \text{ J}) / (9.1 \times 10^{-31} \text{ kg}))^{1/2}$

$$\therefore v_e(\text{Au}) = 1.4 \times 10^6 \text{ ms}^{-1}$$

The density of states is given by

$$g(E) = (8\pi 2^{1/2}) \left(\frac{m_e}{h^2} \right)^{3/2} E^{1/2}$$

For Na,
$$g_{E_F}(\text{Na}) = (8\pi 2^{1/2}) \left[\frac{9.1 \times 10^{-31} \text{ kg}}{6.626 \times 10^{-34} \text{ Js}} \right]^{3/2} (3.15 \times 1.6 \times 10^{-19} \text{ J})^{1/2}$$

$$\therefore g_{E_F}(\text{Na}) = 7.54 \times 10^{46} \text{ m}^{-3} \text{ J}^{-1} = (7.54 \times 10^{46} \text{ m}^{-3} \text{ J}^{-1})(10^{-6} \text{ m}^3 \text{ cm}^{-3})(1.6 \times 10^{-19} \text{ J eV}^{-1})$$

$$= \mathbf{1.2 \times 10^{22} \text{ cm}^{-3} \text{ eV}^{-1}}$$

$$(E_F + \Phi)/2 = (3.15 \text{ eV} + 2.75 \text{ eV})/2 = 2.95 \text{ eV}$$

$$\therefore g_{\text{center}}(\text{Na}) = (8\pi 2^{1/2}) \left[\frac{9.1 \times 10^{-31} \text{ kg}}{6.626 \times 10^{-34} \text{ Js}} \right]^{3/2} (2.95 \times 1.6 \times 10^{-19} \text{ J})^{1/2}$$

$$\therefore g_{\text{center}}(\text{Na}) = 7.3 \times 10^{46} \text{ m}^{-3} \text{ J}^{-1} = (7.3 \times 10^{46} \text{ m}^{-3} \text{ J}^{-1})(10^{-6} \text{ m}^3 \text{ cm}^{-3})(1.6 \times 10^{-19} \text{ J eV}^{-1})$$

$$= \mathbf{1.17 \times 10^{22} \text{ cm}^{-3} \text{ eV}^{-1}}$$

For Au,
$$g_{E_F}(\text{Au}) = (8\pi 2^{1/2}) \left[\frac{9.1 \times 10^{-31} \text{ kg}}{6.626 \times 10^{-34} \text{ Js}} \right]^{3/2} (5.54 \times 1.6 \times 10^{-19} \text{ J})^{1/2}$$

$$\therefore g_{E_F}(\text{Au}) = 10 \times 10^{46} \text{ m}^{-3} \text{ J}^{-1} = (10 \times 10^{46} \text{ m}^{-3} \text{ J}^{-1})(10^{-6} \text{ m}^3 \text{ cm}^{-3})(1.6 \times 10^{-19} \text{ J eV}^{-1})$$

$$= \mathbf{1.6 \times 10^{22} \text{ cm}^{-3} \text{ eV}^{-1}}$$

$$(E_F + \Phi)/2 = (5.54 \text{ eV} + 5.1 \text{ eV})/2 = 5.32 \text{ eV}$$

$$\therefore g_{\text{center}}(\text{Au}) = (8\pi 2^{1/2}) \left[\frac{9.1 \times 10^{-31} \text{ kg}}{6.626 \times 10^{-34} \text{ Js}} \right]^{3/2} (5.32 \times 1.6 \times 10^{-19} \text{ J})^{1/2}$$

$$\therefore g_{\text{center}}(\text{Au}) = 9.8 \times 10^{46} \text{ m}^{-3} \text{ J}^{-1} = (9.8 \times 10^{46} \text{ m}^{-3} \text{ J}^{-1})(10^{-6} \text{ m}^3 \text{ cm}^{-3})(1.6 \times 10^{-19} \text{ J eV}^{-1})$$

$$= \mathbf{1.568 \times 10^{22} \text{ cm}^{-3} \text{ eV}^{-1}}$$

4.9 Fermi energy and electron concentration Consider the metals in Table 4.8 from groups I, II and III in the Periodic Table. Calculate the Fermi energies at absolute zero, and compare the values with the experimental values. What is your conclusion?

Table 4.8

Metal	Group	M_{at}	Density (g cm^{-3})	$E_F(\text{eV})$ [Calculated]	$E_F(\text{eV})$ [Experiment]
Cu	I	63.55	8.96	—	6.5
Zn	II	65.38	7.14	—	11.0
Al	III	27	2.70	—	11.8

Solution

Since Cu is in group I, its valency is also 1. The electron concentration n is then the atomic concentration multiplied by the group number, or:

$$n = (\text{Valency}) \frac{N_A D}{M_{at}} = (1) \frac{(6.022 \times 10^{23} \text{ mol}^{-1})(8.96 \times 10^3 \text{ kg/m}^3)}{63.55 \times 10^{-3} \text{ kg/mol}} = 8.490 \times 10^{28} \text{ m}^{-3}$$

Using Equation 4.22:

$$E_{FO} = \frac{h^2}{8m_e} \left(\frac{3n}{\pi} \right)^{\frac{2}{3}} \left(\frac{1}{q} \right)$$

$$\therefore E_{FO} = \frac{(6.626 \times 10^{-34} \text{ J s})^2}{8(9.109 \times 10^{-31} \text{ kg})} \left(\frac{3(8.490 \times 10^{28} \text{ m}^{-3})}{\pi} \right)^{\frac{2}{3}} \left(\frac{1}{1.602 \times 10^{-19} \text{ J/eV}} \right)$$

$$\therefore E_{FO} = 7.04 \text{ eV}$$

Comparing with the experimental value:

$$|\% \text{ difference}| = \left| \frac{7.04 \text{ eV} - 6.5 \text{ eV}}{6.5 \text{ eV}} \right| \times 100\% = 8.31\%$$

E_{FO} can be calculated for Zn and Al in the same way (remember to take into account the different valencies). The values are summarized in the following table and it can be seen that calculated values are close to experimental values:

Metal	$n \text{ (m}^{-3}) (\times 10^{28})$	$E_{FO} \text{ (eV)}$ (calculated)	$E_{FO} \text{ (eV)}$ (experimental)	% Difference
Cu	8.490	7.04	6.5	8.31
Zn	13.15	9.43	11.0	14.3
Al	18.07	11.7	11.8	0.847

Table 4Q9-1: Summarized values for Fermi energy at absolute zero temperature.

4.10 Temperature dependence of the Fermi energy

- Given that the Fermi energy for Cu is 7.0 eV at absolute zero, calculate the E_F at 300 K. What is the percentage change in E_F and what is your conclusion?
- Given the Fermi energy for Cu at absolute zero, calculate the average energy and mean speed per conduction electron at absolute zero and 300 K, and comment.

Solution

a. The Fermi energy in eV at 0 K is given as 7.0 eV. The temperature dependence of E_F is given by Equation 4.23. Remember that E_{FO} is given in eV.

$$E_F = E_{FO} \left(1 - \frac{\pi^2}{12} \left[\frac{kT}{E_{FO}} \right]^2 \right)$$

$$\therefore E_F = (7.0 \text{ eV}) \left(1 - \frac{\pi^2}{12} \left[\frac{(1.381 \times 10^{-23} \text{ J/K})(300 \text{ K})}{(7.0 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})} \right]^2 \right) = \mathbf{6.999921 \text{ eV}}$$

$$\therefore |\% \text{ difference}| = \left| \frac{6.999921 \text{ eV} - 7.0 \text{ eV}}{7.0 \text{ eV}} \right| \times 100\% = \mathbf{0.00129\%}$$

This is a very small change. The Fermi energy appears to be almost unaffected by temperature.

b. The average energy per electron at 0 K is:

$$E_{av}(0 \text{ K}) = 3/5 (E_{FO}) = \mathbf{4.2 \text{ eV}}$$

The average energy at 300 K can be calculated from Equation 4.26:

$$E_{av}(T) = \frac{3}{5} E_{FO} \left(1 + \frac{5\pi^2}{12} \left[\frac{kT}{E_{FO}} \right]^2 \right)$$

$$\therefore E_{av}(300 \text{ K}) = \frac{3}{5} (7.0 \text{ eV}) \left(1 + \frac{5\pi^2}{12} \left[\frac{(1.381 \times 10^{-23} \text{ J/K})(300 \text{ K})}{(7.0 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})} \right]^2 \right)$$

$$\therefore E_{av}(300 \text{ K}) = \mathbf{4.200236 \text{ eV}}$$

This is a very small change.

Assume that the mean speed will be close to the effective speed v_e . Effective speed at absolute zero is denoted as v_{eo} , and is given by:

$$E_{av}(0 \text{ K}) \times q = \frac{1}{2} m_e v_{eo}^2$$

$$\therefore v_{eo} = \sqrt{2 \frac{qE_{av}(0 \text{ K})}{m_e}} = \sqrt{2 \frac{(1.602 \times 10^{-19} \text{ J/eV})(4.2 \text{ eV})}{(9.109 \times 10^{-31} \text{ kg})}} = \mathbf{1215446 \text{ m/s}}$$

At 300 K, the effective speed is v_e :

$$v_e = \sqrt{2 \frac{qE_{av}(300 \text{ K})}{m_e}} = \sqrt{2 \frac{(1.602 \times 10^{-19} \text{ J/eV})(4.200236 \text{ eV})}{(9.109 \times 10^{-31} \text{ kg})}} = \mathbf{1215480 \text{ m/s}}$$

Comparing the values:

$$|\% \text{ difference}| = \left| \frac{1215480 \text{ m/s} - 1215446 \text{ m/s}}{1215446 \text{ m/s}} \right| \times 100\% = \mathbf{0.002797\%}$$

The mean speed has increased by a negligible amount (0.003%) from 0 K to 300 K.

Note: For thermal conduction this tiny increase in the velocity is sufficient to transport energy from hot regions to cold regions. This very small increase in the velocity also allows the electrons to diffuse from hot to cold regions giving rise to the Seebeck effect.

4.11 X-ray emission spectrum from sodium Structure of Na atom is $[\text{Ne}]3s^1$. Figure 4.59a shows formation of the $3s$ and $3p$ energy bands in Na as a function of internuclear separation. Figure 4.59b shows the x-ray emission spectrum (called the L -band) from crystalline sodium in the soft x-ray range as explained in Example 4.6.

- From Figure 4.59a, estimate the nearest neighbor equilibrium separation between Na atoms in the crystal if some electrons in the $3s$ band spill over into the states in the $3p$ band.
- Explain the origin of X-ray emission band in Figure 4.59b and the reason for calling it the L -band.
- What is the Fermi energy of the electrons in Na from Figure 4.59b?
- Taking the valency of Na to be I, what is the expected Fermi energy and how does it compare with that in (c)?

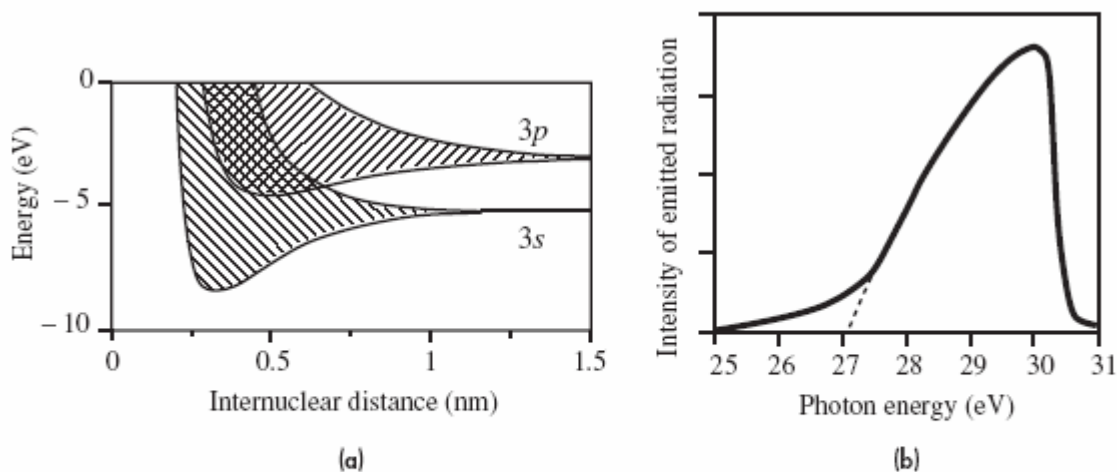


Figure 4.59

(a) Energy band formation in sodium.

(b) L -emission band of X-rays from sodium.

| SOURCE: (b) Data extracted from W. M. Cadt and D. H. Tomboulion, *Phys. Rev.*, **59**, 1941, p. 381.

Solution

a. As represented in Figure 4.59a, the estimated nearest interatomic separation is near the minimum of the $3s$ band, or slightly above it, **0.36 - 0.37 nm**.

b. When an electron, for some reason, is leaving the closed inner L -shell of an atom, an empty state is created there. An electron from the energy band of the metal drops into the L -shell to fill the vacancy and emits a soft X-ray photon in this process. The spectrum of this X-ray emission from metal involves a range of energies, corresponding to transitions from the bottom of the band and from the Fermi level to the L -shell. So all the X-ray photons emitted from the electrons during their transitions to the L -shell will have energies lying in that range (band) and because all of them are emitted due to a transition to the L -shell, this X-ray band is called L -band.

c. As explained in part **b**, the width of the X-ray band corresponds to the distance from the bottom of the energy band to the Fermi level. As shown in Figure 4.59b, the position of the Fermi level with respect to the bottom of the energy band is approximately **3.2 eV**.

d. Theoretically, the position of the Fermi level is given by Equation 4.23. Since the temperature dependence of the Fermi level is really very weak, we can neglect it. Then $E_F(T) = E_{F0}$ and using Equation 4.22, we receive

$$E_F = \frac{h^2}{8m_e} \left(\frac{3n}{\pi} \right)^{\frac{2}{3}}$$

The electron concentration in Na can be calculated, assuming that each Na atom donates exactly one electron to the crystal. Taking from Appendix B the density of Na d (0.97 g cm^{-3}) and its atomic mass M_{at} (22.99 g mol^{-1}), we receive

$$n = \frac{dN_A}{M_{at}} = \frac{(0.97 \text{ g cm}^{-3})(6.022 \times 10^{23} \text{ mol}^{-1})}{(22.99 \text{ g mol}^{-1})} = 2.54 \times 10^{22} \text{ cm}^{-3} = 2.54 \times 10^{28} \text{ m}^{-3}$$

Thus the Fermi level is

$$E_F = \frac{h^2}{8m_e} \left(\frac{3n}{\pi} \right)^{\frac{2}{3}} = \frac{(6.626 \times 10^{-34} \text{ J s})^2}{8(9.1 \times 10^{-31} \text{ kg})} \left[\frac{3(2.54 \times 10^{28} \text{ m}^{-3})}{\pi} \right]^{\frac{2}{3}} = 5.05 \times 10^{-19} \text{ J}$$

or $E_F = 3.15 \text{ eV}$,

which is very close to the value obtained from the X-ray spectrum.

4.12 Conductivity of metals in the free electron model Consider the general expression for the conductivity of metals in terms of the density of states $g(E_F)$ at E_F given by

$$\sigma = \frac{1}{3} e^2 v_F^2 \tau g(E_F)$$

Show that within the free electron theory, this reduces to $\sigma = e^2 n \tau / m_e$, the Drude expression.

Solution

At Fermi energy

$$\frac{1}{2} m_e v_F^2 = E_F$$

$$\therefore v_e = (2E_F / m_e)^{1/2}$$

The density of states

$$g(E_F) = (8\pi 2^{1/2}) \left(\frac{m_e}{h^2} \right)^{3/2} E_F^{1/2}$$

And the Fermi energy is

$$E_F = \frac{h^2}{8m_e} \left(\frac{3n}{\pi} \right)^{\frac{2}{3}}$$

Now,
$$\sigma = \frac{1}{3} e^2 v_F^2 \tau g(E_F)$$

Putting the value of v_F and $g(E_F)$ we get

$$\begin{aligned}\sigma &= \frac{1}{3} e^2 \frac{2E_F}{m_e} \tau \times \left(8\pi 2^{1/2}\right) \left(\frac{m_e}{h^2}\right)^{3/2} E_F^{1/2} \\ &= \frac{2}{3} \frac{e^2 \tau}{m_e} \times \left(8\pi 2^{1/2}\right) \left(\frac{m_e}{h^2}\right)^{3/2} E_F^{3/2} \\ &= \frac{2}{3} \frac{e^2 \tau}{m_e} \times \left(8\pi 2^{1/2}\right) \left(\frac{m_e}{h^2}\right)^{3/2} \left(\frac{h^2}{8m_e}\right)^{3/2} \left(\frac{3n}{\pi}\right)\end{aligned}$$

$$\therefore \sigma = \frac{e^2 n \tau}{m_e}$$

4.13 Mean free path of conduction electrons in a metal Show that within the free electron theory, the mean free path ℓ and conductivity σ are related by

$$\sigma = \frac{e^2}{3^{1/3} \pi^{2/3} \hbar} \ell n^{2/3} = 7.87 \times 10^{-5} \ell n^{2/3}$$

Calculate ℓ for Cu and Au, given each metal's resistivity of 17 nΩ m and 22 nΩ m, respectively, and that each has a valency of I. We are used to seeing $\sigma \propto n$. Can you explain why $\sigma \propto n^{2/3}$?

Solution

Mean free path, $\ell = v_F \tau$

The Fermi energy is,

$$E_F = \frac{1}{2} m_e v_F^2$$

$$\therefore v_e = (2E_F / m_e)^{1/2}$$

The density of states

$$g(E_F) = \left(8\pi 2^{1/2}\right) \left(\frac{m_e}{h^2}\right)^{3/2} E_F^{1/2}$$

And the Fermi energy

$$E_F = \frac{h^2}{8m_e} \left(\frac{3n}{\pi}\right)^{2/3}$$

Now,

$$\begin{aligned}
 \sigma &= \frac{1}{3} e^2 v_F^2 \tau g(E_F) \\
 &= \frac{1}{3} e^2 (v_F \tau) g(E_F) v_F \\
 &= \frac{1}{3} e^2 \ell \times (8\pi 2^{1/2}) \left(\frac{m_e}{h^2} \right)^{3/2} E_F^{1/2} \times (2E_F / m_e)^{1/2} \\
 &= \frac{1}{3} e^2 \ell \times (8\pi 2^{1/2}) \left(\frac{m_e}{h^2} \right)^{3/2} \times \frac{2^{1/2}}{m_e^{1/2}} \times E_F \\
 &= \frac{1}{3} e^2 \ell \times (8 \times 2\pi) \frac{m_e}{h^3} \times \frac{h^2}{8m_e} \frac{n^{2/3}}{\pi^{2/3}} \\
 \therefore \sigma &= \frac{e^2}{3^{1/3} \pi^{2/3} \hbar} \ell n^{2/3} = \frac{(1.602 \times 10^{-19})^2}{3^{1/3} \pi^{2/3} \times 1.055 \times 10^{-34}} \ell n^{2/3} = 7.87 \times 10^{-5} \ell n^{2/3} \\
 \therefore \ell &= \frac{\sigma}{7.87 \times 10^{-5} n^{2/3}} \\
 \therefore \ell_{\text{Cu}} &= \frac{1}{(17 \times 10^{-9}) 7.87 \times 10^{-5} (8.45 \times 10^{28})^{2/3}} = 3.88 \times 10^{-8} \text{ m or } 38.8 \text{ nm} \\
 \text{and } \ell_{\text{Au}} &= \frac{1}{(22 \times 10^{-9}) 7.87 \times 10^{-5} (5.9 \times 10^{28})^{2/3}} = 3.81 \times 10^{-8} \text{ m or } 38.1 \text{ nm}
 \end{aligned}$$

The statement $\sigma \propto n$ would be true if the drift mobility and hence the mean free path ℓ did not change with n at all. ℓ depends on v_F and E_F and hence v_F depends on n .

***4.14 Low-temperature heat capacity of metals** The heat capacity of conduction electrons in a metal is proportional to the temperature. The overall heat capacity of a metal is determined by the lattice heat capacity, except at the lowest temperatures. If δE_t is the increase in the total energy of the conduction electrons (per unit volume) and δT is the increase in the temperature of the metal as a result of heat addition, E_t has been calculated as follows:

$$E_t = \int_0^\infty E g(E) f(E) dE = E_t(0) + \left(\frac{\pi^2}{4} \right) \frac{n(kT)^2}{E_{FO}}$$

where $E_t(0)$ is the total energy per unit volume at 0 K, n is the concentration of conduction electrons, and E_{FO} is the Fermi energy at 0 K. Show that the heat capacity per unit volume due to conduction electrons in the free electron model of metals is

$$C_e = \frac{\pi^2}{2} \left(\frac{nk^2}{E_{FO}} \right) T = \gamma T$$

where $\gamma = (\pi^2/2)(nk^2/E_{FO})$. Calculate C_e for Cu, and then using the Debye equation for the lattice heat capacity, find C_v for Cu at 10 K. Compare the two values and comment. What is the comparison at

room temperature? (Note: $C_{\text{volume}} = C_{\text{molar}}(\rho/M_{\text{at}})$, where ρ is the density in g cm^{-3} , C_{volume} is in $\text{J K}^{-1}\text{cm}^{-3}$, and M_{at} is the atomic mass in g mol^{-1} .)

Solution

$$C_e = \frac{dE_t}{dT} = \frac{d}{dT} \left(E_t(0) + \left(\frac{\pi^2}{4} \right) \frac{n(kT)^2}{E_{FO}} \right) = 0 + \frac{\pi^2}{2} \left(\frac{nk^2}{E_{FO}} \right) T = \gamma T$$

For Cu, $n = 8.45 \times 10^{28} \text{ m}^{-3}$ and $E_{FO} = 7 \times 1.6 \times 10^{-19} \text{ J}$

$$\therefore C_e = \frac{\pi^2}{2} \left(\frac{nk^2}{E_{FO}} \right) T = \frac{\pi^2}{2} \frac{(8.45 \times 10^{28} \text{ m}^{-3})(1.38 \times 10^{-23} \text{ JK}^{-1})^2}{7 \times 1.6 \times 10^{-19} \text{ J}} (10 \text{ K}) = \mathbf{709 \text{ JK}^{-1}\text{m}^{-3}}$$

$$\therefore C_e = \mathbf{7.09 \times 10^{-4} \text{ J K}^{-1}\text{cm}^{-3}}$$

For Cu, Debye temperature, $T_D = 315 \text{ K}$

$$\therefore T/T_D = 0.0317$$

$$\therefore C_m(\text{Phonons}) = 9R(0.0317)^3 \int_0^{31.5} \frac{x^4 e^x dx}{(e^x - 1)^2}$$

Solving this equation using math software, we get

$$C_m(\text{Phonons}) = 0.0622 \text{ JK}^{-1} \text{ mol}^{-1}$$

$$\therefore C_v(\text{Phonons}) = C_m(\text{Phonons}) \times (\rho/M_{\text{at}}) \\ = (0.0622 \text{ J K}^{-1} \text{ mol}^{-1})(8.96 \text{ g cm}^{-3}/63.5 \text{ g mol}^{-1}) = \mathbf{8.77 \times 10^{-3} \text{ J K}^{-1}\text{cm}^{-3}}$$

Comment: At 10 K, the heat capacity C_e due to electrons is roughly 8% of $C_v(\text{Phonons})$ due to lattice vibrations (phonons) only. C_e is not quite negligible. If we decrease the temperature further, C_e will eventually be larger than $C_v(\text{Phonons})$.

At room temperature, $T = 300 \text{ K}$

$$\therefore C_e = \frac{\pi^2}{2} \left(\frac{nk^2}{E_{FO}} \right) T = \frac{\pi^2}{2} \frac{(8.45 \times 10^{28} \text{ m}^{-3})(1.38 \times 10^{-23} \text{ JK}^{-1})^2}{7 \times 1.6 \times 10^{-19} \text{ J}} (300 \text{ K}) = \mathbf{21.3 \times 10^3 \text{ JK}^{-1}\text{m}^{-3}}$$

$$\therefore C_e = \mathbf{21.3 \times 10^{-3} \text{ JK}^{-1}\text{cm}^{-3}}$$

$$\text{And } T/T_D = 0.95$$

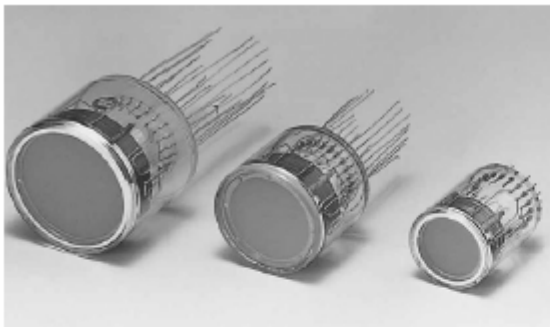
From the Fig. 4.45, we get

$$C_m(\text{Phonons}) = 23.5 \text{ JK}^{-1} \text{ mol}^{-1}$$

$$\therefore C_v(\text{Phonons}) = C_m(\text{Phonons}) \times (\rho/M_{\text{at}}) = (23.5 \text{ JK}^{-1} \text{ mol}^{-1})(8.96 \text{ g cm}^{-3}/63.5 \text{ g mol}^{-1}) \\ = \mathbf{3.316 \text{ JK}^{-1}\text{cm}^{-3}}$$

Comment: At 300 K, the heat capacity C_e due to electrons is negligible (0.6 %) compared to $C_v(\text{Phonons})$ due to lattice vibrations (phonons) only.

***4.15 Secondary emission and photomultiplier tubes** When an energetic (high velocity) projectile electron collides with a material with a low work function, it can cause electron emission from the surface. This phenomenon is called **secondary emission**. It is fruitfully utilized in photomultiplier tubes as illustrated in Figure 4.60. The tube is evacuated and has a photocathode for receiving photons as a signal. An incoming photon causes photoemission of an electron from the photocathode material. The electron is then accelerated by a positive voltage applied to an electrode called a dynode which has a work function that easily allows secondary emission. When the accelerated electron strikes dynode D_1 , it can release several electrons. All these electrons, the original and the secondary electrons, are then accelerated by the more positive voltage applied to dynode D_2 . On impact with D_2 , further electrons are released by secondary emission. The secondary emission process continues at each dynode stage until the final electrode, called the anode, is reached whereupon all the electrons are collected which results in a signal. Typical applications for photomultiplier tubes are in X-ray and nuclear medical instruments



Photomultiplier tubes.
| SOURCE: Courtesy of Hamamatsu.

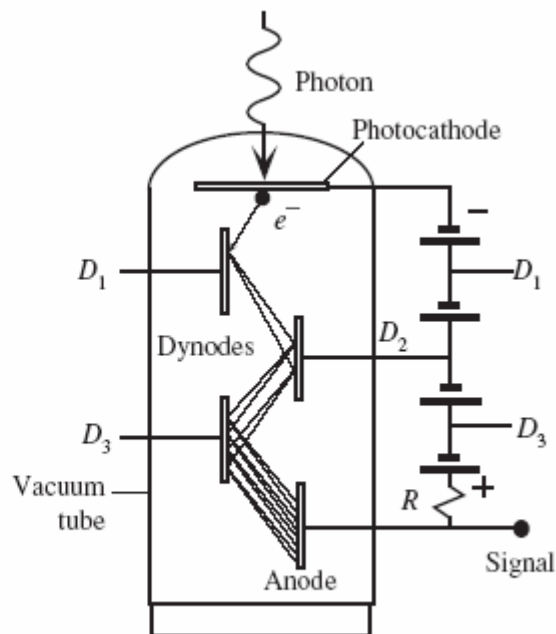


Figure 4.60 The photomultiplier tube.

(X-ray CT scanner, positron CT scanner, gamma camera, etc.), radiation measuring instruments (e.g., radon counter), X-ray diffractometers, and radiation measurement in high-energy physics research.

A particular photomultiplier tube has the following properties. The photocathode is made of a semiconductor-type material with $E_g \approx 1$ eV, an electron affinity χ of 0.4 eV, and a quantum efficiency of 20 percent at 400 nm. *Quantum efficiency* is defined as the number of photoemitted electrons per absorbed photon. The diameter of the photocathode is 18 mm. There are 10 dynode electrodes and an applied voltage of 1250 V between the photocathode and anode. Assume that this voltage is equally distributed among all the electrodes.

- What is the longest threshold wavelength for the phototube?
- What is the maximum kinetic energy of the emitted electron if the photocathode is illuminated with a 400 nm radiation?

- c. What is the emission current from the photocathode at 400 nm illumination per **unit intensity of radiation**?
- d. What is the *KE* of the electron as it strikes the first dynode electrode? It has been found that the tube has a gain of 10^6 electrons per incident photon. What is the average number of secondary electrons released at each dynode?

Solution

- a. For longest threshold wavelength,

$$hc / \lambda_{th} = E_g + \chi$$

$$\therefore \lambda_{th} = (6.626 \times 10^{-34} \text{ J s})(3 \times 10^8 \text{ ms}^{-1}) / (1.4 \times 1.6 \times 10^{-19} \text{ J}) = \mathbf{8.87 \times 10^{-7} \text{ m or } 887 \text{ nm}}$$

With 400 nm radiation,

$$E_{ph} = hc/\lambda = (4.13 \times 10^{-15} \text{ eV s})(3 \times 10^8 \text{ ms}^{-1}) / (400 \times 10^{-9} \text{ m}) = 3.1 \text{ eV}$$

- b. The excess energy over ($E_g + \chi$) goes as kinetic energy

$$\therefore KE_m = E_{ph} - (E_g + \chi) = 1.7 \text{ eV}$$

- c. The intensity, $I = 1 \text{ W m}^{-2}$.

The number of photons arriving per unit area per unit time at the photocathode is

$$\Gamma_{ph} = \frac{I}{E_{ph}} = \frac{(1 \text{ J s}^{-1} \text{ m}^{-2})}{(3.1 \text{ eV} \times 1.6 \times 10^{-19} \text{ J eV}^{-1})} = 2.02 \times 10^{18} \text{ s}^{-1} \text{ m}^{-2}$$

The current density is then simply

$$J = e\Gamma_{ph}QE = (1.6 \times 10^{-19} \text{ C})(2.02 \times 10^{18} \text{ s}^{-1} \text{ m}^{-2})(0.2) = 0.065 \text{ A m}^{-2}$$

$$\therefore \text{The emission current, } I = J \times A = 0.065 \text{ A m}^{-2} \times \pi(9 \times 10^{-3} \text{ m})^2 = \mathbf{1.654 \times 10^{-5} \text{ A or } 16.54 \text{ }\mu\text{A}}$$

- d. The potential difference between cathode and first anode is, $V_d = 1250\text{V}/11 = 113.64 \text{ V}$

$$\therefore KE_{D1} = KE_m + eV_d = 1.7 \text{ eV} + 113.64 \text{ eV} = 115.34 \text{ eV}$$

- e. Quantum efficiency is 20 percent. So, ejected electron is 0.2 per photon

$$\therefore \text{The gain of the tube} = 10^6 / 0.2 = 5 \times 10^6 \text{ per ejected electron from the cathode.}$$

and $\text{Average number of secondary electron released} = (5 \times 10^6)^{1/10} = \mathbf{4.68}$

4.16 Thermoelectric effects and E_F Consider a thermocouple pair that consists of gold and aluminum. One junction is at 100 °C and the other is at 0 °C. A voltmeter (with a very large input resistance) is inserted into the aluminum wire. Use the properties of Au and Al in Table 4.3 to estimate the emf registered by the voltmeter and identify the positive end.

Solution

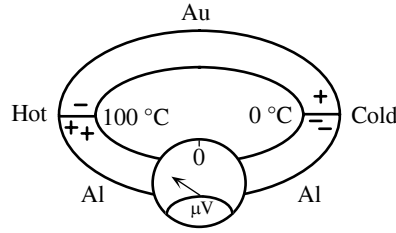


Figure 4Q16-1: The Al-Au thermocouple. The cold end is maintained at 0 °C which is the reference temperature. The other junction is used to sense the temperature. In this example it is heated to 100 °C.

We essentially have the arrangement shown above. For each metal there will be a voltage across it given by integrating the Seebeck coefficient. From the Mott-Jones equation:

$$\Delta V = \int_{T_0}^T S dT = \int_{T_0}^T -\frac{x\pi^2 k^2 T}{3eE_{FO}} dT = -\frac{x\pi^2 k^2}{6eE_{FO}} (T^2 - T_0^2)$$

The emf (V_{AB}) available is the difference in ΔV for the two metals labeled A ($= \text{Al}$) and B ($= \text{Au}$) so that

$$V_{AB} = \Delta V_A - \Delta V_B$$

where in this example, $T = 373 \text{ K}$ and $T_0 = 273 \text{ K}$. We can calculate E_{FAO} and E_{FBO} for each metal as in Example 4.9 by using

$$E_{FO} = \frac{h^2}{8m_e} \left(\frac{3n}{\pi} \right)^{2/3}$$

where n is the electron concentration.

$n = \text{atomic concentration } (n_{at}) \times \text{number of conduction electrons per atom}$

From the density d and atomic mass M_{at} , the atomic concentration of Al is:

$$n_{Al} = \frac{N_A d}{M_{at}} = \frac{(6.022 \times 10^{23} \text{ mol}^{-1})(2700 \text{ kg/m}^3)}{(0.027 \text{ kg/mol})} = 6.022 \times 10^{28} \text{ m}^{-3}$$

so that $n = 3n_{Al} = 1.807 \times 10^{29} \text{ m}^{-3}$

which leads to

$$E_{FAO} = \frac{h^2}{8m_e} \left(\frac{3n}{\pi} \right)^{2/3} = \frac{(6.626 \times 10^{-34})^2}{8(9.109 \times 10^{-31})} \left(\frac{3(1.807 \times 10^{29})}{\pi} \right)^{2/3}$$

i.e. $E_{FAO} = 1.867 \times 10^{-18} \text{ J}$ or 11.66 eV

Similarly for Au, we find $E_{FBO} = 5.527 \text{ eV}$.

Substituting x and E_F values for A (Al) and B (Au) we find,

$$\Delta V_A = -\frac{\pi^2 k^2 x_A}{6eE_{FAO}} (T^2 - T_0^2) = -188.4 \text{ } \mu\text{V}$$

and $\Delta V_B = -\frac{\pi^2 k^2 x_B}{6eE_{FBO}} (T^2 - T_0^2) = 211.3 \text{ } \mu\text{V}$

so that the magnitude of the voltage difference is

$$|V_{AB}| = |-188.3 \mu\text{V} - 1.3 \mu\text{V}| = 399.7 \mu\text{V}$$

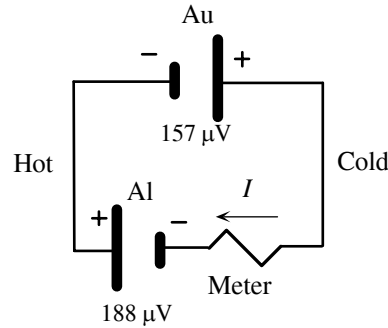


Figure 4Q16-2

To find which end is positive, we put in the resistance of the voltmeter and replace each metal by its emf and determine the direction of current flow as in the figure. For the particular circuit shown, the cold connected side of the voltmeter is positive.

4.17 The thermocouple equation Although inputting the measured emf for V in the thermocouple equation $V = a\Delta T + b(\Delta T)^2$ leads to a quadratic equation, which in principle can be solved for ΔT , in general ΔT is related to the measured emf via

$$\Delta T = a_1 V + a_2 V^2 + a_3 V^3 + \dots$$

with the coefficients a_1 , a_2 etc., determined for each pair of TCs. By carrying out a Taylor's expansion of TC equation, find the first two coefficients a_1 and a_2 . Using an emf table for the K-type thermocouple or Figure 4.33, evaluate a_1 and a_2 .

Solution

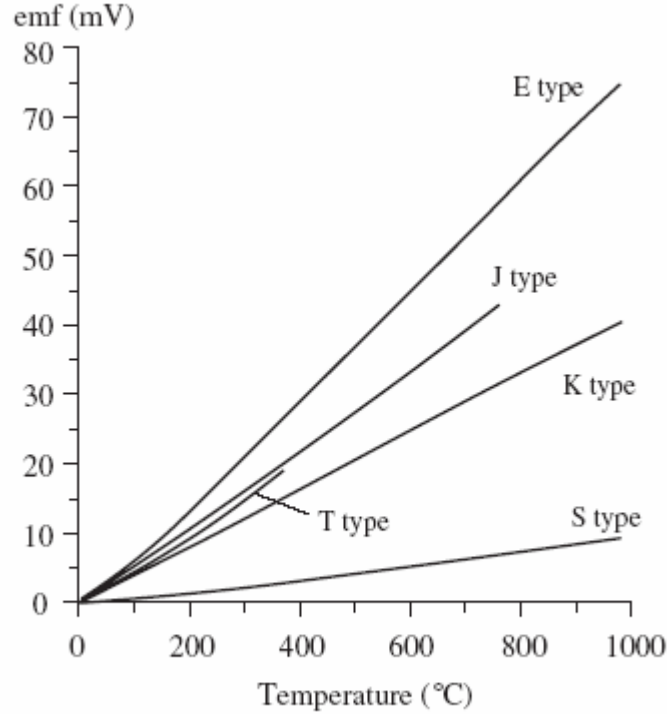


Figure 4.33: Output emf versus temperature (°C) for various thermocouples between 0 to 1000 °C.

From Example 4.11, the emf voltage (V) can be expressed:

$$V = \frac{\pi^2 k^2}{4e} \left(\frac{1}{E_{FAO}} - \frac{1}{E_{FBO}} \right) (T^2 - T_o^2)$$

Since we know that $\Delta T = T - T_o$, and therefore $T = \Delta T + T_o$, we can make the following substitution:

$$V = \frac{\pi^2 k^2}{4e} \left(\frac{1}{E_{FAO}} - \frac{1}{E_{FBO}} \right) ((\Delta T + T_o)^2 - T_o^2)$$

expanding,
$$V = \frac{\pi^2 k^2 T_o \Delta T}{2e E_{FAO}} - \frac{\pi^2 k^2 T_o \Delta T}{2e E_{FBO}} + \frac{\pi^2 k^2 (\Delta T)^2}{4e E_{FAO}} - \frac{\pi^2 k^2 (\Delta T)^2}{4e E_{FBO}}$$

factoring,
$$V = \frac{\pi^2 k^2 T_o}{2e} \left(\frac{1}{E_{FAO}} - \frac{1}{E_{FBO}} \right) \Delta T + \frac{\pi^2 k^2}{4e} \left(\frac{1}{E_{FAO}} - \frac{1}{E_{FBO}} \right) (\Delta T)^2$$

Upon inspection, it can be seen that this equation is in the form of the thermocouple equation, $V = a\Delta T + b\Delta T^2$, and therefore we know that the coefficients a and b are equal to:

$$a = \frac{\pi^2 k^2 T_o}{2e} \left(\frac{1}{E_{FAO}} - \frac{1}{E_{FBO}} \right) \quad b = \frac{\pi^2 k^2}{4e} \left(\frac{1}{E_{FAO}} - \frac{1}{E_{FBO}} \right)$$

Continuing with the thermocouple equation, we can rearrange it as follows to obtain a quadratic equation:

$$-b(\Delta T)^2 - a\Delta T + V = 0$$

This is a quadratic equation in ΔT . The solution is

$$\Delta T = \frac{a - \sqrt{a^2 + 4bV}}{-2b} = -\frac{a}{2b} + \frac{a\sqrt{1 + 4bV/a^2}}{2b}$$

$$\therefore \Delta T = -\frac{a}{2b} + \frac{a}{2b} \left(1 + 4bV/a^2\right)^{\frac{1}{2}}$$

$$\therefore \Delta T = -\frac{a}{2b} + \frac{a}{2b} \sum_{n=0}^{\infty} \binom{-1/2}{n} \left(1 + \frac{4bV}{a^2}\right)^n$$

Taylor expansion:

$$\Delta T = -\frac{a}{2b} + \frac{a}{2b} \left[1 + \frac{1}{2} \left(\frac{4bV}{a^2} \right) + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)}{2!} \left(\frac{4bV}{a^2} \right)^2 + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!} \left(\frac{4bV}{a^2} \right)^3 + \dots \right]$$

$$\therefore \Delta T = \frac{V}{a} - \frac{bV^2}{a^3} + 2\frac{b^2V^3}{a^5} + \dots$$

The positive root is not used because $V = 0$ must give $\Delta T = 0$, and with the positive root the $a/2b$ terms will not cancel out. This equation is of the form given in the question,

$$\Delta T = a_1V + a_2V^2 + a_3V^3 + \dots$$

such that $a_1 = \frac{1}{a}$ and $a_2 = \frac{-b}{a^3}$

$$\therefore a_1 = \frac{1}{a} = \frac{1}{\frac{\pi^2 k^2 T_o}{2e} \left(\frac{1}{E_{FAO}} - \frac{1}{E_{FBO}} \right)} = \frac{2eE_{FAO}E_{FBO}}{\pi^2 k^2 T_o (E_{FBO} - E_{FAO})}$$

and $a_2 = -\frac{b}{a^3} = -\frac{\frac{\pi^2 k^2}{4e} \left(\frac{1}{E_{FAO}} - \frac{1}{E_{FBO}} \right)}{\left[\frac{\pi^2 k^2 T_o}{2e} \left(\frac{1}{E_{FAO}} - \frac{1}{E_{FBO}} \right) \right]^3} = -\frac{2e^2 E_{FAO}^2 E_{FBO}^2}{\pi^4 k^4 T_o^3 (E_{FBO} - E_{FAO})^2}$

From Figure 4.33, at $\Delta T = 200^\circ\text{C}$, $V = 8\text{ mV}$, and at $\Delta T = 800^\circ\text{C}$, $V = 33\text{ mV}$. Using these values and neglecting the effect of a_3 , we have two simultaneous equations we can solve:

$$\Delta T = a_1V + a_2V^2$$

$$\Delta T = 200^\circ\text{C},$$

$$200^\circ\text{C} = a_1(8\text{ mV}) + a_2(8\text{ mV})^2$$

$$\therefore a_1 = (-(64\text{ mV}^2)a_2 + 200^\circ\text{C}) / (8\text{ mV})$$

$$\Delta T = 800^\circ\text{C},$$

$$800^\circ\text{C} = a_1(33\text{ mV}) + a_2(33\text{ mV})^2$$

substitute for a_1 ,

$$800^\circ\text{C} = [(-(64\text{ mV}^2)a_2 + 200^\circ\text{C}) / (8\text{ mV})](33\text{ mV}) + a_2(33\text{ mV})^2$$

isolate a_2 : $a_2 = -0.03030 \text{ }^\circ\text{C/mV}^2$

$\therefore 200 \text{ }^\circ\text{C} = a_1(8 \text{ mV}) + (-0.03030 \text{ }^\circ\text{C/mV}^2)(8 \text{ mV})^2$

$\therefore a_1 = 25.24 \text{ }^\circ\text{C/mV}$

We can check our calculation by calculating ΔT when $V = 20 \text{ mV}$ and comparing the value we obtain from examining Figure 4.33

4.18 Thermionic emission A vacuum tube is required to have a cathode operating at $800 \text{ }^\circ\text{C}$ and providing an emission (saturation) current of 10 A . What should be the surface area of the cathode for the two materials in Table 4.9? What should be the operating temperature for the Th on W cathode, if it is to have the same surface area as the oxide-coated cathode?

Table 4.9		
	$B_e (\text{A m}^{-2} \text{ K}^{-2})$	$\Phi (\text{eV})$
Th on W	3×10^4	2.6
Oxide coating	100	1

Solution

Operating temperature T is given as $800 \text{ }^\circ\text{C} = 1073 \text{ K}$ and emission current I is given as 10 A . The temperature and current of the tube are related to its area by Equation 4.44:

$$J = \frac{I}{A} = B_e T^2 \exp \left[\frac{-(\Phi - \beta_s \sqrt{E})}{kT} \right]$$

$$\therefore A = \frac{I}{B_e T^2 \exp \left[\frac{-(\Phi - \beta_s \sqrt{E})}{kT} \right]}$$

Assuming there is no assisting field emission, the area needed for Th on W is:

$$A = \frac{10 \text{ A}}{(3 \times 10^4 \text{ A m}^{-2} \text{ K}^{-2})(1073 \text{ K})^2 \exp \left[\frac{-(2.6 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})}{(1.381 \times 10^{-23} \text{ J K}^{-1})(1073 \text{ K})} \right]}$$

$$\therefore A = 467 \text{ m}^2 \text{ (large tube)}$$

For the oxide coating:

$$A = \frac{10 \text{ A}}{(100 \text{ A m}^{-2} \text{ K}^{-2})(1073 \text{ K})^2 \exp \left[\frac{-(1 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})}{(1.381 \times 10^{-23} \text{ J K}^{-1})(1073 \text{ K})} \right]}$$

$$\therefore A = 0.00431 \text{ m}^2 \quad \text{(small and practical tube)}$$

To find the temperature that the Th on W cathode would have to work at to have the same surface area as the oxide coated cathode, the area of the oxide cathode can be used in the current equation and the

temperature can be solved for. It is a more difficult equation, but can be solved through graphical methods.

$$I = AB_e T^2 \exp\left[\frac{-\Phi}{kT}\right]$$

The plot of thermionic emission current I versus temperature T is shown below, with $A = 0.00431 \text{ m}^2$, $B_e = 3 \times 10^{-4} \text{ A m}^{-2} \text{ K}^{-2}$, and $\Phi = (2.6 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})$.

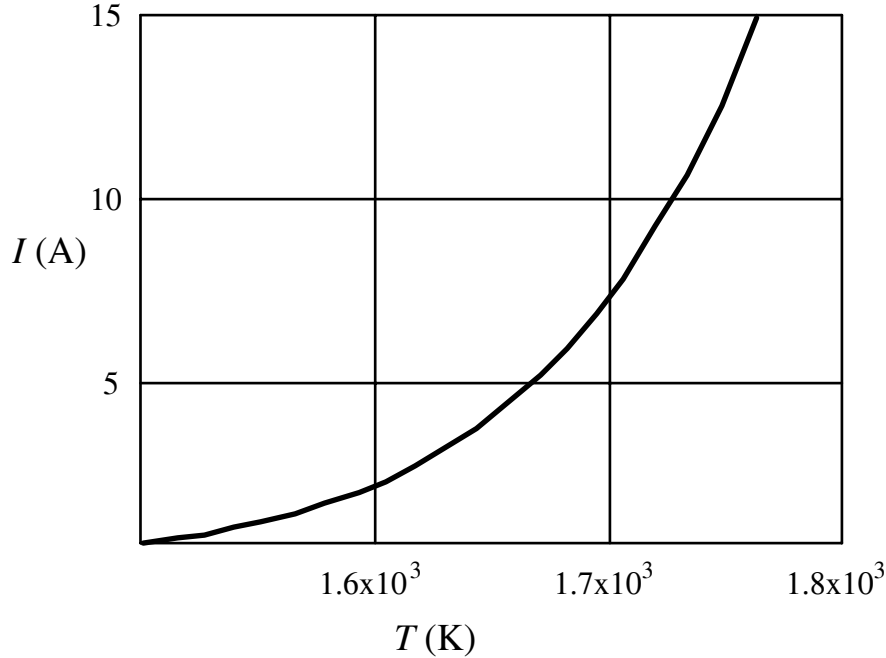


Figure 4Q18-1: Behavior of current versus temperature for the Th on W cathode.

From the graph, it appears that at 10 A of current the cathode will be operating at a temperature of $T = 1725 \text{ K}$ or $1452 \text{ }^\circ\text{C}$.

4.19 Field-assisted emission in MOS device Metal-oxide-semiconductor (MOS) transistors in microelectronics have metal gate on an SiO_2 insulating layer on the surface of doped Si crystal. Consider this as a parallel plate capacitor. Suppose the gate is an Al electrode of area $50 \mu\text{m} \times 50 \mu\text{m}$ and has a voltage of 10 V with respect of the Si crystal. Consider two thicknesses for the SiO_2 , (a) 100 Å and (b) 40 Å, where ($1 \text{ Å} = 10^{-10} \text{ m}$). The work function of Al is 4.2 eV, but this refers to electron emission into vacuum, whereas in this case, the electron is emitted into the oxide. The potential energy barrier Φ_B between Al and SiO_2 is about 3.1 eV, and the field emission current density is given by Equation 4.46a and b. Calculate the field emission current for the two cases. For simplicity take m_e to be the electron mass in free space. What is your conclusion?

Solution

We can begin the calculation of field emission current finding the values of the field independent

constants E_c , $B = \frac{e^3}{8\pi\hbar\Phi_B}$ and the area A of the Al electrode.

Thus,

$$E_c = \frac{8\pi(2m_e\Phi_B^3)^{1/2}}{3eh} = \frac{8\pi[2(9.1 \times 10^{-31} \text{ kg})(3.1 \times 1.602 \times 10^{-19} \text{ J})^3]^{1/2}}{3(1.602 \times 10^{-19} \text{ C})(6.626 \times 10^{-34} \text{ J s})} = 3.726 \times 10^{10} \text{ V m}^{-1}$$

$$B = \frac{e^3}{8\pi h \Phi_B} = \frac{(1.602 \times 10^{-19} \text{ C})^3}{8\pi(6.626 \times 10^{-34} \text{ J s})(3.1 \times 1.602 \times 10^{-19} \text{ J})} = 4.971 \times 10^{-7} \text{ A V}^{-2}$$

$$A = (50 \times 10^{-6} \text{ m}) \times (50 \times 10^{-6} \text{ m}) = 2.5 \times 10^{-9} \text{ m}^2$$

When the thickness of SiO₂ layer d is 100 Å, the field in the MOS device is

$$E = \frac{10 \text{ V}}{100 \times 10^{-10} \text{ m}} = 1 \times 10^9 \text{ V m}^{-1}$$

and the field emission current is

$$\begin{aligned} I &= AJ_{\text{field-emission}} = ABE^2 \exp\left(-\frac{E_c}{E}\right) \\ &= (2.5 \times 10^{-9} \text{ m}^2)(4.971 \times 10^{-7} \text{ A V}^{-2})(1 \times 10^9 \text{ V m}^{-1})^2 \exp\left[-\frac{(3.726 \times 10^{10} \text{ V m}^{-1})}{(1 \times 10^9 \text{ V m}^{-1})}\right] \\ &= 8.18 \times 10^{-14} \text{ A.} \end{aligned}$$

In the second case the SiO₂ layer is 2.5 times thinner (40 Å) and the field in the device is 2.5 times stronger.

$$E = \frac{10 \text{ V}}{40 \times 10^{-10} \text{ m}} = 2.5 \times 10^9 \text{ V m}^{-1}$$

The current in this case is

$$\begin{aligned} I &= AJ_{\text{field-emission}} = ABE^2 \exp\left(-\frac{E_c}{E}\right) = \\ &= (2.5 \times 10^{-9} \text{ m}^2)(4.971 \times 10^{-7} \text{ A V}^{-2})(2.5 \times 10^9 \text{ V m}^{-1})^2 \exp\left[-\frac{(3.726 \times 10^{10} \text{ V m}^{-1})}{(2.5 \times 10^9 \text{ V m}^{-1})}\right] \\ &= 2.62 \times 10^{-3} \text{ A.} \end{aligned}$$

So, as predicted by equation 4.47, the field-assisted emission current is a very strong function of the electric field.

4.20 CNTs and field emission The electric field at the tip of a sharp emitter is much greater than the “applied field,” E_o . The applied field is simply defined as V_G/d where d is the distance from the cathode tip to the gate or the grid; it represents the average nearly uniform field that would exist if the tip were replaced by a flat surface so that the cathode and the gate would almost constitute a parallel plate capacitor. The tip experiences an effective field E that is much greater than E_o , which is expressed by a **field enhancement factor** β that depends on the geometry of the cathode–gate emitter, and the shape of the emitter; $E = \beta E_o$. Further, we can take $\Phi_{\text{eff}}^2 \approx \Phi^{3/2}$ in Equation 4.46. The final expression for the field-emission current density then becomes

$$J = \frac{1.5 \times 10^{-6}}{\Phi} \beta^2 E_o^2 \exp\left(\frac{10.4}{\Phi^{1/2}}\right) \exp\left(-\frac{6.44 \times 10^7 \Phi^{3/2}}{\beta E_o}\right) \quad [4.85]$$

Where J is in A cm^{-2} , E_o is in V cm^{-1} , and Φ is in eV. For a particular CNT emitter, $\Phi = 4.9$ eV. Estimate the applied field required to achieve a field-emission current density of 100 mA cm^{-2} in the absence of field enhancement ($\beta = 1$) and with a field enhancement of $\beta = 800$ (typical value for a CNT emitter).

Solution

In the absence of field enhancement, $\beta = 1$ and current density, $J = 100 \text{ mA cm}^{-2} = 0.1 \text{ A cm}^{-2}$ and $\Phi = 4.9$ eV

Substituting in Equation 4.85, we get

$$0.1 = \frac{1.5 \times 10^{-6}}{4.9} E_o^2 \exp\left(\frac{10.4}{4.9^{1/2}}\right) \exp\left(-\frac{6.44 \times 10^7 \times 4.9^{3/2}}{E_o}\right)$$

or,

$$2.976 \times 10^{-9} = E_o^2 \exp\left(-\frac{69.85 \times 10^7}{E_o}\right)$$

Solving this equation we get, $E_o = 13.32 \times 10^6 \text{ V cm}^{-1}$ (or $13.32 \text{ V}/\mu\text{m}$)

In the presence of field enhancement, $\beta = 800$

$$\therefore 0.1 = \frac{1.5 \times 10^{-6}}{4.9} \times 800^2 \times E_o^2 \exp\left(\frac{10.4}{4.9^{1/2}}\right) \exp\left(-\frac{6.44 \times 10^7 \times 4.9^{3/2}}{800 E_o}\right)$$

or,

$$4.65 \times 10^{-15} = E_o^2 \exp\left(-\frac{69.85 \times 10^7}{800 E_o}\right)$$

Solving this equation we get, $E_o = 16.65 \times 10^3 \text{ V cm}^{-1}$ (or $0.016 \text{ V}/\mu\text{m}$)

Author's Note to the Instructor: The term $\exp(10.4/F)$ is usually ignored.

4.21 Nordheim-Fowler field emission in a FED Table 4.10 shows the results of I-V measurements on a Motorola FED microemitter. By a suitable plot show that the I-V follows the Nordheim-Fowler emission characteristics. Can you estimate Φ ?

Table 4.10 Tests on a Motorola FED micro field emitter

V_G (V)	40.0	42	44	46	48	50	52	53.8	56.2	58.2	60.4
I_{emission} (μA)	0.40	2.14	9.40	20.4	34.1	61	93.8	142.5	202	279	367

Solution

The Nordheim-Fowler emission characteristics is $I_A = aV_G^2 \exp\left(-\frac{b}{V_G}\right)$, where a and b are constants.

So a plot of $\ln(I_A/V_G^2)$ versus $1/V_G$ must be a straight line.

V_G	40.0	42	44	46	48	50	52	53.8	56.2	58.2	60.4
I_{emission}	0.40	2.14	9.40	20.4	34.1	61	93.8	142.5	202	279	367
$\ln(I/V_G^2)$	-8.3	-6.7	-5.3	-4.6	-4.2	-3.7	-3.36	-3.0	-2.75	-2.5	-2.3
$1/V_G$	0.025	0.024	0.023	0.022	0.021	0.02	0.019	0.0185	0.018	0.017	0.0165

Table 4Q21-1

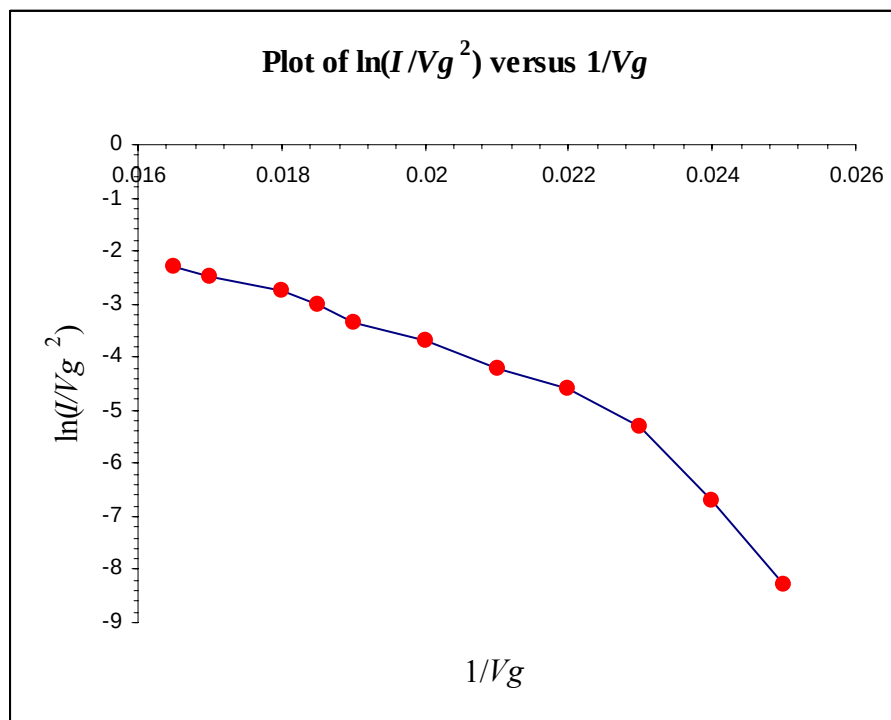


Figure 4Q21-1: Plot of $\ln(I/Vg^2)$ versus $1/Vg$

Note: Nordheim-Fowler field emission is applicable only at high fields so we have to neglect the three points in the low field region.

Φ cannot be calculated. Since we do not have the distance or the electrode separation to calculate electric field.

4.22 Lattice waves and heat capacity

- a.* Consider an aluminum sample. The nearest separation $2R$ ($2 \times$ atomic radius) between the Al-Al atoms in the crystal is 0.286 nm. Taking a to be $2R$, and given the sound velocity in Al as 5100 m s^{-1} , calculate the force constant β in Equation 4.66. Use the group velocity v_g from the actual

dispersion relation, Equation 4.55, to calculate the “sound velocity” at wavelengths of $\Lambda = 1 \text{ mm}$, $1 \mu\text{m}$ and 1 nm . What is your conclusion?

- b. Aluminum has a Debye temperature of 394 K. Calculate its specific heat capacity at 30°C (Darwin, Australia) and at -30°C (January, Resolute Nunavut, Canada).
- c. Calculate the specific heat capacity of a germanium crystal at 25°C and compare it with the experimental value in table 4.5

Solution

a. The group velocity of lattice waves is given by Equation 4.55. For sufficiently small K , or long wavelengths, such that $1/2Ka \rightarrow 0$, the expression for the group velocity can be simplified like in Equation 4.6 to

$$v_g = a \sqrt{\frac{\beta}{M}}$$

From here we can calculate the force constant β

$$\beta = M \left(\frac{v_g}{a} \right)^2$$

The mass of one Al atom is

$$M = \frac{M_{at}}{N_A}$$

and finally for the force constant we receive

$$\beta = \frac{M_{at}}{N_A} \left(\frac{v_g}{a} \right)^2 = \left(\frac{27 \times 10^{-3} \text{ kg mol}^{-1}}{6.022 \times 10^{23} \text{ mol}^{-1}} \right) \left(\frac{5100 \text{ m s}^{-1}}{0.286 \times 10^{-9} \text{ m}} \right)^2 = \mathbf{14.26 \text{ kg s}^{-2}}$$

Now considering the dispersion relation $K = \frac{2\pi}{\Lambda}$ and Equation 4.55 we receive

$$v_g(\Lambda) = a \left(\frac{\beta N_A}{M_{at}} \right)^{\frac{1}{2}} \cos \left(\frac{\pi a}{\Lambda} \right)$$

Performing the calculations for the given wavelengths, we receive the following results:

$$v_g(10^{-3} \text{ m}) = \mathbf{5100 \text{ m s}^{-1}}$$

$$v_g(10^{-6} \text{ m}) = \mathbf{5099.998 \text{ m s}^{-1}}$$

$$v_g(10^{-9} \text{ m}) = \mathbf{3176.22 \text{ m s}^{-1}}$$

It is evident that for the first two wavelengths, $1/2Ka \rightarrow 0$ and we can use the approximation in Equation 4.66. For the third wavelength, this is not true and we have to use the exact dispersion relation when calculating the group velocity.

b. In summer, the temperature is given to be $T = 30^\circ\text{C} = 303\text{ K}$ and T/T_D is $303/394 = 0.769$. The molar heat capacity of Al at 30°C is

$$C_m = 0.92 \times (3R) = \mathbf{22.95\text{ J K}^{-1}\text{ mol}^{-1}}$$

The corresponding specific heat capacity is

$$c_s = \frac{C_m}{M_{at}} = \frac{(22.95\text{ J K}^{-1}\text{ mol}^{-1})}{(27\text{ g mol}^{-1})} = \mathbf{0.85\text{ J K}^{-1}\text{ g}^{-1}}$$

At -30°C , $T = 243\text{ K}$ and T/T_D is $243/394 = 0.62$.

$$C_m = \mathbf{21.94\text{ J K}^{-1}\text{ mol}^{-1}} \text{ and } c_s = \frac{C_m}{M_{at}} = \frac{(22.40\text{ J K}^{-1}\text{ mol}^{-1})}{(27\text{ g mol}^{-1})} = \mathbf{0.81\text{ J K}^{-1}\text{ g}^{-1}}$$

c. We can find the heat capacity of Ge in the way described in part **b**. Alternatively, we can find C_m performing the integration in Equation 4.64 numerically

$$\begin{aligned} C_m &= 9R \left(\frac{T}{T_D} \right)^3 \int_0^{\frac{T_D}{T}} \frac{x^4 e^x}{(e^x - 1)^2} dx = 3R \left[3 \left(\frac{298}{360} \right)^3 \int_0^{\frac{360}{298}} \frac{x^4 e^x}{(e^x - 1)^2} dx \right] = 3R(0.931) \\ &= \mathbf{23.22\text{ J K}^{-1}\text{ mol}^{-1}} \end{aligned}$$

Thus the specific heat capacity is:

$$c_s = \frac{C_m}{M_{at}} = \frac{(23.22\text{ J K}^{-1}\text{ mol}^{-1})}{(72.59 \times 10^{-3}\text{ kg mol}^{-1})} = \mathbf{319.9\text{ J K}^{-1}\text{ kg}^{-1}}$$

From Table 4.5, the specific heat capacity is $23.38\text{ J K}^{-1}\text{ mol}^{-1}$.

4.23 Specific heat capacity of GaAs and InSb

- a.** The Debye temperature T_D of GaAs is 344 K . Calculate its specific heat capacity at 300 K and at -30°C .
- b.** For InSb, $T_D = 203\text{ K}$. Calculate the room temperature specific heat capacity of InSb and compare it with the value expected from the Dulong-Petit rule ($T > T_D$).

Solution

a. $T = 300\text{ K}$, $T_D = 344\text{ K}$

$$\therefore (T/T_D) = 0.87$$

From Figure 4.45, the molar heat capacity, $C_m = 23.1\text{ J K}^{-1}\text{ mol}^{-1}$

The specific heat capacity c_s from the Debye curve is

$$c_s = \frac{C_m}{M_{at}} \approx \frac{23.1\text{ J K}^{-1}\text{ mol}^{-1}}{72.3\text{ g mol}^{-1}} = \mathbf{0.32\text{ J K}^{-1}\text{ g}^{-1}}$$

At -30°C , $T = 243\text{ K}$,

$$\therefore (T / T_D) = 0.706$$

From Figure 4.45, at -30°C the molar heat capacity, $C_m = 22.7 \text{ J K}^{-1} \text{ mol}^{-1}$

The specific heat capacity c_s from the Debye curve is

$$c_s = \frac{C_m}{M_{at}} \approx \frac{22.7 \text{ J K}^{-1} \text{ mol}^{-1}}{72.3 \text{ g mol}^{-1}} = 0.31 \text{ J K}^{-1} \text{ g}^{-1}$$

$$\text{b. } T = 300 \text{ K}, T_D = 203 \text{ K}$$

$$\therefore (T / T_D) = 1.478$$

$$\begin{aligned} \therefore C_m &= 9R \left(\frac{T}{T_D} \right)^3 \int_0^{\frac{T_D}{T}} \frac{x^4 e^x}{(e^x - 1)^2} dx = 9R(1.478)^3 \int_0^{0.677} \frac{x^4 e^x}{(e^x - 1)^2} dx \\ &= 24.37 \text{ J K}^{-1} \text{ mol}^{-1} = 0.977 (3R) \end{aligned}$$

The calculated value is close to $3R$ (from Dulong-Petit rule).

4.24 Thermal conductivity

- Given that silicon has Young's modulus of about 110 GPa and a density of 2.3 g cm^{-3} , calculate the mean free path of phonons in Si at room temperature.
- Diamond has the same crystal structure as Si but has a very large thermal conductivity, about $1000 \text{ W m}^{-1} \text{ K}^{-1}$ at room temperature. Given that diamond has a specific heat capacity c_s of $0.50 \text{ J K}^{-1} \text{ g}^{-1}$, Young's modulus of 830 GPa, and density ρ of 0.35 g cm^{-3} , calculate the mean free path of phonons in diamond.
- GaAs has a thermal conductivity of $200 \text{ W m}^{-1} \text{ K}^{-1}$ at 100 K and $80 \text{ W m}^{-1} \text{ K}^{-1}$ at 200 K. Calculate its thermal conductivity at 25°C and compare with the experimental value of $44 \text{ W m}^{-1} \text{ K}^{-1}$. (Hint: Take $\kappa \propto T^n$ in the temperature region of interest; see Figure 4.48)

Solution

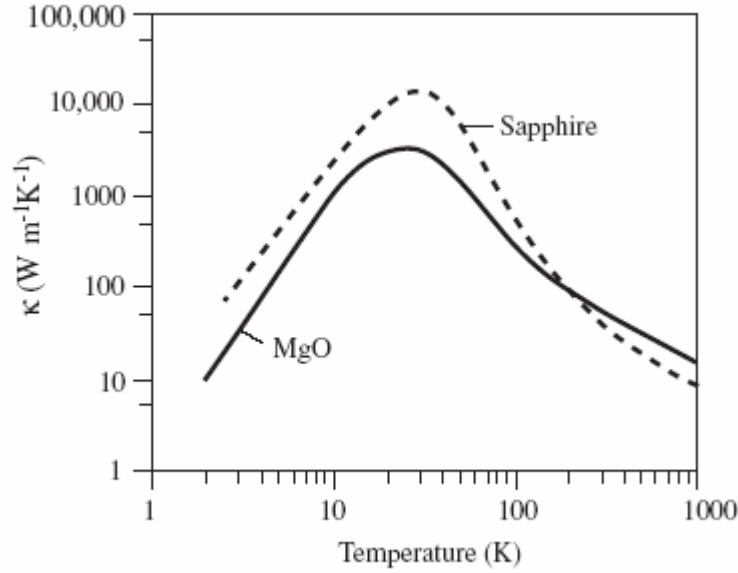


Figure 4.48: Thermal conductivity of sapphire and MgO as a function of temperature.

a. Assume room temperature of 25 °C (298 K). For this temperature from Table 4.5, we can find the thermal conductivity κ ($\kappa = 148 \text{ W m}^{-1} \text{ K}^{-1}$) for silicon and its specific heat capacity C_s ($C_s = 0.703 \text{ J K}^{-1} \text{ g}^{-1}$). We can calculate the phonon mean free path at this temperature from Equation 4.68,

$$\ell_{ph} = \frac{3\kappa}{C_V v_{ph}}$$

where C_V is the heat capacity per unit volume. C_V can be found from the specific heat capacity $C_s = \rho C_s$ and the phonon velocity can be obtained from Equation 4.67, $v_{ph} \approx \sqrt{\frac{Y}{\rho}}$.

Thus the phonon mean free path in Si at 25 °C is

$$\begin{aligned} \ell_{ph} &= \frac{3\kappa}{\rho C_s} \sqrt{\frac{\rho}{Y}} = \frac{3\kappa}{C_s \sqrt{\rho Y}} = \\ &= \frac{3(148 \text{ W m}^{-1} \text{ K}^{-1})}{(0.703 \times 10^3 \text{ J K}^{-1} \text{ kg}^{-1}) \sqrt{(2.3 \times 10^3 \text{ kg m}^{-3})(110 \times 10^9 \text{ Pa})}} = 3.971 \times 10^{-8} \text{ m} \end{aligned}$$

b. The mean free path of phonons in diamond is

$$\begin{aligned} \ell_{ph} &= \frac{3\kappa}{C_s \sqrt{\rho Y}} = \frac{3(1000 \text{ W m}^{-1} \text{ K}^{-1})}{(0.5 \times 10^3 \text{ J K}^{-1} \text{ kg}^{-1}) \sqrt{(3.5 \times 10^3 \text{ kg m}^{-3})(830 \times 10^9 \text{ Pa})}} \\ &= 1.113 \times 10^{-7} \text{ m} \end{aligned}$$

c. The temperatures at which the thermal conductivity κ is given can be considered as relatively high. For this temperature range, we can assume that C_V is almost constant and since the phonon velocity is approximately independent from temperature according to Equation 4.68 the thermal conductivity is proportional to the mean free path of phonons ℓ_{ph} . Since the phonon concentration increases with

temperature, $n_{ph} \propto T$, the mean free path decreases as $\ell_{ph} \propto \frac{1}{T}$. Thus, κ decreases in the same manner with temperature as in Figure 4.48.

We can assume that the temperature dependence of the thermal conductivity is given by:

$$\kappa(T) = \frac{A}{T} + B$$

Then we have two equations and two unknowns

$$200 = \frac{A}{100} + B$$

$$80 = \frac{B}{200} + B$$

and for the coefficients A and B we receive: $A = 2.4 \times 10^4 \text{ W m}^{-1}$ and $B = -40 \text{ W m}^{-1} \text{ K}^{-1}$

The thermal conductivity at 25 °C (298 K) is

$$\kappa = \frac{2.4 \times 10^4 \text{ W m}^{-1}}{298 \text{ K}} - 40 \text{ W m}^{-1} \text{ K}^{-1} = 40.5 \text{ W m}^{-1} \text{ K}^{-1}$$

which underestimates the experimental value of $44 \text{ W m}^{-1} \text{ K}^{-1}$ but, nonetheless, still close.

Alternatively, we can take $\kappa = AT^{-n}$. Then

$$200 = A(100)^{-n} \quad \text{and} \quad 80 = A(200)^{-n}$$

solving we find, $A = 8.8 \times 10^4$ and $n = 1.32$. Thus at $T = 25 + 273 \text{ K}$,

$$\kappa = AT^{-n} = (8.8 \times 10^4)(298)^{-1.32} = 47.2 \text{ W m}^{-1} \text{ K}^{-1}$$

which overestimates the experimental value of $44 \text{ W m}^{-1} \text{ K}^{-1}$ but, nonetheless, still close.

***4.25 Overlapping bands** Consider Cu and Ni with their density of states as schematically sketched in Figure 4.61. Both have overlapping $3d$ and $4s$ bands, but the $3d$ band is very narrow compared to the $4s$ band. In the case of Cu the band is full, whereas in Ni, it is only partially filled.

- In Cu, do the electrons in the $3d$ band contribute to electrical conduction? Explain.
- In Ni, do electrons in both bands contribute to conduction? Explain.
- Do electrons have the same effective mass in the two bands? Explain.
- Can an electron in the $4s$ band with energy around E_F become scattered into the $3d$ band as a result of a scattering process? Consider both metals.
- Scattering of electrons from the $4s$ band to the $3d$ band and vice versa can be viewed as an additional scattering process. How would you expect the resistivity of Ni to compare with that of Cu, even though Ni has two valence electrons and nearly the same density as Cu? In which case would you expect a stronger temperature dependence for the resistivity?

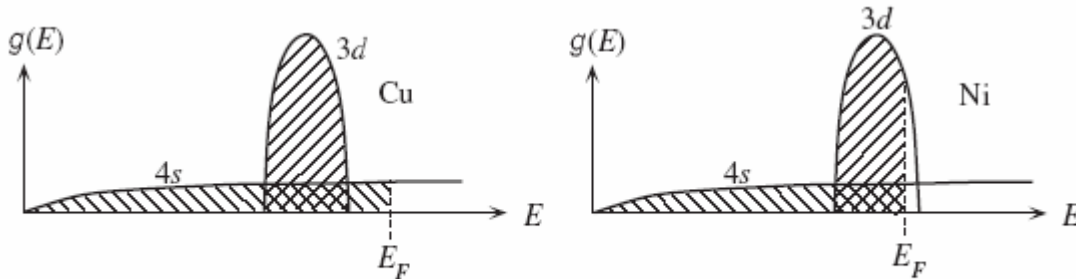


Figure 4.61 Density of states and electron filling in Cu and Ni.

Solution

- a.** In Cu the $3d$ band is full, so the electrons in this band do not contribute to conduction.
- b.** In Ni both the $3d$ and $4s$ bands are partially filled so electrons in both bands can gain energy from the field and move to higher energy levels. Thus both contribute to electrical conductivity.
- c.** No, because the effective mass depends on how easily the electron can gain energy from the field and accelerate or move to higher energy levels. The energy distributions in the two bands are different. In the $4s$ band, the concentration of states is increasing with energy whereas in the $3d$ band, it is decreasing with energy. One would therefore expect different inertial resistances to acceleration, different effective mass and hence different drift mobility for electrons in these bands.
- d.** Not in copper because the $3d$ band is full and cannot take electrons. In Ni the electrons can indeed be scattered from one band to the other, e.g. an electron in the $4s$ band can be scattered into the $3d$ band. Its mobility will then change. Electrons in the $3d$ band are very sluggish (low drift mobility) and contribute less to the conductivity.
- e.** Ni should be more resistive because of the additional scattering mechanism from the $4s$ to the $3d$ band (Matthiessen's rule). This scattering is called s - d scattering. One may at first think that this s - d scattering de-emphasizes the importance of scattering from lattice vibrations and hence, overall, the resistivity should be less temperature dependent. In reality, electrons in Ni also get scattered by magnetic interactions with Ni ion magnetic moments (Nickel is ferromagnetic; Ch. 8 in the textbook) which has a stronger temperature dependence than $\rho \sim T$.

***4.26 Overlapping bands at E_F and higher resistivity** Figure 4.61 shows the density of states for Cu (or Ag) and Ni (or Pd). The d -band in Cu is filled and only electrons at E_F in the s band make a contribution to the conductivity. In Ni, on the other hand, there are electrons at E_F both in the s and d bands. The d band is narrow compared with the s band, and the electron's effective mass in this d band is large; for simplicity, we will assume m_e^* is "infinite" in this band. Consequently, the d -band electrons cannot be accelerated by the field (infinite m_e^*), have a negligible drift mobility and make no contribution to the conductivity. Electrons in the s band can become scattered by phonons into the d band, and hence become relatively immobile until they are scattered back into the s -band when they can drift again. Consider Ni, and one particular conduction electron at E_F starting in the s band. Sketch schematically the magnitude of the velocity gained $|v_x - u_x|$ from the field E_x as a function of time for 10 scattering events; v_x and u_x are the instantaneous and initial velocities, and $|v_x - u_x|$ increases linearly with time, as the electron accelerates in the s band, and then drops to zero upon scattering. If

τ_{ss} is the mean time for s to s -band scattering, τ_{sd} is for s -band to d -band scattering, τ_{ds} is for d -band to s -band scattering, assume the following sequence of 10 events in your sketch: τ_{ss} , τ_{ss} , τ_{sd} , τ_{ds} , τ_{ss} , τ_{sd} , τ_{ds} , τ_{ss} , τ_{sd} , τ_{ds} . What would a similar sketch look like for Cu? Suppose that we wish to apply Equation 4.27. What does $g(E_F)$ and τ represent? What is the most important factor that makes Ni more resistive than Cu? Consider Matthiessen's rule. (Note: There are also electron spin related effects on the resistivity of Ni, but for simplicity these have been neglected.)

Solution

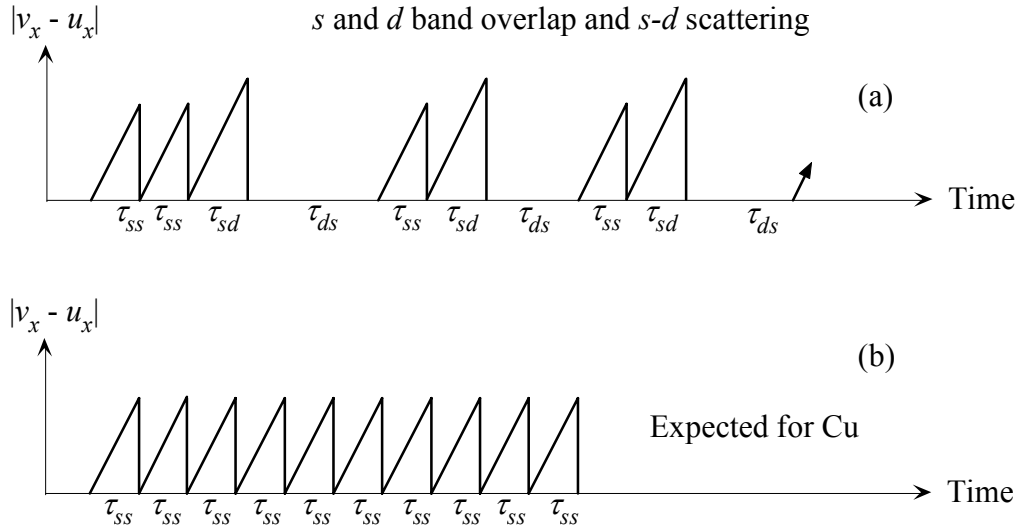


Figure 4Q26-1: (a) Upper for Ni. (b) Lower for Cu

τ in Equation 4.27 represents the scattering time averaged for all possible scattering processes, and $g(E_F)$ is the total density of states at E_F . In the presence of multiple types of scattering, we have to apply the Matthiessen rule to find τ . For example in the presence of two bands, s and d bands,

$$\frac{1}{\tau} = \frac{1}{\tau_{ss}} + \frac{1}{\tau_{sd}} + \frac{1}{\tau_{dd}} + \frac{1}{\tau_{ds}}$$

where τ_{ss} is the scattering time in the s -band, τ_{sd} is the scattering time from the s -band to the d -band, τ_{dd} is the scattering time in the d -band, τ_{ds} is the scattering time from the d -band to the s -band. Thus, the overall τ will be shorter than the usual mean free time τ_{ss} in a single s -band conduction. On the other hand, for Cu, there is only one s -band for conduction, and the mean free time for scattering events is for a single type of scattering process within the s -band (in Figure 4Q26-1b). An intuitive way to look at conduction in Ni is that, time to time, a conduction electron in the s -band gets scattered into the d -band where it is immobile. During this time it does not gain velocity from the field (the effective mass in the d -band is very large), and hence, its overall average velocity is less than it would have been had it stayed in the s -band all the time.

The most important factor is the fact that the conduction electrons in Ni can be scattered into a band where their drift mobility is very low, and this *additional* scattering mechanism makes Ni more resistive than Cu.

4.27 Grüneisen's law Al and Cu both have metallic bonding and the same crystal structure. Assuming that the Grüneisen's parameter γ for Al is the same as that for Cu, $\gamma = 0.23$, estimate the linear expansion coefficient λ of Al, given that its bulk modulus $K = 75 \text{ GPa}$, $c_s = 900 \text{ J K}^{-1} \text{ kg}^{-1}$, and $\rho = 2.7 \text{ g cm}^{-3}$. Compare your estimate with the experimental value of $23.5 \times 10^{-6} \text{ K}^{-1}$.

Solution

Given that, $K = 75 \text{ GPa} = 75 \times 10^9 \text{ J m}^{-3}$

$$c_s = 900 \text{ J K}^{-1} \text{ kg}^{-1}$$

and $\rho = 2.7 \text{ g cm}^{-3} = 2.7 \times 10^3 \text{ kg m}^{-3}$

$$\text{Now } \lambda = 3\gamma \frac{\rho c_s}{K} = 3(0.23) \frac{(2.7 \times 10^3 \text{ kg m}^{-3})(900 \text{ J K}^{-1} \text{ kg}^{-1})}{(75 \times 10^9 \text{ J m}^{-3})} = 22.36 \times 10^{-6} \text{ K}^{-1}$$

So the estimated value is close to the experimental value.

"After a year's research, one realises that it could have been done in a week."

Sir William Henry Bragg